LOW-NOISE TECHNIQUES

CHAPTER 8

In many applications you're dealing with small signals, for which it is essential to minimize the degrading effects of amplifier "noise." Low-noise design is thus an important part of the art of electronics. The extensive detail (decorated with more than our usually paltry quota of equations!) in this chapter reflects the richness of the low-noise design. So does its length – it's the longest chapter in the book. Recognizing that many readers will have a less-than-passionate interest in the various subjects treated here, we offer the following guide.

A quick guide to this chapter The basics of noise are explained in §8.1 ("Noise"), which should be read first. Readers interested primarily in low-noise design with *op-amps* can then skip ahead to the discussion, tables, and graphs in §8.9 ("Noise in operational amplifier circuits"). Those interested in low-noise design with discrete transistors (or interested in gaining a fuller understanding of what's going on "under the hood" of op-amps) should read §8.5 ("Low-noise design with bipolar transistors") and §8.6 ("Low-noise design with JFETs"). Readers working with photodiode circuits and the like will want to read §8.11 ("Noise in transimpedance amplifiers").[1] For a discussion of noise *measurement*, go to §8.12 ("Noise measurements and noise sources") and §8.13 ("Bandwidth Limiting and RMS Voltage Measurement").

An even quicker guide to noise This chapter is lengthy, and it's filled with mathematical details and information about hundreds of transistors and op-amps. But noise doesn't have to be complicated. In a spirit of encapsulating the essence of noise, we offer a one-minute breathless end-of-class "takeaway":

> The random noise you care about is characterized by its *density* (rms noise amplitude in a 1 Hz band of frequency); voltage noise density is called e_n, and has units like $nV/\sqrt{Hz}$. Likewise, the symbol for noise *current* is i_n; a noise current at an amplifier's input flows through the signal's source resistance, creating its own noise voltage $e_n = i_n R_s$. If a noise source is uniform over frequency, it's called "white noise," and the rms noise voltage contained within a bandwidth B is just $v_n = e_n\sqrt{B}$. Knowing that, you can go to Tables 8.3a–8.3c on page 522, which lists e_n and i_n for a wide selection of op-amps, to figure out how much noise is added in an amplifier stage. Multiply by the amplifier's gain and, voilà, you've got the output noise.
>
> Amplifiers aren't the only source of noise. A resistor generates "Johnson noise", eq'n 8.4, and the discrete charges in a flow of current generate "shot noise", eq'n 8.6. Both of these are white noise.[2] Finally (with ten seconds remaining) – to figure the total noise in a circuit with multiple independent noise sources, you take the sum of the squares of each noise density, multiply by the bandwidth, then take the square root. Time's up. End of class.

8.1 "Noise"

In almost every area of measurement the ultimate limit of detectability of weak signals is set by noise – unwanted signals that obscure the desired signal. Even if the quantity being measured is not weak, the presence of noise degrades the accuracy of the measurement. Some forms of noise are unavoidable (e.g., real fluctuations in the quantity being measured), and they can be overcome only with the techniques of *signal averaging* and *bandwidth narrowing*.[3] Other forms of noise (e.g., radiofrequency interference and "ground loops") can be reduced or eliminated by a variety of tricks, including filtering and careful attention to wiring configuration and parts location. Finally, there is noise that arises in the amplification process itself, and it can be reduced through the techniques of low-noise amplifier design. Although the techniques of signal averaging

[1] And also the related discussion of stability and bandwidth in §*4x.3*.

[2] Things get more interesting when noise density varies with frequency, for example the notorious pink "flicker noise" that rises as $e_n \propto 1/\sqrt{f}$ at low frequencies. This fascinating (and infuriating) annoyance does not go unnoticed in this chapter!

[3] See §8.14, and also Chapter 15 of this book's second edition (1989).

can often be used to rescue a signal buried in noise, it always pays to begin with a system that is free of preventable interference and that possesses the lowest amplifier noise practicable.

We begin by talking about the origins and characteristics of the different kinds of noise that afflict electronic circuits. Then we launch into a discussion of bipolar-transistor (BJT) and field-effect transistor (FET) noise, including methods for low-noise design with a given signal source, and we present some design examples. After a short discussion of noise in differential and feedback amplifiers, we continue with low-noise design with op-amps, including transimpedance (current-to-voltage) amplifiers. Sections on noise measurements, bandwidth limiting, and lock-in detection follow, then a short discussion of power-supply noise. We conclude with a section on proper grounding and shielding and the elimination of interference and pickup.

Because the term *noise* can be applied to anything that obscures a desired signal,[4] noise can take the form of another signal (“interference”); most often, however, we use the term to describe “random” noise of a physical (often thermal) origin. Noise can be characterized by its frequency spectrum, its amplitude distribution, and the physical mechanism responsible for its generation. Let’s next look at the chief offenders:

Johnson noise: Random-noise voltage created by thermal fluctuations in a resistor.

Shot noise: Random statistical fluctuations in a flowing current caused by the discrete nature of electrical charge.

Flicker noise: Additional random noise, rising typically as $1/f$ in power at low frequencies, with a multitude of causes.

Burst noise: low-frequency noise typically seen as random jumps between a pair of levels, caused by material device defects.

8.1.1 Johnson (Nyquist) noise

Any old resistor just sitting on the table generates a noise voltage across its terminals known as Johnson noise (or Nyquist noise).[5] It has a flat frequency spectrum, meaning that there is the same noise power in each hertz of frequency (up to some limit, of course). Noise with a flat spectrum is also called “white noise.” The actual open-circuit noise voltage generated by a resistance R at temperature T is given by

$$v_{\text{noise}}(\text{rms}) = v_{\text{n}} = (4kTRB)^{\frac{1}{2}} \quad \text{V(rms)}, \tag{8.1}$$

where k is Boltzmann’s constant, T is the absolute temperature in Kelvins (K=°C + 273.16), and B is the bandwidth in hertz. Thus v_{noise}(rms) is what you would measure at the output if you drove a perfect noiseless bandpass filter (of bandwidth B) with the voltage generated by a resistor at temperature T. At room temperature (68°F = 20°C = 293K),

$$\begin{aligned} 4kT &= 1.62\times10^{-20} && \text{V}^2/\text{Hz}-\Omega, \\ (4kTR)^{\frac{1}{2}} &= 1.27\times10^{-10}R^{\frac{1}{2}} && \text{V}/\text{Hz}^{\frac{1}{2}} \\ &= 1.27\times10^{-4}R^{\frac{1}{2}} && \mu\text{V}/\text{Hz}^{\frac{1}{2}}. \end{aligned} \tag{8.2}$$

For example, a 10k resistor at room temperature has an open-circuit rms voltage of 1.3μV, measured with a bandwidth of 10 kHz (e.g., by placing it across the input of a good audio amplifier and measuring the output with a voltmeter). The source resistance of this noise voltage is just R. If you connect the terminals of the resistor together, you get a (short-circuit) current of

$$i_{\text{noise}}(\text{rms}) = v_{\text{noise}}(\text{rms})/R = v_{\text{nR}}/R = (4kTB/R)^{\frac{1}{2}}. \tag{8.3}$$

As we’ll see in §8.2.1, it’s convenient to express noise voltage (or current) as a *density* e_{n} (rms voltage per square root bandwidth). Johnson noise, with its flat (white) spectrum, has constant noise voltage-density

$$e_{\text{n}}=\sqrt{4kTR} \quad \text{V}/\text{Hz}^{\frac{1}{2}}, \tag{8.4}$$

from which the rms noise voltage in some limited bandwidth B is then simply $v_{\text{n}}=e_{\text{n}}\sqrt{B}$. Likewise, the short-circuit noise-current density is

$$i_{\text{n}}=\sqrt{4kT/R} \quad \text{A}/\text{Hz}^{\frac{1}{2}}. \tag{8.5}$$

Figure 8.1 plots the simple relationship between Johnson-noise voltage density and source resistance; also shown is the short-circuit noise current density. An easy number to remember, when choosing resistor values for low-noise amplifier designs, is that a 1kΩ resistor at room temperature generates an open-circuit noise voltage density of $4\,\text{nV}/\sqrt{\text{Hz}}$; scale by the square root of resistance for other values.[6]

[4] As Lew Branscomb famously noted, “Nature does not ‘know’ what experiment a scientist is trying to do. God loves the noise as much as the signal.” See L. Branscomb, “Integrity in Science,” *Am. Sci.* **73**, 421–23 (1985).

[5] Experiment and formulas by J. B. Johnson, Letter to *Nature*, **119**, 50 (1927), *Phys. Rev.* **32**, “Thermal agitation of electricity in conductors,” 97–109, (1928), subsequent theory by H. Nyquist, *Phys. Rev.*, “Thermal agitation of electric charge in conductors,” **32**, 110–113, (1928).

[6] We find it handy to remember the values of q and of $4kT$ (which keeps

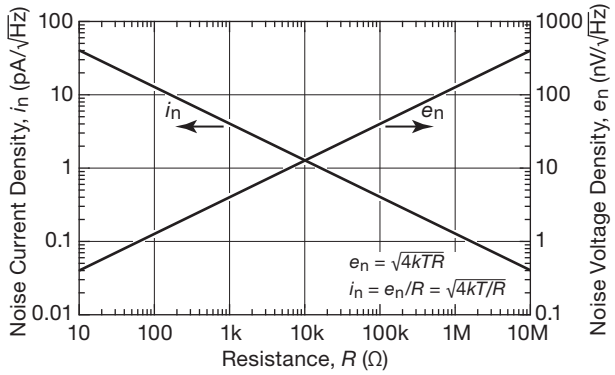


Figure 8.1. Open-circuit thermal noise-voltage and short-circuit thermal noise-current densities versus resistance at 25°C.

Here's a handy Johnson-noise mini-table, listing both voltage and current noise *densities* (units of $V/\sqrt{Hz}$ and $A/\sqrt{Hz}$), and noise within a 10 kHz band, for seven decade-related values of resistance:

	Johnson noise, at T=25°C			
	open circuit		short circuit	
R	e_n ($nV/\sqrt{Hz}$)	$e_n\sqrt{B}$ B=10 kHz (μV)	i_n ($pA/\sqrt{Hz}$)	$i_n\sqrt{B}$ B=10 kHz (pA)
100 Ω	1.28	0.128	12.8	1280
1k	4.06	0.406	4.06	406
10k	12.8	1.28	1.28	128
100k	40.6	4.06	0.406	40.6
1M	128	12.8	0.128	12.8
10M	406	40.6	0.041	4.06
100M	1280	128	0.0128	1.28

The amplitude of the Johnson-noise voltage at any instant is, in general, unpredictable, but it obeys a Gaussian amplitude distribution (Figure 8.2), where $p(V)dV$ is the probability that the instantaneous voltage lies between V and $V+dV$, and $v_n(\text{rms})$ is the rms noise voltage, given earlier.[7]

The significance of Johnson noise is that it sets a lower limit on the noise voltage in any detector, signal source, or amplifier having resistance. The resistive part of a source impedance generates Johnson noise, as do the bias and load resistors of an amplifier. You will see how it all works out presently.

popping up) together, because in SI units they are 1.6×10^{-19} and 1.6×10^{-20}, respectively.

[7] See also Figure 8.115, which plots the odds (over 9 decades) that the instantaneous amplitude exceeds some multiple of the rms amplitude.

It is interesting to note that the physical analog of resistance (any mechanism of energy loss in a physical system, e.g., viscous friction acting on small particles in a liquid) has associated with it fluctuations in the associated physical quantity (in this case, the particles' velocity, manifest as the chaotic Brownian motion). Johnson noise is just a special case of this fluctuation–dissipation phenomenon.

Johnson noise should not be confused with the additional noise voltage created by the effect of resistance fluctuations when an externally applied current flows through a resistor. This "excess noise" has a $1/f$ spectrum (approximately) and is heavily dependent on the actual construction of the resistor. We will talk about it later.

8.1.2 Shot noise

An electric current is the flow of discrete electric charges, not a smooth fluidlike flow. The finiteness of the charge quantum results in statistical fluctuations of the current. If the charges act independently of each other, the fluctuating current's noise density is given by

$$i_n = \sqrt{2qI_{dc}} \quad A/Hz^{\frac{1}{2}}, \tag{8.6}$$

where q is the electron charge (1.60×10^{-19} coulomb). This noise, like resistor Johnson noise, is white and Gaussian. So its amplitude, taken over a measurement bandwidth B, is just

$$i_{noise}(\text{rms}) = i_{nR}(\text{rms}) = i_n\sqrt{B} = (2qI_{dc}B)^{\frac{1}{2}} \quad A(\text{rms}). \tag{8.7}$$

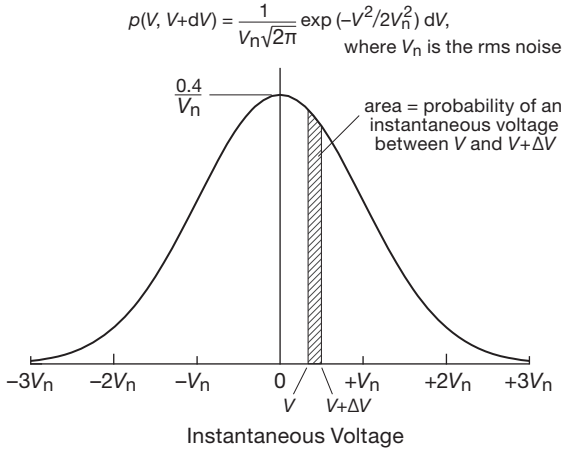


Figure 8.2. Johnson noise obeys a Gaussian distribution of amplitudes. The normalizing factor $0.4/V_n$ ensures dimensionless unit area under the bell curve (the "0.4" is actually $1/\sqrt{2\pi}$, about 0.3989).

For example, a “steady” current of 1 A actually has an rms fluctuation of 57 nA, measured in a 10 kHz bandwidth; i.e., it fluctuates by about 0.000006%. The relative fluctuations are larger for smaller currents: a “steady” current of 1 μA actually has an rms current-noise fluctuation, measured over a 10 kHz bandwidth, of 0.006%, i.e., −85 dB. At 1 pA dc, the rms current fluctuation (same bandwidth) is 57 fA, i.e., a 5.7% variation! Shot noise is “rain on a tin roof.”

Here’s a handy minitable listing shot-noise current density and shot noise current in a 10 kHz band for decadal currents spanning 12 orders of magnitude:

	Shot noise current $B = 10$ kHz		
I_{dc}	i_n	$i_n\sqrt{B}$ (10 kHz)	$\frac{i_n\sqrt{B}}{I_{dc}}$
1 fA	18 aA/$\sqrt{\text{Hz}}$	1.8 fA	+5 dB
1 pA	0.57 fA/$\sqrt{\text{Hz}}$	57 fA	−25 dB
1 nA	18 fA/$\sqrt{\text{Hz}}$	1.8 pA	−55 dB
1 μA	0.57 pA/$\sqrt{\text{Hz}}$	57 pA	−85 dB
1 mA	18 pA/$\sqrt{\text{Hz}}$	1.8 nA	−115 dB

An important point: the shot-noise formula given earlier assumes that the charge carriers making up the current act independently. That is indeed the case for charges crossing a barrier, for example the current in a junction diode, where the charges move by diffusion; but it is not true for the important case of metallic conductors, where there are long-range correlations between charge carriers. Thus the current in a simple resistive circuit has far less noise than is predicted by the shot-noise formula. Another important exception to the shot-noise formula is provided by our standard transistor current-source circuit (Figure 2.32); we discuss this further in §8.3.5.

Exercise 8.1. A resistor is used as the collector load in a low-noise amplifier; the collector current I_C is accompanied by shot noise. Show that the output noise voltage is dominated by shot noise (rather than Johnson noise in the resistor) as long as the quiescent voltage drop across the load resistor is greater than $2kT/q$ (50mV, at room temperature).

8.1.3 1/*f* noise (flicker noise)

Shot noise and Johnson noise are irreducible forms of noise generated according to physical principles. The most expensive and most carefully made resistor has exactly the same Johnson noise as the cheapest carbon resistor of the same resistance. Real devices have, in addition, various sources of “excess noise.” Real resistors suffer from fluctuations in resistance, generating an additional noise voltage (which adds to the ever-present Johnson noise) proportional to the dc current flowing through them. This noise depends on many factors having to do with the construction of the particular resistor, including the resistive material and especially the end-cap connections. Here is a listing of typical excess noise for various resistor types, given as rms microvolts per volt applied across the resistor, measured over one decade of frequency: (eq’n 8.13)

Carbon composition	0.10 μV to 3.0 μV
Carbon film	0.05 μV to 0.3 μV
Metal film	0.02 μV to 0.2 μV
Wire wound	0.01 μV to 0.2 μV

This noise has approximately a $1/f$ power spectrum (equal power per decade of frequency) and is sometimes called “pink noise.” When plotted against voltage or current (rather than power) its *amplitude* falls as $1/\sqrt{f}$, as shown in Figure 8.3. Figure 8.4 shows how it looks in comparison with a sample of white noise and of what’s sometimes called “red noise” ($1/f^2$ power spectrum); if you want to make your own, look ahead to Figure 8.93 to see how.

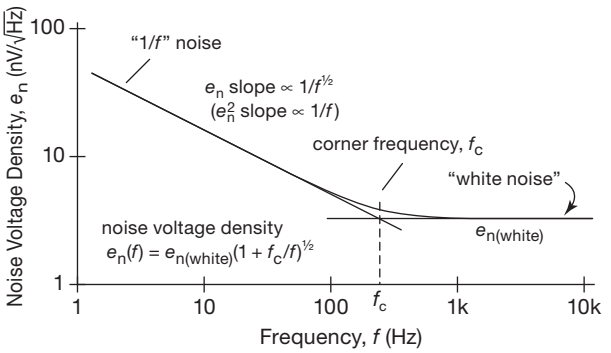


Figure 8.3. When plotted on log axes as noise voltage versus frequency, $1/f$ noise slopes downward with a slope of 1/2, i.e., as $1/f^{\frac{1}{2}}$ (it is noise *power* that goes as $1/f$).

You often see the notation f_c for the corner frequency at which the $1/f$ noise is the same as an underlying white-noise component.[8] The combined noise voltage density is

$$e_n(f) = e_{n(\text{white})}\sqrt{1 + f_c/f}, \tag{8.8}$$

from which the rms integrated noise voltage in a band extending from f_1 to f_2 can be calculated; see eq’n 8.59 on page 565.

[8] You can estimate f_c with the help of eq’n 8.27 on page 491.

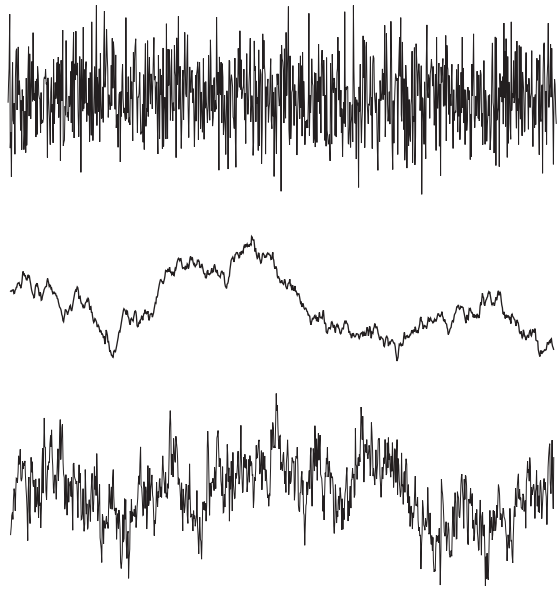


Figure 8.4. Three noises: top, "white noise" (uniform power per Hz); middle, "red noise" (power per Hz proportional to $1/f^2$); bottom, "pink noise" (or $1/f$ noise, power per Hz proportional to $1/f$).

Other noise-generating mechanisms often produce $1/f$ noise, examples being base-current noise in transistors and cathode current noise in vacuum tubes. Curiously enough, $1/f$ noise is present in nature in unexpected places, e.g., the speed of ocean currents, the flow of sand in an hourglass, the flow of traffic on Japanese expressways, and the yearly flow of the Nile measured over the last 2,000 years.[9] If you plot the loudness of a piece of classical music versus time, you get a $1/f$ spectrum! No unifying principle has been found for all the $1/f$ noise that seems to be swirling around us, although particular sources can often be identified in each instance.

8.1.4 Burst noise

Not all noise sources are characterized by a Gaussian (or even *smooth*) distribution of amplitudes. Most notorious among the exceptions is *burst noise* (also variously called *popcorn* noise, *bistable* noise, or *random telegraph signal* noise), seen occasionally in semiconductor devices (particularly in parts dating back to the 1970s and earlier). It consists of random jumps between two (usually) voltage levels, taking place on time scales of tens of milliseconds; when played out on a loudspeaker it sounds like the birth throes of popcorn. Figure 8.5 shows a typical waveform, the output of a vintage[10] 741 op-amp wired as a noninverting amplifier with G=100.

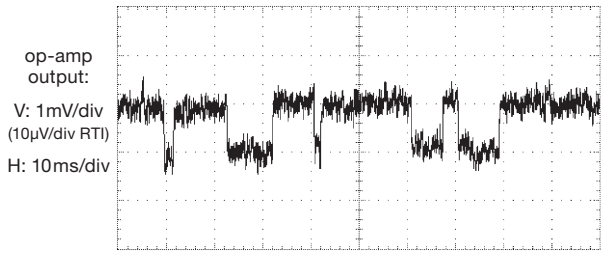


Figure 8.5. Burst noise from a 1973-vintage 741 op-amp, configured as an $\times$100 noninverting amplifier with grounded input. The output was bandpass filtered to 0.1 Hz–3 kHz, with 6 dB/octave rolloffs.

Viewed in the frequency domain, the effect of burst noise is a raised low frequency portion, without any obvious spectral peaks. You can see this in Figure 8.6, where the voltage noise spectrum of the noisy and quiet op-amp specimens are plotted.[11]

8.1.5 Band-limited noise

All circuits operate within some limited frequency band. So, although it's nice to talk about (and calculate with) noise-*density* quantities, what you usually care about is the rms noise voltage contained within some signal band of interest (called B in eq'n 8.1). In many cases you are dealing with a white noise source (e.g., Johnson noise or shot noise). If that is passed through a perfectly sharp bandpass filter (a "brick-wall" filter), the band-limited rms amplitude is simply $v_{n(rms)}=e_n\sqrt{B}$. But analog brick-wall filters aren't practical – so what you want to know is the equivalent bandwidth of a real filter, say a simple *RC* lowpass. It turns out that the equivalent brick-wall bandwidth is given by

$$B = \frac{\pi}{2} f_{3dB} = 1.57 f_{3dB} = \frac{1}{4RC} \quad \text{Hz}, \tag{8.9}$$

[9] A delightful reference is W. H. Press, "Flicker noises in astronomy and elsewhere," *Comm. on Astrophys.* **7**, 103–119 (1978). Available at `http://www.nr.com/whp/Flicker_Noise_1978.pdf`.

[10] Semiconductor manufacturers have worked hard to alleviate this problem (believed to be caused by intermittent trapping of charge carriers at defects and interfaces), and popcorn noise is largely a thing of the past. We tested ten samples of 741s, from six different manufacturers, before we found this one. A second sample from the same manufacturer and with the same date code showed no evidence of popcorn noise, as seen in the pair of spectra of Figure 8.6.

[11] It is possible that a related mechanism is responsible for similarly shaped noise plateaus seen in some JFETs; see for example the LSK389 in Figure 8.47.

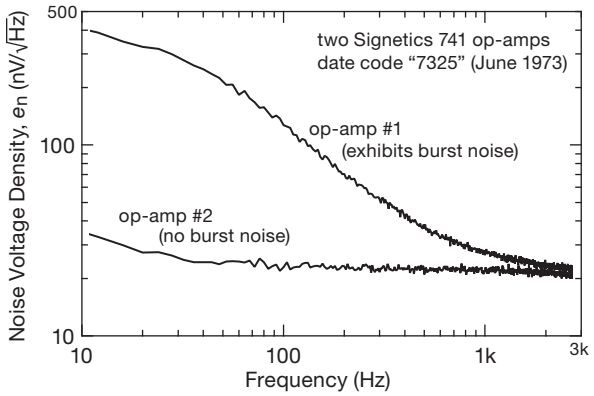


Figure 8.6. Spectrum of the burst noise produced by the same op-amp as used for Figure 8.5, along with that of a second op-amp from the same batch that exhibited no burst noise. The vertical scale shows RTI (referred to the input) rms noise densities.

where $f_{3\text{dB}}=1/2\pi RC$. You can use higher-order filters, of course, for example a 2-pole Butterworth lowpass filter; its equivalent brick-wall bandwidth is $B=1.11f_{3\text{dB}}$. For still higher-order filters (including bandpass) see the expressions in Table 8.4 on page 564. For slowly varying (or dc) signals you can instead do a simple averaging (as, for example, with an integrating ADC, see §13.8.3); in that case the equivalent noise bandwidth is $B=1/2T$, where T is the duration of the (uniform) averaging of the input signal. We’ll have a bit more to say about this in §8.13.1.

Of course, the noise spectrum may be other than white (e.g., it may be $1/f$ noise or a combination of white noise with a $1/f$ rising tail at low frequencies). In such a case you cannot simply multiply the noise density by the square root of the bandwidth. Instead you have to integrate the (changing) noise density over the bandpass. For an ideal brick-wall bandpass that is just $v_\text{n}^2=\int e_\text{n}^2(f)df$ from the lower to upper frequency cutoff of the filter. For a realizable filter you have to integrate the noise density, multiplied by the filter’s spectral response $H(f)$, over the bandpass: $v_\text{n}^2=\int |e_\text{n}(f)H(f)|^2df$. For an arbitrary noise spectrum this is what you would need to do. But life is simpler if you’re dealing with classic noise spectra like $1/f$ flicker noise, in which case the noise integrals can be expressed analytically. We’ve gathered these together in Table 8.4 on page 564, which includes results for white-, pink-, and red-noise spectra, when band limited by brick-wall, single-pole, 2-pole Butterworth, and m-pole Butterworth bandpass filters (response from f_1 to f_2). Those tabulated formulas let you get the results for lowpass ($f_1=0$) or highpass filters ($f_2=\infty$), which are just special cases of the more general bandpass filter.

We discuss this in plenty of detail later in the chapter, at §8.13.

8.1.6 Interference

As we mentioned earlier, an interfering signal or stray pickup constitutes a form of noise. Here the spectrum and amplitude characteristics depend on the interfering signal. For example, 60 Hz powerline pickup has a sharp spectrum and relatively constant amplitude, whereas car ignition noise, lightning, and other impulsive interferences are broad in spectrum and spiky in amplitude. Other sources of interference are radio and television stations (a particularly serious problem near large cities), nearby electrical equipment, motors and elevators, subways, switching regulators, and television sets. Cellphones often dwarf all other sources of RF interference. Even when not in use, the cellphone transmits periodically to tell the cell tower its location, generating interference with a distinctive galloping rhythm.[12] The same goes for mobile computers that use the cellular network for Internet access.

In a slightly different guise you have the same sort of problem generated by anything that puts a signal into the parameter you are measuring. For example, an optical interferometer is susceptible to vibration, and a sensitive RF measurement (e.g., nuclear magnetic resonance, NMR or MRI) can be affected by ambient RF. Many circuits, as well as detectors and even cables, are sensitive to vibration and sound; they are *microphonic*, in the terminology of the trade.

Many of these noise sources can be controlled by careful shielding and filtering, as we discuss later in the chapter. At other times we are forced to take Draconian measures, involving massive stone tables (for vibration isolation), constant-temperature rooms, anechoic chambers, and electrically shielded (“Faraday cage”) rooms.

8.2 Signal-to-noise ratio and noise figure

Before getting into the details of amplifier noise and low-noise design, we need to define a few terms that are often used to describe amplifier performance. These involve ratios of noise voltages, measured at the same place in the circuit. It is conventional to refer noise voltages to the input of an amplifier (although the measurements are usually made at the output), i.e., to describe source noise and amplifier noise in terms of microvolts *at the input* that would

[12] Which you can hear at `www.covingtoninnovations.com/michael/blog/0506/050622-cellnoise.mp3`.

generate the observed output noise. This makes sense when you want to think of the relative noise added by the amplifier to a given signal, independent of amplifier gain; it's also realistic, because most of the amplifier noise is usually contributed by the input stage. Unless we state otherwise, noise voltages are referred to the input (RTI).

8.2.1 Noise power density and bandwidth

In the preceding examples of Johnson noise and shot noise, the noise voltage you measure depends both on the measurement bandwidth B (i.e., how much noise you see depends on how fast you look) and on the variables (R and I) of the noise source itself. So it's convenient to talk about an rms noise-voltage "density" e_n:

$$v_{n(rms)} = e_n B^{\frac{1}{2}} = (4kTR)^{\frac{1}{2}} B^{\frac{1}{2}} \quad \text{Vrms}, \tag{8.10}$$

where v_n is the rms noise voltage you would measure in a bandwidth B. White-noise sources have an e_n that doesn't depend on frequency, whereas pink noise, for instance, has a e_n that drops off at 3 dB/octave. You'll often see e_n^2, too, the mean squared noise density. Since e_n always refers to rms and e_n^2 always refers to mean square, you can just square e_n to get e_n^2! Sounds simple (and it is), but we want to make sure you don't get confused.

Note that B and the square root of B keep popping up. Thus, for example, for Johnson noise from a resistor R,

$$\begin{aligned} e_{nR}(\text{rms}) &= (4kTR)^{\frac{1}{2}} && \text{V/Hz}^{\frac{1}{2}}, \\ e_{nR}^2 &= 4kTR && \text{V}^2/\text{Hz}, \\ v_{n(rms)} &= v_{nR} B^{\frac{1}{2}} &= (4kTRB)^{\frac{1}{2}} & \text{ V}, \\ v_n^2 &= v_{nR}^2 B &= 4kTRB & \text{V}^2. \end{aligned}$$

On datasheets you may see graphs of e_n or e_n^2, with units like "nanovolts per root Hz" or "volts squared per Hz." The quantities e_n and i_n that will soon appear work just the same way.

When you add two signals that are uncorrelated (two noise signals, or noise plus a real signal), you add their noise *power*; that is, their *squared* amplitudes add:

$$v = (v_s^2 + v_n^2)^{\frac{1}{2}},$$

where v is the rms signal obtained by adding together a signal of rms amplitude v_s and a noise signal of rms amplitude v_n. The rms amplitudes[13] *do not* add.

[13] Which, we emphasize, are the convenient and familiar quantites found on datasheets, etc. For example, we're used to thinking of a $3\,\text{nV}/\sqrt{\text{Hz}}$ amplifier as quiet; it's hard to recognize a $0.9\times10^{-17}\,\text{V}^2/\text{Hz}$ amplifier as the same thing.

8.2.2 Signal-to-noise ratio

Signal-to-noise ratio (SNR) is simply defined as

$$\text{SNR} = 10\log_{10}(v_s^2/v_n^2) = 20\log_{10}(v_s/v_n) \quad \text{dB} \tag{8.11}$$

where the voltages are rms values, and some bandwidth and center frequency are specified; i.e., it is the ratio, in decibels, of the rms voltage of the desired signal to the rms voltage of the noise that is also present.[14] The "signal" itself may be sinusoidal, or a modulated information-carrying waveform, or even a noiselike signal itself. It is particularly important to specify the bandwidth if the signal has some sort of narrowband spectrum, because the SNR will decrease as the bandwidth is increased beyond that of the signal: the amplifier keeps adding noise power, while the signal power remains constant.

8.2.3 Noise figure

Any real signal source or measuring device generates noise because of Johnson noise in its source resistance (the real part of its complex source impedance). There may be additional noise, of course, from other causes. The *noise figure* (NF) of an amplifier is simply the ratio, in decibels, of the output of the real amplifier to the output of a "perfect" (noiseless) amplifier of the same gain, with a resistor of value R_s connected across the amplifier's input terminals in each case. That is, the Johnson noise of R_s is the "input signal":

$$\text{NF} = 10\log_{10}\left(\frac{4kTR_s + v_n^2}{4kTR_s}\right) \tag{8.12}$$

$$= 10\log_{10}\left(1 + \frac{v_n^2}{4kTR_s}\right) \quad \text{dB}, \tag{8.13}$$

where v_n^2 is the mean squared noise voltage per hertz contributed by the amplifier, with a noiseless (cold) resistor of value R_s connected across its input. This latter restriction is important, as you will see shortly, because the noise voltage contributed by an amplifier depends very much on the source impedance (Figure 8.7).

Noise figure is handy as a figure of merit for an amplifier when you have a signal source of a given source impedance and want to compare amplifiers (or transistors, for which the NF is often specified). The NF varies with frequency

[14] The expression in terms of squared amplitudes suggests a ratio in terms of *power*, which is the origin of the decibel ratio definition. But the "$20\log_{10}$" form is widely used, even when there is no actual power, for example with an open-circuit load (or, more confusingly, when the result is at odds with the actual power ratio, for example when expressing the ratio of amplitudes created by a signal transformer).

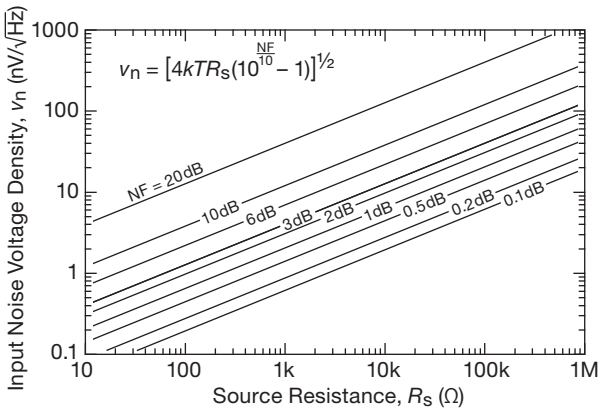


Figure 8.7. Effective input-noise-voltage density versus noise figure and source resistance.

and source impedance, and it is often given as a set of contours of constant NF versus frequency and R_s(we'll see examples later, in Figures 8.22 and 8.27). It may also be given as a set of graphs of NF versus frequency, one curve for each collector current, or a similar set of graphs of NF versus R_s, one for each collector current. *Note*: the foregoing expressions for NF assume that the amplifier's input impedance is much larger than the source impedance, i.e., $Z_{in} \gg R_s$. However, in the special case of RF amplifiers, you usually have $R_s = Z_{in} = 50\,\Omega$, with the NF defined accordingly. For this special case of matched impedances, simply remove the factor "4" from eq'ns 8.12 and 8.13.

Big fallacy: don't try to improve things by adding a resistor in series with a signal source to reach a region of minimum NF. All you're doing is making the source noisier to make the amplifier look better! Noise figure can be very deceptive for this reason. To add to the deception, the NF specification (e.g., NF = 2 dB) for a transistor or FET will always be for the optimum combination of R_s and I_C. It doesn't tell you much about actual performance, except that the manufacturer thinks the noise figure is worth bragging about.

In general, when evaluating the performance of some amplifier, you're probably least likely to get confused if you stick with a SNR calculated for that source voltage and impedance. Here's how to convert from NF to SNR:

$$\mathrm{SNR} = 10\log_{10}\left(\frac{v_s^2}{4kTR_s}\right) - \mathrm{NF(dB)(at}\,R_s) \quad \mathrm{dB}, \qquad (8.14)$$

where v_s is the rms signal amplitude, R_s is the source impedance, and NF is the noise figure of the amplifier for source impedance R_s. See §§8.3.1 and 8.5.6 for some noise figure calculation examples.

8.2.4 Noise temperature

Rather than noise *figure*, you sometimes see noise *temperature* used to express the noise performance of an amplifier. Both methods give the same information, namely the excess noise contribution of the amplifier when driven by a signal source of impedance R_s; they are equivalent ways of expressing the same thing.

Look at Figure 8.8 to see how noise temperature works: we first imagine the actual (noisy) amplifier connected to a *noiseless* source of impedance R_s (Figure 8.8A). If you have trouble imagining a noiseless source, think of a resistor of value R_s cooled to absolute zero. There will be some noise at the output, even though the source is noiseless, because the amplifier has noise. Now imagine constructing Figure 8.8B, where we magically make the amplifier noiseless and bring the source R_s up to some temperature T_n such that *the output noise voltage is the same as in Figure 8.8A*. T_n is called the noise temperature of the amplifier, for source impedance R_s.

As we remarked earlier, noise figure and noise temperature are simply different ways of conveying the same information. In fact, you can show that they are related by the following expressions:

$$T_n = T\left(10^{\mathrm{NF(dB)}/10} - 1\right) \quad \mathrm{Kelvin}, \qquad (8.15)$$

$$\mathrm{NF(dB)} = 10\log_{10}\left(\frac{T_n}{T} + 1\right), \qquad (8.16)$$

where T is the ambient temperature, usually taken as 293°K.

Generally speaking, good low-noise amplifiers have noise temperatures far below room temperature (or,

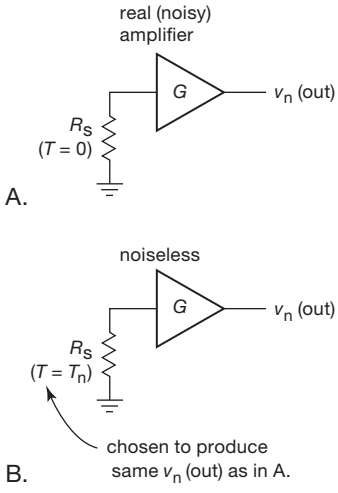


Figure 8.8. Noise temperature.

equivalently, they have noise figures far less than 3 dB). Later in the chapter we explain how you go about measuring the noise figure (or temperature) of an amplifier. First, however, we need to understand noise in transistors and the techniques of low-noise design. We hope the discussion that follows will clarify what is often a murky subject.

8.3 Bipolar transistor amplifier noise

The noise generated by an amplifier is easily described by a simple noise model that is accurate enough for most purposes. In Figure 8.9, e_n and i_n represent the transistor's internal noise, modeled as a noise voltage e_n in series with the input, combined with a noise current i_n injected at the input. The transistor symbol itself (or amplifier, in general) is assumed noiseless, and it simply amplifies the input-noise voltage it sees (caused by its own e_n, combined with its i_n flowing through the input signal's source impedance R_s). That is, the amplifier contributes a total noise voltage e_a, referred to the input, of

$$e_a(\text{rms}) = [e_n^2 + (R_s i_n)^2]^{\frac{1}{2}} \qquad \text{V/Hz}^{\frac{1}{2}}. \tag{8.17}$$

The two terms are simply the squared values of the amplifier input-noise voltage, and the noise voltage generated by the amplifier's input noise current passing through the source resistance.[15] Because the two noise terms are usually uncorrelated, their squared amplitudes add to produce the effective noise voltage seen by the amplifier. For low source resistances the noise voltage e_n dominates, whereas for high source impedances the noise current i_n generally dominates.

Just to get an idea of what these look like, look at Figure 8.10, which shows a graph of e_n and i_n versus I_C and f for the excellent (but, sadly, discontinued[16]) low-noise *npn* 2SD786. We go into some detail now, describing these and showing how to design for minimum noise. It is worth noting that voltage noise and current noise for a bipolar transistor are in the range of nanovolts and picoamps per root hertz (i.e., $\text{nV}/\sqrt{\text{Hz}}$ and $\text{pA}/\sqrt{\text{Hz}}$); for FETs the current noise is lower, down in the $\text{fA}/\sqrt{\text{Hz}}$ range.

8.3.1 Voltage noise, e_n

The equivalent voltage noise looking in series with the base of a bipolar transistor arises from collector-current shot noise generating a noise voltage across the intrinsic emitter resistance[17] r_e, Johnson noise generated in the base resistance $r_{bb'}$, and (as we'll see later) base current shot noise through that resistance. Deferring for the moment the base current term (which most often does not contribute significantly to a BJT's voltage noise), the input noise voltage density looks like this:

$$e_n^2 = 2qI_C r_e^2 + 4kTr_{bb'} \tag{8.18}$$

$$= 4kT\left(\frac{V_T}{2I_C} + r_{bb'}\right) \tag{8.19}$$

$$= 4kT\left(\frac{r_e}{2} + r_{bb'}\right) \quad \text{V}^2/\text{Hz}, \tag{8.20}$$

where we've eliminated r_e in the second form (eq'n 8.19) and I_C in the third form (eq'n 8.20) by remembering that $r_e = V_T/I_C = kT/qI_C$.

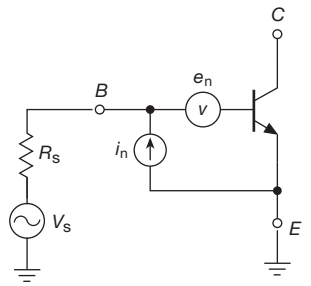


Figure 8.9. Noise model of a transistor.

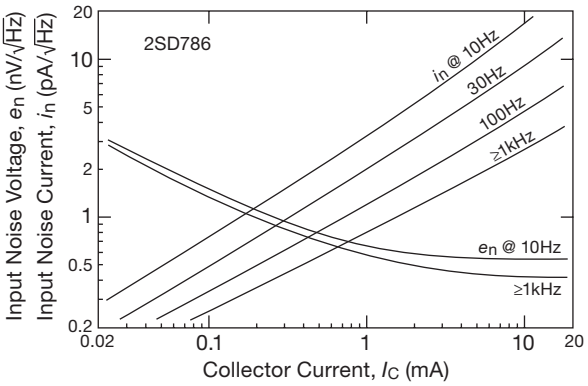


Figure 8.10. Equivalent rms input noise voltage (e_n) and noise current (i_n) versus collector current for a 2SD786 *npn* transistor, adapted from the datasheet.

[15] An additional term, important at higher frequencies or when i_n is small, is the noise current i_n generated by e_n in combination with the input capacitance: $i_n = e_n \omega C_{in}$. See §§8.11.3 and *4x.3.4*.

[16] The 2SD786 was for many years the standard to which low-noise transistors aspired. We feature it here because it is well characterized, with a wealth of information and excellent noise performance. Our nomination for its successor is the ZTX851 (with ZTX951 *pnp* complement), characterized and used in numerous examples later in the chapter.

[17] Properly speaking, there is no "resistance" r_e; rather, it represents the inverse transconductance of the transistor: $r_e = v_b/i_c = 1/g_m$ (or the voltage change at the base corresponding to changes in collector current). It's really just a convenient way of speaking. Be careful, though, when thinking of it loosely as a resistor: it is a "noiseless" resistance; that is, it has no Johnson noise.

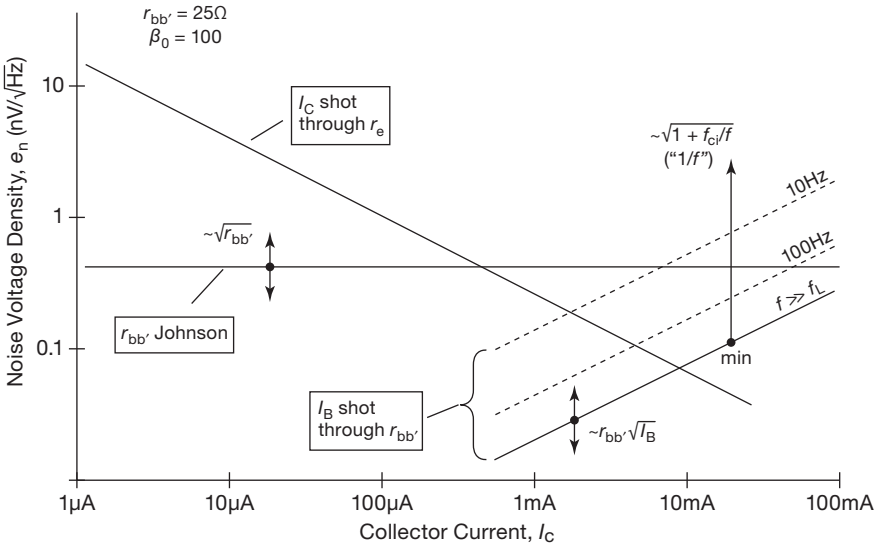


Figure 8.11. Input noise voltage e_n in a BJT. At low currents the shot noise in I_C through r_e dominates; otherwise the Johnson noise in $r_{bb'}$ is the dominant term. However, at low frequencies and high currents, the base-current shot noise through $r_{bb'}$ causes e_n to rise again. These curves assume typical values of flicker-noise breakpoint ($f_{ci} \sim 1$ kHz) and base resistance ($r_{bb'} \sim 25\,\Omega$), with double-sided arrows indicating variation with $r_{bb'}$, and the arrow upward from "min" indicating $1/f$ rise with decreasing frequency.

The third form is handy because it lets you think about the noise voltage as if arising from combining the separate Johnson noise from two resistors. To get a sense of the scale of BJT noise voltage, it's helpful to know that the base resistance $r_{bb'}$ for typical BJTs (see Table 8.1a on page 501) goes from a few ohms to a few hundred ohms; thus the second term's noise voltage contribution is typically in the range of $0.2\,\text{nV}/\sqrt{\text{Hz}}$ to $2\,\text{nV}/\sqrt{\text{Hz}}$ (we like to remember that a $100\,\Omega$ resistor has $e_n = 1.28\,\text{nV}/\sqrt{\text{Hz}}$, and that e_n scales as the square root of R).

As for the first term in eq'n 8.20, it's telling us that the *collector shot-noise current through r_e produces the same noise voltage as the Johnson noise of a fictional resistor of value $R = r_e/2$*. For example, in a BJT running at $100\,\mu$A (thus $r_e = 250\,\Omega$) this amounts to $e_n = \sqrt{4kT \cdot 125\,\Omega}$, or $1.4\,\text{nV}/\sqrt{\text{Hz}}$. This is a useful thing to know, for example when choosing resistor values so as not to compromise the noise performance of a low-noise amplifier. Lest we leave the wrong impression, we hasten to emphasize again that the intrinsic emitter resistance r_e is *not* a "real" resistor – it does not have Johnson noise; the noise-voltage term we're describing arises solely from the noise voltage generated by collector shot-noise current flowing through a noiseless r_e.[18]

Both voltage-noise terms in eq'ns 8.18 and 8.20 have a flat (white) spectrum, with a Gaussian distribution of instantaneous amplitude. Because r_e falls inversely with collector current, a BJT's e_n falls as $1/\sqrt{I_C}$ with rising collector current, ultimately reaching a minimum that depends on $r_{bb'}$ (Figure 8.11). For this reason it's usually best to run at relatively high collector currents if the objective is to minimize noise voltage; the price you pay is increased base current and increased heating. Taking again the excellent 2SD786 as an example, at frequencies above 1 kHz it has an e_n of $1.5\,\text{nV}/\sqrt{\text{Hz}}$ at $I_C = 100\,\mu$A and $0.6\,\text{nV}/\sqrt{\text{Hz}}$ at $I_C = 1$ mA (Figure 8.10). This transistor uses special geometry to achieve an unusually low $r_{bb'}$ of $4\,\Omega$, which is needed to realize the lowest values of e_n.

Of course, if you need to operate at a low collector current (where the effect of r_e dominates), it's of no particular benefit to have a low value of $r_{bb'}$. To make this point graphically, we plotted in Figure 8.12 the predicted and measured e_n versus collector current for six low-noise *npn* transistors, using a simple one-parameter model based upon $r_{bb'}$ alone. Selecting a transistor with low $r_{bb'}$ really matters at higher currents, though, which is where you have to operate if you want to achieve e_n values southward of $1\,\text{nV}/\sqrt{\text{Hz}}$.[19] We illustrate that point dramatically in §8.5.9, in an input stage with $e_n < 0.1\,\text{nV}/\sqrt{\text{Hz}}$.

The voltage-noise density e_n at 1 kHz, as plotted in

[18] See §8.3.5 for some further discussion of this subtle point.

[19] The effective $r_{bb'}$ is said to increase slightly at low currents. For example, if the 2N5089 excess noise at $10\,\mu$A ($5.9\,\text{nV}/\sqrt{\text{Hz}}$ instead of $4.5\,\text{nV}/\sqrt{\text{Hz}}$) is due to $r_{bb'}$, we can calculate that $r_{bb'}$ increased from

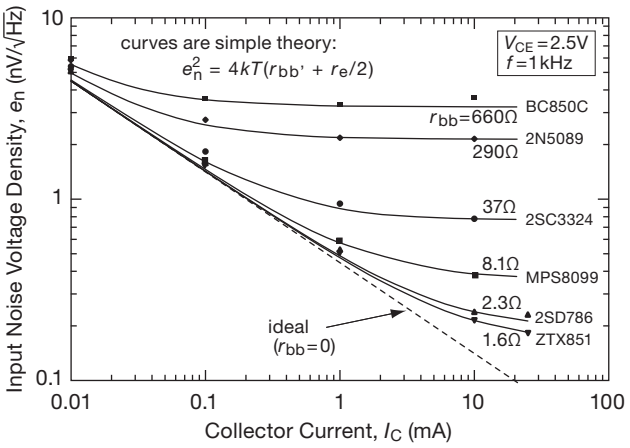


Figure 8.12. A simple model of BJT noise voltage (shot noise current through r_e, combined with Johnson noise of base resistance, solid curves) provides a good first approximation to measured noise values (datapoints at four currents), shown here for six selected low-noise BJTs. The dashed line shows the theoretical minimum noise voltage at current I_C, as given in eq'n 8.18. The measured noise is 10%–20% high at collector currents below 100μA. See Figure 8.17 for the noise spectra of these six BJTs at I_C=10 mA.

Figure 8.12, does not tell the whole story, of course. If you care about the noise at lower frequencies, you have to worry about a component of the transistor's noise that is frequency dependent, usually in the form of a rising $1/f$ flicker-noise tail (characterized by a $1/f$ corner frequency f_c). For BJTs, the $1/f$ noise in their e_n comes from one additional noise term that we've ignored thus far – the noise voltage produced by the base's noise current i_n flowing through the transistor's own $r_{bb'}$, i.e., e_n=$i_n r_{bb'}$. This latter term becomes a significant contributor to e_n only at low frequencies and at relatively high collector currents. But it is quite important for another reason: it flows through the input signal's source resistance, generating an additional noise voltage v_n=$i_n R_{sig}$. Let's look at the business of input noise current, after which we briefly revisit transistor e_n with this effect included.

8.3.2 Current noise i_n

The transistor's input noise current generates an additional noise voltage across the input signal source impedance. The main source of base-current noise is shot-noise fluctuation in the steady base current,

$$i_n = \sqrt{2qI_B} = \sqrt{2qI_C/\beta_0} \quad \text{A/Hz}^{\frac{1}{2}}, \tag{8.21}$$

which is Gaussian noise exhibiting a flat frequency spectrum (i.e., white noise); we've used the symbol β_0 here to emphasize that it's the beta *at dc*.

There is in addition a flicker-noise component that rears its ugly head at low frequencies. The latter exhibits the typical $1/f$ frequency dependence, with a corner frequency that we'll call f_{ci}; i.e., it contributes a term $2qI_B f_{ci}/f$ to the overall i_n^2. It rises somewhat more rapidly with I_C, owing to a rising $1/f$ breakpoint, typically with f_{ci} going as $I_B^{1/3}$ to $I_B^{1/4}$. See Figures 8.13 and 8.14. Typical current-noise corner frequencies are f_{ci}~50–300 Hz at 1–10 μA, and 200 Hz–2 kHz at 1 mA.

Note that this simple base-current shot noise is *not* collector-current shot noise divided by beta. If it were, it would be $\sqrt{2qI_C}/\beta_0$ rather than the correct (and larger) i_n=$\sqrt{2qI_C}/\sqrt{\beta_0}$ (as given by eq'n 8.21). In fact, to first order the base-current shot noise is uncorrelated with the collector-current shot noise.

However, at high frequencies this is no longer true, leading to one final current-noise term: at frequencies approaching the transistor's f_T (i.e., as the current gain approaches unity) the decreasing beta makes the collector-current's shot noise visible at the base.

Putting these terms together, we have, finally

$$i_n^2 = 2q\frac{I_C}{\beta_0}\left(1+\frac{f_{ci}}{f}\right) + 2qI_C\left(\frac{f}{f_T}\right) \quad \text{A}^2/\text{Hz}, \tag{8.22}$$

where the last term effectively represents a frequency-dependent beta.

Taking the example of the 2SD786 again (Figure 8.10), above 1 kHz i_n is about 0.25 pA/$\sqrt{\text{Hz}}$ at I_C=100 μA and 0.8 pA/$\sqrt{\text{Hz}}$ at I_C=1 mA. The noise current increases and the noise voltage decreases as I_C is increased. In the next subsection (§8.3.3) we see how this dictates operating current in low-noise design. Figure 8.15 shows graphs of i_n versus frequency and current for a pair of low-noise transistors.

Current-noise times input impedance The input-noise current flowing through the input signal's source impedance generates a noise-voltage density of magnitude v_n=$i_n Z_s$, which combines (as sum of squares) with the transistor's input noise-voltage density.[20] Usually the source impedance is resistive, in which case you need to

290 Ω at 10 mA to about 900 Ω at 10 μA (threefold over 3 decades), or approximately as $(I_c)^{-1/6}$. (SPICE transistor noise models include a parameter for this effect.)

[20] The primary complicating issues are increases of i_n with temperature, and with decreasing frequency ($1/f$ behavior).

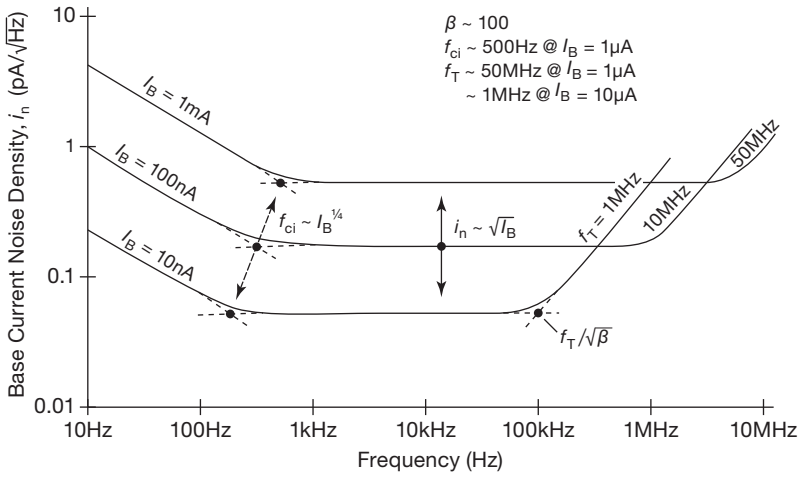


Figure 8.13. Input noise-current density i_n versus frequency in a BJT. At mid-frequencies it is entirely base-current shot noise, proportional to $\sqrt{I_B}$. At low frequencies i_n rises as $1/\sqrt{f}$ (it is the noise *power* that goes as $1/f$); however, the $1/f$ breakpoint (f_{ci}) rises with increasing current, so that at a given low frequency below f_{ci} the noise current rises faster than the square root of base current, as shown. At high frequencies the falling beta ($\beta \to 1$ at f_T) causes the noise density to rise $\propto f$. These curves assume a typical value of flicker noise breakpoint (f_{ci} ~500 Hz at I_B=1 μA), with the solid arrow indicating variation of i_n with base current, and the dashed arrow indicating variation of $1/f$ breakpoint with base current.

add its Johnson noise as well; in other words, the total input-referred squared noise voltage is

$$v_n^2 = e_n^2 + 4kTR_s + (i_n R_s)^2 \quad \mathrm{V^2/Hz}. \tag{8.23}$$

If the source impedance is *reactive*, however, the current noise contribution (the last term in the equation) will be frequency dependent. A common situation in which this matters is an ac-coupled input signal. If you aren't thinking about noise, you'd normally choose the blocking capacitor value based on the downstream load impedance (transistor input impedance and bias resistor), to set the low-frequency rolloff somewhat below the lowest frequency of interest. That could be a rather small capacitance (if the amplifier has a high input impedance, or if the low-frequency cutoff isn't terribly low), through which the noise current would generate substantial noise voltage (and, of course, increasing proportional to the capacitor's reactance $X_C = 1/2\pi fC$, i.e., proportional to $1/f$).[21]

So, you need to turn the process around: first, choose the capacitor value to keep its noise-voltage contribution $v_n = i_n X_C$ small enough at the lowest frequency of operation (and remember that base-current noise often exhibits a $1/f$ tail); then choose the transistor's input bias resistor value to get the desired low-frequency rolloff. We've been bitten by this issue, surprised by a $1/f$ noise voltage ($1/f$ in *amplitude*, not power) caused by the blocking capacitor's reactance versus frequency. To fix the problem we had to increase the blocking capacitor's value by a factor of 50!

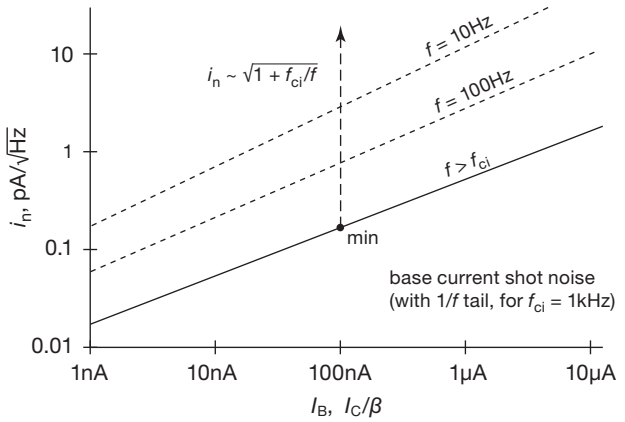


Figure 8.14. Input noise current in a BJT is shot noise, scaling as the square root of base current for frequencies above the $1/f$ corner frequency f_{ci}. At lower frequencies the curve is steeper, because f_{ci} itself rises with increasing current (see Figure 8.15).

8.3.3 BJT voltage noise, revisited

As suggested in §8.3.1, the transistor's input *current* noise i_n can become a significant contributor to the voltage noise e_n seen at its input terminals, owing to the noise voltage

[21] Note that an ideal capacitor (or inductor, for that matter) generates no Johnson noise.

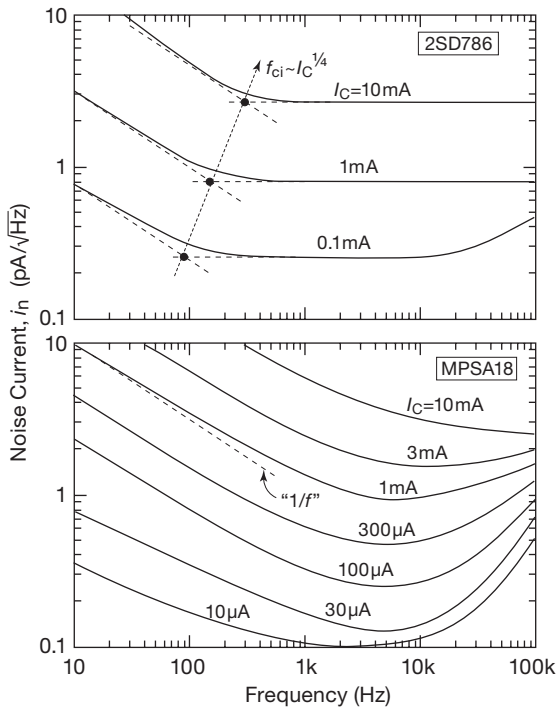


Figure 8.15. Noise current (i_n) versus frequency for two low-noise *npn* transistors. The low-frequency noise current exhibits a somewhat greater rise with collector current, owing to a rising $1/f$ corner frequency.

generated across its internal $r_{bb'}$ by its own input-noise current i_n. Because i_n is shot noise in the dc base current, it increases as the square root of collector current (for constant beta); and, because i_n of a BJT is seriously afflicted by a $1/f$ low-frequency tail, its contribution to the overall e_n is seen primarily (and often *only*) at low frequencies. A more detailed noise model of the bipolar transistor lets us incorporate this effect.

Adding this noise term to our original (incomplete) BJT noise voltage formula, eq'n 8.18, we have

$$e_n^2 = 2qI_C r_e^2 + 4kTr_{bb'} + 2q\frac{I_C}{\beta_0}r_{bb'}\left(1+\frac{f_{ci}}{f}\right),\ \mathrm{V^2/Hz}, \quad (8.24)$$

where, as before, the first term can be replaced by $2kTr_e$, and, as in eq'n 8.22, the current-noise corner frequency f_{ci} rises approximately as the fourth root of I_C.

To see how this works out, look at Figure 8.16, where we've plotted separately (for three choices of collector current) the e_n from the first two terms (as dashed lines) and the e_n from the third term (as thin solid lines). The thick solid lines represent the total voltage noise, i.e., including all three terms of eq'n 8.24. For these calculated curves we used typical BJT noise parameters (β=200, $r_{bb'}$=50Ω, f_{ci}=1 kHz at I_C=1 mA).

This shows how the $i_n r_{bb'}$ noise-voltage term competes with the other two e_n noise terms, dominating at high collector current and low frequency. Although the noise-current $1/f$ breakpoint f_{ci} increases only mildly with I_C, its contribution acts to create a rapidly rising noise-*voltage* breakpoint f_c. Compare this with Figure 8.11, where the same effect can be seen: there, however, e_n is plotted versus collector current, with a family of several spot frequencies.

To maintain some perspective here, note that additional noise voltage is evident only at very low frequencies (Figure 8.16 goes down to 0.01 Hz!) and at relatively high current densities. It's not something over which to lose sleep.[22] This can be seen in the measured noise-voltage curves plotted in Figure 8.17, where the effect (at the substantial collector current of 10 mA) is serious above 10 Hz only for BC850 with its unusually high $r_{bb'}$≈750Ω. In fact, for transistors with attractively low $r_{bb'}$ the e_n improvement gained from running at high collector current more than compensates for the rise at low frequencies that comes from the base-current shot-noise contribution, as seen in Figure 8.18.

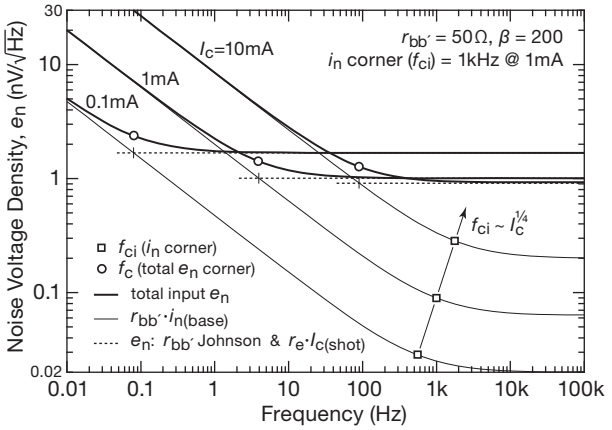


Figure 8.16. BJT noise-voltage density versus frequency (from eq'n 8.24) for a hypothetical transistor with the listed parameters, showing the effect of current noise i_n at low frequencies and high collector currents.

[22] But if you *are* losing sleep over $1/f$ problems in a high-I_C low-noise amplifier, there's a cute trick, namely to stack a pair of transistors so that their $1/f$ noise fluctuations cancel. See Broderson, Chenette, and Jaeger, "A superior low-noise amplifier," 1970 IEEE ISSCC, p 164. Suppression of low-frequency noise by a factor of five is possible. The penalty for using their two-transistor trick is a 3 dB increase in the higher-frequency (white noise) component of e_n.

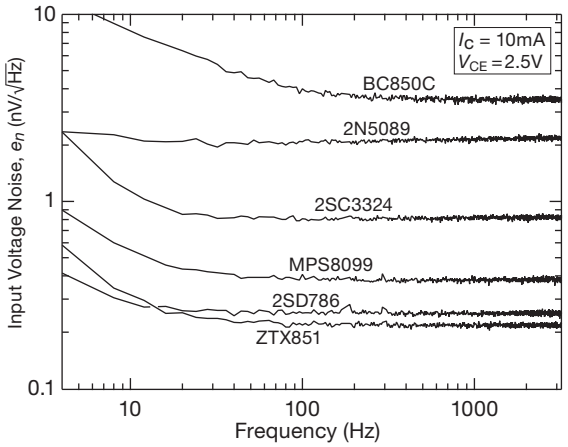


Figure 8.17. Measured input noise-voltage spectra (e_n versus frequency at I_C=10 mA) for the six BJTs of Figure 8.12.

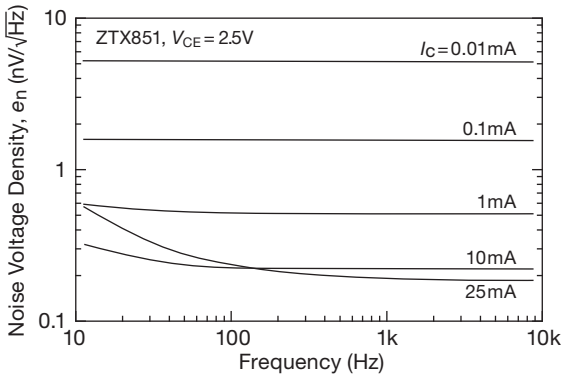


Figure 8.18. Measured input noise-voltage spectra for a ZTX851 at five choices of collector current. Base noise current adds noise voltage at low frequencies and high current, but you're still ahead at collector currents up to 10 mA.

8.3.4 A simple design example: loudspeaker as microphone

Let's put this noise theory into practice by designing a simple and inexpensive ac-coupled audio amplifier that runs on low current from a single +9 V supply (which could be a battery, or a "wall-wart" ac adapter), with input noise of just a few nV/$\sqrt{\text{Hz}}$. If you want to have an application in mind, think about a front-door intercom, where the small loudspeaker also serves reciprocally as a microphone.[23]

To get a handle on the noise specifications, we put a few small "8 Ω" loudspeaker samples on our lab bench, spoke to them in a normal voice, and measured output audio voltages in the range of 30–100 μVrms. Let's aim for a signal-to-noise ratio of 40 dB or better; that translates to an input noise voltage of 0.5 μVrms or less. And let's keep the component cost *really* low – we'll aim for 25¢.

The basic circuit Figure 8.19 shows the circuit, an ac-coupled common-emitter stage (with bypassed emitter resistor) driving an emitter follower. We're not worried about distortion (with output swings in the tens of millivolts, from ∼50 μV input levels), but we want plenty of voltage gain, and bypassing the emitter resistor eliminates it as a noise source. We provisionally chose operating currents of 100 μA, to be revisited (as with any circuit design) after an initial evaluation of the circuit's performance.

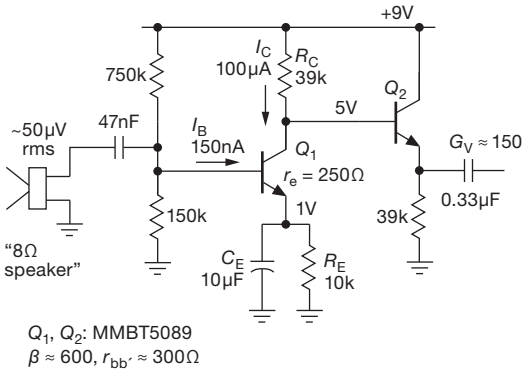


Figure 8.19. Low noise, and even lower cost: a simple audio preamp for running a loudspeaker "backward."

Noise calculation We chose the '5089 for this example because it's widely available (half a million in stock at Mouser and DigiKey, four different manufacturers), it's inexpensive ($0.026 in qty 100), and it promotes itself for low-noise amplifier applications.[24] It has lots of current gain (400 minimum, 600 typical at 100 μA), but it suffers from relatively high base resistance $r_{bb'}$≈300 Ω.

From eq'n 8.24 it's easy to calculate the various noise voltage density (e_n) contributions (and the reader is encouraged to do so), referred to the input (RTI):

Shot noise of I_C through r_e	$r_e\sqrt{2qI_C}$	1.4 nV/$\sqrt{\text{Hz}}$
Johnson noise in $r_{bb'}$	$\sqrt{4kTr_{bb'}}$	2.2 nV/$\sqrt{\text{Hz}}$
Johnson noise in R_C	$\sqrt{4kTR_C}/G_V$	0.17 nV/$\sqrt{\text{Hz}}$
Shot noise of I_B through $r_{bb'}$	$r_{bb'}\sqrt{2qI_B}$	0.066 nV/$\sqrt{\text{Hz}}$

[23] To those who suspect that "the walls have ears," this may suggest interesting possibilities for surveillance.

[24] For example, ON Semiconductor explicitly calls it a "Low Noise Transistor"; Fairchild and NXP call it a "General Purpose Transistor," but promote it for "low noise, high gain, general purpose amplifier applications" and "low-noise input stages in audio equipment," respectively.

The dominant noise source is Johnson noise in Q_1's internal base resistance, followed by collector shot noise flowing through Q_1's intrinsic emitter resistance r_e of 250Ω at its collector current of 0.1 mA. The last two terms are negligible by comparison: there's plenty of Johnson noise in Q_1's collector resistor, but owing to that stage's voltage gain of ∼150 it is an insignificant contribution when referred to the input (RTI). Even smaller is the voltage noise produced by base-current shot noise flowing through $r_{bb'}$. There are other noise possibilities, but they are vanishingly small by comparison: as an example, the Johnson-noise current in the bias divider flowing through the speaker's 8 Ω source impedance generates a noise voltage density of just $0.003\,\text{nV}/\sqrt{\text{Hz}}$.

Performance From these noise-voltage *densities* we arrive at the total rms noise voltage over the signal bandwidth: $v_n = e_n\sqrt{\Delta f} = 142\,\text{nVrms}$, where e_n is the combined noise-voltage density of $2.6\,\text{nV}/\sqrt{\text{Hz}}$, and Δf is the audio bandwidth (for which we've used the "telephone bandwidth" of 3 kHz). Compared with the nominal 50 μVrms audio input, that amounts to a signal-to-noise ratio of 51 dB – not bad for the price!

Speaking of which, component cost is a "performance" parameter, of sorts. A quick search of the usual distributors (DigiKey, Mouser, Newark, Future) turned up the following prices (SMT components – chip resistors and capacitors, SOT-23 transistors):

	qty 100	qty 1k
Q_1,Q_2	$0.026	$0.023
C_{in}	$0.032	$0.012
C_E	$0.068	$0.039
C_{out}	$0.034	$0.017
resistors	$0.012	$0.004
total parts	$0.246	$0.134

Variations We exceeded our targets of SNR and cost, so we could leave it at that. But it's always good to review an initial design, looking to improve things. Here the noise budget is dominated by the relatively high base resistance of these "low-noise" transistors. We could improve the noise performance by choosing an input-stage transistor with lower $r_{bb'}$ – look ahead to Table 8.1a on page 501, and at Figure 8.12, to see some candidates. However, once you've driven down the $r_{bb'}$ Johnson noise, the collector shot noise through r_e becomes the dominant term, and you're forced to increase the collector current to reap further improvements in v_n.[25] In fact, as Figure 8.12 shows, down at $I_C = 100\,\mu\text{A}$ the '5089 is hardly worse (not even a factor of 2) than transistors with the lowest $r_{bb'}$.

Looked at another way, we could instead stick with the inexpensive and widely available MMBT5089, and reduce somewhat the collector current (say, to 50 μA), with very little increase in overall noise: the noise density would increase from $2.6\,\text{nV}/\sqrt{\text{Hz}}$ to $3.0\,\text{nV}/\sqrt{\text{Hz}}$, which would reduce the audio–band SNR by just 1 dB (from 50.9 dB to 49.8 dB). Certainly worth it, if you're running from a battery.

8.3.5 Shot noise in current sources and emitter followers

We remarked earlier (in §8.1.2) that shot noise is suppressed in the classic BJT current source (redrawn in Figure 8.20A). You might at first think this is obvious, because the collector current is just the emitter current (apart from a small base current contribution), in which shot noise is absent (as it always is in metallic conductors). But at the collector you've got just the sort of barrier in which shot noise is unavoidable.

OK, you might say, perhaps the current source *does* exhibit shot noise, but surely at the emitter terminal you've got a quiet current (and thus an absence of noise voltage) – just look at that resistor!

Paradoxically, that intuition is wrong. It turns out that the current source is "quiet," but the emitter follower's output *does* indeed include a noise voltage just equal to a shot-noise current (calculated from I_C) flowing through r_e, i.e., just the $r_e\sqrt{2qI_C}$ we saw in eq'n 8.24.[26]

It's actually rather easy to test this. Look at Figure 8.20C, where a ∼10 μA current source is loaded with a 1 MΩ collector resistor. Absent collector shot noise, we should see only a Johnson-noise voltage density of $181\,\text{nV}/\sqrt{\text{Hz}}$ at the collector,[27] to be compared with a noise voltage 10× higher if the collector current has the statutory shot noise $i_n = \sqrt{2qI_C}$. Meanwhile, at the emitter, that shot-noise current, flowing through r_e, would produce a noise-voltage density of $4.8\,\text{nV}/\sqrt{\text{Hz}}$; in the absence of emitter shot noise the follower would be quiet (apart from the other sources of BJT noise in eq'n 8.24).

We built circuit C, and...(drumroll)...we measured $190\,\text{nV}/\sqrt{\text{Hz}}$ at the collector (consistent with Johnson

[25] To see how far you can take this, visit the "extreme challenge" in §8.5.9.

[26] To which you must of course add the additional terms, Johnson noise and base current shot noise in $r_{bb'}$.

[27] This is $\sqrt{2}$ times the $128\,\text{nV}/\sqrt{\text{Hz}}$ Johnson noise of a 1 MΩ resistor. The $\sqrt{2}$ arises because this is a unity-gain inverter, so R_E's Johnson noise contributes an equal uncorrelated noise voltage.

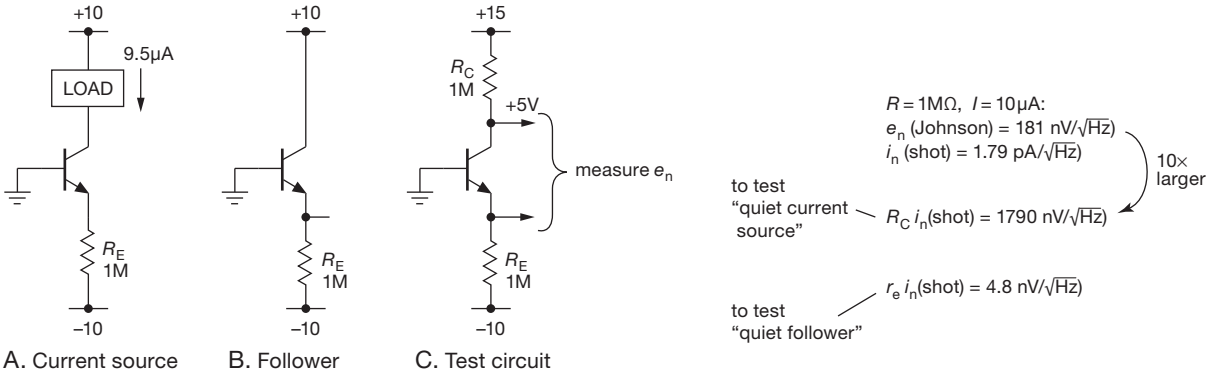


Figure 8.20. Shot noise is suppressed in the transistor current source, A, but not in the emitter follower, B. This is easily confirmed by measuring the noise-voltage density in the test circuit C.

noise alone, i.e., a quiet current source); and we measured $4.93\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$ at the emitter (consistent with collector-current shot noise through r_e). Evidently shot noise is suppressed in the current source, but not in the follower. Interesting! And, to test the possibility that collector current *never* exhibits shot noise, we put a bypass capacitor between the base and emitter terminals. The noise voltage at the collector jumped up, to $1679\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$, which agrees well with the shot-noise formula's predicted[28] $1664\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$. How can we understand this?

To get an understanding of this puzzle, it's best to start with what *must* be true and see where it leads. Look at Figure 8.21, as we take it in steps.

1. We know for sure that the emitter current is quiet, because it's the current through a metallic conductor. And any worry we might have that this current could become noisy because of the fluctuations in the transistor's base-to-emitter *voltage* (i.e., input-voltage noise) is easily addressed: we can reduce any such effect (already negligible, because noise variations $v_{n(BE)}$ are down in the microvolts) by as much as we like by simply increasing both R_E and the negative supply voltage proportionally.

2. Somewhere within the transistor there must be a diffusive collector current that is subject to classic uncorrelated charge-flow shot noise, $i_n = \sqrt{2qI_{C(dc)}}$, and over which the transistor has no control.

3. But this is a three-terminal device with negligible base current, so we know that the current flowing into the collector terminal equals the current flowing out of the emitter terminal (in the limit of large beta); and the latter is "quiet."

4. So it must be the case that the controllable portion of the BJT's collector current is fluctuating in just such a way as to cancel the shot noise fluctuations.

For that to be so, the input voltage V_1 to the BJT transconductance model must be such that $g_m V_1 = I_E - i_n$, and so the input noise-voltage portion of V_1 is just

$$e_n = -i_n/g_m \tag{8.25}$$

which (recall $r_e = 1/g_m$) is the familiar $i_n r_e$ form from eq'n 8.24.

This may seem like a circular argument; but in a real sense this is how this component of the base-to-emitter noise voltage is (and must be) created. This line of reasoning also explains nicely the presence of noise voltage e_n at the emitter of our test circuit (and, of course at the output of an emitter follower, or at the input of a common-emitter amplifier, etc.). And it explains also why bypassing the emitter liberates the full shot noise at the collector of our current source: such a connection invalidates the starting assumption (#1, above) because the capacitor diverts

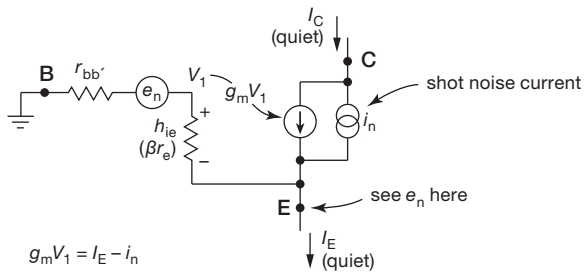


Figure 8.21. Three-terminal simplified "hybrid-π" model of the current-source–follower of Figure 8.20C. The noise voltage e_n reflects the unavoidable internal shot-noise current i_n. If the emitter current is quiet, so will be the resulting collector current I_C.

[28] For this test the collector current was $8.65\,\mu$A, constrained by operation from batteries, thus the slightly smaller shot-noise prediction.

entirely the emitter current at signal frequencies. In fact, it creates a grounded-emitter amplifier (rather than a current source or follower), in which the full input-noise voltage as given above (eq'n 8.25) is amplified by the transconductance g_m to produce an output-noise current equal to $g_m e_n$. And that is exactly the full (unsuppressed) shot noise corresponding to the dc collector current I_C (rewind to step #2, above), which is what we measured in our test setup.

A final (and important) point: the shot-noise suppression in the current source requires a quiet emitter current sink, satisfied in this case by a large (compared with r_e) pull-down resistor R_E (here 1M versus 2.5k). But that's not the case for, say, an undegenerated current mirror. This raises the quantitative question, how large must R_E be, to be "large" in this sense?

This is easy enough to figure out: at the emitter we have a noise voltage e_n equivalent to the Johnson noise of a resistor of value $r_e/2$ and with source impedance equal to r_e; i.e., a noise current equivalent to the Johnson-noise current of a resistor of value $2r_e$. This is diluted by the quiet (i.e., shot-noise-free) pull-down resistor R_E, so that the reduction of current noise at the collector is in the ratio $\sqrt{2r_e/R_E}$. For example, at 1 mA collector current we have an impedance seen at the emitter of r_e=25 Ω, with a noise voltage equivalent to that of a 12.5 Ω resistor, or a noise current equivalent to that of a 50 Ω resistor. So a 50 Ω pull-down resistor reduces the noise by 3 dB, and a 4.95 kΩ pulldown reduces it by 20 dB, etc. Stated more generally, the reduction in collector noise current i_n is in the ratio $\sqrt{50\text{mV}/V_{R_E}}$, where V_{R_E} is the dc voltage across the emitter pull-down resistor R_E.

8.4 Finding e_n from noise-figure specifications

Transistor datasheets traditionally provided some tabulated values for e_n, and (often) graphs at selected collector currents of e_n and i_n versus frequency (or e_n and i_n versus collector current at selected frequencies), as we've seen for the 2SD786 and the MPSA18.

That was *then*! Now you see instead tabulations and graphs of noise figure (NF) – see for example the contours of constant NF plotted against I_C and R_s for the low-noise 2SC3324 *npn* bipolar transistor[29] from Toshiba (Figure 8.22).

There's a lot of information in these plots, even though they tell the story only at two frequencies (10 Hz and 1 kHz; it would sure be nice to see a plot of noise parameters versus frequency). Let's see how much we can squeeze out of these two plots.

8.4.1 Step 1: NF versus I_C

We can make a plot of NF versus collector current, for each of a set of source resistances, by simply reading off values along a horizontal line in Figure 8.22. It's helpful to use a spreadsheet program with plotting capabilities such as Microsoft Excel, or (if you want to show off) a more sophisticated mathematical package such as MATLAB® or Mathematica®. Figure 8.23 shows what you get by reading across the 1 kHz plot of Figure 8.22 for six values of source resistance. Here we estimated NF values for half-decade steps of I_C (thus 10 μA, 30 μA, 100 μA, and so on), plotted them in Excel, then smoothed them with Bézier curves in Adobe Illustrator®. Noise-figure values below 1 dB should not be considered reliable, because Figure 8.22 (from which the data were derived) shows no contours below NF=1 dB.[30] The dashed line indicates NF=3 dB, where the noise contributed by the transistor is equal to the Johnson noise in the source resistor.

Before moving on, let's get some intuition about these curves. For low values of R_s the source has low Johnson-noise voltage (e.g., R_s=100 Ω has e_n=1.3 nV/$\sqrt{\text{Hz}}$), quite a bit less than the transistor's input e_n at small collector currents.[31] That's why the noise figure improves with increasing collector current. On the other hand, for large values of R_s the transistor's voltage noise is unimportant compared with the resistor's much larger Johnson noise. But now when we operate at large collector currents the transistor's input *current* noise (base current shot noise: $i_n \approx \sqrt{2qI_B}$) generates a substantial noise voltage across that larger R_s, such that the noise voltage at the input terminal is much larger than Johnson noise alone, hence a large (poor) noise figure.

8.4.2 Step 2: NF versus R_s

Figure 8.24 is an analogous plot, this time however showing NF versus *source resistance* (instead of I_C) for each of

[29] The *pnp* complement is the 2SA1312.

[30] The datasheet lists a value of 0.2 dB (typical) in the tabulated data, at 1 kHz, 100 μA collector current, and 10 kΩ source resistance, a set of conditions that puts you nicely in the "eye" of the contour plot. But the datasheet also lists a worst-case (maximum) value of 3 dB! You'll have to do some cherry picking if you need NF=0.2 dB.

[31] Recall that the latter is approximately collector-current shot noise through r_e: $e_n \approx \sqrt{2qI_C} \cdot (kT/qI_C)$. That's equal to the Johnson noise of a fictitious resistor of value $R=r_e/2$, which is 100 Ω at I_C=125 μA. So the transistor's e_n rises with falling collector current, as $e_n \propto 1/\sqrt{I_C}$.

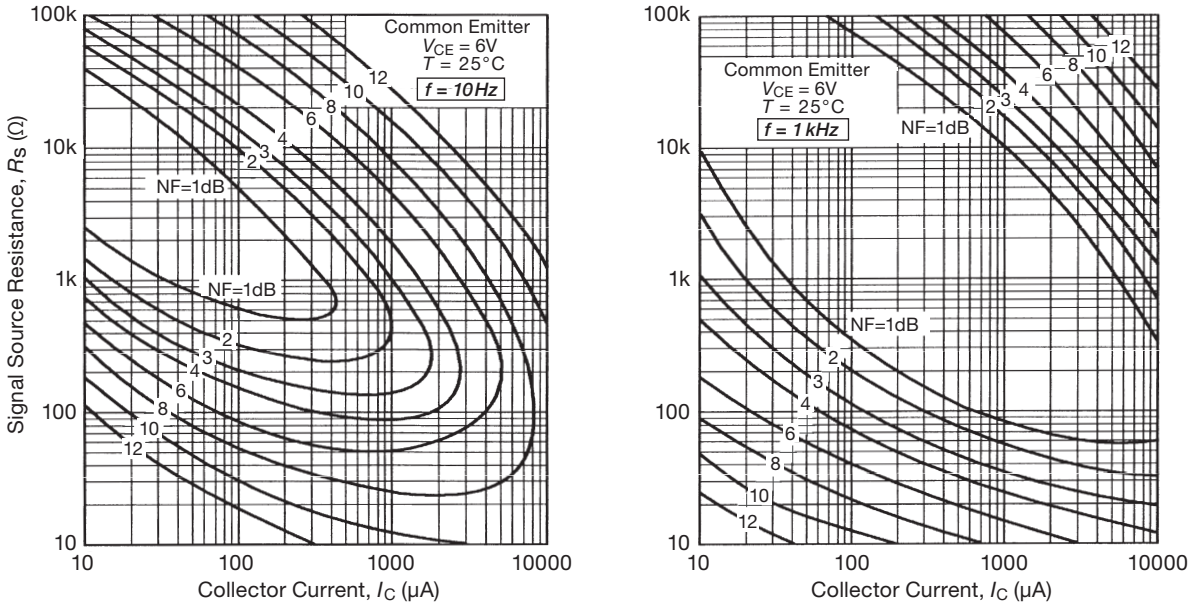


Figure 8.22. Toshiba's datasheet for its low-noise 2SC3324 *npn* transistor does not provide values or graphs for input voltage noise e_n. Instead you get these graphs of noise figure (NF) versus collector current and source resistance.

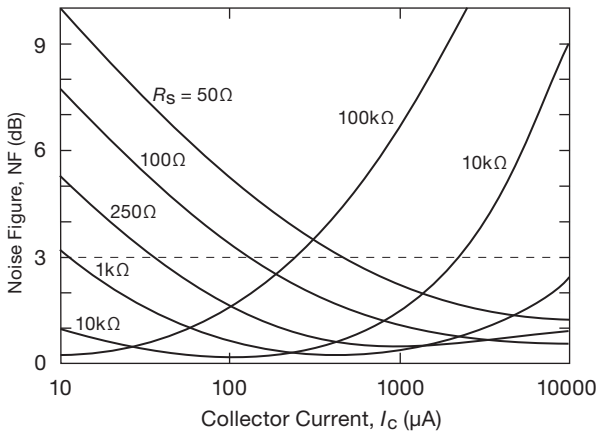


Figure 8.23. 2SC3324 noise figure versus collector current, from the datasheet contour plots shown in Figure 8.22.

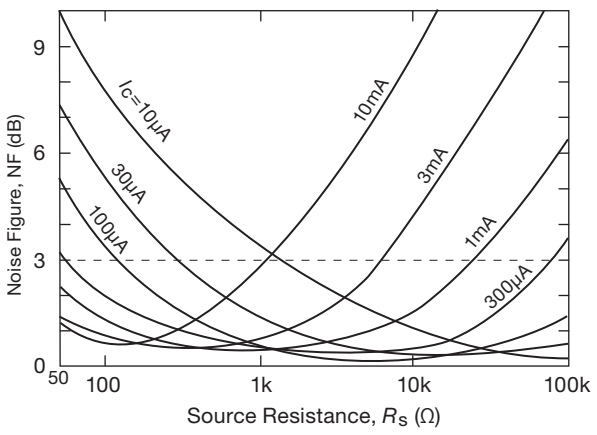


Figure 8.24. 2SC3324 noise figure versus source resistance, from the data of Figure 8.22.

a set of collector currents; it is taken from the same 1 kHz contour curves of Figure 8.22, this time reading off values along vertical lines of constant I_C. It's a useful graph for determining, approximately, the optimum operating current for a signal of a given source impedance.

8.4.3 Step 3: getting to e_n

The previous two graphs (NF versus I_C, NF versus R_s) are simply reorganizations of the manufacturer's 1 kHz noise figure contour plot (Figure 8.22). We didn't need to calculate anything.

That's not the situation with noise voltage, e_n. Here we need to invert the defining equation (eq'n 8.13) for noise figure in order to find e_n:

$$e_n = \sqrt{4kTR_s}\,\sqrt{10^{NF/10} - 1}. \tag{8.26}$$

The first term is the signal source resistor's Johnson-noise voltage density, and the second term is the multiplying

factor attributable to the transistor's noise contribution, as given by its noise figure. The latter includes contributions from both e_n and i_n; so we must use the NF values for small source resistance so that the contribution from the transistor's i_n is negligible.[32]

Figure 8.25 shows what you get for e_n versus collector current, beginning with the noise-figure values corresponding to R_s=50 Ω (from Figure 8.23 or 8.22). In common with most devices, this transistor exhibits the familiar "flicker-noise" excess below its "$1/f$ noise corner-frequency" (see §8.1.3, and also §5.10.6).

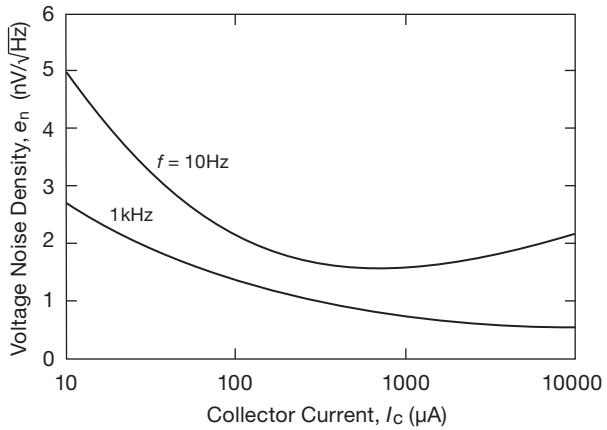


Figure 8.25. 2SC3324 noise voltage versus collector current, derived from the data of Figure 8.22.

8.4.4 Step 4: the spectrum of e_n

We complained earlier about the lack of noise data versus frequency – it's given only for 10 Hz and 1 kHz. But we can fill in the details, on the reasonable assumption that the noise power at low frequencies goes approximately as $1/f$ (that is, $e_n \propto 1/\sqrt{f}$), and that the two datapoints straddle the $1/f$ noise corner-frequency.

To do this, we first find the $1/f$ corner frequency f_c from the e_n values that straddle it. To a good approximation,

$$f_c = f_L \left(\frac{e_{nL}^2}{e_{nH}^2} - 1 \right), \tag{8.27}$$

where e_{nL} is the noise density at a frequency f_L below the corner frequency, and e_{nH} is the noise density well above f_c (see the discussion on page 566). Once we've got f_c, we can find e_n versus frequency:

$$e_n(f) = e_{nH}\sqrt{1 + f_c/f}. \tag{8.28}$$

Because the corner frequency depends on collector current, we read off e_{nL} and e_{nH} from Figure 8.25 for each of four decade values of I_C, and then plot the corresponding e_n versus f. Figure 8.26 shows the resulting noise-voltage spectra.

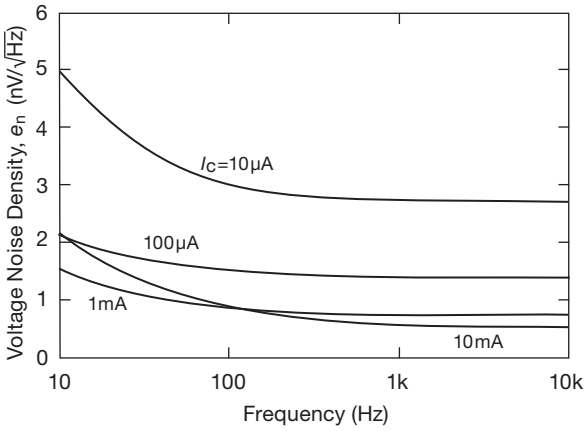


Figure 8.26. 2SC3324 noise voltage versus frequency, deduced from the curves of Figure 8.25.

8.4.5 The spectrum of i_n

By a similar procedure we could extract curves of current noise i_n versus frequency. Above the $1/f$ corner we would see base-current shot noise ($i_n = \sqrt{2qI_b}$), with the characteristic $1/f$ rise at low frequencies.

Once you've gotten plots of e_n and i_n versus frequency, you have all the basic information contained within a noise-figure contour plot. Look ahead to §8.9.1E, and especially Figure 8.58, to see how you can use e_n and i_n to predict the total effective input noise density versus source resistance. The graphic shown in the upper left of that figure shows how to find the breakpoints that separate the e_n-dominated (low-R_s) region, the source Johnson-noise-dominated (mid-R_s) region, and the i_n-dominated (high-R_s) region. We prefer that simpler approach, and we'll be using e_n and i_n extensively in the rest of the chapter.

8.4.6 When operating current is not your choice

Knowing how e_n and i_n vary with collector (or drain) current helps in setting the operating point for optimum noise performance, as we'll see in the next section.

[32] Similarly, we could extract i_n from the NF values for large source resistance, where the transistor's e_n contribution is negligible.

Sometimes, though, the choice has already been made, by person or persons unknown. In such a case you live with what you're given. Figure 8.27 shows an example, with published noise-figure contours for the Stanford Research Systems model SR560 preamplifier (§8.6.4). It's clear from the high input noise resistance (∼200 kΩ) that this is a JFET-input amplifier, and that its noise performance is far from optimum with a source of low impedance: NF=15 dB for R_s=50 Ω(!); compare that with the ∼2 dB noise figure of the 2SC3324 running at I_C=5 mA.

What *not* to do OK, you say, I'll put a 200k resistor in series with the input signal, and, voilà, I'll get a 0.05 dB noise figure.

Don't do that! What you've done is to add lots of noise to the input signal, so that it dominates over the amplifier's noise. An amplifier like this is optimized for high-Z signal sources and is far from optimum for 50 Ω systems. What you *can* do, if you insist on using it with a 50 Ω source, is to transform the source impedance upward by a modest factor with a signal transformer, for example a North Hills HB0904 (50 Ω:1200 Ω, with 1 kHz–6 MHz bandpass) or a Mini-Circuits T16-6T-X65 ((50 Ω:800 Ω, with 30 kHz–75 MHz bandpass). This puts you in the ∼3 dB territory of noise figure, quite an improvement over the NF=15 dB you get with a bare 50 Ω source. For more discussion of signal transformers, see §8.10.

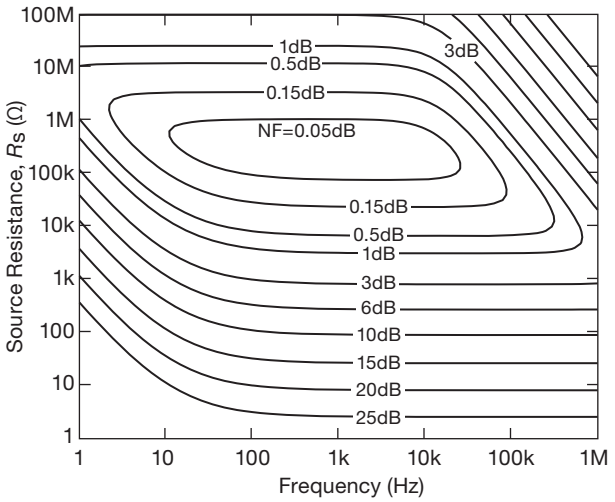


Figure 8.27. Noise figure versus frequency and source resistance for the SR560 low-noise preamp. (Courtesy of Stanford Research Systems.)

8.5 Low-noise design with bipolar transistors

The fact that e_n falls and i_n rises with increasing I_C provides a simple way to optimize transistor operating current to give lowest noise with a given source. Look at the model again (Figure 8.28). The noiseless signal source v_s has added to it an irreducible noise voltage from the Johnson noise of its source resistance:

$$e_R^2(\text{source}) = 4kTR_s \quad (\text{V}^2/\text{Hz}). \tag{8.29}$$

The amplifier adds noise of its own, namely,

$$e_a^2(\text{amplifier}) = e_n^2 + (i_n R_s)^2 \quad (\text{V}^2/\text{Hz}). \tag{8.30}$$

Thus the amplifier's noise voltage is added to the input signal, and, in addition, its noise current generates a noise voltage across the source impedance. These two are uncorrelated (except at very high frequencies), so you add their squares. The idea is to reduce the amplifier's total noise contribution as much as possible. That's easy, once you know R_s, because you just look at a graph of e_n and i_n versus I_C in the region of the signal frequency, picking I_C to minimize $e_n^2 + (i_n R_s)^2$. Alternatively, if you are lucky and have a plot of noise-figure contours versus I_C and R_s, you can quickly locate the optimum value of I_C.

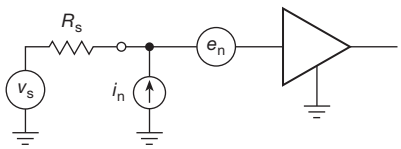


Figure 8.28. Amplifier noise model.

8.5.1 Noise-figure example

As an example, suppose we have a small signal in the region of 1 kHz with source resistance of 10k, and we wish to make a low-noise common-emitter amplifier with a 2N5087. From the datasheet graphs of e_n and i_n versus collector current (Figure 8.29), we see that the sum of voltage and current terms (with 10k source) is minimized for a collector current of about 20–40 μA. Because the current noise is falling faster than the voltage noise is rising as I_C is reduced, it might be a good idea to use slightly less collector current, especially if operation at a lower frequency is anticipated (i_n rises rapidly with decreasing frequency). We can estimate the noise figure using i_n and e_n at 1 kHz:

$$\text{NF} = 10\log_{10}\left(1 + \frac{e_n^2 + (i_n R_s)^2}{4kTR_s}\right) \text{ dB}. \tag{8.31}$$

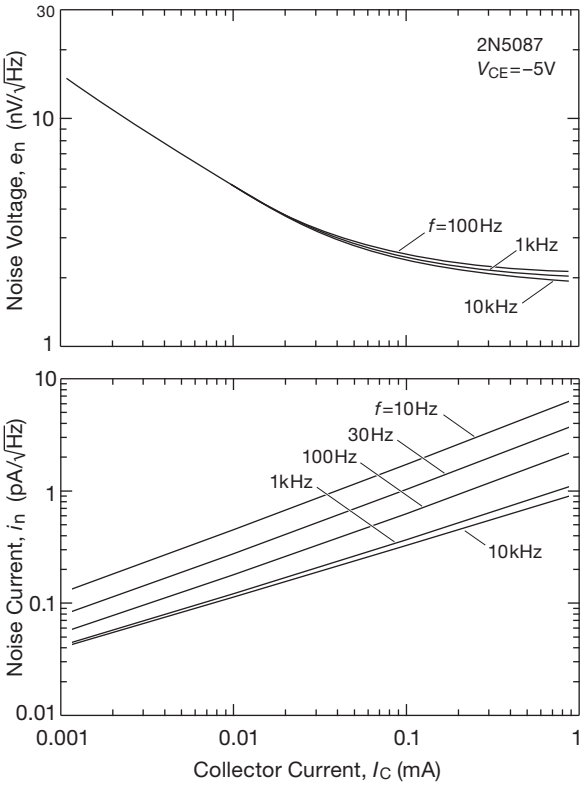


Figure 8.29. Voltage and current noise versus frequency for the 2N5087 *pnp* transistor.

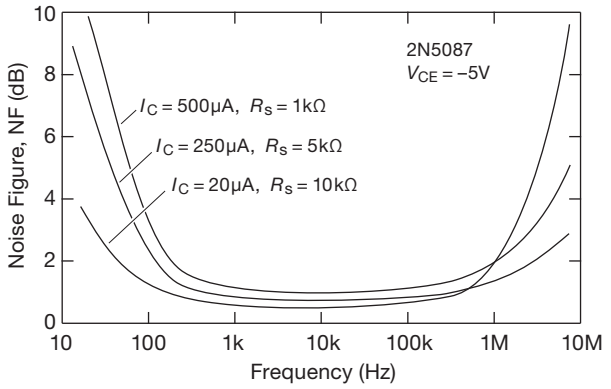


Figure 8.30. Noise figure (NF) versus frequency, for three choices of I_C and R_s, for the 2N5087.

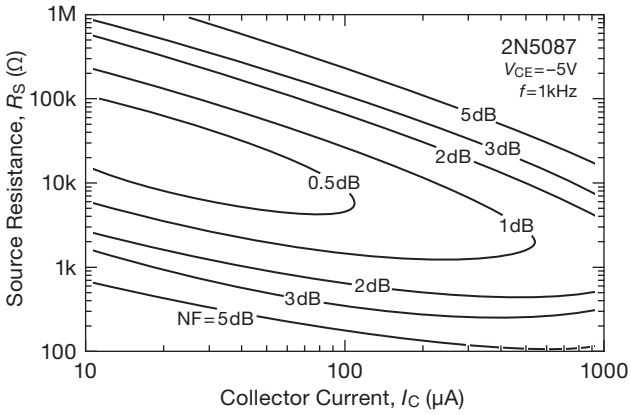


Figure 8.31. Contours of constant narrowband noise figure for the 2N5087 transistor. (From the ON Semiconductor datasheet.)

Reading from Figure 8.29, at I_C=20 μA, e_n=3.7 nV/$\sqrt{\text{Hz}}$ and i_n=0.17 pA/$\sqrt{\text{Hz}}$; and $4kTR_s$=1.65 × 10^{-16} V^2/Hz for the 10k source resistance. The calculated noise figure is therefore 0.42 dB. This is consistent with the datasheet's graph (Figure 8.30) showing NF versus frequency, in which they have chosen this operating current for R_s=10k. This choice of collector current is also roughly what you would get from the graph in Figure 8.31 of noise-figure contours at 1 kHz, although the actual noise figure can be estimated only approximately from that plot as being slightly less than 0.5 dB.

Exercise 8.2. Find the optimum I_C and corresponding noise figure for R_s=100k and f=1 kHz, using the graph in Figure 8.29 of e_n and i_n. Check your answer from the noise-figure contours (Figure 8.31).

For the other amplifier configurations (follower, grounded base) the noise figure is essentially the same, for given R_s and I_C, since e_n and i_n are unchanged. Of course, a stage with unity voltage gain (a follower) may just pass the problem along to the next stage,[33] since the signal level hasn't been increased to the point that low-noise design can be ignored in subsequent stages.

8.5.2 Charting amplifier noise with e_n and i_n

The noise calculations just presented, although straightforward, make the whole subject of amplifier design appear somewhat formidable. If you misplace a factor of Boltzmann's constant, you suddenly get an amplifier with a 10,000 dB noise figure! In this section we present a simplified noise-estimation technique of great utility.

The method consists of first choosing some frequency of interest in order to get values for e_n and i_n versus I_C from the transistor datasheets. Then, for a given collector current, you can plot the total noise contributions from e_n and

[33] "Kicking the can down the road," in the political parlance of our time.

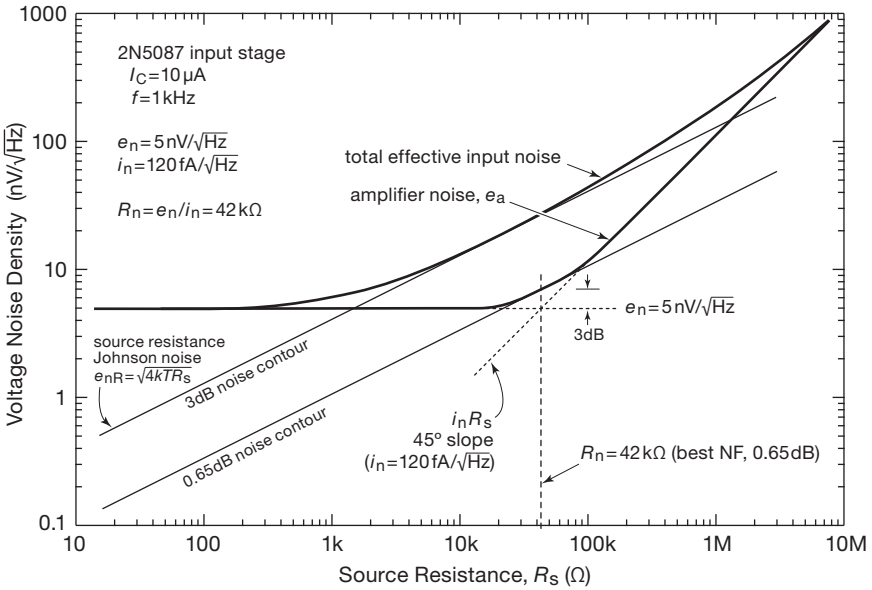


Figure 8.32. Total amplifier input voltage noise (e_a) plotted from the e_n and i_n parameters. The "total effective input noise" curve includes the Johnson noise from a resistive input source, i.e., $Z_s=R_s$.

i_n as a graph of e_a versus source resistance R_s. Figure 8.32 shows what that looks like at 1 kHz for an input stage using a 2N5087 *pnp* transistor running at 10 µA of collector current. The e_n noise voltage is constant, and the i_nR_s voltage increases proportional to R_s, i.e., with a 45° slope. The amplifier noise curve is drawn as shown, with care being taken to ensure that it passes through a point 3 dB (voltage ratio of 1.4) above the crossing point of individual voltage- and current-noise contributions. Also plotted is the noise voltage of the source resistance, which is also the 3 dB NF contour. The other lines of constant noise figure are simply straight lines parallel to this line, as you will see in the examples that follow.

The best noise figure (0.65 dB) at this collector current and frequency occurs for a source resistance of 42kΩ, and the noise figure is easily seen to be less than 3 dB for all source resistances between 2kΩ and 1MΩ, the points at which the 3 dB NF contour intersects the amplifier noise curve.

The next step is to draw a few of these noise curves on the same graph, using different collector currents or frequencies, or maybe a selection of transistor types, in order to evaluate amplifier performance. Before we go on to do that, let's show how we can talk about this same amplifier using a different pair of noise parameters, the noise resistance R_n and the noise figure NF(R_n), both of which pop right out of the graph.

We'll have some fun with this technique later (§8.7), after we've learned about noise in JFETs, by refereeing over a world championship low-noise shootout between a best-in-class BJT (2SD786) and a comparably excellent JFET (2SK170).

8.5.3 Noise resistance

The lowest noise figure in this example occurs for a source resistance R_s=42k, which equals the ratio of e_n to i_n. That defines the noise resistance:

$$R_n = \frac{e_n}{i_n} \quad \text{ohms.} \tag{8.32}$$

You can find the noise figure for a source of that resistance from our earlier expression, eq'n 8.31, for noise figure:

$$\text{NF(at } R_n) = 10\log_{10}\left(1 + 1.23\times10^{20}\frac{e_n^2}{R_n}\right) \text{ dB} \quad \approx \quad 0.31\,\text{dB}.$$

Noise resistance isn't actually a real resistance in the transistor, or anything like that. It is a tool to help you quickly find the value of source resistance for minimum noise

figure, ideally so that you can vary the collector current to shift R_n close to the value of source resistance you're actually using. R_n corresponds to the point where the e_n and i_n lines cross.

The noise figure for a source resistance equal to R_n then follows simply from the preceding equation.

8.5.4 Charting comparative noise

It's easy to compare candidate transistors with this charting technique, plotting total amplifier noise for each of a selection of possible collector currents. We've done this in Figure 8.33, where we compare total amplifier noise (which includes source resistance Johnson noise) versus source resistance for the high-beta 2N5962 and the low-$r_{bb'}$ ZTX851 *npn* transistors, using measured values of beta and $r_{bb'}$ listed in Table 8.1a on page 501. You can see that the high-beta part operated at low collector current is the clear winner for high source resistances, where its relatively high $r_{bb'}$ (480 Ω) is harmless, being swamped by the Johnson noise of the signal source. By contrast, for low source resistances (say 1k and below) the ZTX851's admirably low $r_{bb'}$ ($\sim$1.7Ω) achieves the lowest noise, particularly when operated at a relatively high collector current to minimize the "r_e" noise term (recall that collector-current shot noise through $1/g_m$ produces a noise voltage density e_n equivalent to the Johnson noise of a resistor of value $r_e/2$, eq'n 8.20).

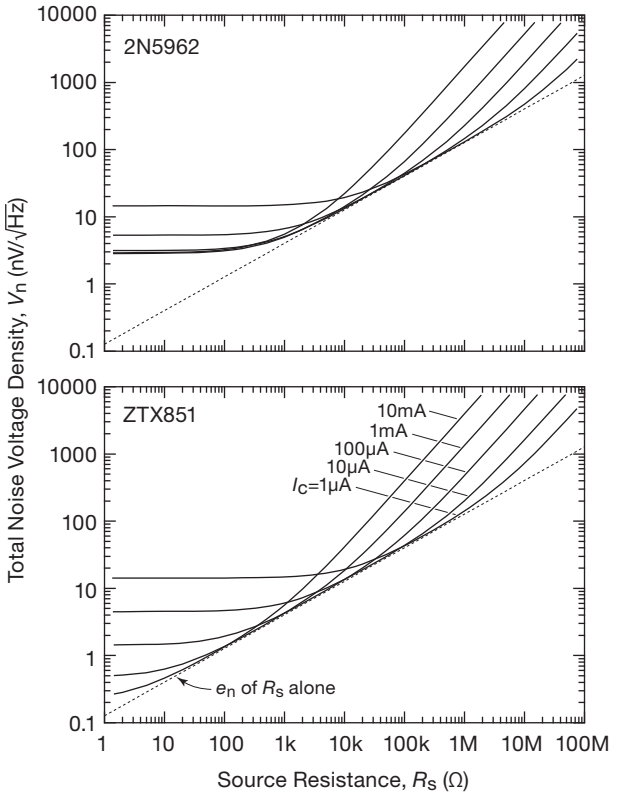


Figure 8.33. Comparing the total amplifier input-noise voltage versus source resistance of two candidate low-noise BJTs. The curves plot five decade values of collector current. For low source resistances the ZTX851's low $r_{bb'}$ yields low noise at high collector currents; by contrast, the 2N5962's higher $r_{bb'}$ limits the ultimate noise voltage, but its higher beta (thus lower base-current shot noise, $\sqrt{2qI_C/\beta}$) results in improved performance with high source resistances. These curves include the Johnson noise of the source resistance.

8.5.5 Low-noise design with BJTs: two examples

Let's put these ideas and equations to work, first by looking at a simple single-ended low-noise audio preamp that achieved popularity beginning in the 1980s, then by comparing with a classic differential design that addresses many of the shortcomings of the single-ended circuit.

A. The Naim preamp

Figure 8.34 shows the input stage used for many years in low-level preamps from the British manufacturer Naim Audio. It's an ac-coupled single-ended two-stage series feedback design (Figure 2.92), tailored for low input-noise voltage. The overall voltage gain is $G_V = 1 + R_f/R_E$ (here 30 dB), with R_E chosen rather small to keep its Johnson noise-voltage contribution down below a nanovolt per root hertz (15 Ω has e_n=0.5 nV/$\sqrt{\text{Hz}}$). The other significant noise term is the transistor's contribution from $r_{bb'}$, which is variously specified on datasheets either as an e_n value (at some specified collector current) or graph, or as a noise-figure value or graph, or (rarely) as a value for $r_{bb'}$ itself; or (worst of all) perhaps none of these. For the "Low Noise" ZTX384C used in this preamp, the datasheet is rather, uh, *taciturn*, revealing only that NF=4 dB (max) from 30 Hz–15 kHz for R_s=2k and I_C=0.2 mA. This is not terribly helpful, because that corresponds to a seriously large Johnson-noise voltage of 6.9 nV/$\sqrt{\text{Hz}}$.

The fact is that these amplifiers were pretty quiet. One reason is that worst-case specifications of difficult-to-measure parameters tend to be overly pessimistic. For example, the contemporary low-noise *npn* 2SC3324 specifies (for a particular test condition) NF=3 dB (max), but 0.2 dB (typ). Naim may have selected parts for low noise. The other reason is that Naim's circuit actually uses five transistors in parallel (each with a 15 Ω emitter ballasting resistor) for Q_1, which drives down its noise voltage by a factor

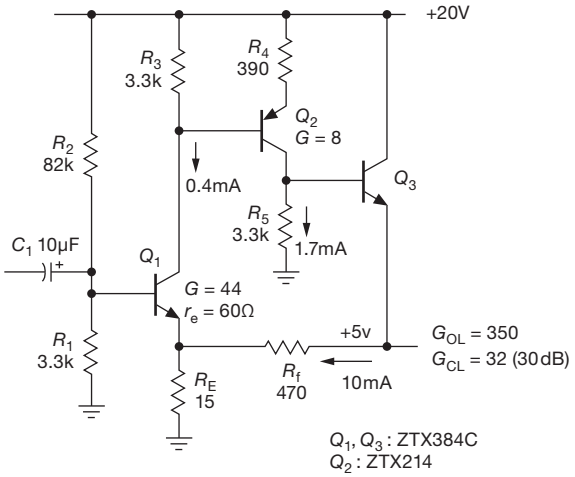


Figure 8.34. Low-level audio preamp, similar to that used in the NAIM NA323. Q_1 is made from a parallel connection of four or five selected transistors.

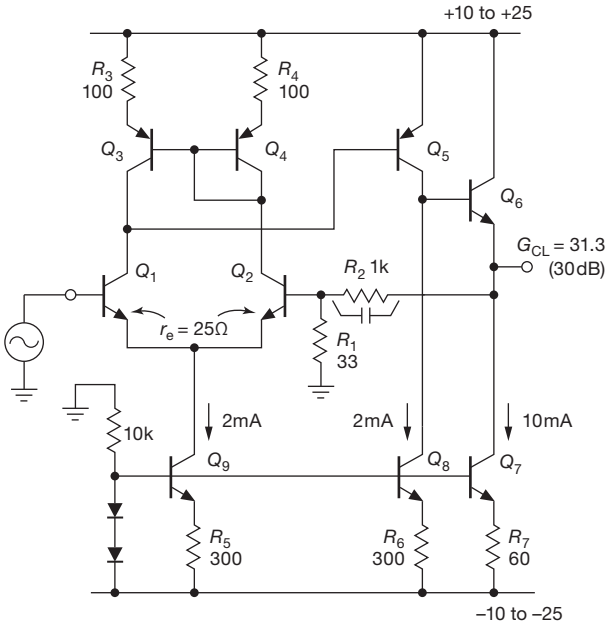


Figure 8.35. A low-noise dc-coupled audio preamplifier with predictable biasing.

of $\sqrt{5}$. Bottom line: this preamp's noise was likely in the neighborhood of $1\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$, as befits a preamp designed for the low signal levels characteristic of the moving-coil phono cartridges for which it was optimized.

This circuit is simple; but there are tradeoffs caused by the small R_E that was chosen for low noise. It makes the bias scheme somewhat problematical, with the bias divider R_1R_2 acting more as a current source (compared with the classic voltage biasing, for example as in Figure 2.35), thus making the quiescent operating point rather dependent on Q_1's beta. The low value of R_E also results in a low input impedance, owing to the required stiff bias divider's impedance of 3.2 kΩ.[34]

B. A better design: dc-coupled differential preamp

Many of the shortcomings of the Naim preamp are nicely addressed in the dc-coupled differential preamp shown in Figure 8.35, which retains the former's desirable low-noise performance. It's basically the classic 2-stage op-amp configuration (e.g., Figures 4.43 or 2.91), here simplified with a single-ended class-A output stage with current-sink active load (with the assumption that the next stage will not require substantial drive current).

This circuit eliminates electrolytic blocking capacitors, both at the input and in the gain-setting feedback divider R_1R_2; it minimizes Early effect nonlinearity in the input stage; and its use of active (current-source) loads both improves its overall linearity (through improved single-stage linearity, plus higher loop gain) and provides stable and predictable biasing. The price you pay in exchange is small: increased circuit complexity, and the 3 dB increase in noise (owing to Q_2's contribution). At these collector currents the noise is dominated by the Johnson noise of the base resistance $r_{bb'}$. So, as with the Naim amplifier, the input transistors could be implemented as a parallel array of matched-V_{BE} transistors (even if single transistors are used, the pair should be matched to ∼10 mV or so), or (better) implemented with larger die-size transistors with lower $r_{bb'}$.[35]

8.5.6 Minimizing noise: BJTs, FETs, and transformers

Bipolar transistor amplifiers can provide very good noise performance over the range of source impedances from about 200 Ω to 1 MΩ, with corresponding optimum collector currents generally in the range of several milliamps down to a microamp. (With low source impedances you want to minimize e_n, whereas with high source impedances

[34] Which dominates the impedance seen looking into the base ($\beta R_E G_{loop}$), the latter about 50 kΩ.

[35] For example the inexpensive 2SD2653, specified to 2A continuous collector current, but with high beta even at low currents: $\beta \approx 500$ at I_C=1 mA. See §8.5.9B.

you want to minimize i_n; as we've seen, that dictates high and low collector current, respectively).

If the source impedance is high, say greater than 100k or so, transistor *current* noise dominates, and the best device for low-noise amplification is a FET. Although their voltage noise is usually greater than that of bipolar transistors, the gate current (and its noise) can be exceedingly small, making them ideally suited for low-noise high-impedance amplifiers. We'll dive into FET noise shortly (§8.6), after an entertaining "challenge" interlude.

For very low source impedances (say 50 Ω), transistor voltage noise will always dominate, and noise figures will be poor. One approach in such cases is to use a signal transformer to raise the signal level (and impedance), treating the signal on the secondary as before. Transformers have their drawbacks, of course: they're ac coupled; they operate over only a few decades of bandwidth (and never to dc); those intended for low-frequency operation are bulky and expensive and exhibit nonlinearities; and they are susceptible to magnetic pickup. Nevertheless, they can be the magic sauce when you're dealing with a low impedance signal (say less than 100 Ω); see §8.10.

8.5.7 A design example: 40¢ "lightning detector" preamp

Here's an interesting design challenge, and a chance to exercise some of our noise theory. Imagine we want to manufacture an inexpensive photodiode amplifier that runs on low current from a 9V battery, with input noise of just a few nV/$\sqrt{\text{Hz}}$, and with response time of a few microseconds. You can think of this as a simple "lightning detector," because lightning is almost unique in delivering microsecond-scale pulses of light in the outdoor environment.[36] In a fully refined implementation this could serve as a useful early warning device to install in locations where you'd want to be indoors when the sky is becoming electric – golf courses, harbors, soccer fields, etc. Here we explore the basics, a launching point to inspire the electronic hobbyist. It will turn out that we can do the job with standard parts, for a total component cost (in 1000-piece quantities) of just $0.40. It will be a "throwaway" device, manufactured in the millions, with commensurate profits and fame (dream on!). Let's take it in easy stages, as shown in Figures 8.36A–E (similarly labeled in the following paragraphs).

A. Block diagram We back-bias the PIN photodiode to lower its capacitance (thus lower noise and faster response), and we use a blocking capacitor into the summing junction of a transimpedance stage to eliminate the dc level from ambient light and leakage current. The feedback resistor R_f sets the gain ($G = -R_f$ volts/amp).

B. Discrete design, first iteration We need inverting feedback, so we begin with a grounded-emitter stage (Q_1), with an emitter follower (Q_3) to create a low-impedance output and feedback source.

C. Add cascode This circuit will be running at low currents (0.1 mA or so), thus relatively high impedances at which Miller effect causes a significant bandwidth reduction. So we add a cascode (Q_2) atop the gain stage's collector.

D. Inverted cascode A 9 V battery is down to 6 V at end-of-life, so now we're running out of headroom! We want to keep plenty of dynamic range, so we fix this problem by turning the cascode around, creating an "inverted cascode," in which collector current variations in Q_1 pass through *pnp* cascode transistor Q_2, while the latter continues to clamp Q_1's collector voltage. The output follower plays a dual role here, with its V_{BE} setting Q_2's collector current: $I_{C2} = V_{BE3}/R_3$.

E. Front-end follower and input current noise At the ∼50 μA collector current in Q_1 that we'll need to get sufficient bandwidth, there would be too much current noise at the input. (Remember that the input signal is a *current*.) So we add a high-beta follower (Q_4) and choose a collector current of 1 μA as an initial trial value. We also choose a large feedback resistor R_2 to minimize its noise current as seen at the input, with shunt capacitance C_c to limit the bandwidth to 100 kHz. The idea is to calculate the input noise to see where the dominant contribution comes from, then iterate the design toward an optimum. We've filled in component values for an initial design.

[36] Roughly 90% of lightning activity is cloud-to-cloud, stretching over large distances and exhibiting relatively long rise- and fall-times; their signals look not unlike typical background "cultural" noise. The remaining 10% are lightning strikes to ground (which matter to us!), with unique short rise-time signatures, typically less than 5 μs in duration (even for distant ground strikes). This allows us to create filters to distinguish weak distant cloud-to-ground lightning strikes from nearby transient interference of other origins. The circuit described here is a simple low-noise preamp suitable for use with such filters and discriminators.

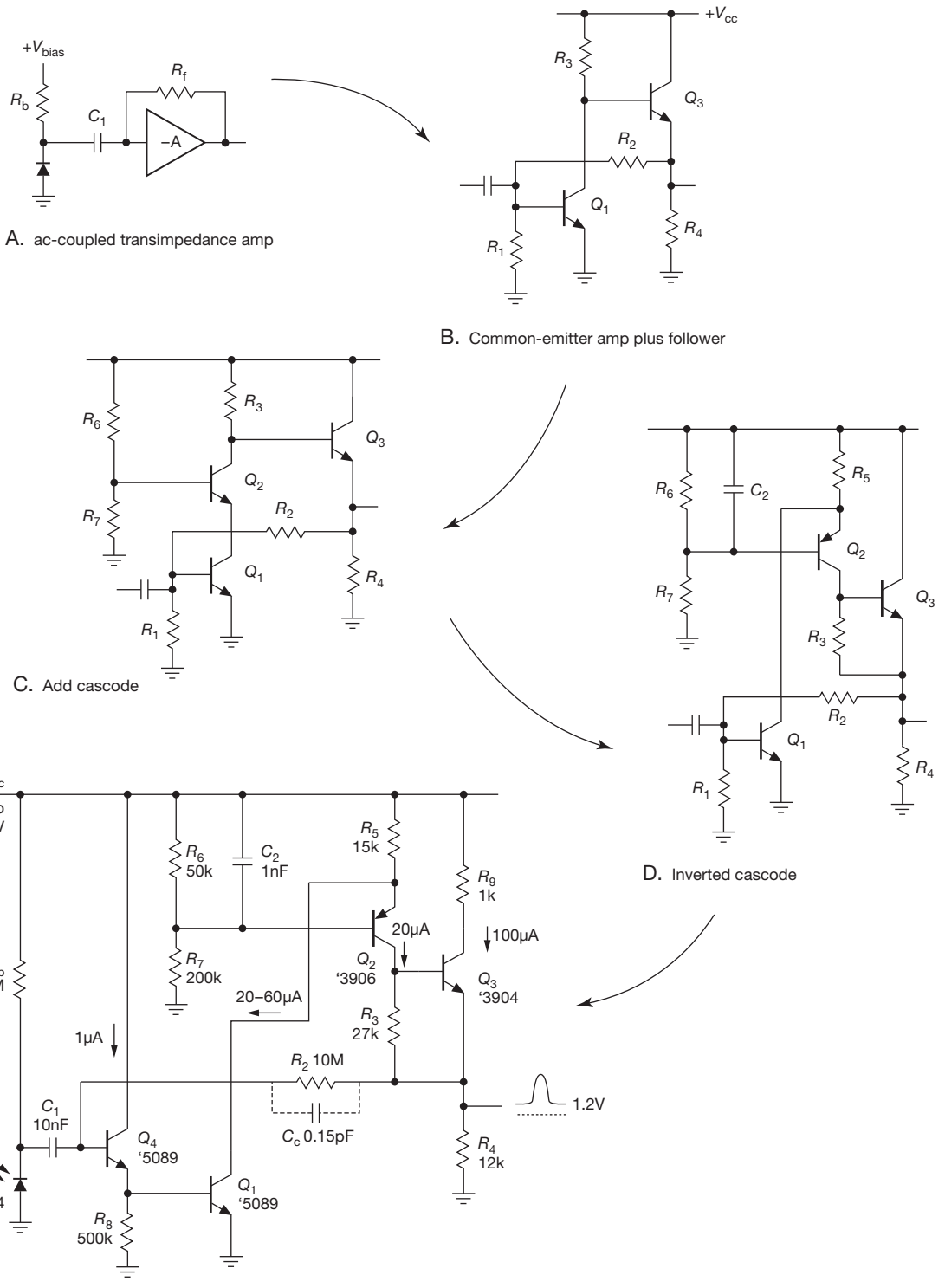


Figure 8.36. Tutorial evolution of a low-noise and low-power photodiode amplifier built with low-cost discrete components. Apart from Q_1 and the photodiode, all parts are available in surface-mount packages (prepend "MMBT" to transistor part numbers, e.g., MMBT5089).

Calculating the noise performance Let's estimate the noise current at the input; that's what competes with the (current) signal from the photodiode. There's shot noise in Q_4's base current, and there's Johnson-noise current from the feedback resistor R_2 and the photodiode bias resistor R_b. In addition, we have to worry about noise *voltage* at the input: in combination with the input capacitance C_{in} it creates an input noise current $i_n = e_n \omega C_{in}$ (as we'll see in detail in §8.11). Input noise voltage is the combined e_n contributions of Q_1 and Q_4, each having Johnson noise in their $r_{bb'}$ and collector shot noise through their r_e. Let's take these in turn.

Noise current Estimating Q_4's $\beta \approx 350$ at its 1 μA collector current, we find that its base current of 3 nA creates a shot-noise density $i_{n(shot)} = \sqrt{2qI_B} = 30\,\text{fA}/\sqrt{\text{Hz}}$. Feedback resistor R_2 and bias resistor R_b create a Johnson-noise current $i_{n(R)} = \sqrt{4kT/R} = 57\,\text{fA}/\sqrt{\text{Hz}}$. These combine (square root of sum-of-squares) to create $i_n(\text{total}) = 65\,\text{fA}/\sqrt{\text{Hz}}$.

Noise voltage Emitter follower Q_4, running at 1 μA, generates a shot noise voltage across its r_e of $e_{n(shot4)} = r_e\sqrt{2qI_C} = 14.3\,\text{nV}/\sqrt{\text{Hz}}$. Its internal base resistance of $\sim 300\,\Omega$ adds a relatively insignificant Johnson-noise voltage $e_{n(J4)} = \sqrt{4kTr_{bb'}} = 2.2\,\text{nV}/\sqrt{\text{Hz}}$, for a combined $e_{n4} = 14.5\,\text{nV}/\sqrt{\text{Hz}}$.

Grounded-emitter amplifier Q_1, running at nominal 40 μA and with an $r_{bb'}$ of $\sim 300\,\Omega$, generates corresponding noise voltages $e_{n(shot1)} = r_e\sqrt{2qI_C} = 2.3\,\text{nV}/\sqrt{\text{Hz}}$ and $e_{n(J1)} = \sqrt{4kTr_{bb'}} = 2.2\,\text{nV}/\sqrt{\text{Hz}}$, for a combined $e_{n1} = 3.2\,\text{nV}/\sqrt{\text{Hz}}$.

Combining the noise voltages of Q_1 and Q_4, we find[37] $e_n(\text{total}) = 14.8\,\text{nV}/\sqrt{\text{Hz}}$. Evidently Q_4 is the big noise gorilla in the room here; but let's continue with the analysis.

In combination with the input capacitance of ~ 10 pF (5 pF for the photodiode, 2.5 pF for Q_4's C_{cb}, and 2.5 pF for wiring capacitance) that noise voltage creates an effective input-current noise (§8.11.3)of $i_n = e_n \omega C_{in} = 90\,\text{fA}/\sqrt{\text{Hz}}$, if we take for ω a characteristic frequency of 100 kHz.

Verifying feedback stability This is a feedback circuit, with the ever-present potential for oscillation. And, in a transresistance configuration like this, the shunt capacitance at the input combines with the large-value feedback resistor to introduce an additional lagging phase shift. We deal with this in detail in §§*4x.3* and 8.11; the criterion for stability is that the open-loop amplifier's unity-gain bandwidth must satisfy

$$f_T(\text{open-loop}) > f^2_{R_2C_c}/f_{R_2C_{in}}. \tag{8.33}$$

(Stated differently, the feedback network's -3 dB rolloff frequency must be less than the geometric mean of (a) the amplifier's open-loop unity-gain frequency and (b) the -3 dB rolloff frequency of the feedback resistor in combination with the input capacitance.)

So, for the circuit of Figure 8.36E we require $f_T(\text{open-loop}) > 106\text{kHz}^2/1.6\text{kHz}$, or 7 MHz. We estimate the amplifier's open-loop unity gain as follows.

(a) The low-frequency voltage gain is $G = g_{m1}R_{load}$, where R_{load} is the impedance seen at cascode transistor Q_2's collector.
(b) The latter is $R_{load} = R_4 \cdot g_{m3}R_3 \approx 1.3\,\text{M}\Omega$, so the low-frequency open-loop gain G_{OL} ranges from 1000 to 3000 for Q_1 collector currents from 20 μA to 60 μA, (battery voltage from 6 V to 9 V), respectively.
(c) This gain rolls off at 6 dB/octave, starting at a frequency set by R_{load} in combination with the capacitance seen at Q_2's collector; taking $C_{cb} \sim 2.5$ pF for each of Q_2 and Q_3, that's $f_{3dB} = 25$ kHz.
(d) That brings the amplifier's open-loop gain down to unity at about 25 MHz (for Q_1's minimum collector current), a good margin of stability given our 7 MHz requirement.

The cascode was essential for getting this bandwidth at these low currents. Without it the gain stage (Q_1) would see its load capacitance multiplied by the voltage gain of 1000, owing to Miller effect.[38]

Optimization The contributions from amplifier noise currents ($65\,\text{fA}/\sqrt{\text{Hz}}$) and from noise voltage generating noise current through the input capacitance ($90\,\text{fA}/\sqrt{\text{Hz}}$) are in the same ballpark. The latter rises proportional to frequency, and dominates, slightly, at the high-frequency end of the bandpass; but when integrated over the amplifier's passband, the two noise terms contribute comparable integrated noise, totaling approximately $I_n = 30$ pArms (therefore 0.3 mVrms at the amplifier's output).

The lion's share of the noise current, then, comes from the 10M gain-setting and biasing resistors. These could be raised in value, though R_b must not be so large that the photodiode saturates in nighttime ambient light. The latter problem could be circumvented by sensing the voltage at the photodiode and using it to control a quiet biasing current source. If the resistor noise were brought down

[37] At these low collector currents we found actual measured noise voltages to be about 25% higher than those predicted by simple theory.

[38] See also the section "Analysis by Equivalent Circuits" in Chapter *2x* for a discussion of classical feedback analysis in circuits like this.

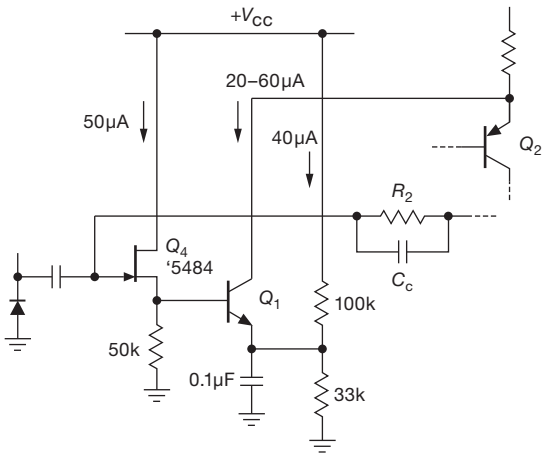


Figure 8.37. Replacing Figure 8.36's BJT input transistor with a low-capacitance JFET reduces input noise voltage and noise current by a factor of 3.

enough, the dominant remaining noise source would be Q_4, whose voltage noise (multiplied by ωC_{in}) and current noise are comparable. Raising its operating current reduces its voltage noise, but increases its current noise. We seem to be stuck.

But no, we're *not*! If we replace the BJT input follower with a low-capacitance JFET (Figure 8.37), we can score a significant improvement. The voltage noise of a 2N5484 ($C_{in}\approx 2.2$ pF) running at 50 μA is about $e_n = 5\,\text{nV}/\sqrt{\text{Hz}}$ (based on measurements at 100 μA), and its current noise is negligible; so we reduce the transistor noise contribution by a factor of 3 (for an additional cost of 5–10 cents). Of course, this makes sense only if the R_b and R_2 Johnson-noise contributions can be reduced by at least a factor of 4. With those improvements the amplifier's total integrated input noise becomes $I_n \approx 10$ pArms. That's comparable to shot noise in the photodiode's leakage current (3 nA, typ, at -10 V and 25°C) – no point in grooming the amplifier further.

We take up the subject of low-noise design with JFETs shortly, in §8.6.

8.5.8 Selecting a low-noise bipolar transistor

It's important to select the right transistors, and appropriate operating currents, in order to optimize a circuit's low-noise performance. This task has been made more difficult as some of the finest low-noise favorites have been recently discontinued by one manufacturer after another (most of these came in through-hole packages, not favored for machine assembly). Often this means you can't simply choose a part used in someone else's good design. The task is made difficult also by the lack of relevant information in datasheets – noise-voltage curves, or even a mention of noise voltage is the exception in most contemporary BJT datasheets. We come to your rescue with Tables 8.1a,b, where we've listed a range of good candidate transistors,[39] along with measured noise parameters such as e_n, $r_{bb'}$, V_A (Early voltage), and plots of beta versus collector current. Herewith a guide to the table, and to the task of selecting and using a low-noise transistor.[40]

The **beta** or h_{FE} column shows the manufacturer's stated minimum values. While the *typical* beta is often much higher, you have to accept the reality that a minimum-beta part may end up in your circuit (unless you are willing to screen incoming parts). The manufacturer's beta specs are often at higher currents than you'll be using, and the beta may be severely degraded at lower currents, see the discussion of recombination and Gummel plots in §*2x.13*. To see which transistors are susceptible to this, we measured beta versus collector current, going from 1 μA to 100 mA, with the results shown in Figure 8.39 (with corresponding labels in the **Beta plot** column of Tables 8.1a,b) . The falloff of beta at low current is summarized in the **Linearity** column in the table, rating the transistors on a scale of 1 (large beta falloff at microamp currents) to 5 (beta does not change with current). Note how some parts run out of steam at currents above 1mA; these are small-die parts running at high current density. Keep in mind our measurements were taken at $V_{CE}=5$ V, and that the high-current beta of some BJTs falls rapidly at lower collector voltages (a saturation effect).[41] Our measured plots go to 50 mA, but the datasheets of some larger-die parts (like the ZTX851) provide curves of beta going as high as 10 A.

Some BJT types are measured by the manufacturer and

[39] In many cases these transistors are not promoted by the manufacturer as "low noise." We made some guesses, then ordered and tested samples of some 100 different types.

[40] Some advice about finding parts. It's helpful to search using the inner (numeric) portion of the part name, because there may be versions with differing prefixes or suffixes. You'll have an easier time finding surface-mount parts than through-hole parts – see the notes in the SOT-23 column of Table 8.1a. Some parts are stocked only at specialty suppliers, such as B&D Enterprises, MCM, Donberg Electronics, Encompass, and Littlediode. Discontinued parts may also be available on eBay, in small quantities suitable for making specialized laboratory instruments. Some large Pacific rim distributors have good older-part inventory and sell from their websites, through Alibaba, or on eBay. The Sanyo four parts near the bottom of the table have been discontinued, but they make other parts that may offer similar performance.

[41] We've not provided measurements of beta at lower voltages, but we are confident that the low-noise winners (the *npn* ZTX851 and *pnp* ZTX951) do very well in this respect, having been designed as high-current transistors with low saturation voltage.

Table 8.1a Low-noise BJTs[a]

Part #	Manuf	npn	pnp	Names & Packages: TO-92	SOT-23	Small SMT	SOT-223	TO-126	TO-220	Pinout[u]	V_{CEO} max (V)	f_T typ (MHz)	C_{ob} typ (pF)	beta @ I_C min	(I_C mA)	Linearity[e]	Beta plot	Early V_A meas (V)	Noise parameters: measured e_n @ I_C (nV/◊)	(I_C mA)	measured $r_{bb'}$[a] (Ω)	"optimum"[o] I_C (mA)	e_n (nV/◊)	i_n (pA/◊)	R_n (Ω)
BCX70J	m	•	-	-	•	-	-	-	-	D	45	250	1.7	250	2	4	37	215	3.54	10	**760**	0.015	5.1	0.14	37k
BC850C	I	•	-	b	•	a	-	-	-	D	45	250	2.5	420	2	4	37	220	3.32	1	**650**[r]	0.017	4.7	0.11	41k
BC860C	I	-	•	b	•	a	-	-	-	D	45	250	2.5	420	2	3	17	30	3.09	1	**590**[r]	0.02	4.5	0.38	12k
MPSA18	m	•	-	•	-	-	-	-	-	A	45	160	1.7	500	0.1	4	41	-	6.5[d]	0.1	**-**	-	-	-	-
2SC3624A	H	•	-	-	•	-	-	-	-	D	50	250	3	1000	1	-	-	60[d]	-	-	**-**	-	-	-	-
2N5962	m	•	-	•	m	-	-	-	-	A,D	45	100	1.5	500	0.1	2	42	60	2.77	10	**480**	0.02	4.0	0.38	10k
MMBT6429	O	•	-	-	m,p	-	-	-	-	D	45	400	1.3	500	0.1	5	39	140	2.27	10	**310**	0.04	3.2	0.15	21k
2N5089	m	•	-	•	m,p,c	-	-	-	-	A,D	25	50	1.3	450	1	5	38	200	2.15	10	**290**	0.04	3.1	0.17	19k
2N5088	m	•	-	•	m,p	-	-	-	-	A	30	50	1.3	300	0.1	5	38	180	1.97	10	**240**	0.05	2.9	0.22	13k
MPS8098	m	•	-	•	-	-	-	-	-	A	60	150	2.6	100	1	4	27	435	1.79	10	**195**	0.06	2.6	0.43	6.0k
2N5087	m	-	•	•	m	-	-	-	-	A,D	50	40	2.4	250	0.1	4	15	55	0.81	10	**40**	0.28	1.2	0.60	2.0k
2N5210	C	•	-	•	m	-	-	-	-	A,D	50	50	1.3	250	1	5	35	270	1.92	10	**230**	0.05	2.8	0.25	11k
2SC2412-R	R	•	-	a	•	a	-	-	-	D	50	180	2	180	0.1	5	32	360	1.66	10	**165**	0.07	2.4	0.35	6.8k
2SA1037-R	R	-	•	a	•	a	-	-	-	D	50	140	4	180	0.1	-	-	60[d]	-	-	**-**	-	-	-	-
2SC2712-GR	T	•	-	-	•	-	-	-	-	D	50	80	2	200	2	5	32	390	1.57	10	**150**	0.07	2.3	0.34	6.6k
2SA1162-GR	T	-	•	-	•	-	-	-	-	D	50	80	4	200	2	4	10	50	1.40	10	**120**[y]	0.09	2.0	0.38	5.2k
2SC3906K	R	•	-	a	•	a	-	-	-	D	120	140	2.5	180	2	5	35	280[d]	-	-	**-**	-	-	-	-
2SA1514K	R	-	•	a	•	a	-	-	-	D	120	140	3.2	180	2	4	12	165	1.35	10	**110**	0.10	1.9	0.42	4.6k
2N5401		-	•	•	m	-	-	-	-	A,D	160	150	6	50	1	5	2	-	-	-	**-**	-	-	-	-
2N5551[s]		•	-	•	m	-	-	-	-	A,D	160	150	2.7	80	1	5	25	-	-	-	**105**	0.11	1.3	0.65	2.0k
2N3904	m	•	-	•	m	-	a	-	-	A,D	40	300	2.5	70	1	5	31	340	1.35	10	**110**	0.10	1.9	0.68	2.8k
2N3906	m	-	•	•	m	-	a	-	-	A,D	40	250	3	80	1	1	3	25	0.74	10	**32**	0.35	1.0	1.18	890
2SC3324[g]	T	•	-	-	•	-	-	-	-	D	120	100	3	200	2	5	35	560	0.78	10	**35**	0.32	1.1	0.71	1.5k
2SA1312	T	-	•	-	•	-	-	-	-	D	120	100	4	200	2	5	13	180	0.58	10	**20**	0.56	0.8	0.94	880
2SB1197K-Q[b]	R	-	•	-	•	-	-	-	-	D	32	200	12	120	100	5	6	110	0.60	10	**20**	0.56	0.8	1.22	680
2N4401	m	•	-	•	m	a	-	-	-	A,D	40	250	4	80	10	5	31	410	0.84	10	**40**	0.28	1.2	1.05	1.1k
2N4403	m	-	•	•	m	a	-	-	-	A,D	40	200	5.5	50	10	1	1	-	0.55	10	**17**	0.65	0.8	2.05	370
2SD2653	R	•	-	-	•	-	-	-	-	D	12	360	20	270	200	4	37	65	0.54	10	**17**	0.65	0.8	0.88	860
2SB1690K	R	-	•	-	•	-	-	-	-	D	12	360	15	270	200	5	19	10	0.52	10	**15**	0.74	0.7	0.94	760
2SB1424	R	-	•	a	-	•	-	-	-	F	20	240	35	120	100	4	5	30[d]	0.35	10	**9.4**	1.2	0.6	1.78	320
ZXTN19020D	Z	•	-	-	•	-	-	-	-	D	20	160	33	300	100	3	36	100	0.40	10	**11**	1.0	0.6	1.04	590
2SD1898	R	•	-	a	-	-	d	-	a	F	80	100	20	120	500	-	-	580	0.40	10	**8.3**	1.3	0.5	1.89	280
2SB1260	R	-	•	a	-	-	d	-	a	F	80	100	20	120	100	-	-	-	0.36	10	**7**	1.6	0.5	2.06	240
MPS8599	O	-	•	•	m	-	-	-	-	A,D	80	150	2.9	100	1	5	6	170	0.47	10	**12**	0.9	0.6	1.72	370
MPS8099	O	•	-	•	m	-	-	-	-	A,D	80	150	2.5	100	1	4	27	540	0.37	10	**8**	1.4	0.5	2.11	250
DSS20201L[b]	D	•	-	-	•	-	-	-	-	D	20	150	16	250	10	1	30	150	0.38	10	**8.0**	1.4	0.5	1.34	390
ZTX450	Z	•	-	•	-	-	-	-	-	A	45	150	15 m	100	150	3	26	730	0.39	10	**8.5**	1.3	0.5	2.05	260
ZTX550	Z	-	•	•	-	-	-	-	-	A	45	150	-	100	150	5	6	-	0.38	10	**7.7**	1.4	0.5	2.15	240
ZTX618	Z	•	-	•	f	-	-	-	-	A,D	20	140	23	200	10	3	36	90	0.41	10	**9.3**	1.2	0.6	1.38	410
ZTX718	Z	-	•	•	f	-	-	-	-	A,D	20	180	21	300	10	3	18	25	0.38	10	**7.3**	1.5	0.5	1.27	390
ZXTN19100C	Z	•	-	-	•	-	-	-	-	D	100	150	16	200	100	3	34	480	0.36	10	**6.8**	1.6	0.48	1.61	300
2SD1684[b,x]	S	•	-	-	-	-	-	•	-	E	100	120	11	100	100	2	30	380	0.29	10	**4.0**	2.8	0.37	2.98	125
2SB1243Q[b]	R	-	•	-	-	-	-	-	v	G2	50	70	50	120	500	3	7	180[d]	0.28	10	**3.7**	3.0	0.35	2.83	125
BD437	m	•	-	-	-	-	-	•	-	E	45	3	65	30	10	2	21	1100	0.27	25	**3.9**	2.8	0.36	5.51	65
BU406	O	•	-	-	-	-	-	-	•	G	200	10	80	45[t]	100	1	z	150[d]	0.26	10	**3.1**	3.5	0.33	5.02	65
2SC3955[v,x]	S	•	-	-	-	-	-	•	-	E	200	300	1.9	40	10	5	22	600[d]	0.24	10	**2.3**	4.8	0.28	6.20	45
2SD786-S[x]	R	•	-	•	-	-	-	-	-	B	40	100	13	270	10	4	33	320	0.24	10	**2.3**	4.8	0.28	2.39	120
2SB737[x]	R	-	•	•	-	-	-	-	-	B	40	100	25	270	10	4	7	140	0.21	10	**1.7**	6.5	0.24	2.78	85
ZXTN2018F	Z	•	-	-	•	-	-	-	-	D	60	130	28	100	10	2	28	4600	0.27	10	**3.3**	3.4	0.33	3.28	100
ZXTP2027F	Z	-	•	-	•	-	-	-	-	D	60	165	44	100	10	4	7	100	0.21	10	**1.46**	7.6	0.22	4.93	45
2SC6102[b,x]	S	•	-	-	-	-	-	•	-	E	30	290	40	200	500	5	32	280	0.23	10	**2.0**	5.5	0.26	2.98	88
2SC3601[v,x]	S	•	-	-	-	-	-	•	-	E	200	400	2	40	10	5	25	500[d]	0.22	10	**1.61**	6.9	0.23	7.43	31
ZTX851	Z	•	-	•	-	-	z	-	-	A,F	60	130	45	100	10	2	23	410	0.18	25[f]	**1.67**	6.7	0.24	4.61	52
ZTX951	Z	-	•	•	-	-	z	-	-	A,F	60	120	74	100	10	3	4	120	0.20	10	**1.24**	9.0	0.21	5.35	38

Manufacturers: **A** - Analog Devices, **C** - Central, **D** - Diodes, Inc., **H** - Renesas, **I** - Infineon, **In** - Intersil, **L** - Linear Integrated Systems, **m** - many, **N** - NSC, **O** - ON Semi, **R** - Rohm, **S** - Sanyo (ON Semi), **Th** - THAT, **T** - Toshiba, **Z** - Zetex (Diodes, Inc.).

Names & Packages: **a** - available, see datasheet for part numbers, most are pinout B. **b** - for TO-92 try BC550 and BC560, both pinout C. **c** - CMPTxxx, **d** - SOT-89, **f** - FMMTxxx, **m** - MMBTxxx, **p** - PMBTxxx or PMSTxxx, **v** - ATV, **z** - FZTxxx.

Notes: ◊=√Hz. **(a)** listed in order of decreasing $r_{bb'}$. **(b)** complement available, see datasheet. **(c)** complements grouped together. **(d)** from datasheet. **(e)** ranking of beta constancy over collector current, ranked on a scale from 1 to 5 (best). **(f)** lower 1/*f* noise and e_n=0.21nV/√Hz at I_C=10mA. **(g)** discussed extensively in this chapter's section "Finding e_n from noise figure." **(m)** maximum. **(o)** collector current I_C at which the r_e noise contribution raises the irreducible $r_{bb'}$ Johnson noise voltage by 50%; corresponding values of e_n, i_n, and R_n=e_n/i_n are listed. **(r)** ON Semi BC550C datasheet has $r_{bb'}$=170Ω plot (drops 25% from 0.1 to 10mA). **(s)** 2N5550 is lower-beta version. **(t)** typical. (u) see figure. **(v)** video transistor, included for its low C_{cb}. **(x)** discontinued, included for comparison. **(y)** measured on a 2SA1162-Y. **(z)** measured beta≈25.

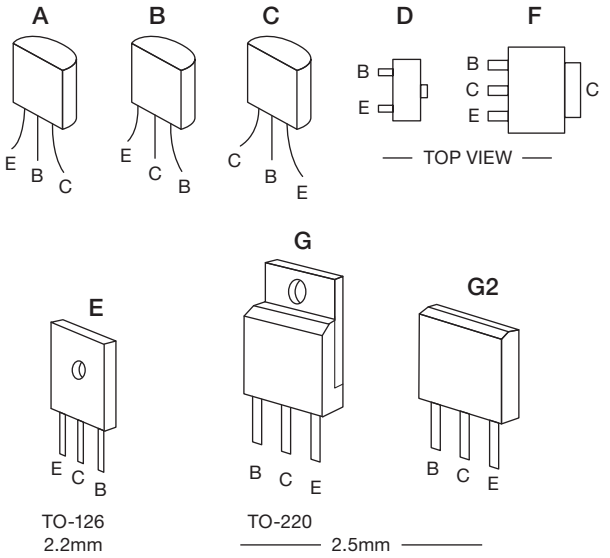


Figure 8.38. Pinouts for the BJTs in Table 8.1a.

binned into beta categories. For example the BC850 has A, B, and C grades, with typical betas at 2 mA of 180, 290, and 520, respectively. It's tempting to go for the highest-grade part in your design, but often you'll find no availability at the distributors. The higher beta grades also suffer from degraded noise performance, lower V_{CEO} ratings and lower Early voltages (i.e., lower output impedances).[42]

Looking at the BC850 again, you see that it's a small-area transistor that you'd probably use at much lower currents than the 2 mA at which its beta is specified; so it's nice to see in Figure 8.39 that its beta drops by less than 10% at 1 μA, compared, say, with the 2N5962, whose beta drops by 50%. The popular *npn* 2N3904 holds steady, but the beta of its *pnp* sibling 2N3906 drops by a factor of 3 over four decades of current, a common situation with *pnp* parts. However, there's plenty of good news: the betas of the *pnp* BC860 and ZTX718 drop less than 20% at 1 μA and the beta of the lower-gain 2N5087 is flat clear down to 1 μA.

Before discussing the all-important noise measurements, let's look at the **Early voltage**, or V_A column. Look back to §§2.3.2 and *2x.8* for a discussion of the Early effect. The Early voltage V_A provides an estimate of the BJT's output conductance (and output impedance): $g_{oe}=1/r_o=I_C/(V_A+V_{CE})$. We can also estimate a maximum possible single-stage gain (that is, with $R_L=\infty$), $G_{max}=g_m/g_{oe}$. By substituting $g_m=I_C/V_T$, we get (ignoring V_{CE} compared with the usually much larger V_A) $G_{max}=V_A/V_T$.

While it's possible to overcome the drawbacks of a low Early voltage, for example by adding a cascode stage (see §§2.4.5B and *3x.4*) or by adding emitter degeneration, it's frequently convenient not to have to do so. For such applications it's best to restrict your choices to BJTs with relatively high V_A. As the entries in Table 8.1a demonstrate, though, the *pnp* member of an *npn–pnp* complementary pair (they're grouped together in the table) usually suffers from dramatically lower Early voltages; e.g., the *pnp* BC860C has $V_A=30$ V compared with the *npn* BC850C's value of 220 V. This is unfortunate, because *pnp* transistors are useful in current mirrors for *npn* differential amplifiers, where their low Early voltage greatly reduces the stage gain. The problem can be solved in several ways, for example by adding emitter degeneration in the *pnp* mirror, as in Figure 8.35, or by using a Wilson mirror.

Looking at the table's **maximum collector voltage** (V_{CEO}) values, we see many low-noise BJTs with ratings

Table 8.1b Dual Low-noise BJTs[a]

Dual part #	Manuf[b]	Single part #, etc.	npn/pnp	V_{CEO} (V)	Beta plot	Noise[x] e_n @ I_C (nV/◊)	(mA)	x $r_{bb'}$ (Ω)	Match[m] V_{OS} (mV)	h_{FE} (%)
BCM847	I	BC850B	N	45	37	3.3	1	650	2	10
BCM857	I	BC860B	P	50	17	3.1	1	590	2	10
LS301	L	-	N	18	-	-	-	-	1	5
IT124	L	-	N	2	-	-	-	-	5	10
CMKT5089M	C	2N5089	N	25	38	2.2	10	291	5	10
DMMT3904	D	2N3904	N	40	31	-	-	-	1	2
DMMT3906	D	2N3906	P	40	3	-	-	-	1	2
DMMT5551	D	2N5551	N	160	25	-	-	-	1	2
DMMT5401	D	2N5401	P	160	2	-	-	-	1	2
LM394C	N	obsolete	N	20	40	1.8	0.1	60	0.2	5
MAT12	A	-	N	40	-	0.85	1	28	0.2	5
HFA3134	In	f_T=8GHz	N	12	-	0.8	1	40	6	8
HFA3135	In	f_T=7GHz	P	12	-	1.3	1	105	6	8
SSM2212	A	-	N	40	-	0.85	1	28	0.2	5
SSM2220	A	-	P	36	-	0.7	1	25	0.2	6
THAT 300	Th	four npn	N	36	-	0.9	1	30	3	4
THAT 320	Th	four pnp	P	36	-	0.75	1	25	3	5
HN3C51F	T	2SC3324	N	120	-	0.78	10	35	-	-
HN3A51F	T	2SA1312	P	120	-	0.28	10	20	-	-

Notes: (◊) square-root Hz. (a) see Table 8.2a for single BJTs. (b) see footnotes in Table 8.2a. (m) maximum. (x) measured.

[42] For example, the 2N5089 has a minimum beta of 450, compared with 300 for the '5088; its maximum voltage is reduced to 25 V from 30 V, and, most significantly, our samples showed somewhat (~10%) higher measured e_n noise. Similarly, the D version of the ZXTN19020 has a beta of 300 (versus 200 for the C version), with 10% higher e_n. As another example, measuring the Q grade of the 2SB1197K (min beta of 120) gave us an Early voltage of $V_A=110$ V, whereas for the R grade (min beta of 180) the Early voltage dropped to 70 V.

of 120 V or more, leading us to wonder why they have such a high voltage rating. Could it be to achieve a high Early voltage? Perhaps, but it's been observed that high-voltage "video-output" transistors often exhibit admirably low noise voltage. For example, Sanyo's 2SC3601 has a very low measured e_n of $0.22\,nV/\sqrt{Hz}$, with an $r_{bb'}$ of $1.7\,\Omega$.[43]

In general, high voltage transistors have lower betas; for example, the 300 V MPSA42 (#20 on our graph), has the lowest gain. A contender for the part with highest beta, the Linear Integrated Systems' IT124 (based on an old Intersil part), suffers from a V_{CEO} rating of just 2 V (!) in exchange for its superbeta gain. Happily, they've morphed the design into their LS301 with an 18 V rating, and it wins the beta contest (curve #44) with an astonishing $\beta=3000$ at $1\,\mu A$.[44]

A significant task is choosing the transistor's operating current. It's tempting to scan down the table's "Noise parameters" columns and pick a part near the bottom with an e_n under $0.5\,nV/\sqrt{Hz}$. But watch out for the high base-current noise that these transistors create, when operated at the relatively high collector currents needed for those low e_n values. For example, the winning ZTX851 has a very attractive $e_n=0.5\,nV/\sqrt{Hz}$ at $I_C=10\,mA$. But, with its β of 220 (see Figure 8.39), its base current of $45\,\mu A$ creates an input noise current i_n $(=\sqrt{2qI_C})$ of $3.8\,pA/\sqrt{Hz}$. That sounds small – but it creates an additional voltage noise of $0.19\,nV/\sqrt{Hz}$ across even a low $50\,\Omega$ source impedance, adding 3 dB to the transistor's input noise voltage. The problem only gets worse with higher source impedances, because the added i_nZ_s noise voltage grows linearly with source impedance, compared with Johnson noise from a resistive source (which grows as the square root of R_s).[45] For example, this base current noise creates $2.3\,nV/\sqrt{Hz}$ with a $600\,\Omega$ source impedance.

We took the table's noise measurements at high-enough currents to eliminate the I_C term in the BJT noise equation $e_n=[4kT(r_{bb'}+0.5V_T/I_C)]^{\frac{1}{2}}$ (see Figure 8.10), because we wanted to expose their $r_{bb'}$ values. But usually you'll want to operate at lower currents. For example, reducing the ZTX851's current to 1 mA raises the noise voltage by a factor of 2.3 (to $0.48\,nV/\sqrt{Hz}$), but the decrease in base current (to $5\,\mu A$) lowers the noise current by a factor of 3.2, to $1.2\,pA/\sqrt{Hz}$. At this lower current the i_nR_s noise-voltage contribution drops to $0.72\,nV/\sqrt{Hz}$ for a $600\,\Omega$ source impedance, for a total amplifier noise of $0.86\,nV/\sqrt{Hz}$, considerably lower than the $2.3\,nV/\sqrt{Hz}$ at 10 mA collector current, and nicely lower than the $3.1\,nV/\sqrt{Hz}$ of $600\,\Omega$ resistive Johnson noise. Sometimes you may want to operate at a really low current, where the r_e $(=V_T/I_C)$ term dominates and the contribution of the $r_{bb'}$ term is insignificant.

The measured noise that is listed in Table 8.1a is based on a small sampling of contemporary parts. Because some of these parts are not specifically intended for low-noise applications, the manufacturer does not specify (or control) their noise properties. It is important to keep this in mind, because you may encounter batches of parts with degraded noise. Figure 8.40 shows an example: four parts from a single batch of power transistors of a type that have been favored by audio enthusiasts, but which exhibited disappointingly wide variation in low-frequency noise voltage.

These tradeoff considerations were discussed at length in §8.5, and they're illustrated in Figure 8.32, where the concept of a transistor's noise resistance $R_n=e_n/i_n$ is shown. Look back at Figure 8.33, where we compared the voltage noise of the small-die high-beta 2N5962 with that of the large-die low-$r_{bb'}$ ZTX851, at collector currents ranging from $1\,\mu A$ to 10 mA.[46] The '851 outperforms the '5962 for source impedances of less than $1\,k\Omega$ when operated at a collector current of $100\,\mu A$ or more. Conversely, the '5962 beats the '851 for source impedances above 100k for almost any value collector current.

Figure 8.41 compares the noise voltage versus source resistance for six exemplary low-noise transistors at operating currents of $10\,\mu A$ and 1 mA. The parts are ranked by increasing e_n at 1 mA collector current and with low source resistance, i.e., they are ranked by increasing $r_{bb'}$. While it's impressive to see $0.5\,nV/\sqrt{Hz}$ performance of the lowest-noise parts (like the ZTX851), it's sobering to realize that this holds only if the resistive component of the source impedance is below $10\,\Omega$. Note the generally inverse relationship of the ranking at high R_s relative to that at low R_s. Note also how at a collector current of $10\,\mu A$ all of the transistors have comparable noise voltages for source resistances less than 10k, although the high beta

[43] It also has a low $C_{ob}=2\,pF$, much lower than the other low-e_n BJTs listed near the bottom of Table 8.1a, demonstrating that it's possible to combine low capacitance with low $r_{bb'}$.

[44] We ordered some IT124s, but they sent us LS301 parts, tested to pass the easier IT124 specs!

[45] And, of course, you can have a high-impedance signal source that is not resistive and therefore has little or no Johnson noise.

[46] These curves were prepared with a spreadsheet, using as input our measured beta and $r_{bb'}$ data of Table 8.1a.

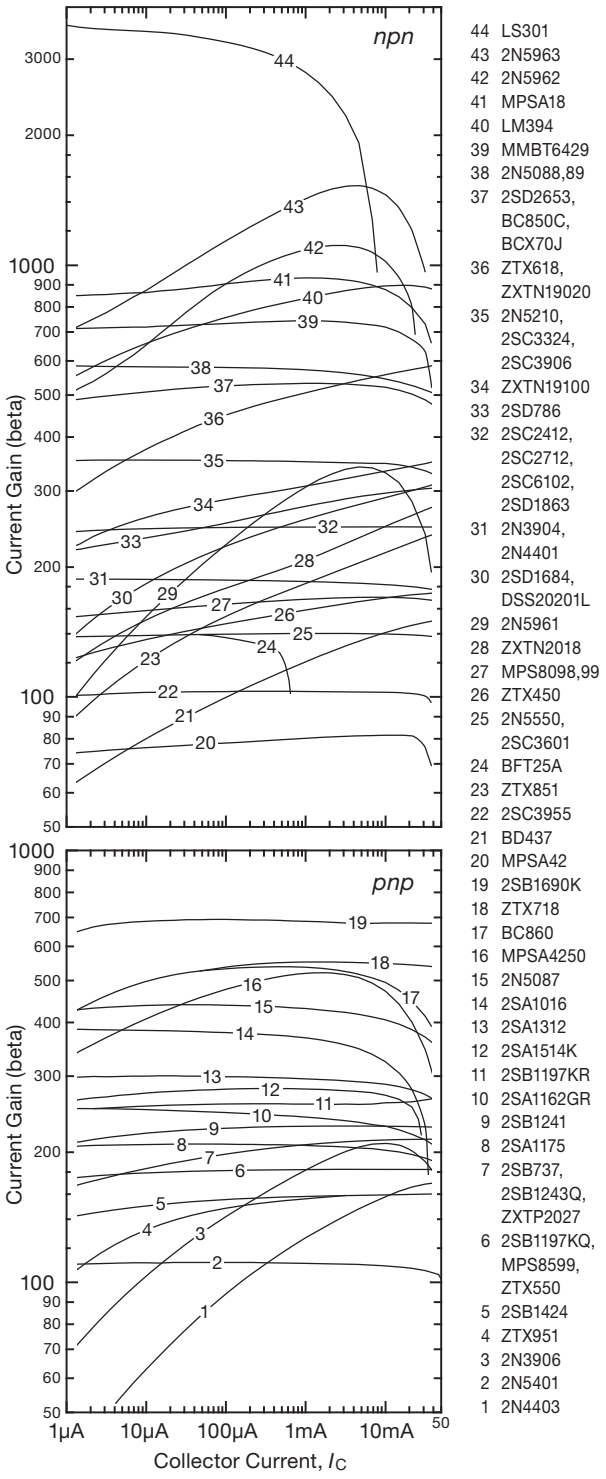


Figure 8.39. Measured beta versus collector current, at $V_{CE}=5$V, for the transistors in Table 8.1a. Use these data with $i_n=\sqrt{2qI_C/\beta}$ to find the base-current noise at the chosen collector current.

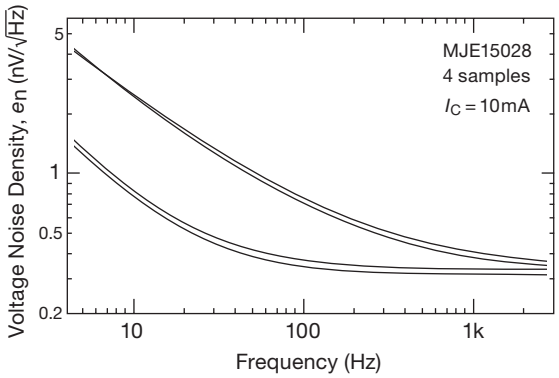


Figure 8.40. Because of their very low base resistance ($r_{bb'}$), the MJE15028-33 family of 8A bipolar power transistors have been used by audio experimenters to make low-noise audio amplifiers. That sounded good, to us. . . but we discovered >10 dB variations in their (unspecified) low-frequency noise, even within a single batch of parts.

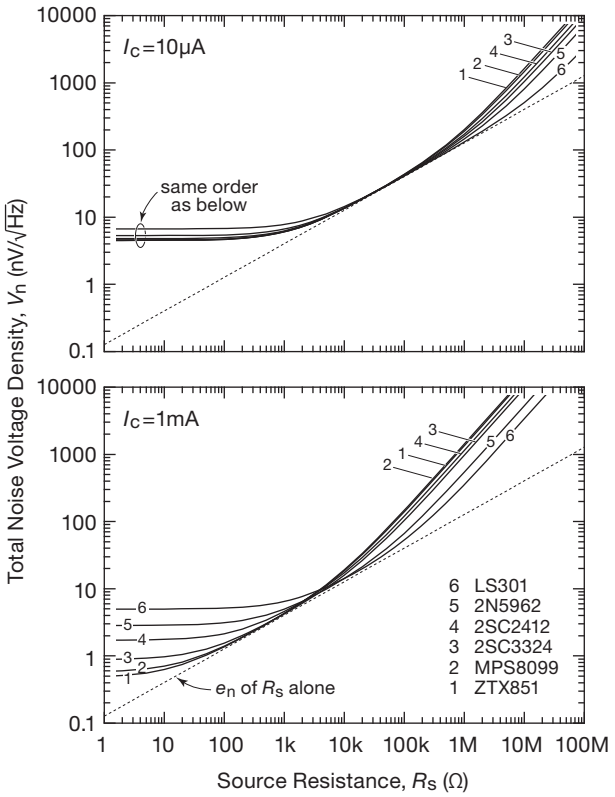


Figure 8.41. Comparing total input noise of six BJT candidates, plotted as in Figure 8.33. For low source resistance you want a transistor with low $r_{bb'}$, operated at high current; for high source resistance you want a high-beta transistor, operated at low current.

parts shine above 1 MΩ.[47] These data and analysis illustrate how low-noise performance can be optimized for a given source impedance by selecting the right transistor and running it at the right current for the job.

The curves in Figures 8.33 and 8.41 include the Johnson noise from a resistive source, R_s, and would not be valid for reactive high-impedance sources such as capacitive sensors. In that case the e_n and the i_nZ_s traces meet, similar to the 0.65 dB line in Figure 8.32, without being pushed up by an intervening R_s region.

The typical values of **capacitance** listed in Table 8.1a are from the datasheet specifications, or sometimes from plots, and generally at V_{CE}=10 V. Most of the listed values are C_{ob}, although some are C_{cb}. These are related by $C_{ob}=C_{cb}+C_{ce}$, but the C_{ce} contribution is usually small, often well under 25%.

The four columns of optimum **noise parameters** deserve some explanation. They experiment with the concept of reducing I_C until the noise-voltage contribution from r_e (i.e., collector-current shot noise times $1/g_m$) raises the overall voltage noise by 50% over its high-current (i.e., $r_{bb'}$ Johnson noise alone) value. Put another way, the idea is to throttle down the collector current in order to reduce the transistor's noise *current*, but only to the extent that it increases the transistor's noise *voltage* by a modest amount. That collector current is listed, along with the corresponding e_n and i_n (conservatively calculated from the specified minimum beta), and the transistor's noise resistance $R_n=e_n/i_n$ at that current. If you buy into the idea of not running the BJT at an "excessive current" and are willing to compromise e_n to reduce i_n, these values give an upper limit for I_C and a lower limit for e_n.[48]

Looking at the bottom quarter of the table, now with this "optimum" perspective, it's sobering to see the low R_n values, ranging from 30 Ω to 125 Ω. And much of the next quartile has trouble getting above 400 Ω. This means that for commonly encountered signal impedances above 100–400 Ω, the amplifier's noise contribution when operated at the "optimum" collector current would be dominated by the i_nZ_s contribution (i.e., input-noise current flowing through the signal's source impedance). We'd be strongly motivated to drive down i_n, by finding parts with higher beta, and by lowering I_C. In this sense the "optimum" current is really a maximum practical current. Looking out of the corner into which we've painted ourselves, the alternative of a quiet JFET (instead of a BJT) looks seriously worth considering.

And that is the subject we take up in the next major section (§8.6), after taking a brief (but challenging) interlude.

8.5.9 An extreme low-noise design challenge: transformerless ribbon microphone preamp

We looked around for a challenging application, one that requires unusually low input noise voltage. A traditional application is a "moving-coil" phonograph cartridge, with its very low signal voltage (∼1 mV at maximum groove modulation, thus a mere 0.1 μV for an 80 dB dynamic range). Problem is, it's *too* traditional![49]

A closely related technology is used in the *ribbon microphone*, in which an outrageously delicate strip of metal foil (typically 2 μm thick – that's a mere 4 wavelengths of light!) is suspended in a magnet's gap, there wafted about by the ambient sound field. You can think of it as a single-turn generator, propelled by the sound's subtle vibrations. Ribbon mics were the first truly high fidelity microphones: they're the clunky big studio mics you see in old movies. These things go back to the 1930s, with models from RCA (the classic 44B) and BBC–Marconi (named, with British understatement, the "Type A"). They're clunky because they have a large magnet inside; they weigh typically 5–10 pounds. They continue to be built and used today,[50] favored by some for their "smooth sound," variously described (in the inimitable language of the audiophile) as "intimate, warm and detailed, yet never harsh."

The output signal from a ribbon mic is tiny by comparison even with the low-level moving-coil phono cartridge: at the standard sound pressure level of 1 Pa you

[47] You may be tempted to use a high-gain low-$r_{bb'}$ part at 10 μA, because its current noise would be no worse than that of a part with higher $r_{bb'}$; but notice the dramatically higher capacitances associated with low $r_{bb'}$, which can really degrade performance with input signals of high impedance.

[48] BJTs with high beta at low currents tend to have poor (i.e., high) $r_{bb'}$ values. Recognizing that high R_n and low $r_{bb'}$ are both desirable, we can define a figure of merit FOM = $R_n/r_{bb'}$. Going from the top to the bottom of Table 8.1a, the transistors have FOM values ranging from 20 to 70. The discontinued low-e_n favorites (2SD786 and 2SB737) stand out with FOM = 51. Others with FOMs greater than 50 are the ZTX718, DSS2020, ZXTN19020, 2SB1690, 2SD2653, 2N5088 and '5089. The MMBT6429 takes first place with FOM = 69, and the 2N5089 and BC850C take the second and third places. This would support the notion that the parts at the top of the table are the winners, rather than those at the bottom. This does reinforce the concept that you need to pick the right part for the job.

[49] Although vinyl records have a faithful base of devotees, who favor the "silky smoothness" and "warmer" sound, said to be best captured with a moving-coil cartridge's "clarity and transparency of tone."

[50] See, for example, Wes Dooley's website ribbonmics.com.

get 50–100 μV directly from the ribbon. That sounds like plenty of signal – until you realize that this reference level of one pascal corresponds to +94 dB SPL. That's *loud* – a jackhammer at 5 feet! A sensitive microphone needs to get down another 80 dB or so to capture the quietest sounds in a concert.[51] At those levels the signal directly from the ribbon is a mere 5–10 nVrms. To set the scale, an op-amp with the lowest voltage noise density (LT1028, e_n=0.85 nV/$\sqrt{\text{Hz}}$) has an input-noise voltage of about 100 nVrms, integrated over the audio band of 20 Hz–20 kHz; that's 20–25 dB larger than the quiet-level audio output directly from the ribbon.

So ribbon mics invariably include an audio step-up transformer, with a turns ratio typically of 1:30. That increases the signal amplitude by the same ratio, so that a well-designed low-noise audio amplifier does not compromise the low-level performance. But transformers can be problematical, both in terms of linearity and in maintaining a flat response over the 1000:1 frequency-range of high-quality audio. We could eliminate the transformer entirely by designing an amplifier whose input-noise voltage is at least 20 dB better than that of an LT1028, i.e., with $e_n \le 0.1\,\text{nV}/\sqrt{\text{Hz}}$.

A. A simple 70 *pico*volt per-root-hertz preamp test design

To get to this kind of noise level you've got to use a BJT (or, as we'll see, a bunch of them in parallel). You pay a price in terms of input current (i.e., relatively low input impedance); but we're aided here by the very low native source impedance of the ribbon, which is less than 1 Ω over the whole audio band. We seek a transistor with low base resistance ($r_{bb'}$), which we'll operate at relatively high collector current in order to drive down the r_e noise contribution (collector shot noise flowing through r_e); recall that the latter is numerically the same as the Johnson noise created by a "real" resistor of value $R = r_e/2$.

To set the scale, note that our target noise voltage of $e_n = 0.1\,\text{nV}/\sqrt{\text{Hz}}$ corresponds to the Johnson-noise voltage created by a 0.6 Ω resistor! In other words, we need a transistor whose $r_{bb'}$ is significantly less than that, and we need to run it at a collector current of at least ∼50 mA (where the "r_e" noise contribution is equivalent to that of a 0.25 Ω resistor). And we need a circuit configuration that delivers on that low-noise promise.

There are several possible circuit configuration choices. We could try something like the Naim circuit of Figure 8.34 – but we'd have to reduce R_E to a fraction of an ohm, forcing Q_3 to drive a very stiff load. The same problem afflicts the "better" design of Figure 8.35. They have in common the need to keep the impedance of the feedback signal very low.

Two approaches we settled on are shown, in simplified form, in Figure 8.42. Both circuits dispense with feedback, arguing that a class-A preamp (of moderate gain, say $G \approx 100$) for microvolt-level signals is inherently linear. The single-ended common-emitter amplifier is simplest, and has a 3 dB noise advantage compared with a differential configuration. But it requires a *huge* input blocking capacitor (≲150,000 μF to preserve the signal's very low source impedance down to a few hertz), which causes a long startup settling time; it's esthetically ugly, too. We

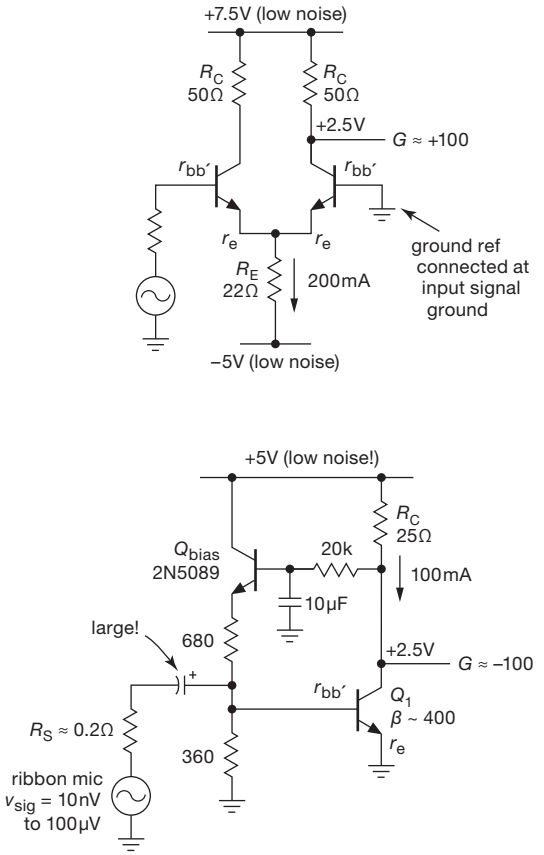


Figure 8.42. Ribbon microphone preamp configurations. Compared with the dc-coupled differential circuit, the single-ended circuit is quieter, but requires a very large blocking capacitor at the input. In either case it's necessary to use multiple paralleled transistors so that the effective base resistance $r_{bb'}$ is in the range of 0.1 Ω or less. See Figure 8.45.

[51] A *classical* music concert, that is. No problem of low levels in a rock concert.

did use this circuit to characterize a slew of candidate low-noise BJTs (look ahead to Figure 8.92 and associated discussion), but for our entry to the ribbon-mic preamp challenge we went with the dc-coupled differential open-loop configuration shown. No giant input capacitors here; but to deal with the doubled noise power we had to reduce $r_{bb'}$ and r_e by a factor of 2 (i.e., we doubled the number of paralleled transistors used for each side of the input pair).

B. Choosing a low-noise BJT

For a circuit like this it's essential to use a transistor (or several in parallel) that exhibits very low input-noise voltage. That requires a base resistance $r_{bb'}$ down in the range of just a few ohms. Sadly, $r_{bb'}$ is rarely specified; and the excellent low-$r_{bb'}$ transistors of yesteryear have largely gone extinct. A case in point is the fine Toyo–Rohm 2SD786, with a specified typical $r_{bb'}$ of 4 Ω (it's *npn*; the 2SB737 *pnp* complement is better still, with a specified typical $r_{bb'}$ of 2 Ω). Actual samples of each performed even better, with measured $r_{bb'}$ values around 2.3 Ω and 1.2 Ω, respectively.

You can't get these anymore (and we won't part with any of our precious and dwindling "lifetime" stock! Even if you beg us.). What about the "low-noise" small-signal BJTs you *can* get, such as the 2SC3324? The good news is that they *do* specify their noise performance, as we saw in §8.4. The bad news is that they don't promise a whole lot ("NF=0.2 dB typ, 3 dB max"); and these small-geometry transistors are generally optimized for operation at low current, where the r_e noise contribution is so large that there's no need to keep $r_{bb'}$ small. For example, the 2SC3324 achieves its minimum noise figure around $I_C = 30\,\mu$A; at that current r_e is 830 Ω, and thus its $r_{bb'}$ noise contribution is insignificant as long as $r_{bb'}$ is kept below, say, 200 Ω. Our samples measured $r_{bb'} \approx 40\,\Omega$ – plenty good enough for these transistors' intended low-current application, but not helpful here.

We embarked on a search for low-$r_{bb'}$ transistors and found that the situation is not bleak. It's just unspecified. In the circuit design community you'll find mention of good noise properties with large-geometry transistors (i.e., power transistors).[52] We measured the noise voltage (details later) of a few dozen promising candidates (most of which are listed in Table 8.1a on page 501), and found that, indeed, some of the large geometry ("power") transistors have $r_{bb'}$ values down below 10 Ω, and deliver quite respectable input-referred e_n when operated at currents of 10 mA or so (where the r_e contribution is equivalent to Johnson noise from a 1.25 Ω resistor). Figure 8.17 (back on page 486) shows that the ZTX851 (a 5 A, 60 V transistor, lacking any official low-noise endorsement) performs on a par with the legendary 2SD786; you can see, also, that the larger $r_{bb'}$ of the smaller geometry "low noise" parts like the 2N5089 and 2SC3324 makes them non-competitive in this high-I_C/low-e_n arena.

As we remarked earlier (§8.3.1), where we showed a plot (Figure 8.12) of e_n versus collector current for six candidate low-noise transistors, there's not much benefit in having a low $r_{bb'}$ if you are operating at low current (where the effect of r_e dominates); but it really matters at higher currents, which is where you have to operate if you want to achieve e_n values southward of 1 nV/$\sqrt{\text{Hz}}$.

To emphasize this latter point, we show in Figure 8.43 a plot of the voltage noise spectra of the ZTX851 over a large range of collector currents (seen earlier in Figure 8.18, and here extended down to 4 Hz). The lesson here is that you have to run plenty of current to get really low e_n. Don't overdo it, though: as seen in the earlier Figure 8.18, at still higher currents the low-frequency noise rises, owing to a rapidly increasing base-current flicker-noise contribution. For this transistor the "sweet spot" is at a collector current of ∼5 mA.

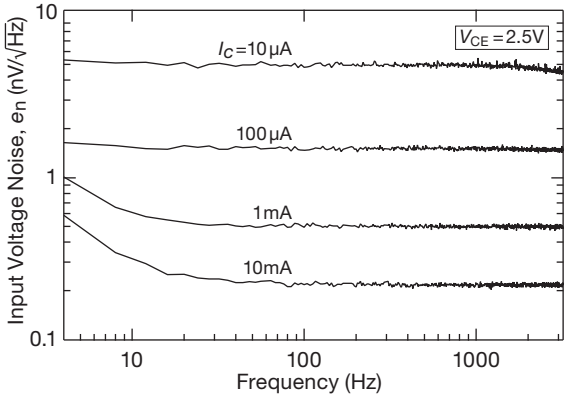


Figure 8.43. To reap the benefits of a transistor with desirable low base spreading resistance $r_{bb'}$ (thus low Johnson noise), you must operate at a high enough collector current to make the r_e noise contribution comparably small, as shown in these measured noise spectra of a ZTX851 *npn* transistor.

C. Beating the 0.1 nV/$\sqrt{\text{Hz}}$ goal

Among our candidates, we found that the quietest *pnp* transistors were somewhat better than their *npn* cousins. For example, at $I_C = 10$ mA a typical (average of six samples) *pnp* ZTX951 measured $e_n = 0.20\,\text{nV}/\sqrt{\text{Hz}}$, compared with 0.21 nV/$\sqrt{\text{Hz}}$ for the ZTX851 *npn* complement

[52] Check out the website of the remarkable Uwe Beis (www.beis.de), who has used the MJE13007 (8 A 400 V); or Ovidiu's low-noise designs at www.synaesthesia.ca, with reported $r_{bb'}$ values below 2 Ω for the 2SC3601 and 2SC2547.

(corresponding to $r_{bb'}$ values of 1.2 Ω and 1.5 Ω, respectively). We're not sure why this is so, although as we remarked earlier the noise specifications of the legendary *npn* 2SD786 (with *pnp* 2SB737 complement) indicate the same trend.[53]

We used the differential circuit variant of Figure 8.42, with a parallel array of transistors for each of the input pair, followed with a G=30 quiet op-amp stage (made with the tried-and-tested LT1128, whose input noise voltage is $\lesssim$1 nV/$\sqrt{\text{Hz}}$ all the way down to 10 Hz). Ordinarily you'd include a small "emitter-ballasting" resistor in series with each emitter of the parallel array, chosen to drop ∼50 mV or so in order to equalize the currents and prevent current hogging; at 10 mA per transistor that would amount to 5 Ω, adding an unacceptable amount of noise voltage. It's often said that transistors from a single production batch are inherently matched in their base-emitter voltage drops; out of curiosity we measured V_{BE} for a batch of 100 each of ZTX851 and ZTX951, with the results shown in the histogram of Figure 8.44. The matching is quite good, enough so that no emitter ballasting resistors are needed.[54] The same is *not* true for JFETs, by the way; see Figure 3.17's histogram of three hundred representative *n*-channel parts, painstakingly made[55] from three batches of 100 JFETs, in the popular 2N5457–59 series of *n*-channel parts (graded by I_{DSS}).

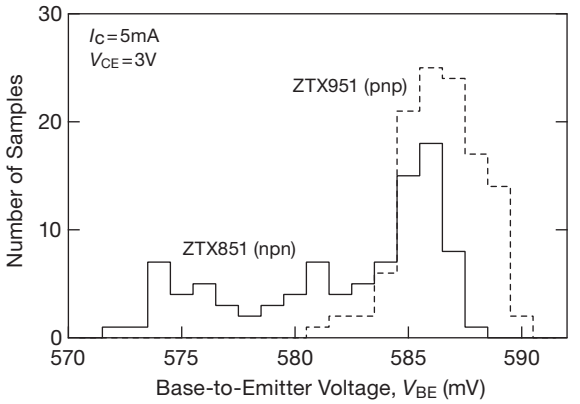


Figure 8.44. Distribution of measured V_{BE} in a batch of 100 each of *npn* and *pnp* parts.

For a circuit like this, with resistive collector loads and emitter pulldowns, the dc supply rails must be quiet, because noise on the rails appears unattenuated at the output. Here we used a so-called *capacitance multiplier* to eliminate supply noise; it's a useful subcircuit – see the description in §8.15.

The final results?...(drumroll)...here are measured noise voltages for several transistor configurations (with some graphs in Figure 8.45):

transistor	qty	I_C (mA)	e_n (nV/$\sqrt{\text{Hz}}$) @1 kHz	@100 Hz
ZTX951	2×16	2×100	0.085	0.10
	2×32	2×200	0.070	0.09
BD437	2×24	2×100	0.095	0.17
	2×24	2×200	0.080	0.13
2SC3601E	2×12	2×200	0.093	0.18

This is impressively low input noise voltage – you rarely see figures below 0.5 nV/$\sqrt{\text{Hz}}$ – but it's only fair to point out the unusual nature of this challenge, with its ultralow source impedance of less than an ohm, thus tolerant of the low input impedance (and relatively high input current) of a grounded-emitter amplifier stage running at a collector current of 100 mA. Getting to much lower input currents, though, is best accomplished with JFETs, as we'll see presently.

D. Measuring BJT noise

How did we make these transistor noise measurements? There are fancy commercial instruments you can buy to measure noise parameters (e_n, i_n, NF) of discrete transistors (BJT or FET); these are generally aimed at RF and microwave characterization (although the discontinued

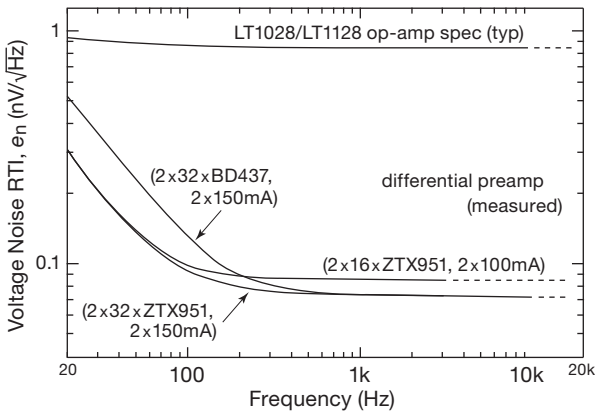


Figure 8.45. Measured noise voltage spectra for the amplifier of Figure 8.42, with three choices of input transistor and operating current. The preamp with 64 ZTX951 transistors is the winner.

[53] This may be related to the larger die size needed for a similarly-performing *pnp* transistor.

[54] We found similarly good matching for a batch of BD437s (an inexpensive and widely available BJT for "medium-power linear and switching applications").

[55] *Literally*! One's fingertips were rendered quite tender, after lead-forming and inserting three hundred transistors.

HP/Agilent 4470A measured e_n, i_n, and NF at eleven spot frequencies, two per decade from 10 Hz to 1 MHz). We took a more modest route, building our own simple circuit. It's basically the device under test (DUT) configured as a grounded emitter amplifier stage, with settable collector current and collector-to-emitter voltage, and provision for determining its voltage gain. Knowing the latter, you measure the output noise voltage spectrum, first with the input bypassed (to get e_n), then with a series resistor at the input (to get i_n). It's described in detail in §8.12.2.

8.6 Low-noise design with JFETS

Picking up where we left off (at §8.5.6), for high source impedances it's transistor *current* noise that dominates, thus favoring FETs, usually in the form of a JFET. Compared with BJTs, JFETs generally have greater voltage noise, but very much lower gate current (and current noise), and so they're the universal choice for low-noise high-impedance amplifiers. In this context it is sometimes useful to think of Johnson-noise voltage, in combination with the signal's source resistance, as a *current* noise $i_n = v_n/R_s$ (as plotted earlier in Figure 8.1). This lets you compare source-noise contributions with amplifier current noise.

We can use the same amplifier noise model for FETs, namely a series noise-voltage source and a parallel noise-current source. You can analyze the noise performance with exactly the same methods used for bipolar transistors. For example, see the graphs in Figure 8.53 in §8.7 where we referee the "bipolar/FET shootout."

8.6.1 Voltage noise of JFETs

For JFETs the voltage noise e_n is essentially the Johnson noise of the channel resistance, given approximately by

$$e_n^2 = 4kT\left(\frac{2}{3}\frac{1}{g_m}\right) \quad \mathrm{V^2/Hz}, \tag{8.34}$$

where the inverse transconductance term takes the place of resistance in the Johnson-noise formula; in other words, the noise voltage is the same as the Johnson noise produced by a resistor of value $R = \frac{2}{3}\frac{1}{g_m}$ (analogous to the input noise voltage of a BJT, whose e_n is equivalent to the Johnson noise of a resistor of value $R = \frac{1}{2}\frac{1}{g_m}$). You can see from Figure 8.46 (measured noise voltage versus transconductance[56]) that this sets a reliable lower bound to the actual noise voltage, which can be somewhat greater in practice.

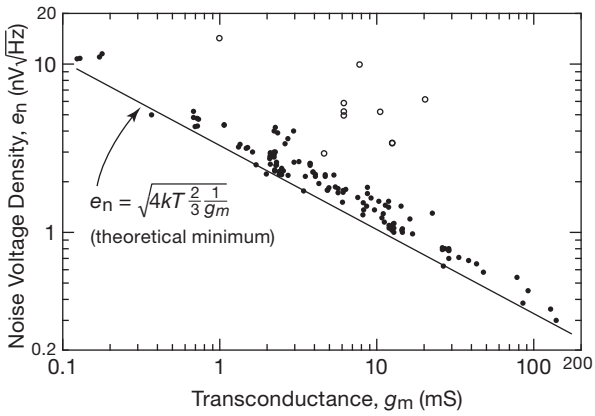


Figure 8.46. Testing the JFET noise formula: scatterplot of measured noise voltage density (well above the $1/f$ corner frequency) versus measured transconductance for a selection of 50 different JFET types, operated at various drain currents. Open circles indicate JFETs whose noise had not flattened to the white-noise floor at our maximum frequency.

Because the transconductance rises with increasing drain current (as $\sqrt{I_D}$, see §3.3.3, Figure 3.54, and Table 3.7), it is generally best to operate FETs at high drain current for lowest voltage noise. However, because the e_n is Johnson noise, which goes only as $1/\sqrt{g_m}$, and that in turn goes as $\sqrt{I_D}$, e_n is finally inversely proportional to the fourth root of I_D. With such a mild dependence of e_n on I_D it doesn't pay to run at a drain current so high that other properties of the amplifier are degraded. In particular, a FET running at high current gets hot, which (a) decreases g_m, (b) increases offset voltage drift and CMRR, and (c) raises gate leakage dramatically; the latter effect can actually *increase* voltage noise, because there is some contribution to e_n from flicker noise associated with the gate leakage current.

There is another way to increase g_m and therefore decrease JFET voltage noise: by paralleling a pair of JFETs, you get twice the g_m, but of course this is at twice the I_D. But now if you run the combination at the previous I_D,

[56] More than 100 measurements on some 65 transistors chosen from 50 different types, and at various currents ranging from 75 μA to 50 mA, creating many different g_m test values. We followed the declining e_n with rising frequency until we found the high-frequency white-noise floor. This yielded a wide range of conditions under which to test the formula. No parts were better, but many were 10% to 30% worse. Others were much worse, up to $\times 2$. For some the noise continued to fall with increasing frequency, so we couldn't find the noise floor; these are indicated with open circles on the graph. The takeaway lesson: get g_m as high as possible for low e_n. Remember, paralleling or using larger-die JFETs is a good way to increase g_m. But watch out for high capacitance and leakage. And keep in mind that $1/f$ noise and low-frequency noise plateaus can dominate your design. Examine our plots and the data for JFETs in Table 8.2 on page 516. For critical projects, take your own measurements.

you still improve g_m by a factor of $\sqrt{2}$ over the single-JFET value, without increasing total drain current; so you decrease e_n by a factor of the 4th root of 2, to 84% of the single-JFET value.[57] In practice you can simply parallel a number of matched JFETs, or look for a large-geometry JFET like the IFN146 ($0.7\,\text{nV}/\sqrt{\text{Hz}}$) or IF3601 ($0.3\,\text{nV}/\sqrt{\text{Hz}}$) mentioned earlier.[58]

There is a price to pay, however. All the capacitances scale with the number of paralleled JFETs. As a result, high-frequency performance (including noise figure) is degraded. In practice you should stop paralleling additional transistors when the circuit's input capacitance roughly matches the source's capacitance. If you care about performance at high frequencies, choose JFETs with high g_m and low C_{rss}; you might consider the ratios g_m/C_{rss} and g_m/C_{iss} to be high-frequency figures of merit (recall that $f_T = g_m/2\pi C$, where C is the input capacitance or Miller capacitance, depending on the circuit configuration). Note that circuit configurations can also play an important role; e.g., the cascode circuit can be used to eliminate the Miller effect (gain multiplication) on C_{rss}.

In Figure 8.52 we've plotted an extensive collection of measured noise voltage for many of the JFETs in Table 8.2. See also Table 3.7, with additional JFET characteristics (not related to noise), and the "Measured transconductance" graph in §*3x.2* for plots of JFET transconductance versus drain current.

A. JFET 1/*f* voltage noise

Like their BJT brothers, most JFETs suffer also from rising noise voltage density at low frequencies, as seen in the measured data of Figure 8.47. In some cases (such as the 2SK170B) the $1/f$ noise is quite subdued and can be characterized by a single noise-corner frequency f_c, similar to BJTs. In such cases the noise corner can be estimated with eq'n 8.27, and, once f_c is known, the noise density can be calculated with eq'n 8.28. The plot shows 1 kHz noise voltage density ranging from $1\,\text{nV}/\sqrt{\text{Hz}}$ for the 2SK170B, to $2.8\,\text{nV}/\sqrt{\text{Hz}}$ for the low-capacitance 2N5457, to $11\,\text{nV}/\sqrt{\text{Hz}}$ for the small-die PN4117 (for which the drain current is I_{DSS}, here 75 μA).

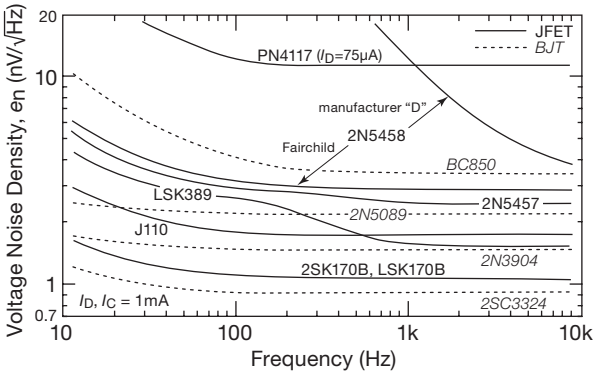


Figure 8.47. Measured voltage noise density versus frequency for several BJTs and JFETs, illustrating the $1/f$-like rise at low frequencies. See Figure 8.52 for an extensive set of JFET noise plots.

But some JFETs suffer from an elevated low-frequency noise plateau region,[59] like that of the LSK389. This part has an attractively low-noise density of $1.8\,\text{nV}/\sqrt{\text{Hz}}$ at 1 kHz and above, but it rises, in a snake-like curve, to twice that value in the 100 Hz region. This is not a terribly serious drawback for this otherwise excellent part, and it still enjoys low wideband noise levels of about 70 nVrms and 180 nVrms for 1 kHz and 10 kHz total bandwidths, respectively.

Sometimes this effect can get out of hand. For example, compare Fairchild's quiet version of the 2N5486, with $e_n = 3\,\text{nV}/\sqrt{\text{Hz}}$, with the noisy sample we measured from "manufacturer D": the latter's $4.5\,\text{nV}/\sqrt{\text{Hz}}$ at 10 kHz climbs steeply at lower frequencies, going off-scale at 700 Hz. It climbs to $50\,\text{nV}/\sqrt{\text{Hz}}$ at 100 Hz, some 15× that of the quiet Fairchild JFET with the same part number. As a result the noisy part has about 1 μV of wideband noise (to 1 kHz). For a stunning comparison of noise variation

[57] The 4th root noise-to-current relationship holds for JFETs operated in their high current density region, say above $I_{DSS}/100$ (see the "Measured transconductance" plots of g_m versus I_D in §*3x.2*), but at low current densities (where $g_m \propto I_D$) the decrease in noise will approach a square-root relationship, just as with BJTs. For example, the measured noise in an IF3601 increased by a factor of 2.8 (nearly $\sqrt{10}$) when the drain current was reduced from 1 mA to 0.1 mA.

[58] Paralleling JFETs lowers the current density, pushing the g_m versus I_D plot up toward its maximum, where it would match that of a BJT, see Figure *3x.10*, which shows a given JFET's potential for improvement. Some parts (such as the IF3601) are already close to this limit, so that little further improvement can be expected.

[59] Called "roller-coaster" noise by Motchenbacher and Connelly. The mechanism for the excess $1/f$ noise was identified in the early 1960s as a bulk rather than surface effect, involving random capture or emission of electrons trapped in the depletion region. See, for example, P. O. Lauritzen, "Low-frequency generation noise in junction field effect transistors," *Solid-State Electron.*, **8**, 1, 41–58 (1965), or C. T. Sah, "Theory of low-frequency generation noise in junction-gate field-effect transistors," *Proc. IEEE*, **52**, 7, 795–814 (1994). In the time domain this noise can take the form either of discrete steps between two or more voltage levels ("popcorn" or "telegraph" noise) or random sudden voltage excursions (burst noise); see Figures 8.5 and 8.6 for examples of popcorn noise in a vintage 741 BJT op-amp.

in 2N5486s made by multiple manufacturers, see the measured spectra in Figure 8.51. The noisiest part in that group has a 210 nV/$\sqrt{\text{Hz}}$ plateau, resulting in about 2.5 μVrms low frequency integrated voltage noise. In its defense, the 2N5486 "RF Amplifier" datasheet lists noise-figure specifications only at 100 MHz and 400 MHz. Note also that this low-frequency-noise excess does not affect operation at much higher frequencies (where "$e_n \cdot C_{in}$" noise, discussed in §8.11, rears its ugly head).

Although it's common to talk about $1/f$ noise and a corresponding corner frequency f_c, the reality is that many JFETs do not conform to such a neat model. In their datasheets, manufacturers deal with the issue of excess low-frequency noise in several ways. First, they may avoid a noise specification altogether. Second, they may give the e_n noise-density spec at relatively high frequencies – 10 kHz, 100 kHz, or even higher. Third, they may provide instead an rms noise voltage specification at some specified bandwidth. Fourth, they may give an intentionally high (conservative) specification, such as 115 nV/$\sqrt{\text{Hz}}$ max, or perhaps a 3 dB noise figure for R_s=1 MΩ (whose source noise-voltage density is a high 126 nV/$\sqrt{\text{Hz}}$). Fifth, they may give an e_n spec at say 100 Hz (where it's useful), but give a *typical* value only. Finally, for parts intended for high-frequency amplification, they may give their noise spec at *radio* frequencies – 100 MHz or higher.

8.6.2 Current noise of JFETs

At low frequencies the current noise i_n is extremely small, arising from the shot noise in the gate leakage current (Figure 8.48):[60]

$$i_n = \sqrt{2qI_G} = (3.2 \times 10^{-19} I_G)^{\frac{1}{2}} \quad \text{A/Hz}^{\frac{1}{2}}. \tag{8.35}$$

The noise current rises with increasing temperature, as the gate leakage current rises. Watch out for the rapidly increasing gate leakage in *n*-channel JFETs that occurs for operation at high V_{DG} and/or high I_D (see §3.2.8). A dose of reality: it's hard to estimate with any accuracy the level of current noise in JFETs, because the gate leakage is poorly specified. Too often you see unrealistically high worst-case specifications, for example I_G=1 nA (max), whereas at normal ambient temperatures it's more typically down in the 1–10 pA range. With such low bias currents the input noise-current density is very small, e.g., 1.8 fA/$\sqrt{\text{Hz}}$ for a gate leakage current of 10 pA. That noise current would generate an e_n just 1.8 nV/$\sqrt{\text{Hz}}$ through a signal source

[60] In addition, there is a flicker ($1/f$) current-noise component in some FETs.

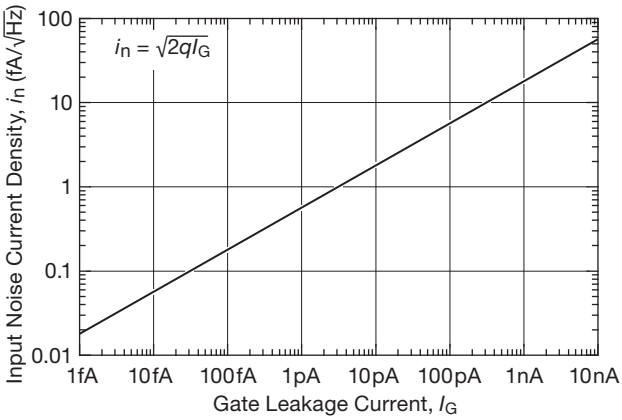


Figure 8.48. Input shot-noise current density versus FET gate leakage current.

impedance of 1 MΩ (which, by the way, would itself have a Johnson-noise voltage of 128 nV/$\sqrt{\text{Hz}}$; the source resistance would have to reach 5 GΩ before the noise voltage generated by the JFET's noise current would match the Johnson noise of the source).

At moderate to high frequencies there are additional noise and resistive terms.

(a) If there's a summing junction (for example in a transimpedance amplifier) then the input noise voltage e_n driving the input capacitance C_{in} generates a noise current of $i_n = e_n \omega C_{in}$, see §8.11 for details.

(b) In a common-source amplifier without cascode, there is an effective resistive part of the (normally capacitive) input impedance seen looking into the gate. This comes from the effect of feedback capacitance (Miller effect) when there is a phase shift at the output caused by load capacitance; i.e., the part of the output signal that is shifted 90° couples through the feedback capacitance C_{rss} to produce an effective resistance at the input, given by

$$R = \frac{1 + \omega C_L R_L}{\omega^2 g_m C_{rss} C_L R_L^2} \text{ ohms} \tag{8.36}$$

Both effects rise linearly with frequency above a breakpoint, and both have similar breakpoint frequencies, typically in the range of 2–100 kHz for low-leakage JFETs.

As an example, the 2N5486 *n*-channel JFET has a noise current of ∼5 fA/$\sqrt{\text{Hz}}$ and a noise voltage e_n of 2.5 nV/$\sqrt{\text{Hz}}$, both at I_{DSS} and 10 kHz. These figures are roughly 200 times better in i_n and 2 times worse in e_n than the corresponding figures for the 2N5087 BJT used in connection with the noise figure plots in §8.5.1 (Figure 8.29). If we assume a 470 Ω load with 5 pF of shunt capacitance

(68 MHz drain rolloff), the noise current begins climbing at about 30 kHz, and reaches levels of concern to the RF designer in the 10–100 MHz region.[61] It's not uncommon to see a cascode transistor added to suppress the effect and to improve the high-frequency gain.

With FETs you can achieve good noise performance for input impedances in the range of 10k to 100M. As we've seen (Figure 8.27), and whose secrets will be revealed presently (§8.6.4), the SRS model SR560 preamp has a noise figure of 1 dB or better for source impedances from 5kΩ to 20MΩ in the frequency range from 100 Hz to 10 kHz. Its performance at moderate frequencies corresponds to a noise voltage of $4\,\text{nV}/\sqrt{\text{Hz}}$ and a noise current of $0.013\,\text{pA}/\sqrt{\text{Hz}}$.

8.6.3 Design example: low-noise wideband JFET "hybrid" amplifiers

You can improve on the noise performance of commercial JFET preamps (and op-amps) by combining the best discrete JFETs with an op-amp in a "hybrid" design. Several nice circuit examples are shown in progressive examples in other chapters: (a) an ac-coupled low-noise wideband amplifier (to ∼20 MHz) with $e_n \approx 1\,\text{nV}/\sqrt{\text{Hz}}$ in Figure 3.34; (b) an analogous dc-coupled amplifier with $e_n \approx 2\,\text{nV}/\sqrt{\text{Hz}}$ in Figure 3.37; and (c) a dc-coupled low-noise wideband differential amplifier with $e_n \approx 2\,\text{nV}/\sqrt{\text{Hz}}$ in §*3x.4*.

The reader is encouraged to spend a few minutes (or even a half hour!) reviewing those examples, which illustrate important techniques in low-noise amplifier design with discrete JFETs (and also when combined with an op-amp second stage). Here we complete the progression of such hybrid designs with a dc-coupled *instrumentation amplifier* (INA; see §5.15) of comparably low-noise and wide bandwidth.

The circuit in Figure 8.49 is an evolution of the single-ended dc-coupled hybrid amplifier of Figure 3.37, two of which here form the differential amplifier input of the classic three-amplifier INA. The voltage gain (difference input to single-ended output) is $G=100$, set by the ratio $1+2R_f/R_g$. The ±2% trim of R_{f2} is used to maximize the common-mode rejection ratio, a desirable feature that is responsible for much of the INA's fame. JFETs need help to achieve low offset voltage (Q_1 specifies 20 mV, worst case), so we've added an offset balance in $Q_{1a,b}$'s drain.

As for noise, the LSK389B[62] specifies a typical e_n of $0.9\,\text{nV}/\sqrt{\text{Hz}}$ at 1 kHz and $I_D=2$ mA. At our 5 mA drain current we can expect slightly better; but the noise of each differential pair (e.g., $Q_{1a,b}$) is larger by a factor of $\sqrt{2}$, with another factor of $\sqrt{2}$ to account for the combined noise of the upper and lower differential pairs. As with the single-ended amplifier of Figure 3.37, the op-amp's noise ($e_n \sim 12\,\text{nV}/\sqrt{\text{Hz}}$) is reduced by a factor of the JFET pair's gain (∼20) when referred to the input. The overall noise voltage of this instrumentation amplifier is thus $\sim 2\,\text{nV}/\sqrt{\text{Hz}}$. Combined with its bandwidth (∼20 MHz, for signal source impedance ≤1kΩ), this is better performance than can be had in available instrumentation amplifiers.

Can we do better? The easiest way to beat down the JFET voltage noise is to add a second pair of JFETs at each input, paralleling the existing gate and drain terminals, but with each pair pulled down with its own 10 mA current sink (and with the drain resistors now 500 Ω). That reduces the input voltage noise by $\sqrt{2}$, and there is no need to worry about V_{GS} matching between the separate JFET matched pairs (figure out why). Finally, if you want to minimize input current, add a pair of cascode transistors to keep the V_{DS} of Q_1 below 5 V; see §3.2.8 and Figures 5.61 and 8.67.

8.6.4 Designs by the masters: SR560 low-noise preamplifier

It's always instructive to look at the innards of a good commercial product. Stanford Research Systems (SRS) has a nice line of scientific instruments, among which you'll find their SR560 low-noise preamplifier, in continuous production since 1989. It uses a differential JFET front-end for high input impedance (100 MΩ‖25 pF), and it has a nice set of panel controls that let you select things like the voltage gain (calibrated ×1 to ×50,000, in a 1–2–5 sequence or uncalibrated over the same range), frequency response (via switchable lowpass and highpass sections), ac or dc coupling, phase inversion, and so on. It's a good performer – noise-voltage density better than $4\,\text{nV}/\sqrt{\text{Hz}}$, response flat to ±0.5 dB to 1 MHz, 0.01% typical distortion, and output swing to 10 Vpp.

We showed this puppy's specified noise-figure contours in Figure 8.27. Let's look under the hood to see how they did it. Figure 8.50 shows the front-end amplifier circuit, in full detail (Figure 8.50A) and in simplified forms (circuit B, block diagram C).

[61] We made bench measurements that confirmed (within a factor of 2) the resistive input component predicted by eq'n 8.36. In a JFET common-source amplifier without cascode this effect could greatly lower the Q of an RF-tuned circuit.

[62] An available substitute for the excellent but discontinued 2SK389 from Toshiba.

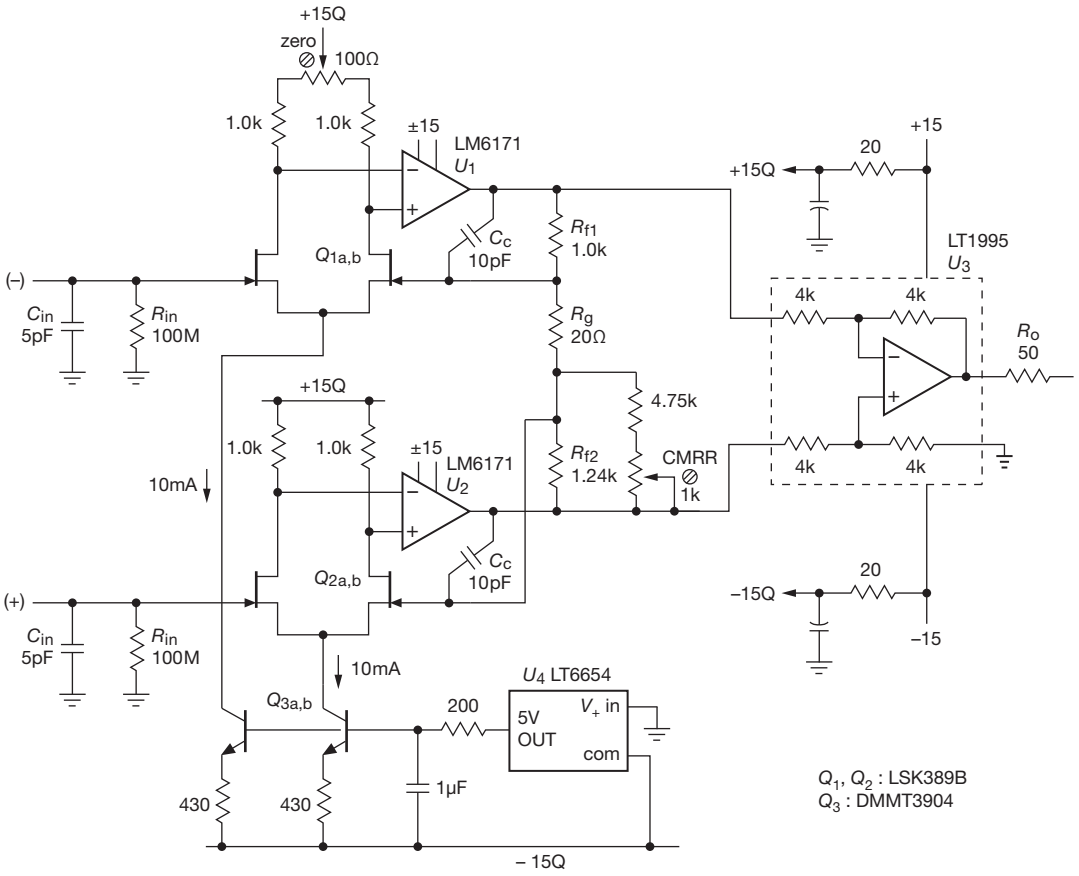


Figure 8.49. Hybrid JFET-input low-noise wideband instrumentation amplifier: G=100, $e_n \approx 2\,\text{nV}/\sqrt{\text{Hz}}$, BW≈ 20 MHz.

Overall topology Matched low-noise JFET pair Q_{1ab}, running at 4.6 mA (close to its guaranteed minimum I_{DSS}, see below), forms the fully differential first stage, in a hybrid configuration in which the JFET differential output drives an op-amp (analogous to Figure 3.37), but here configured with feedback to Q_{1a}'s source terminal so that both JFET gates are available as external inputs. The overall configuration is therefore a differential "current-feedback amplifier" (CFB, see §*4.x.6*), analogous to the single-ended hybrid circuit of Figure 3.34, and instrumentation amplifier configuration "C" in Figure 5.88. The first-stage native voltage gain is selectable as ×10 or ×2; for the latter a 499 Ω series resistor is switched in, and thus unity net gain is seen at the second stage's input. The second stage is an LT1028 low-noise op-amp ($e_n \sim 1\,\text{nV}/\sqrt{\text{Hz}}$ at 10 Hz), G=10, with $\lesssim$100 Ω source impedance at both inputs to preserve low noise voltage. With the input divider the overall second-stage gain is ×2, ×5, or ×10.

JFET bias The designer nicely finessed the business of uncertain I_{DSS} by sensing the average drain voltage, comparing it with a +6.2 V reference, and closing the loop via integrator U_2. This is a reliable way to deal with loose I_{DSS} specs, for example those of the original NPD5564 dual JFET (I_{DSS}=5 mA min, 30 mA max). That JFET is no longer available from the original manufacturer,[63] but fortunately there's a superior replacement (LSK389B), with pleasantly tighter I_{DSS} specs (6 mA min, 12 mA max), and, as a bonus, lower noise voltage (at I_D=2 mA the NPD5564's typical noise-voltage densities at 10 Hz and at 1 kHz are $12\,\text{nV}/\sqrt{\text{Hz}}$ and $3.8\,\text{nV}/\sqrt{\text{Hz}}$, respectively; the LSK389B's corresponding figures are $2.5\,\text{nV}/\sqrt{\text{Hz}}$ and $0.9\,\text{nV}/\sqrt{\text{Hz}}$).

[63] Currently InterFET (direct sales only) is offering their similarly-spec'd IFN5564

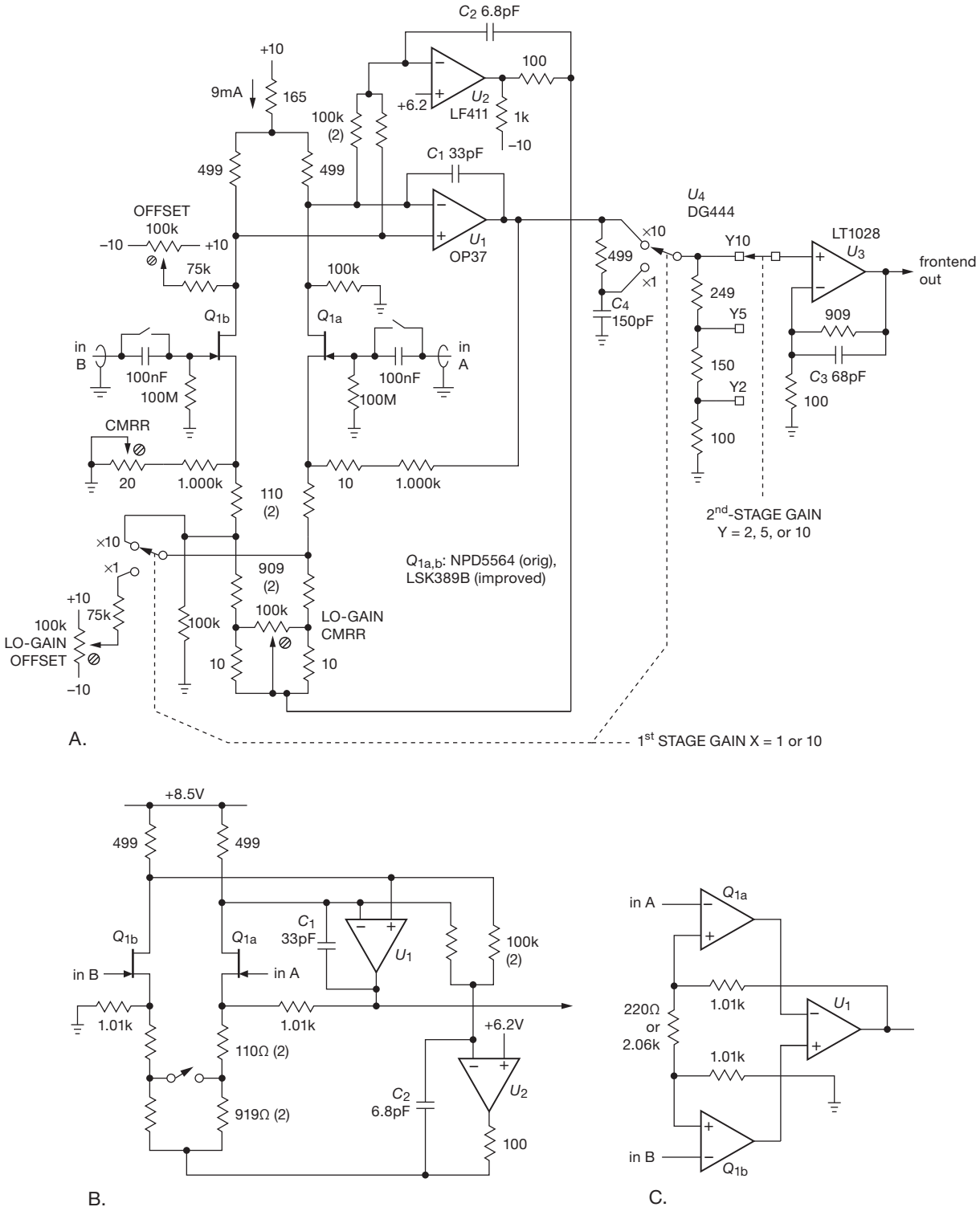


Figure 8.50. Front-end amplifier stages of the Stanford Research Systems SR560 Low-Noise Preamplifier (A: detailed; B and C: simplified). Some details of input signal switching have been omitted.

Bandwidth In the *signal path*, amplifier U_1 is a decompensated low-noise ($e_n \sim 3.5\,nV/\sqrt{Hz}$ typ at 10 Hz) wideband ($f_T \sim 63$ MHz) op-amp, here configured as an integrator with $f_T = 5$ MHz (33 pF and 1.0 kΩ). Its slew rate of 17 V/μs translates to a first-stage output of 5 Vpp at the instrument's rated 1 MHz bandwidth. In the second stage, capacitor C_3 limits the bandwidth to ∼2 MHz. In the first-stage *bias path* (which matters for common-mode rejection), integrator U_2 has an output slew rate of 10 V/μs (min), so it can follow a 3 Vpp common-mode input (the instrument's rated maximum) at the full 1 MHz rated bandwidth. The integrator's unity-gain bandwidth is 230 kHz (6.8 pF and 50 kΩ), thus a bias-setting loop gain of 230 at 1 kHz. That puts an increasing CMRR burden on U_1 at higher frequencies, evident in the instrument's CMRR specification: > 90 dB to 1 kHz, decreasing by 6 dB/octave (i.e., as $1/f$) above 1 kHz.[64]

Offset voltage These dual JFETs are not in the same league as precision-matched BJTs, and they don't claim to be. The NPD5564 (the best grade in its family) has a specified maximum offset voltage (i.e., gate-source mismatch, $|V_{GS1} - V_{GS2}|$) of 5 mV at $I_D = 2$ mA; the LSK389B, though excelling in noise performance, is considerably poorer, with a maximum mismatch of 20 mV (specified at $I_D = 1$ mA). The SR560 deals with this problem by throwing pots at it: there's a 10-turn OFFSET trimmer in Q_{1b}'s drain, with enough range (±0.13 mA) to balance a worst-case offset, and a second 10-turn trimmer down in the source resistor string to independently balance the offset when switched to low gain. There is a pair of analogous CMRR trimmers, 20-turns each. We've done the calibration routine (complete with some iteration of the not-very-orthogonal adjustments), and can attest to the fussiness of getting (and sustaining) deep offset and CMRR nulls.

Noise voltage To minimize noise voltage, the JFETs are operated at relatively high current, and the resistor values (in the source, drain, and signal path generally) are intentionally small. With the original noise budget of $\sim 4\,nV/\sqrt{Hz}$ it was acceptable, for example, to use 110 Ω resistors in the JFET sources. But a series pair of that resistance produces a Johnson noise-voltage density of $2\,nV/\sqrt{Hz}$, a bit more than we'd like when paired with these quieter JFETs. An optimized design would reduce those values by perhaps a factor of 2.

Miscellany Low-level signal switching (e.g., ac/dc/gnd input coupling, ×1/×10 gain selection, and A/B input interchange) is done with electromechanical relays. Not just any old relay, though – these guys use relays with gold-cladded twin contacts (intended for stable low resistance, even with low-level signals where there is no contact cleaning; see §*1x.6.1*). And these relays are the *latching* type, where a momentary pulse switches the state, thus eliminating noise-inducing coil currents in the steady state; this further eliminates local heating as a source of drift. In this instrument a microcontroller runs all relays and indicator lights, sensing a front-panel command (or external digital control), and appropriately pulses relays as needed to change their state.

8.6.5 Selecting low-noise JFETS

As we mentioned earlier, bipolar transistors offer the best noise performance with low source impedances, owing to their lower input-voltage noise. Voltage noise, e_n, is reduced by choosing a transistor with low base resistance, $r_{bb'}$, and operating at high collector current (as long as h_{FE} remains high). For higher source impedances the current noise can be minimized instead by operating at lower collector current.

At high values of source impedance, FETs are the best choices. Their voltage noise can be reduced by operating at higher drain currents, where the transconductance is highest. FETs intended for low-noise applications have high κ values (see §3.1.4), which generally means high input capacitance. For example, the low-noise 2SK170 has a typical C_{iss} of 30 pF, whereas the PN4117–9 series of low-current FETs has a *maximum* C_{iss} of just 3 pF.

Table 8.2 on the following page presents a selection of JFET candidates for low-noise circuits. It's meant to be used in conjunction with Table 3.7, which has more extensive tabular data, including single-sample measurements. Also refer to Figure *3x.10*'s transconductance plots, and Figure 8.52's noise plots. The entries near the bottom of Table 8.2 are dual JFETs, suitable for differential stages; but note that their V_{os} is not impressive – generally far inferior to even the poorest op-amps (the interdigitated LS840, whose layout is shown in Figure 3.18, is the best of the lot, with 2 mV typical, 5 mV worst-case offset voltage).

We favor JFETs with high transconductance and low gate-cutoff voltages, but sadly many of the old favorites with superior performance have been discontinued. NXP's BF862 has become a "go-to" low-noise part, both quiet and with attractively low capacitance. Toshiba's 2SK170 has been single-handedly kept in stock by large factory purchases by Mouser, and the second-source LSK170 is

[64] The high-frequency CMRR could likely be improved by configuring U_1's output to control a pair of current-sink transistors biasing Q_{1ab}.

Table 8.2 Low-noise JFETs[a]

Part number		Polarity	V_{GS}	I_{DSS}		$V_{GS(off)}$ (10nA)		g_m		@ I_D	R_{on}	C_{iss}	Noise curve	Measured noise voltage, e_n (nV/√Hz)			
SOT-23	TO-92, etc		max (V)	min (mA)	max (mA)	min (V)	max (V)	min (mS)	max (mS)	(mA)	typ (Ω)	typ (pF)		f=100Hz I_D=1mA	f=10kHz, @I_D 100µA	1mA	5,10mA
MMBF4117	**PN4117**	N	40	0.03	0.09	-0.6	-1.8	0.07	0.21	m	-	1.2	S	12	11nV/◊ at 0.07mA		
MMBF4118	**PN4118**	N	40	0.08	0.24	-1	-3	0.08	0.25	m	-	1.2	U	48	11nV/◊ at 0.2mA		
MMBF4119	**PN4119**	N	40	0.2	0.6	-2	-6	0.10	0.33	m	-	1.2	V	45	11.5nV/◊ at 0.3mA		
BFT46	**-**	N	25	0.2	1.5	-	-1.2	1	-	m	-	3.5	-	-	-	-	-
BF511	**-**	N	20	2.5	7	-1.5	-	4	-	m	-	5[m]	-	20	-	3.8	-
MMBF5457	**2N5457**	N	25	1	5	-0.5	-6	1	5	m	-	4.5	P	2.8[f]	-	2.4	2.2
MMBF5458	**2N5458**	N	25	2	9	-1	-7	1.5	5.5	m	-	4.5	-	11	-	3	-
MMBF5459	**2N5459**	N	25	4	16	-2	-8	2	6	m	-	4.5	-	-	-	-	-
MMBF5460	**2N5460**	P	40	1	5	0.75	6	1	4	m	-	5	-	-	-	-	-
MMBF5461	**2N5461**	P	40	2	9	1	7.5	1.5	5	m	-	5	-	-	-	-	-
MMBF4416	**-**	N	30	5	15	-	-6	4.5	7.5	m	-	2.2	R	3.9	4.7	2.8	2.6
MMBF5484	**2N5484**	N	25	1	5	-0.3	-3	3	6	m	-	2.2	-	5	4.3	2.5	2.2
MMBF5485	**2N5485**	N	25	4	10	-0.4	-4	3.5	7	m	-	2.2	-	2.8	4.3	2.5	2.3
MMBF5486	**2N5486**[n]	N	25	8	20	-2	-6	4	8	m	-	2.2	Q	3n	5	3[f]	2.5
MMBF4392[p]	**PN4392**	N	40	25	75	-2	-5		16[t]	10	60	12	-	-	-	-	-
MMBF4393[p]	**PN4393**[o]	N	40	5	30	-0.5	-3		13[t]	10	100	12	J	1.8[f]	-	1.8	1.4
PMBF4393	**-**	N	*	*	*	*	*		*	*	*	*	-	22	14	5.2	3.4
MMBF5103	**PF5103**	N	40	10	40	-1.2	-2.7	7.5	-	2	30	16[m]	L	2	3.2	1.9	1.5
-	**J107**	N	25	100	-	-0.5	-4.5		75[t]	5	8	35	F	1.4	-	1.3	0.75
PMBFJ109	**J109**	N	25	40	-	-2	-6		26[t]	5	12	15	-	7.8	-	2.2	-
MMBFJ110[p]	**J110**	N	25	10	-	-0.5	-4		20[t]	5	18	15	-	1.6	3.2	1.5	0.9
MMBFJ112[p]	**J112**	N	35	5	-	-1	-5		17[t]	25	50[m]	6	-	2	3.3	1.8	1.5
MMBFJ113[p]	**J113**	N	35	2	-	-0.5	-3		12[t]	5	100[m]	6	H	1.8	3.2	1.7	1.4
MMBFJ309[p]	**J309**	N	25	12	30	-1	-4	13	-	10	50	4	G	18	-	2.6	1.3
MMBFJ310[p]	**J310**	N	25	24	60	-2	-6.5	13	-	10	50	4	K	2	4.8	2	1.5
BF861A	**-**	N	25	2	6.5	-0.2	-1	12	20	m	-	7.5	-	9	-	2.7	1.8
BF861B	**-**	N	25	6	15	-0.5	-1.5	16	25	m	-	7.5	-	9	-	1.5	1.2
BF862[b]	**-**	N	20	10	25	-0.3	-1.2	35	45[t]	m	-	10	E	1.3	2.2	1.1	0.8
-	**2SK147B**[d]	N	40	8	16	-0.3	-1.2	40	-	5	-	75	-	1.9	2.5	1.3	0.85
-	**IFN147**	N	40	5	30	-0.3	-1.2	30	40[t]	m	-	75[m]	-	1.5	2.4	1.1	0.75
-	**2SK170BL**[c]	N	40	6	12	-0.2	-1.5	10	-	1	-	30	B	1.2	2.3	1.1	0.78
-	**LSK170B**	N	40	6	12	-0.2	-2	10	-	1	-	20	-	1.3	2.3	1.0	0.78
-	**IF3601**	N	20	30	-	-0.04	-3	750	-	m	-	300	A	0.65	1.8	0.6	0.38
dual JFETs																	
dual p/n	single p/n	V_{os}[e]															
PMBFJ620[k]	**'J310**[k]	-	N	25	24	60	-2	-6.5	13	-	10	50	3	Y	33	*noisy*		5.2
LS840[g]	**-**	5	N	60	0.5	5	-1	-4.5	2	-	0.2	-	4	-	3.2	-	2.7	-
LS5912	**-**	15	N	25	7	40	-	-	7	-	5	-	5	-	3.1	-	2.7	-
LSK389B	**LSK170**	20	N	40	6	12	-0.15	-2	20	-	3	-	25	M	3.2	3.2	1.7	1.3
2SK146GR[d]	**2SK147**	20	N	40	5	10	-	-	-	-	-	-	75	-	1.3	-	1.1	0.85
IFN146	**IFN147**	20	N	40	-	30	-0.3	-1.2	30	40[t]	m	-	75[m]	C	1.3	-	1.1	0.85
IFN860	**2N6550**	25	N	20	10	-	-0.3	-3	25	40[t]	10	-	30	D	1.3	2.6	1.1	0.83
IF3602	**IF3601**	100	N	20	30	-	-0.04	-3	750	-	m	-	300	-	0.65	1.8	0.63	0.38

PREFIXES: IF = interFET, LS = Linear Integrated Systems, PMB = NXP.

NOTES: ◊ = √Hz. * = same as row above. (a) sorted by decreasing voltage noise, and also by part number within a family. (b) preferred part. (c) complement 2SJ74. (d) discontinued, shown for comparison. (e) mV, max. (f) Fairchild part, others are much noisier. (g) LS840 has an interdigitated die. (k) dual unmatched J310. (m) maximum, or g_m@I_{DSS}. (n) RF JFET, high 1/*f* noise from several manufacturers. (o) also MPF4393 from ON Semi. (p) also PMBF- from NXP, very noisy. (t) typical.

A selection of currently available low-noise discrete JFETs. See also Table 3.7, which includes single-sample measurements of I_{DSS}, $V_{GS(off)}$, g_m, and g_{os}. The "Noise curve" column refers to corresponding measured noise spectra plotted in Figure 8.52. Some JFET types (e.g., the 2N5484–6 series) are intended for RF applications, and may exhibit high noise voltage density at low frequencies, as seen in the measured spectra of Figure 8.51. See also §3x.2 for plots of measured transconductance versus drain current.

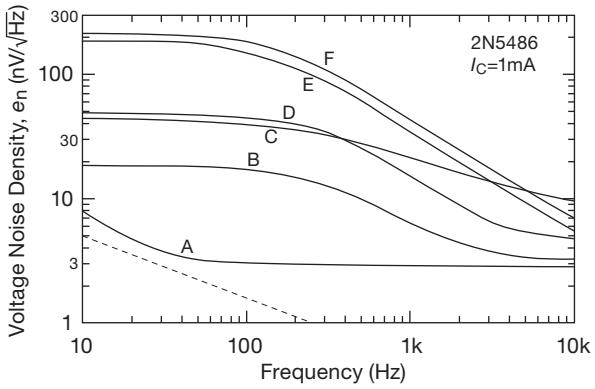


Figure 8.51. Beware unspecified noise parameters: measured voltage noise spectra of six 2N5486 JFET samples from five manufacturers (Fairchild's produced the winning "A" curve). Befitting its intended application as a "High-Frequency Amplifier," some datasheets specify noise performance only at high frequencies (e.g., ≥100 MHz). The slope of $1/f$ noise is indicated by the dashed line. JFETs "E" and "F" have 2.5 μVrms low-frequency noise (to 1 kHz).

available directly from Linear Integrated Systems, who also offer a matched dual version of the venerable '170, the LSK389. The excellent IFN146 and IFN147 are available directly from InterFET. Also consider the Fairchild J107, a low-cost switch that can moonlight as a low-noise JFET; with luck it will remain available.

Unspecified "low-noise" transistors *An important caution:* It is risky to rely on unspecified noise parameters, for instance the assumption (absent a manufacturer's e_n specification) that all transistors of a given part number will exhibit similar noise voltage to that measured in a sample batch. To see what can happen in practice, look at Figure 8.51, where we've plotted the measured noise-voltage spectra of six samples of a 2N5486 *n*-channel JFET. In fact, transistors even from the same manufacturer and production run may show large variations, as seen in Figure 8.40's measured noise spectra of four power transistors taken from a single batch. Figure 8.52 shows comparisons of the noise characteristics of a number of popular and useful JFETs.

Be aware of JFET weaknesses When we discussed the many considerations in choosing a low-noise *bipolar* transistor (§8.5.8), one factor of importance was the transistor's output resistance r_o (or, if you prefer, its output conductance, g_{oe}). For a BJT this is described by its Early voltage[65] V_A, whose measured values are listed in Table 8.1a on page 501. Most *npn* BJTs have fairly high V_A values, which is to say they have a high output resistance, thus ordinarily not a serious concern.

But that's not the case for JFETs. If you tried measuring an analogous "Early voltage" for JFETs, you would be disappointed in their low values. The parameter commonly used to describe the variation of drain current with drain voltage is output conductance, g_{os}, which varies with both drain voltage and drain current. For many JFETs the output conductance (ideally zero) is high enough to have a serious affect on amplifier gain. Because the value of a JFET's g_{os} is roughly proportional to its transconductance (g_m), we've defined a parameter $G_{max}=g_m/g_{os}$, which represents the maximum voltage gain of a common-source amplifier with high-impedance drain load.[66] This parameter is not listed in Table 8.2, but you can find it in the JFET listing of Table 3.7. You need to take it into account when designing a JFET amplifier stage, perhaps circumventing its gain-killing effect by adding a cascode; see §*3x.4*.

Some JFETs suffer also from a severe drop in transconductance at drain voltages of less than 2 V, especially at high drain currents; see the plots of transconductance versus drain voltage in §*3x.2*. If you're thinking of doing anything unusual in a low-noise JFET design, you'd do well to study first the JFET-specific material in Chapters 3 and *3x*.

8.7 Charting the bipolar–FET shootout

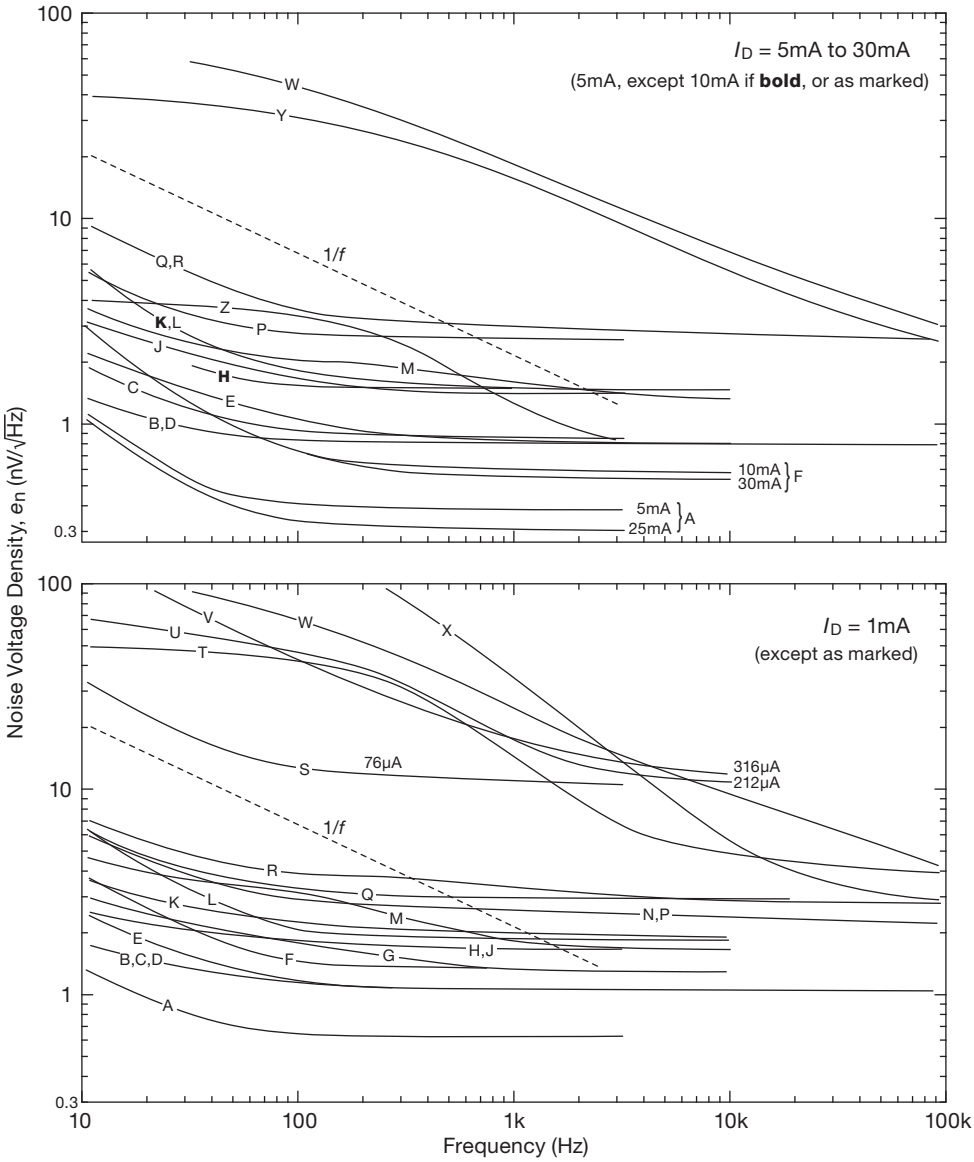
Let's have some fun with the graphical noise-charting technique we introduced in §8.5.2. A perennial bone of contention among engineers is whether FETs or bipolar transistors are "better." We dispose of this issue with characteristic humility by matching two of the best contenders and letting them deliver their best punches.

In the bipolar corner we have the magnificent 2SD786, whose vital statistics are displayed in Figure 8.10 and listed in Table 8.1a on page 501. Its noise voltage is $\sim$0.5 nV/$\sqrt{\text{Hz}}$ at 1 mA, with an admirably low $1/f$ noise corner well below 10 Hz; and its very low $r_{bb'}$ (we measured 2.3 Ω) allows us to get down to the 0.25 nV/$\sqrt{\text{Hz}}$ territory at higher collector currents. It has plenty of beta, too: above 200 at currents all the way down to 1 μA, which is helpful in achieving a low noise-*current*. It has been a perennial favorite among audio enthusiasts.

The FET entry is the 2SK170 *n*-channel JFET, known far and wide for its stunning low-noise performance, reputed to exceed that of bipolar transistors. Its very low

[65] Introduced in §2.3.2, and discussed in detail in §*2x.8*.

[66] We like G_{max} because it is largely independent of drain current, and varies in a predictable way (linearly) with drain voltage.



		g_m @ I_D (mS) 1mA	5mA
F	J107	8.2	30
W	pmbfJ108	7.8	20
H	J113	5.7	11
C	IFN146	13	30*
Z	2SK147V	15	38
B	2SK170BL	13	29
G	J309	4.2	8.7
Y	pmbfJ309	-	10.6
K	J310	4.3	9.1
M	LSK389	9.3	23
D	IFN860	13	35*
E	BF862	12.1	26
A	IF3601	27	86
S	PN4117	0.13 @ 0.07mA	
U	PN4118	0.17 @ 0.2mA	
V	PN4119	0.18 @ 0.3mA	
N	PN4391	4.7	8.8
J	PN4393	4.9	12
R	mmbf4416	2.2	3.8
L	mmbf5103	5.5	10.4
P	2N5457	2.3	3.8
Q	2N5486 (FSC)	2.0	3.5**
T	2N5486 (Vish)	2.3	3.9
X	2N5486 (Cen)	2.2	4.0

(*) estimated
(**) at I_D=3.5mA (I_{DSS})

Figure 8.52. Measured noise voltage density versus frequency for a selection of JFETs listed in Table 8.2. All data taken at V_{DS}=5 V.

measured noise voltage can be seen in Figure 8.54, and its vital statistics are listed in Table 8.2 on page 516. According to its datasheet, it was trained only for drain currents ranging from 200 μA to 10 mA; but we've conservatively extrapolated its noise parameters down to 100 μA, so that we can evaluate it with drain currents ranging from 100 μA to 10 mA. We have high hopes for this fighter, given its very low input-current noise ($\sim$1 fA/$\sqrt{\text{Hz}}$, the shot noise corresponding to its gate leakage current of $\sim$3 pA). In Figure 8.53 we've charted their total amplifier noise voltage versus source resistance at 1 kHz, just as we did for the 2N5087.[67]

[67] In our previous edition the contenders were the LM394 self-described "ultra well-matched monolithic NPN transistor pair" (e_n=1 nV/$\sqrt{\text{Hz}}$ at I_C=1 mA, versus 0.6 nV/$\sqrt{\text{Hz}}$ for this edition's 2SD786 championship contestant) and the 2N6483 matched monolithic *n*-channel JFET pair (e_n=4 nV/$\sqrt{\text{Hz}}$ at I_C=100 μA, versus 1.7 nV/$\sqrt{\text{Hz}}$ for this edition's 2SK170 entry). This edition's fighters are single transistors, rather than matched monolithic duals. Both contenders from the previous edition are long retired, having fought the good fight for several decades.

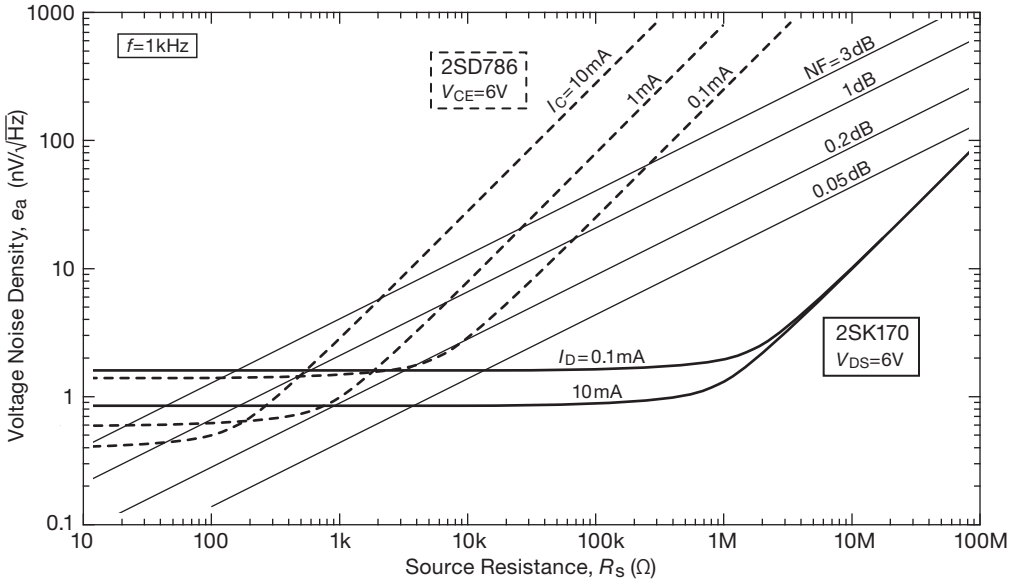


Figure 8.53. Comparison of total amplifier input voltage noise density (e_a) at 1 kHz, for the 2SD786 npn bipolar transistor (dashed lines) and the 2SK170 n-channel JFET (solid lines).

And the winner? Well, it's a split decision. The FET won points on lowest minimum noise figure, NF(R_n), reaching a phenomenal 0.0005 dB noise figure (that's a noise temperature of just 33 *milli*kelvin!), and dipping below 0.2 dB from 1k to 100M source impedance. For high source impedances, FETs remain unbeaten. The bipolar transistor is best at low source impedances, particularly below 5 kΩ, and it can reach a 0.2 dB noise figure for source resistances R_s from 1–10 kΩ, with a suitable choice of collector current. Although its victory in terms of noise alone is slim, it has other important virtues, notably its superior predictability of V_{BE} compared with a JFET's loosely characterized V_{GS}.

Just as in boxing, where the best fighters of yesteryear have retired from the rigors of competing, we must remind our readers, with some sadness, that the 2SD786 has been discontinued by Toyo–Rohm; and Toshiba's 2SK170 is sometimes available, sometimes not. However, also as in boxing, there are some younger contenders for the best low-noise transistor who haven't yet had a chance to compete in a world championship. The replacement LSK170 JFET from Linear Integrated Systems performs almost as well as its mentor; and the ZTX851 BJT from Zetex, though unspecified in noise performance, appears to outperform even the champion 2SD786 (as seen in the measured noise spectra of Figures 8.12 and 8.17).

And, if it's *really* low-noise voltage you want, consider the IF3601 from InterFET, with its 0.35 nV/$\sqrt{\text{Hz}}$ typical noise voltage *even down at 30 Hz*! And this is a *JFET*, with low input current (100 pA typ, hence low i_n, about 6 fA/$\sqrt{\text{Hz}}$), and thus the noise resistance is about 60k. When used as an amplifier with a source impedance equal to its noise resistance (i.e., R_s=60k), its performance is unbeatable – the noise figure is 0.001 dB. InterFET and Linear Integrated Systems JFETs must be purchased directly from the manufacturer, but in our experience they're happy to accept small orders.

Before you go out and buy a bushel of these remarkable JFETs, consider the remarks of the critics, who claim it is muscle-bound – it has high input and feedback capacitance (650 pF and 80 pF, respectively), which limits its usefulness at high frequencies. Its relative, the IFN146, is better in this regard, at the expense of higher e_n (75 pF and 15 pF, respectively, with e_n=0.7 nV/$\sqrt{\text{Hz}}$). These same critics argue that a bipolar complementary pair like the ZTX851 and ZTX951, with e_n as low as 0.3 nV/$\sqrt{\text{Hz}}$, can offer even better performance at moderate source impedances and frequencies.

8.7.1 What about MOSFETs?

The dominant FET species on the planet, by a demographic ratio of at least 10^{12}:1, is the *MOS*FET, of which the JFET is the poor (and neglected) sister. *We* haven't neglected

JFETs, though, because they're the best choice for discrete low-noise design, particularly when low noise at low frequencies is important (e.g., audio). As will be evident in §8.9, JFETs also dominate as input stages within op-amps where both low input current and low noise voltage are important.

So, what about MOSFETs for low-noise design? One problem is that you cannot get small MOSFETs as discrete parts. And MOSFETs tend to have much higher voltage noise than JFETs, particularly at low frequencies where the $1/f$ noise dominates – the $1/f$ knee can be as high as 10 kHz to 100 kHz. Figure 8.54 makes this point dramatically, plotting measured noise-voltage density versus frequency for a selection of representative MOSFETs, JFETs, and BJTs. The quietest BJTs are the winners (with low-noise JFETs not far behind), whereas at these audio frequencies the MOSFETs aren't even in the ballpark (and perhaps not even in the same city). For this reason you wouldn't normally choose a MOSFET for best-in-class low-noise amplifiers below 1 MHz.

MOSFETs *are* used as low-noise linear amplifiers, however, in the form of integrated circuits intended for radiofrequency applications. At those frequencies their noise performance is good enough, and the CMOS process allows convenient integration and low cost.

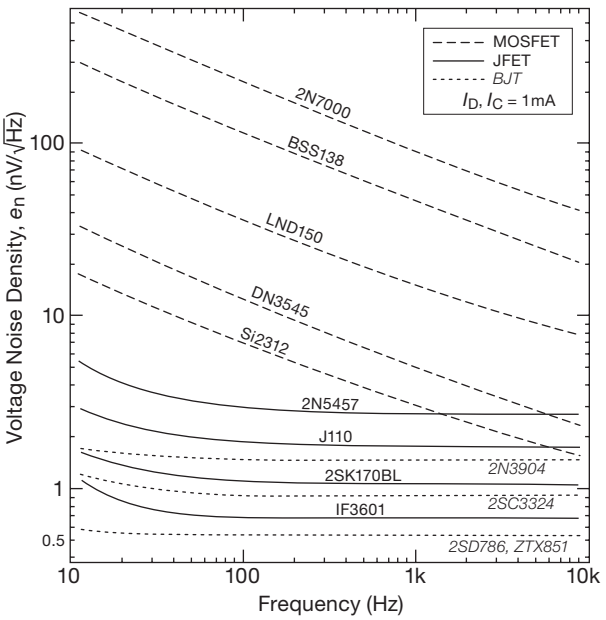


Figure 8.54. Comparison of measured noise-voltage spectra for some popular transistors: MOSFETs, JFETs, and BJTs. We included within each species both a "quietest" part and an inexpensive (<$0.05 in quantity) and popular "jellybean."

MOSFETs are used also in "zero-drift" op-amps: their noise voltage density e_n is not impressive – typically in the range of 25–100 $\text{nV}/\sqrt{\text{Hz}}$ – but they exhibit no $1/f$ rise at low frequencies, so they compete as the quietest amplifiers for very low-frequency applications. This is seen clearly in Figure 5.54.[68]

8.8 Noise in differential and feedback amplifiers

Low-noise amplifiers are often differential, to obtain the usual benefits of low drift and good common-mode rejection. When you calculate the noise performance of a differential amplifier, there are three points to keep in mind: (a) be sure to use the individual collector currents, not the sum, to get e_n and i_n from data sheets; (b) the i_n seen at each input terminal is the same as for a single-ended amplifier configuration; and (c) the e_n seen at one input, with the other input grounded, say, is 3 dB larger than the single-transistor case, i.e., it is multiplied by $\sqrt{2}$.

In amplifiers with feedback, you want to take the equivalent noise sources e_n and i_n out of the feedback loop, so you can use them as previously described when calculating noise performance with a given signal source. Let's call the noise terms brought out of the feedback loop e_A and i_A, for *amplifier* noise terms. Thus the amplifier's noise contribution to a signal with source resistance R_s is

$$e^2 = e_A^2 + (R_s i_A)^2 \quad \text{V}^2/\text{Hz}. \tag{8.37}$$

Let's take the two feedback configurations separately.

Noninverting For the noninverting amplifier (Figure 8.55) the input noise sources become[69]

$$i_A^2 = i_n^2 \quad \text{A}^2(\text{rms}), \tag{8.38}$$

$$e_A^2 = e_n^2 + 4kTR_{\|} + (i_n R_{\|})^2 \quad \text{V}^2(\text{rms}), \tag{8.39}$$

where e_n is the "adjusted" noise voltage for the differential configuration, i.e., 3 dB larger than for a single-transistor stage. The additional noise-voltage terms arise from Johnson noise and input-stage noise current in the feedback resistors. Note that the effective noise voltage and current are now not completely uncorrelated, so calculations in which their squares are added can be in error by a maximum factor of 1.4.

[68] The OPA188 and OPA2188 high-voltage CMOS autozero op-amps are among the best you may find, with an e_n of $9\,\text{nV}/\sqrt{\text{Hz}}$ and a noise corner at 0.4 Hz. We won't vouch for their input *current* noise.

[69] See C. D. Motchenbacher and J. A. Connelly, *Low-noise Electronic System Design*, Wiley (1993), who show the complete expression for a full differential amplifier.

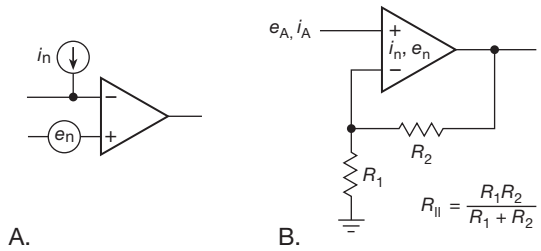


Figure 8.55. A. Noise model of the op-amp. B. Noise sources in the noninverting amplifier.

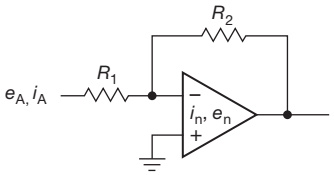


Figure 8.56. Noise sources in the inverting amplifier.

For a follower, R_2 is zero, and the effective noise sources are just those of the differential amplifier alone. Note that these drawings and formulas assume that the signal source resistance is zero (or at least small compared with $R_{||}$, so as not to add Johnson noise).[70]

Inverting For the inverting amplifier (Figure 8.56) the input noise sources become

$$i_A^2 = i_n^2 + 4kT\frac{1}{R_1} \quad A^2(\text{rms}), \tag{8.40}$$

$$e_A^2 = e_n^2\left(1+\frac{R_1}{R_2}\right)^2 + 4kTR_1\left(1+\frac{R_1}{R_2}\right) + (i_n R_1)^2 \quad V^2(\text{rms}). \tag{8.41}$$

For differential amplifiers and ordinary op-amps the two input terminals have comparable input noise voltages, and noise currents. This is not true, however, for current-feedback (CFB) op-amps (§*4x.6*), where the noise current at the inverting input is generally much larger than at the non-inverting input.

Operational amplifiers These have differential inputs, so they play by the same rules. Op-amps dominate most analog designs, and for good reason: they are highly evolved and deliver excellent performance. There are thousands of op-amp choices, optimized for various combinations of parameters that include accuracy, speed, noise, power consumption, and supply voltage. If you can solve a design problem with op-amps, you probably should[71]. Following the theme of this chapter, low-noise design with op-amps is the subject of the next section.

8.9 Noise in operational amplifier circuits

Op-amp datasheets specify input noise in terms of e_n and i_n, just as with transistors and FETs. You'll see tabulated values, and usually plots of e_n and (sometimes) i_n versus frequency. Unlike design with discrete transistors, however, you don't get to adjust internal operating currents and component values – you only get to use them.

There are thousands of op-amps out there, with a good selection of parts aimed at low-noise applications. We've collected some 150 favorites in Tables 8.3a–8.3c, many of whose noise voltage and current density are plotted in Figures 8.60 and 8.61. How to choose among these? There's e_n and i_n, of course (and the corresponding $1/f$ low-frequency corner frequency f_c). But there are also all the usual performance tradeoffs: accuracy, speed, input current, power dissipation, and price. As an example, at $0.85\,\text{nV}/\sqrt{\text{Hz}}$ (and f_c=3.5 Hz) the LT1028 is among the lowest e_n op-amps available (curve A in the upper plot of Figure 8.60). But it draws 8.5 mA of supply current, with ∼1.8 mA devoted to the input stage alone. For all the reasons discussed in §8.3–§8.6, you can't match that kind of low-noise voltage in an op-amp whose total supply current is, say, 0.1 mA. And you may not want to: the LT1028's BJT inputs have plenty of bias current, with correspondingly high noise-current density (∼1 pA/√Hz – curve A again, this time in the lower plot of Figure 8.60). Put another way, the high-noise *current* of this low-e_n champ wipes out its noise advantage for an input signal of source impedance greater than 1 kΩ (its noise resistance, $R_n \equiv e_n/i_n$).[72]

Let's look more closely at low-noise design with op-amps. We begin with a guide to the extensive listing in Table 8.3.

[70] If an additional series resistor is added at the noninverting input (to balance input-current offsets, see Figure 4.55), consider shunting it with a capacitor to quiet its Johnson noise (e.g., C_1 in Figure 8.78).

[71] Look at it this way: an entire op-amp in a SOT-23 package is the same size as a single transistor in a SOT-23 package, and it may not cost much more. Table 8.3 lists over 60 op-amps that are available in SOT-23 packages, with prices as low as $0.72 (qty 25).

[72] While we're complaining about the LT1028, we might add that it's a bit pricey, about $6 apiece in modest quantity.

Table 8.3a Low-noise BJT-input Op-amps[a]

Table 5.5	Part #	# per pkg	Supply range (V)	Supply I_Q^t (mA[p])	Input I_B @25°C typ (nA)	Input I_B @25°C max (nA)	Bias cancel	V_{OS} max (mV)	CMRR min (dB)	Noise $V_{n(pp)}$ 10Hz[b] (μV)	Noise e_n 1kHz ($nV/\sqrt{Hz}$)	Noise f_c[c] (Hz)	Noise i_n^t 1kHz[J] ($pA/\sqrt{Hz}$)	Noise plots	R_n (e_n/i_n) 1kHz (kΩ)	GBW (MHz)	C_{in} (pF)	Swing to Supply? IN + −	Swing to Supply? IN − +	Swing to Supply? OUT + −	Offset trim	Overcomp pin	Pkg DIP	Pkg SOIC-8	Pkg SOT-23	Cost[q] qty 25 ($US)	Comments
	high-voltage BJT																										
	LT1495	1,**2**,4	2.2-36	0.001	0.3	1	-	0.38	90	4	185	10	0.01	b6	20M	0.003	-	•	•	•	-	-	•	•	-	4.15	**A**
❑	LT1490A	**2**,4	2.5-44	0.04	1	8	-	0.5	84	1	50	5.6	0.02	-	3.3M	0.2	4.6	•	•	•	-	-	•	•	-	3.25	**B**
❑	LT1077A	**1**	2.2-44	0.05	7	9	-	0.04	97	0.5	27	0.7	0.07	-	415	0.23	low	-	•	-	•	-	•	•	-	3.84	-
	ADA4096	**2**	3-36	0.06	10	15	-	0.3	73	0.7	27	3	0.2	-	135	0.6	7	•	•	•	-	-	-	•	-	3.81	**C**
❑	LT6010A	**1**,2,4	2.7-40	0.14	0	0.11	•	0.04	107	0.4	14	3.6	0.1	34	140	0.35	4	-	-	•	•	-	-	•	-	2.22	**D**
	LT6011A	**2**,4	2.7-40	0.14	0	0.3	•	0.06	107	0.4	14	3.6	0.1	34	140	0.33	4	-	-	•	•	-	-	•	-	2.36	-
	LT6013A	**1**,2	2.7-40	0.15	0.1	0.25	•	0.06	115	0.2	9.5	2	0.15	-	63	1.6	4	-	-	•	•	-	-	•	-	2.36	**E**
	OP07C	**1**	8-44	2.7	1.8	7	•	0.15	97	0.38	9.8	0.7	0.13	-	75	0.6	-	-	-	-	-	•	-	•	-	***0.62***	**F**
	ISL28107	**1**,2	4.5-42	0.21	0.02	0.3	•	0.08	115	0.34	13	2	0.05	-	245	1	-	-	-	-	-	-	-	•	-	2.73	-
❑	AD8622A	**2**,4	4-36	0.22	0.05	0.2	-	0.13	125	0.2	11	2	0.15	-	73	0.56	5.5	-	-	•	•	-	-	•	-	5.33	-
	LT1013A	**2**,4	3-44	0.35	12	20	-	0.15	100	0.55	22	1.3	0.02	19	1.5M	0.9	-	-	•	-	•	-	•	x	-	2.08	-
	LT1097	**1**	2.4-40	0.35	0	0.25	•	0.05	115	0.5	14	2.5	0.05	-	304	0.7	-	-	-	-	-	•	•	•	-	2.32	-
❑	LT1012AC	**1**	2.4-40	0.37	0.03	0.1	•	0.03	114	0.5	14	2.5	0.05	M	304	0.5	low	-	-	-	-	•	•	•	-	5.11	-
❑	OP97E	**1**,2,4	4.5-40	0.40	0	0.1	•	0.03	114	0.5	14	2.5	0.04	-	333	0.9	low	-	-	-	-	•	•	•	-	5.52	-
	LM741A	**1**,2,4	8-44	1.7	30	80	-	3	80	1.8	28	20	0.22	22	127	1.5	-	-	-	-	-	•	-	•	•	***0.23***	**G**
	LM358	1,**2**,4	4-32	0.7	45	250	-	7	70	1.8	40	10	0.12	24	333	0.7	-	-	•	-	•	-	-	•	•	***0.14***	**H**
	OP2177	1,**2**,4	5-36	0.40	0.5	2	•	0.08	120	0.4	8	5	0.2	-	40	1.3	-	-	-	-	-	-	-	•	-	3.51	-
❑	OPA2188	1,**2**,4	4.0-40	0.42	0.2	0.85	•	0.03	120	0.25	8.8	0.4	0.75	J	12	2	9.5	-	•	•	•	-	-	-	•	•	3.43	**I**
❑	OPA277P	**1**,2,4	4-36	0.79	0.5	1	•	0.02	130	0.22	8	20	0.2	13	40	1	low	-	-	-	-	•	-	•	•	-	3.17	**J**
	ISL28218	1,**2**	3-44	0.85	230	600	-	0.15	102	0.3	5.6	16	0.36	-	16	4	-	-	•	•	•	-	-	-	•	-	4.25	-
	ISL28127	**1**,2	4.5-42	0.85	1	10	•	0.07	115	0.085	2.5	6	0.4	-	6.3	10	-	-	-	-	-	-	-	-	•	-	2.50	-
❑	LT1468	**1**	7-36	3.9	3	10	•	0.08	96	0.3	5	27	0.6	9	8.3	90	4	-	-	-	-	•	-	•	•	-	4.26	**K**
	NE5534	**1**,2	10-44	4	0.5	1.5	-	4	70	0.4	3.5	50	0.4	G	8.8	10	-	-	-	-	-	•	•	•	•	-	0.85	**L**
❑	LT1677	**1**	3-44	2.8	2e	20[e]	•	0.06	109	0.09	3.2	13	1.2	-	2.7	7.2	4.2	•	•	•	•	•	-	•	•	-	3.07	**M**
❑	OP-27A[d1]	**1**	8-44	3	10	40	-	0.03	114	0.08	3	2.7	1	8	3	8	-	-	-	-	-	•	-	•	-	-	2.10	**N**
❑	OPA227[d2]	**1**,2,4	5-36	3.7	2.5	10	•	0.08	120	0.09	3	4	0.4	F	7.5	8	12	-	-	-	-	•	-	•	•	-	1.67	-
	LME49710	**1**	5-36	4.8	7	72	-	0.7	110	0.35	2.5	70	1.6	-	1.6	55	-	-	-	-	-	-	-	•	•	-	1.97	**O**
	LM4562	**2**	5-36	5	10	72	-	0.7	110	0.4	2.7	75	1.6	b4	1.7	55	-	-	-	-	-	-	-	•	•	-	2.45	**O**
	ADA4075-2	**2**	9-40	1.8	30	100	-	1	110	0.06	2.8	3.6	1.2	-	2.3	6.5	2.4	-	-	-	-	-	-	-	•	-	1.80	**P**
❑	AD8675	**1**,2	9-36	2.5	0.5	2	•	0.08	114	0.1	2.8	6	0.3	-	9.3	10	-	-	-	•	•	•	-	-	•	-	2.22	**Q**
	AD8671	**1**,2,4	8-36	3	3	12	•	0.08	100	0.077	2.8	5	0.3	-	9.3	10	7.5	-	-	-	-	-	-	-	•	-	1.97	**Q**
	LT1124	**2**,4	8-44	2.75	8	30	•	0.1	108	0.07	2.7	2.3	0.3	-	9.0	12.5	low	-	-	-	-	-	-	-	•	•	4.91	**R**
❑	LT1007[g,d4]	**1**	4-44	2.7	10	35	•	0.03	117	0.06	2.5	2		7,D	6.3	8	-	-	-	-	-	•	-	•	•	-	2.48	-
❑	OPA209	**1**,2,4	4.5-40	2.2	1	4.5	•	0.15	120	0.13	2.2	16	0.5	-	4.4	18	4	-	-	•	•	-	-	-	•	•	2.27	**S**
❑	ADA4004	**1**,2,4	9-36	2.2	40	90	•	0.13	110	0.15	1.8	2.5	1.2	b1	1.5	12	-	-	-	-	-	-	-	-	•	•	4.20	-
	OPA211	**1**,2	4.5-36	3.6	60	175	-	0.05	114	0.08	1.1	10	1.7	b2	0.65	45	8	-	-	•	•	-	-	-	•	-	12.87	**T**
❑	AD8597	**1**,2	9-36	4.8	40	210	•	0.12	120	0.08	1.1	22	2.3	B	0.48	10	12	-	-	-	-	-	-	-	•	•	3.71	**U**
	MAX9632	**1**	4.5-40	3.9	30	180	•	0.13	120	0.065	0.94	22	3.8	-	0.25	55	-	-	-	•	•	-	-	-	•	-	5.76	**V**
	ADA4898-1	**1**,2	9-36	8.1	100	400	•	0.13	103	0.05[u]	0.9	14	2.4	4	0.38	65	3.2	-	-	-	-	-	-	-	•	-	5.37	-
	AD797	**1**	9-36	5.2	250	1500	-	0.08	114	0.05	0.9	30	2.0	-	0.45	110	-	-	-	-	-	•	•	•	•	-	8.03	-
	LT1115	**1**	9-44	8.5	50	380	•	0.2	104	0.04	0.9	3.5	1.2	3,A	0.75	70	5	-	-	-	-	•	•	•	-	-	3.73	**W**
	LME49990	**1**	10-38	9	30	500	•	1	118	0.03	0.88	10	2.8	-	0.31	110	-	-	-	-	-	-	-	-	•	-	3.98	**X**
❑	LT1028A[h]	**1**	8-44	7.4	25	90	•	0.04	108	0.035	0.85	3.5	1.0	3,A	0.85	75	5	-	-	-	-	•	•	•	•	-	6.48	-
	low-voltage BJT																										
	TLV2401	**1**,2,4	2.7-17	880nA	0.1	0.3	-	1.2	63	35	400	30	0.01	30	50M	0.005	3	w	•	•	•	-	-	-	•	•	2.05	-
❑	LT6003	**1**,2,4	1.5-18	850nA	0	0.14	-	0.5	73	3	325	-	0.01	b5	27M	0.003	6	•	•	•	•	-	-	-	•	•	1.60	-
	TLV2242	1,**2**,4	2.5-16	0.001	0.1	0.5	-	3	60	35	400	30	0.01	31	50M	0.005	3	•	•	•	•	-	-	-	•	•	2.42	-
	OP196	**1**,2,4	3-15	0.045	10	50	-	0.3	65	0.8	26	15	0.19	-	137	0.35	-	•	•	•	•	•	-	-	•	-	3.81	-
	EL8176	**1**	2.4-6	0.055	0.5	2	•	0.1	90	1.5	28	6	0.16	-	175	0.4	-	•	•	•	•	-	-	-	•	•	2.30	**Y**
	LMV358	**1**,2,4	2.7-6	0.11	15	250	-	7	65	-	39	8	0.21	28	186	1	-	-	•	-	-	-	-	-	•	•	***0.74***	**Z**
	LT1783	**1**	2.5-18	0.23	45	80	r	0.8	90	1	20	7.5	0.14	-	143	1.3	5	k	•	•	•	-	-	-	-	•	3.05	**AA**
	TLV2460	**1**,2,4	2.7-6	0.5	4.4	14	-	2	70	2.2	11	150	0.13	-	85	5.2	7	•	•	•	•	-	-	•	•	•	1.55	**BB**
	LT6220	**1**,2,4	2.2-13	0.9	15	150	•	0.35	85	0.5	10	30	0.8	-	12.5	60	2	•	•	•	•	-	-	-	•	•	1.75	-
❑	LMP7731	**1**,2	1.8-6	2.2	1.5	30	•	0.5	101	0.08	2.9	1.4	1.1	b3	2.64	22	-	•	•	•	•	-	-	-	•	•	1.98	-
	MAX410	**1**,2,4	4.8-12	2.5	80	150	•	0.25	115	0.2	1.5	90	1.2	-	1.25	28	4	-	-	-	-	•	-	•	•	-	4.06	-
	LT6230	**1**,2,4	3-12.6	3.3	5μA	10μA	•	0.5	96	0.18	1.1	350	2.4	-	0.46	215	6.5	-	-	•	•	-	-	-	•	•	2.72	-

Notes: (a) see also op-amp tables in Chapters 4, 4x, and 5. (b) 0.01Hz or 0.1Hz to 10Hz. (c) calculated. (d) ***italics*** are lower-voltage JFET types. (d1) OP-37 decomp. (d2) OP228 decomp. (d3) OPA637 decomp. (d4) LT1037 decomp. (d5) OPA657 decomp. (d6) ADA4637 decomp. (d7) MAX4237 decomp. (d8) LMP7717 decomp. (g) LTC suggests LT1677, RRIO. (h) LTC suggests LT6200, 6230. (J) at 1kHz or 10kHz, except 10Hz for chopper types. (k) beyond the rail. (p) per amplifier. (q) ***italics*** are jellybeans. (s) SC70. (t) typical. (u) datasheet specs are 10x higher. (v) V_{OS} vs V_{CM} stable. (w) to 5V above the rail. (x) SOIC is non-std pinout. (y) DIP discontinued, NRND. **A:** specs at 5V. **B:** over-the-top, low I_B for $V_{CM}<V_{CC}-0.7$. **C:** OVP to +32V. **D:** LT1012 replacement, w/RRO. **E:** decomp, $G>5$. **F:** original classic. **G:** noisy classic. **H:** noisy, cheap; LM321 single. **I:** zero-drift, CMOS(!). **J:** improved OP177. **K:** 0.7ppm dist. **L:** classic audio. **M:** "RRIO LT1007."

Table 8.3b Low-noise FET-input Op-amps[a]

Table 5.5	Part #[d]	# per pkg	Supply range[d] (V)	Supply I_Q[t] (mA[p])	Input I_B @25°C typ (pA)	Input I_B @25°C max (pA)	V_{OS} max (mV)	TCV_{OS} typ (µV/°C)	CMRR min (dB)	Noise $V_{n(pp)}$ 10Hz[b] (µV)	Noise e_n 1kHz (nV/√Hz)	Noise f_C[c] (Hz)	Noise $e_n \cdot C_{in}$ 1kHz[J] (nV·pF)	Noise plots	GBW (MHz)	C_{in} (pF)	Swing to Supply? IN +−	Swing to Supply? IN −−	Swing to Supply? OUT +−	Swing to Supply? OUT −−	Offset trim	Pkg DIP	Pkg SOIC-8	Pkg SOT-23	Cost[q] qty 25 ($US)	Comments
	JFET																									
❑	**OPA129B**	**1**	10-36	1.2	30fA[z]	0.1	2	3	80	4	17	310	34	32,U	1	2	-	-	-	-	-	•	•	-	11.28	**CC**
❑	**AD549KH**	**1**	10-36	0.6	75fA[z]	0.1	0.25	2	90	4	35	100	35	-	1	1	-	-	-	-	•	-	-	-	28.00	-
	ISL28110	**1**,2	9-42	2.6	0.3	2	0.3	1	88	0.6	6	45	72	-	12.5	12	-	-	-	-	-	-	-	-	2.34	**DD**
❑	**OPA124PB**	**1**	10-36	2.5	0.35	1	0.25	1	100	1.6	8	195	24	33,P	1.5	3	-	-	-	-	•	•	•	-	6.40	**EE**
❑	**OPA140**	**1**,2,4	4.5-40	1.8	0.5	10	0.12	0.35	126	0.25	5.1	12	51	H	11	10	-	•	•	•	-	-	-	•	3.89	**FF**
	AD8076	**1**	***5-24***	6.4	0.5	5	1	1	89	6	6.6	2000	17	-	350	2.5	-	•	•	•	-	-	-	•	5.32	-
❑	**AD795**	**1**	8-36	1.3	1	2	0.5	3	90	1	8	50	18	-	1.6	2.2	-	-	-	-	-	-	•	-	7.97	**GG**
❑	**OPA141**	**1**,2,4	4.5-40	1.8	2	20	3.5	2	120	0.25	6.5	13	52	-	10	8	-	•	•	•	-	-	•	•	2.36	**FF**
❑	**OPA1641**	**1**,2,4	4.5-40	1.8	2	20	3.5	-	120	0.2	5.1	7	41	f1	20	8	-	•	•	•	-	-	•	-	2.76	**FF**
❑	***AD8620A***	1,**2**	***10-27***	2.5	2	10	0.25	0.5	90	1.8	6	1200	90	-	25	15	-	-	-	-	•	-	•	-	11.86	-
❑	***OPA656***[d5]	**1**	***9-13***	14	2	20	1.8	2	80	4.5	7[x]	1100	20	T	230	2.8	-	-	-	-	-	-	•	•	5.59	-
	LF412C	**1**,2,4	10-44	3.6	50	200	3	7	80	2.5	25	45	75	21	3	3	-	-	-	-	•	•	•	-	***1.32***	**HH**
	OPA171	**1**,2,4	2.7-40	0.48	8	15	1.8	0.3	104	3	14	170	42	18	3	3	-	•	•	•	-	-	•	•	0.91	**ZZ**
	OPA604	1,2	9-50	5.3	50	-	5	8	80	1.3	11	60	110	16	20	10	-	-	-	-	•	•	•	-	2.57	**JJ**
❑	**OPA134**	**1**,2,4	5-36	4	5	100	2	2	86	1.1	8	80	40	f3	8	5	-	-	-	-	•	•	•	-	***1.60***	**KK**
	OPA2132	**1**,**2**,4	5-36	4	5	50	0.5	2	96	1.1	8	80	48	-	8	6	-	-	-	-	•	•	•	-	3.81	-
	LME49880	**2**	10-36	14	5	-	10	3	90	1.9	7	250	-	-	25	-	-	-	-	-	-	-	•	-	2.43	**O**
	ADA4627B[d6]	**1**	9-36	7	5	20	0.2	1	106	0.7	6	60	42	-	19	7	-	-	-	-	•	-	•	-	15.05	**LL**
	LT1793	**1**	9-40	4.2	4	20	0.9	8	81	0.5[u]	5.8	30	9	10	4.3	1.5	-	-	-	-	•	•	•	-	3.15	-
❑	**OPA627A**[d3]	**1**	10-36	7	5	20	0.28	0.5	106	0.6	4.5	90	32	L	16	7	-	-	-	-	•	•	•	-	20.82	-
	LT1792	**1**	9-40	4.2	300	800	0.8	7	82	0.35[u]	4.0	30	56	f2	4.3	14	-	-	-	-	•	•	•	-	3.15	-
❑	**OPA827**	**1**	8-40	4.8	15	50	0.15	1.5	104	0.25	3.8	25	34	-	22	9	-	-	-	-	-	-	•	-	9.22	**MM**
	AD743	**1**	9.6-36	8.1	150	250	1	2	80	0.38	2.9	35	58	-	4.5	20	-	-	-	-	•	y	y	-	7.83	-
	CMOS																									
	ISL28194	**1**	1.8-5.8	330nA	15	80	2	1.5	70	10	265	7	-	-	0.004	-	•	•	•	•	-	-	-	•	1.69	**NN**
	LPV521	**1**	1.6-5.5	470nA	10fA	1	1	0.4	75	22	259	7	-	c3	0.006	-	•	•	•	•	-	-	-	s	1.66	-
	LMC6442A	**2**	2.2-11	950nA	5fA	4	3	0.4	102	3	170	0.5	33	29	0.01	4.7	-	•	•	•	-	•	•	-	2.43	-
	ISL28195	**1**	1.8-5.8	0.001	15	80	2	1.5	70	4	150	3	-	-	0.01	-	•	•	•	•	-	-	-	•	1.57	-
	MAX9911	**1**,2	1.8-6	0.004	1	10	1	5	70	40	400	45	-	-	0.2	-	•	•	•	•	-	-	-	•	0.72	**OO**
	ICL7612	**1**	2-16	0.006	0.5	30	5	15	70	40	100	500	-	c4	0.044	-	•	•	•	•	-	•	•	-	2.87	**PP**
❑	**LMP2232A**	1,**2**,4	1.8-6	0.01	20fA	1	0.15	0.3	81	2.3	60	20	-	25	0.13	-	•	•	•	•	-	-	•	•	3.29	-
	ISL28158	**1**,2	2.4-5.8	0.034	5	35	0.3	0.3	75	1.4	64	26	-	-	0.2	-	•	•	•	•	-	-	•	•	1.34	**QQ**
❑	**AD8603**	**1**,2,4	1.8-6	0.04	0.2	1	0.05	1	85	2.3	25	50	18	20	0.4	2.5	•	•	•	•	-	-	•	•	1.36	-
❑	**LTC6078**	**2**,4	2.7-6	0.055	0.2	1	0.03	0.2	95	1.0	18	25	126	-	0.75	18	•	•	•	•	-	-	•	-	3.54	**RR**
	LMV358	**1**,2,4	2.7-6	0.1	<1nA	<1nA	7	6	50	55	33	7000	-	-	1.4	-	-	•	-	-	-	-	•	s	***0.45***	**Z**
	TLV2221	**1**	2.7-12	0.11	1	150	3	1	70	3	19	100	42	-	0.51	6	-	•	•	•	-	-	-	•	1.20	-
❑	**MAX4236A**[d7]	**1**	2.4-6	0.35	1	500	0.02	0.6	84	0.9	14	17	53	c2	1.7	7.5	-	•	•	•	-	-	•	•	1.78	-
❑	**LMC6001A**	**1**	5-16	0.45	10fA	25fA	0.35	2.5	83	6	22	270	-	-	1.3	-	-	•	•	•	-	•	•	-	12.19	-
❑	**LMC6482A**	**2**,4	3-16	0.5	20fA	4	0.75[v]	1	70	20	37	900	21	17	1.5	3	•	•	•	•	-	•	•	-	***1.88***	**SS**
	LMV751	**1**	2.7-5.5	0.5	1.5	100	1		85	0.7	6.5	45	-	-	4.5	-	-	•	•	•	-	-	-	•	1.82	-
❑	**OPA376**	**1**,2,4	2.2-7	0.76	0.2	10	0.03	0.26	76	0.8	7.5	50	91	c1	5.5	13	•	•	•	•	-	-	•	•	1.32	**TT**
❑	**OPA364**	**1**,2	1.8-5.5	0.85	1	10	0.5	3	74	10	17	2200	21	-	7	3	•	•	•	•	-	-	•	•	2.18	**UU**
❑	**AD8692**	1,**2**,4	2.7-6	0.85	0.2	1	2	0.3	68	1.6	8	215	35	-	10	5	-	•	•	•	-	-	•	•	1.36	**Q**
	LMV791	**1**,2	1.8-6	0.95	0.05	1	1.35	1	80	1.3	6.2	170	105	-	14	15	-	•	•	•	-	-	•	•	1.42	-
❑	**TLC4501A**	**1**,2	4-7	1	1	60	0.04	1	90	1.5	12	500	56	-	4.7	8	-	•	•	•	-	-	•	-	3.06	**VV**
❑	**OPA743**	**1**,2,4	3.5-13	1.1	1	10	7	8	66	11	30	2400	28	26	7	4	•	•	•	•	-	-	•	•	***1.58***	-
	LM6211	**1**	5-24	1.1	0.5	5	2.5	2	85	2.5	6	500	39	-	17	5.5	-	•	•	•	-	-	-	•	2.52	-
❑	**LMP7715**[d8]	**1**,2	1.8-6	1.15	0.05	1	0.15	1	85	1	5.8	110	105	-	17	15	-	•	•	•	-	-	•	•	2.05	-
	LMV710	**1**	2.5-5.5	1.2	4	-	3	-	50	14	20	1300	119	-	5	17	•	•	•	•	-	-	-	•	1.21	**WW**
❑	**LMP7721**	**1**	1.8-6	1.3	3fA	20fA	0.15	1.5	83	2.3	7	360	-	-	17	-	-	•	•	•	-	-	x	-	11.89	-
	OPA320	**1**,2	1.8-6	1.45	0.2	0.9	0.15	1.5	100	2.8	8.5	640	35	-	20	5	•	•	•	•	-	-	•	•	2.16	**XX**
❑	**AD8616**	1,**2**,4	2.7-6	1.7	0.2	1	0.5	1.5	80	2.4	7	1300	49	11	24	7	•	•	•	•	-	-	•	•	***1.52***	-
	TLC2272A	**2**,4	4.5-16	2.4	1	60	1.0	2	75	1.4	9	325	56	S	2.25	8	-	•	•	•	-	•	•	-	1.57	-
	OPA365	**1**,2	2.2-5.5	4.6	0.2	10pA	0.2	1	100	5	4.5	5600	42	V	50	6	•	•	•	•	-	-	•	•	1.75	**XX**
❑	**OPA350**	**1**,2,4	2.5-7	5.2	0.5	10	0.5	4	74	14	7	9000	46	12	38	6.5	•	•	•	•	-	-	•	•	1.67	-
	LMP2021	**1**,2	2.5-5.8	1.1	25	100	0.005	0.004	105	0.26	11	flat	84	##	5	12	-	•	•	•	-	-	•	•	2.92	**YY**

N: classic industrial. **O:** 0.3ppm dist. **P:** cross-coupled. **Q:** 0.6ppm dist. **R:** replaces OP-27, OP270. **S:** 0.25ppm dist. **T:** precision. **U:** 1ppm dist. **V:** 0.4ppm dist. **W:** audio version of LT1028. **X:** 0.1ppm dist. **Y:** charge pump. **Z:** LMV321 single; BJT& CMOS versions (same p/n!). **AA:** above the rail to +18V. **BB:** 80mA output drive. **CC:** lowest I_B for an HV part. **DD:** bootstrapped JFETs. **EE:** substrate pin. **FF:** 0.5ppm dist. **GG:** replaces OPA111. **HH:** AoE jellybean. **JJ:** pro audio. **KK:** 0.8ppm dist. **LL:** replaces OPA627, 637. **MM:** costs less than OPA627. **NN:** <1µW. **OO:** 7.5µW. **PP:** programmable quiescent current. **QQ:** low V_{OS} beyond rails. **RR:** V_{OS} degrades near V+. **SS:** popular; LMC7101 (single) is similar. **TT:** e-trim. **UU:** charge pump, 20ppm dist. **VV:** 0.3sec self-cal at power-on. **WW:** BiCMOS; LMV711 has shdn. **XX:** charge-pump, RRI. **YY:** 2nd best chopper. **ZZ:** CMOS (among several, check out OPA170, 172, 192; also OPA2188).

Table 8.3c High-speed Low-noise Op-amps[a]

Part #	# per pkg	Supply range (V)	Supply I_Q[t] (mA[p])	I_{BIAS} 25°C typ (μA)	CMRR min (dB)	Noise ("◊"=√Hz) e_n[t] 1kHz (nV/◊)	Noise f_c[c] (Hz)	Noise i_n[t] 1kHz[J] (pA/◊)	C_{in} (pF)	FOM e_nC_{in} ("◊"=√Hz) 0pF (pF•nV/◊)	FOM +5pF	Min gain	GBW[t] (MHz)	Slew Rate (V/μs)	Distortion plot[d]	Iout min (mA)	Swing to Supply? IN +	IN −	OUT +	OUT −	Offset trim	Overcomp pin	Shutdown pin	Pkg DIP	Pkg SOIC-8	Pkg SOT-23	Cost[q] qty 25 ($US)	Comments
HV bipolar VFB																												
LT1222	1	8-36	8	0.1	100	3	130	2	2	6.0	21	10	500	250	-	25	-	-	-	-	•	•	-	•	•	-	4.17	-
AD8021	1	4.5-26	7	7.5	86	2.1	2300	2.1	1	2.1	13	1	490	420	•	60	-	-	-	-	-	•	•	-	•	-	2.42	-
AD829	1	9-36	5	3.3	100	1.7	45	1.5	5	8.5	17	1	750[r]	150[g]	-	25	-	-	-	-	-	•	-	•	•	-	4.82	-
THS4031	1,2	9-33	8.5	3	85	1.6	390	1.2	2	3.2	11	1	200	100	-	60	-	-	-	-	•	-	-	•	-	5.60	-
THS4021	1,2	9-33	7.8	3	-	1.5	570	2	1.5	2.3	10	10	3500	470	-	80	-	-	-	-	•	-	-	-	•	-	8.20	-
MAX9632	1	4.6-40	3.9	0.03	120	0.94	16	3.8	-	-	-	1	55	30	-	53	-	-	•	•	-	-	•	-	•	-	5.76	-
AD797B	1	10-36	8.2	0.25	114	0.9	30	2	20	18	23	1	110	20	-	60	-	-	-	-	•	•	-	•	•	-	8.50	-
ADA4898	1,2	9-36	8.1	0.1	102	0.9	14	2.4	3.2	2.9	7.4	1	65	55	-	40	-	-	-	-	-	-	•	-	•	-	5.37	-
LME49990	1	10-38	9	0.03	110	0.88	10	2.8	-	-	-	1	110	22	•	50	-	-	-	-	-	-	-	-	•	-	4.16	A
LT1028A	1	8-44	7.4	0.03	108	0.85[v]	3.5[v]	1	5	4.3	8.5	1	75	15	-	25	-	-	-	-	•	•	-	•	•	-	6.48	B
HV bipolar CFB																												
THS3120	1	9-33	7	1	60	2.5	215	1	0.4	1.0	14	1	525	1500	-	425	-	-	-	-	-	-	•	-	•	-	5.24	C
AD844	1	9-36	6.5	0.2	-	2.0	-	10	2	4.0	14	2	330	2000	-	60	-	-	-	-	•	•	-	•	•	-	5.23	-
THS3091	1	10-33	9.5	4	62	2.0	800	14	1.2	2.4	12	1	950	7300	-	175	-	-	-	-	-	-	•	-	•	-	5.93	D
THS3001	1	9-33	6.6	2	65	1.6	250	13	7.5	12	20	1	1750	6500	-	85	-	-	-	-	-	-	-	-	•	-	6.93	-
HV JFET VFB																												
THS4631	1	10-33	11.5	50pA	95	7	340	20fA	3.9	27	62	1	210	900	-	80	-	-	-	-	-	-	-	-	•	-	8.81	-
AD8067	1	5-24	6.4	0.5	5	6.6	1650	0.6	2.5	17	50	8	540	640	-	25	-	-	-	-	-	-	-	-	-	•	4.63	-
ADA4637	1	10-36	7	1pA	100	6.1	80	1.6fA	8	49	79	5	80	170	•	45	-	-	-	-	•	-	-	-	•	-	10.12	E
LV bipolar VFB																												
AD8045	1	3-12.6	15	2	83	3	3800	3	1.3	3.9	19	1	800	1350	•	55	-	-	-	-	-	-	-	-	•	-	2.59	F
LT6202	1,2,4	3-12.6	3.5	1	83	2.8	940	1.1	1.8	5.0	19	1	95	22	-	30	•	•	•	•	-	-	-	-	•	•	1.90	-
LMH6628	2	5-13	9	0.7	57	2.0	4200	2	1.5	3.0	13	1	200	550	-	50	-	-	-	-	-	-	-	-	•	-	4.39	-
LMH6622	2	4-13	4.3	4.7	80	1.6	1000	1.5	1	1.6	10	1	180	85	-	100	-	-	-	-	-	-	-	-	-	-	3.20	-
LT6230	1,2,4	3-12.6	3.2	5	95	1.1	500	1	6.5	7.2	13	1	100	70	-	30	-	-	•	•	-	-	-	-	•	•	2.73	G
LT6230-10	1	3-12.6	3.2	5	95	1.1	500	1	6.5	7.2	13	10	1450	320	-	30	-	-	•	•	-	-	-	-	•	•	2.72	-
ISL28190	1,2	2.5-5.5	8.5	10	78	1.0	20	2.1	-	-	-	1	120	50	-	90	-	•	•	•	-	-	•	-	•	•	1.37	-
ADA4897	1,2	2.7-11	3	11	92	1.0	50	2.8	11	11.0	16	1	150	120	-	50	-	n	•	•	-	-	•	-	•	•	6.45	-
ADA4895	2	2.7-11	3	11	92	1.0	50	2.8	11	11.0	16	10	230	500	-	50	-	n	•	•	-	-	•	-	•	-		-
LT6200	1,2	3-12.6	16.5	10	75	0.95	4500	2.2	4	3.8	9	1	150	50	•	60	•	•	•	•	-	-	•	-	•	•	2.99	H
LT6200-10	1	3-12.6	16.5	10	75	0.95	4500	2.2	4	3.8	9	10	1600	450	•	60	•	•	•	•	-	-	•	-	•	•	2.99	H
AD8099	1	5-12.6	15	6	98	0.95	2200	5.2	2	1.9	6.7	2[x]	1200	1350	-	100	-	-	-	-	-	•	•	-		-	3.69	-
LMH6624	1,2	5-13	12	13	90	0.9	2200	2.3	2	1.8	6.3	10	1900	400	•	100	-	-	-	-	-	-	-	-	•	•	3.98	K
LMH6629	1	2.7-6	15.5	15	82	0.69	700	2.6	1.7	1.2	4.6	4	3200	530	-	200	-	•	-	-	-	u	•	-	•	•	4.00	-
LMH6629	1	2.7-6	15.5	15	82	0.69	700	2.6	1.7	1.2	4.6	10	9000	1600	-	200	-	•	-	-	-	u	•	-	•	•	4.00	-
LV bipolar CFB																												
AD8007	1,2	5-12	9	4	56	2.7	1300	2	1	2.7	16	1	800	1000	-	70	-	-	-	-	-	-	-	-	•	s	3.05	-
OPA694	1,2	7-13	5.8	5	60	2.1	600	22	1.2	2.5	13	1	2000	1700	-	80	-	-	-	-	-	-	-	-	•	•	3.38	-
AD8001	1,2	6-12.6	5	5	50	2.0	15	2	1.5	3.0	13	1	880	1200	-	85	-	-	-	-	-	-	-	•	•	•	2.72	-
MAX4224	1	6-12	6	2	55	2.0	-	3	0.8	1.6	12	2	1200	1400	-	60	-	-	-	-	-	-	•	-	•	•	6.34	-
AD8009	1	5-12.6	14	50	50	1.9	270	46	2.6	4.9	14	1	3500	5500	-	175	-	-	-	-	-	-	-	-	•	•	3.19	-
LMH6702	1	9-13.5	12.5	8	47	1.83	4600	18.5	1.6	2.9	12	1	1400	3100	-	50	-	-	-	-	-	-	-	-	•	•	3.11	-
OPA695	1,2,4	5-13	13	5	56	1.8	2100	18	1.2	2.2	11	1	5600	4300	-	90	-	-	-	-	-	-	•	-	•	•	3.42	-
OPA691	1,2,3	5-13	5.1	15	52	1.7	750	3.1	2	3.4	12	1	1050	2100	-	140	-	-	-	-	-	-	•	-	•	•	2.41	-
AD8000	1,3	4.5-13	13.5	5	52	1.6	3100	3.4	3.6	5.8	14	1	1300	4100	-	100	-	-	-	-	-	-	•	-	•	-	3.58	-
LV JFET																												
OPA656	1	9-13	14	2pA	80	7	1200	1.3fA	2.8	20	55	1	230	290	•	50	-	-	-	-	-	-	-	-	•	•	5.59	-
OPA657	1	9-13	14	2pA	80	4.8	670	1.3fA	4.5	22	46	7	1600	700	•	50	-	-	-	-	-	-	-	-	•	•	10.01	-
ADA4817	1,2	5-10.6	19	2pA	77	4	7000	2.5fA	1.5	6.0	26	1	410	870	-	70	-	•	-	-	-	-	-	-	-	-	4.93	-
LV CMOS																												
LTC6268	1,2	3-5.5	16	3fA	63	5.5[w]	150k	5.5fA	0.45[w]	1.8[w]	22	1	500	400	-	50	-	•	•	•	-	-	•	-	•	•	7.50	-
LMP7717	1,2	1.8-6	1.2	50fA	85	6.2	330	10fA	15	93	124	10	88	28	-	36	-	•	•	•	-	-	-	-	•	•	2.18	-
OPA380	1,2	2.7-7	7.5	3pA	100	5.8	600k	10fA	3	17	46	1	90	80	-	75	-	•	-	•	-	-	•	-	•	-	5.39	L
OPA365	1,2	2.2-5.5	4.6	0.2pA	100	4.5	8000	4fA	6	27	50	1	35	25	-	50	•	•	•	•	-	-	-	-	•	•	1.75	M

Notes: (a) listed within each category by decreasing e_n. (c) calculated. (d) see plots in Chapter 5. (g) for G=10 or G=20. (J) at 1kHz or 10kHz (above the 1/f corner), except 10Hz for auto-zero op-amps. (n) within 0.1V of V–. (p) per amplifier. (r) without C_{comp}. **(s)** SC70 available. (t) typical. (u) a pin sets G_{min}. (v) the LT1028 has a nasty ~10dB noise peak from 200kHz--600kHz. (w) at 100kHz. (x) G=1 with ext comp.

Comments: A: 0.1ppm distortion. **B:** lowest e_n; LTC suggests LT6200, LT6230. **C:** Hill favorite. **D:** use DDA pkg; Larkin favorite. **E:** OPA637 for DIP. **F:** improved SOIC pinout. **G:** LT6230-10 decomp. **H:** 1% dist at 50MHz. **K:** 0.03% dist at 10MHz. **L:** auto-zero. **M:** zero crossover.

8.9.1 Guide to Table 8.3: choosing low-noise op-amps

Table 8.3 lists a rich selection of op-amps suitable for low-noise applications. Here we provide a guide to the table entries, and advice for selecting op-amps. Because readers interested primarily in low-noise design with op-amps (rather than with discrete transistors) may have skipped over the earlier quantitative discussions of noise in connection with BJTs and JFETs, we restate below many of the basic ideas, with references to the relevant sections and figures. Readers whose route here (to what might be called "the promised land of op-amps") has taken them through the details of discrete design ("the valley of complexity") will find much of this discussion to be a welcome review.

A. Table sorting

In Table 8.3a we've listed BJT-input op-amps by increasing supply current I_Q, which correlates roughly with decreasing input noise voltage e_n. Once the latter falls below $5\,nV/\sqrt{Hz}$, the entries are roughly ordered by decreasing e_n. For JFET-input op-amps (Table 8.3b) the entries are listed by increasing input bias current (up to 1 pA), then by decreasing e_n. CMOS op-amps are sorted by increasing I_Q, which correlates approximately with decreasing e_n. Table 8.3c is devoted to high-speed op-amps of all types, listed within each category by decreasing e_n.

B. Supply voltage, high voltage, low voltage

The table presents a good collection of low-noise op-amps, with a number of representative popular op-amps (some of which might well be called "noisy") for comparison. The latter have other attractive features, such as low cost, high slew rate, low input bias current, low capacitance, low power, multiple manufacturers, and so on. The table separates "high-voltage" op-amps, capable of working with ± 15 V supplies (or up to 36–44 V total supply), from op-amps that are limited to low voltages like 5–20 V maximum.

Note that many of the high-voltage op-amps (especially the bipolar types) can work also at very low supply voltages, as low as 2.2 V to 3 V total. So, for example, a precision "high-voltage" op-amp like the LT1677, which works fine with 3 V total supply, and which allows inputs and outputs to swing to either rail,[73] is also clearly an excellent low-voltage part. Op-amps whose inputs and outputs can swing to the negative rail qualify to be called "single-supply" op-amps, regardless of whether you run them from a single positive supply or from dual supplies.

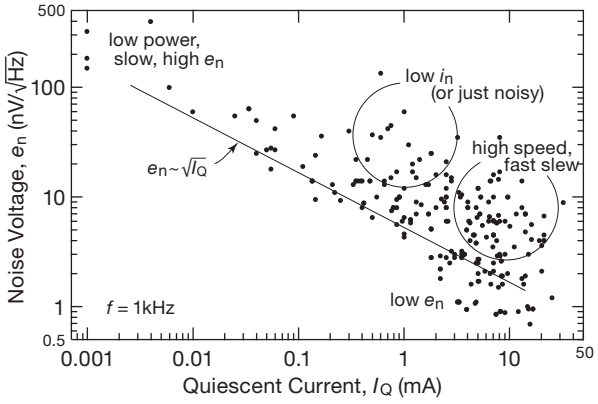


Figure 8.57. To minimize input noise voltage you have to run both BJTs and JFETs at relatively high quiescent currents. That trend extends to integrated op-amps, as seen here in a scatterplot of e_n versus I_Q for most of the op-amps in Table 8.3 and in Figures 8.60 and 8.61.

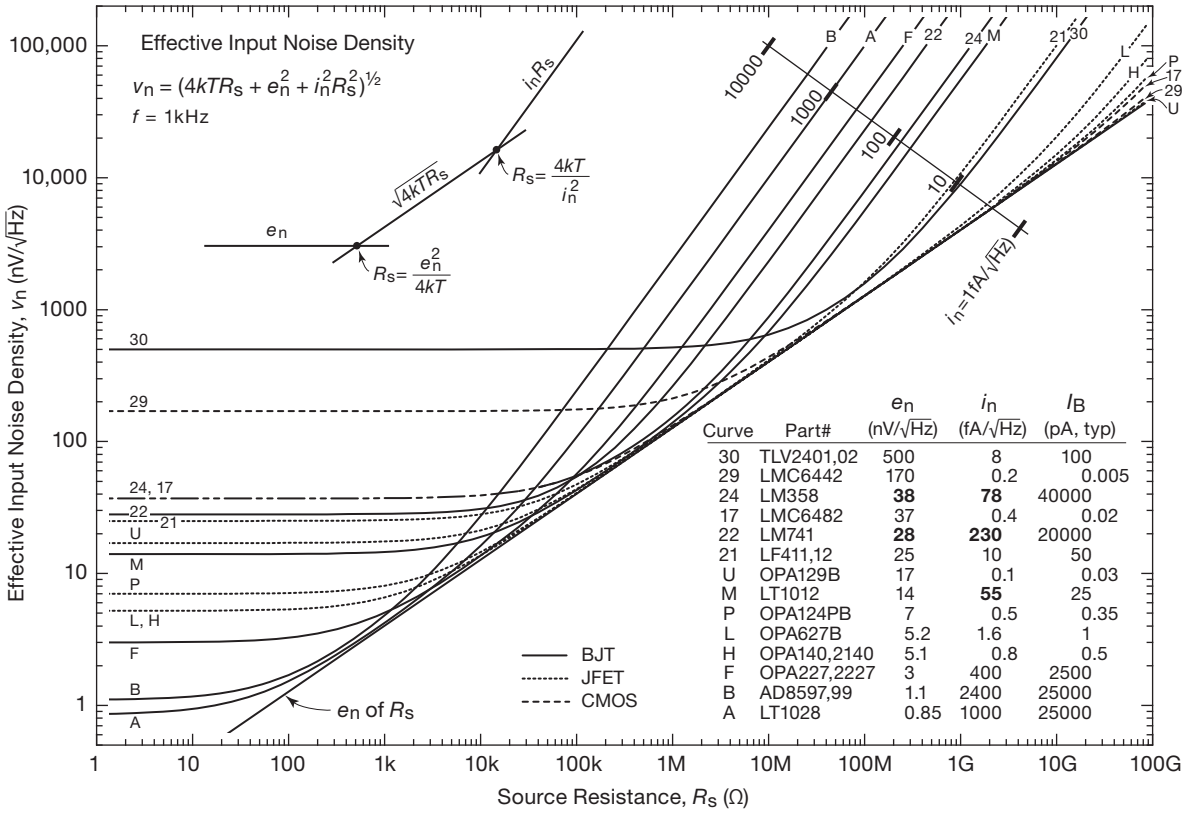
C. Supply current I_Q, and voltage noise

High supply current is also a major consideration for low noise voltage, e_n. Table 8.3 includes op-amps with very low I_Q, even though these op-amps have much higher noise. For example, the ISL28194 uses only 330 nA but has a high $265\,nV/\sqrt{Hz}$ noise density. By comparison, the quiet LT1028 has $0.85\,nV/\sqrt{Hz}$ of noise density, but requires 7.4 mA, or ∼20,000 times as much operating current.[74]

Figure 8.57 is a scatterplot of e_n versus I_Q for several hundred op-amps, showing the voltage-noise and supply-current tradeoff. You can see how the noise falls by approximately the square root of the operating current. Any part that's within a factor of two of the best-available part at a given supply current might be considered a low-noise op-amp. The correlation with supply current is far from perfect. That's because op-amp designers are balancing tradeoffs among many performance parameters, for example high slew rate, low input current, small die size (for small package size and low cost), and so on, at the expense of low noise voltage.

[73] The table has columns labeled "swing to supply" for inputs and outputs to both supply rails.

[74] Recall that collector [or drain] shot noise through r_e [or $1/g_m$] in BJTs [or JFETs] can be described equivalently as resistive Johnson noise created by a resistor of value $\frac{1}{2}r_e$ [or $\frac{2}{3}\frac{1}{g_m}$], according to eq'ns 8.20 or 8.34. Because higher collector [or drain] currents produce lower r_e [or higher g_m], high operating currents are needed to lower an op-amp's input-transistor noise voltage.



Curve	Part#	e_n (nV/√Hz)	i_n (fA/√Hz)	I_B (pA, typ)
30	TLV2401,02	500	8	100
29	LMC6442	170	0.2	0.005
24	LM358	**38**	**78**	40000
17	LMC6482	37	0.4	0.02
22	LM741	**28**	**230**	20000
21	LF411,12	25	10	50
U	OPA129B	17	0.1	0.03
M	LT1012	14	**55**	25
P	OPA124PB	7	0.5	0.35
L	OPA627B	5.2	1.6	1
H	OPA140,2140	5.1	0.8	0.5
F	OPA227,2227	3	400	2500
B	AD8597,99	1.1	2400	25000
A	LT1028	0.85	1000	25000

Figure 8.58. Total noise (source resistor plus amplifier, at 1 kHz) versus source resistance for a selection of low-noise op-amps (letter labels); we've included a few "ordinary" op-amps (number labels) for comparison. The curve labels correspond to those in Table 8.3 and in Figures 8.60 and 8.61. The curves are generated from datasheet values of e_n and i_n, except for measured values (shown in bold font) for the noise parameters of the classic LM741 and LM358 (unspecified) and the LT1012 (datasheet value in error); the current noise of the LMC6482 is an estimate, as the datasheet value cannot be correct.

D. Voltage-noise density, current-noise density: e_n, i_n

Just as with discrete transistors, op-amp noise performance is specified by *voltage* noise, e_n, and *current* noise, i_n, and their variation with frequency; see §§8.3.1 and 8.3.2. Figure 8.58 and its embedded formula show how to combine e_n and i_n with the Johnson noise of your circuit's effective source resistance, R_s, to find an effective total input noise density for your circuit. Stated simply, there are three sources of noise: the op-amp's input voltage noise, the op-amp's current noise flowing through the signal's source impedance, and Johnson noise in the source impedance. These are uncorrelated, so to find the total noise-voltage density you add their squares (units of V^2/Hz) and take the square root. Fourteen representative op-amps are shown as examples in Figure 8.58, with the BJT types best at low source resistances and the JFET and CMOS types best at high resistances. Table 8.3 has datasheet specs for e_n and i_n at 1 kHz, from which you can determine what kind of trace a candidate part will make in such a plot.

E. Noise plots, e_n and i_n

It's one thing to look at reams of e_n and i_n values in datasheets, or to scan those values as organized in the three pages of Table 8.3; but it's another thing to compare noise-density plots, which include $1/f$ breakpoints and other unique information. Figures 8.60 and 8.61 show plots of datasheet e_n and i_n versus frequency for 60 of the op-amps in Table 8.3, where numbers and letters identify the corresponding traces in the plots.[75] Note how the voltage-noise plots range over a factor of 1000:1 (though most op-amps

[75] Figures 8.110 and 8.111 show *measured* noise spectra for a selection of these op-amps.

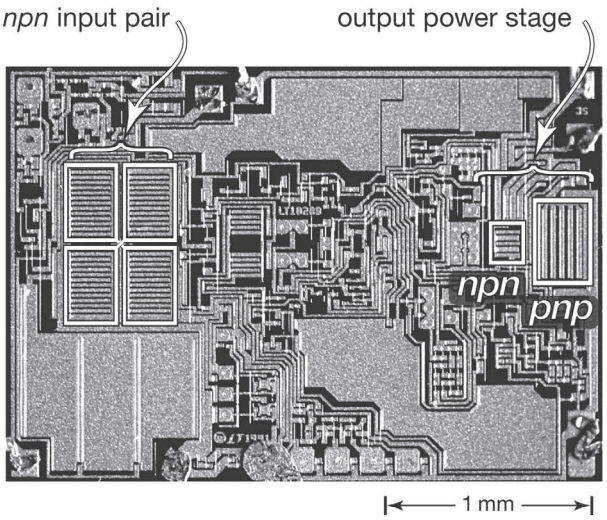


Figure 8.59. The LT1028 op-amp, introduced in 1981, remains the low-noise winner, despite repeated attempts on its title. The input stage runs at high current (1.8 mA total) for low e_n; and the large-input transistors keep the current density low, to achieve an impressively low $1/f$ corner frequency of 3.5 Hz.

are in the range of 1–100 $\text{nV}/\sqrt{\text{Hz}}$), whereas the current-noise plots range over a factor of nearly 10^6:1. The latter reflects the very large range of dc input bias currents, from femtoamps (for some CMOS parts) to tens of microamps (for high-speed BJT parts), a ratio of 10^{10}:1. The *noise* current goes only as the square root of the *dc* current, but hey, the square root of 10^{10} is still pretty large.

An op-amp with very low voltage noise e_n, such as trace A or 3 for the exemplary LT1028 (whose portrait we present in Figure 8.59) in the top panels, generally correlates with a high current noise i_n, as seen in the lower panels. This is a common e_n versus i_n tradeoff, which you have to take into account when selecting parts from Table 8.3 and the corresponding noise plots. As an inducement to learn about low-noise design with discrete transistors, look at traces 1 and 2 of Figure 8.61, the noise voltage of a best-in-class BJT and JFET. With parts like these you can achieve lower noise voltage than with any op-amp, creating a "hybrid low-noise op-amp" in which a discrete front-end (for lowest e_n) combines with an op-amp second stage (to provide gain, and the output stage). We illustrate this in Figures 8.66 and 8.67 in §8.9.5. For a full understanding of discrete low-noise design, see §§8.5 and 8.6.

In general, BJT op-amps have low $1/f$ noise-voltage corner frequencies, typically in the range of 1–30 Hz; but beware their generally much higher noise-*current* corner frequencies, 30 Hz–1 kHz or more. This can seriously affect designs with high-value feedback resistors or high signal source impedances.

If you are considering only op-amps with the lowest noise voltage (less than, say, 1.1 $\text{nV}/\sqrt{\text{Hz}}$), no others have a $1/f$ noise corner as low as the LT1028 and LT1128, at 3.5 Hz. These impressive parts also have a lower noise current than the other contenders. However, be warned that these op-amps have a nasty broad 15 dB noise peak (shown with admirable candor on the datasheet) starting at 150 kHz, and peaking at 400 kHz before fading away above 600 kHz. Other op-amps don't have this problem, but they have much higher corner frequencies (f_c), as high as 5 kHz for some of the high-speed op-amps.

F. I_B and i_n for BJT op-amps; I_B bias cancellation

Table 8.3 has typical and maximum input bias-current, or I_B, values.[76] In general, the listed current noise i_n comes from shot noise in the base current of a BJT op-amp or from gate leakage in a JFET or CMOS op-amp; so we'd expect the noise spec to be closely related to the bias current by the shot-noise equation $i_n = \sqrt{2qI_B}$. But many BJT op-amps use an input bias cancellation scheme (see §*4x.10*), to reduce greatly the dc bias current, I_B. However, the dc current cancellation in this scheme does not reduce the op-amp's i_n noise (in fact, it typically increases it, by $\sqrt{2}$). These op-amps are identified in the "bias cancel" column of Table 8.3a, and they typically have 10× to 40× more i_n noise than you'd expect from shot noise based on the specified bias current I_B.[77] This increased i_n noise is reflected in a lower R_n value, but ideally you'll not find the higher current noise and lowered feedback-resistor values to be a problem.

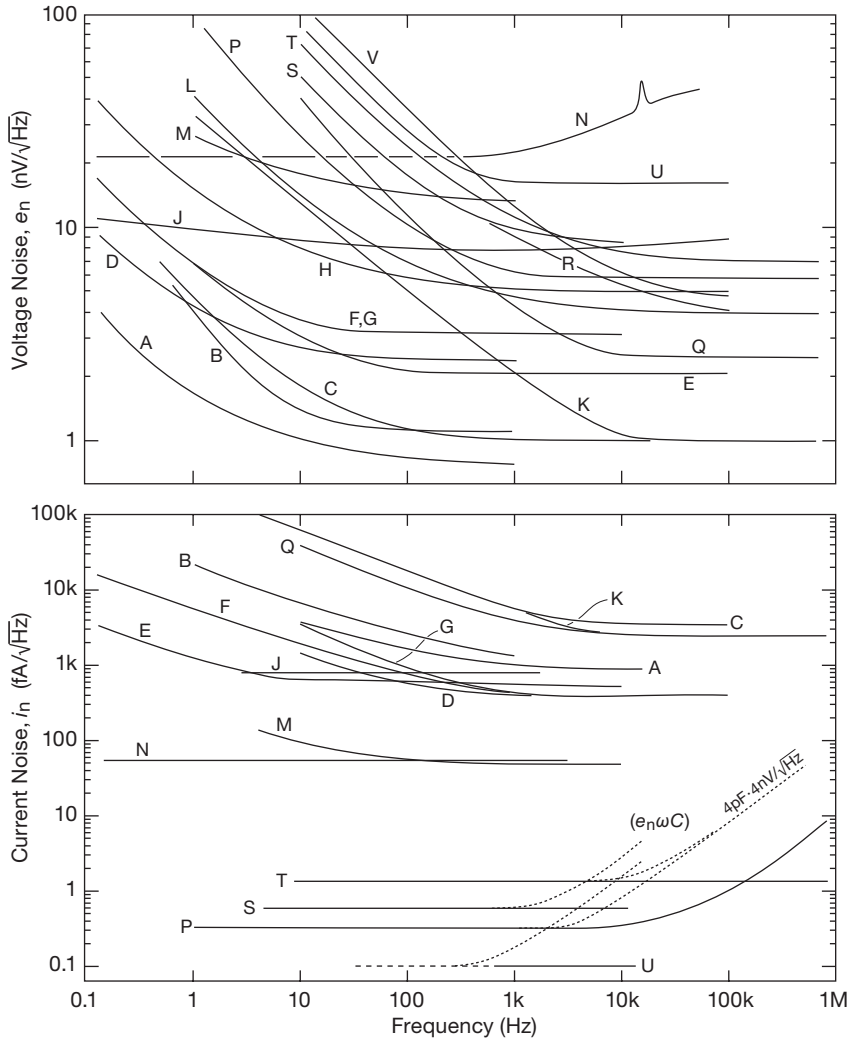
Bias cancellation near the rails Be careful when relying on listed bias currents with bias-cancelled rail-to-rail input op-amps, because their input bias current rises dramatically as the inputs approach the supply rails (see §5.7.2). Datasheets show this in graphical form, but they generally do not reveal this undesirable behavior in the listed performance data.

G. Current-feedback op-amps

Unlike voltage-feedback (VFB) op-amps, current-feedback (CFB) op-amps (used in wideband applications) generally have much higher dc bias currents and current noise at their "− input" than at their corresponding

[76] For BJT op-amps (Table 8.3a) the I_B values are in nanoamps, whereas for the JFET-input op-amps (Table 8.3b) the units are *pico*amps.

[77] In a few cases the manufacturers have misstated the i_n noise; for those the table provides a measured value, indicated with italics.



Bipolar (BJT)		$I_Q^{p,t}$ (mA)	GBW^t (MHz)
	high voltage[a]		
J	OPA188^z, 2188^z	0.4	2
E	OPA209, 2209	2.2	18
F	OPA227, 2227	3.7	8
D	LT1007	2.6	7
M	LT1012	0.4	0.7
A	LT1028	7.4	75
G	NE5534A	4	10
Q	AD8021	7	1000
B	AD8597, 99	5	10
	low voltage		
K	AD8099	15	4000
C	ISL28190, 290	8.5	100
JFET			
	high voltage		
P	OPA124	2.5	2
U	OPA129	1.2	1
H	OPA140, 2140	1.8	11
L	OPA627	7	16
	low voltage		
T	OPA656	14	230
R	ADA4817	19	400
CMOS			
	medium voltage		
S	TLC2272	1.2	2.2
	low voltage		
V	OPA365^x	4.6	50
N	AD8628^z, 29^z	0.8	2.2

(a) all have bias cancellation, except for types G and Q.
(p) per channel.
(t) typical.
(x) zero crossover.
(z) zero drift.

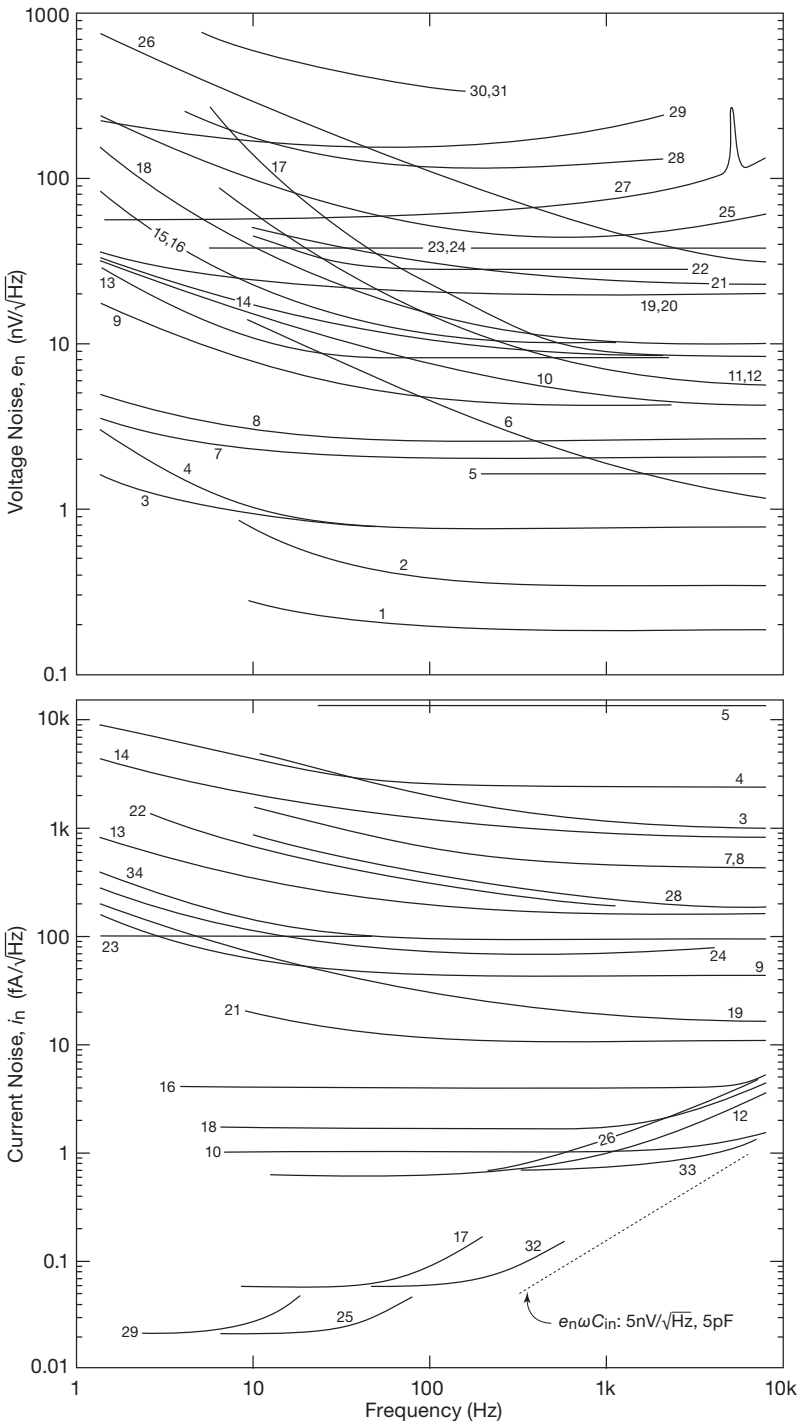
Figure 8.60. Voltage and current noise versus frequency for a selection of "low-noise" op-amps featured in Tables 8.3a–c on pages 522–524. All are adapted from datasheet plots, with the exception of measured i_n for op-amps G and J. The labels are in order of ascending e_n at 10 Hz. See Figures 8.110 and 8.111 for measured noise spectra of selected op-amps.

"+input". For CFB op-amps, Table 8.3c gives the current-noise value for the input with the lowest current and noise. The difference can be as large as a factor of ten. As always, study the datasheet, *carefully!*

H. Noise resistance R_n

The noise-resistance parameter $R_n \equiv e_n/i_n$ has its own column. This is the value of source resistance that corresponds to the minimum possible amplifier noise figure; see for example §8.5.1 and Figure 8.31. But its real usefulness is to let you see at a glance the maximum feedback resistor values you can use: the impedance R_s seen by the op-amp at its inputs should be 5× to 30× smaller than the op-amp's R_n value if you wish to ensure that e_n is the dominant noise source in your circuit; see Figure 8.58. The idea here is that you can't do anything about the e_n value you're stuck with, but you can lower your feedback resistor values to reduce the effect of the op-amp's i_n. But don't forget that you've got to keep the *signal's* source impedance small compared with R_n also. (Also, quite apart from the *noise* current, you may have to worry about the effects of the *dc* bias current flowing through the signal's source, which may be a sensor of relatively high dc resistance.) So in the end you may have to give up on your favorite low-e_n op-amp, and select one with lower i_n even if it has higher e_n.

There are no R_n values listed for the JFET and CMOS



	Bipolar (BJT)	$I_Q^{p,t}$ (mA)	GBW^t (MHz)
	high voltage[a]		
8	OP27	4.6	8
18	OPA171	0.5	3
13	OPA277,2227	0.8	1
24	LM358	1	1
22	LM741	1.7	1.5
5	AD844	6.5	60
7	LT1007	2.7	8
19	LT1013	0.35	0.7
3	LT1028	7.4	75
9	LT1468	3.9	90
4	ADA4898-1,-2	8.1	65
34	LT6010,11	0.14	0.33
14	LM6171,72	2.5	100
	medium voltage		
31	TLV2242	1 µA	5.5 kHz
30	TLV2401,02	0.9 µA	5.5 kHz
6	LT6200,01	16.5	165
	low voltage		
28	LMV321,358	0.13	1
	JFET		
	high voltage		
33	OPA124	2.5	1.5
32	OPA129	1.2	1
21	LF411,12	1.8	4
10	LT1793	4.2	4.3
16	OPA604,2604	5.3	20
	CMOS		
	medium voltage		
17	LMC6482	0.5	1.5
27	AD8638[z]	1	1.4
	low voltage		
12	OPA350,2350	5.2	38
26	OPA743	1.1	7
25	LMP2231,32	0.01	0.13
15	LTC6081	0.33	3.6
29	LMC6442	1 µA	10 kHz
20	AD8603	0.04	0.4
11	AD8615,16	1.7	24
23	MAX9617[z]	0.06	1.5
	Discrete		
1	ZTX851 (npn, 10mA)		
2	IF3601 (nJFET, 5mA; InterFET)		

(a) all have bias cancellation, except for #5, 14, 22, and 24.
(p) per channel. (t) typical. (z) zero drift.

Figure 8.61. Voltage and current noise versus frequency, for a selection of "popular" op-amps featured in Table 8.3 on pages 522–524; a low-noise JFET and BJT are included for comparison. All are adapted from datasheet plots, with the exception of measured e_n for op-amps #22 and #24 and BJT #1, and measured i_n for op-amp #23. The labels are in order of ascending e_n at 10 Hz.

op-amps in Table 8.3. That's because i_n is essentially the shot noise of the input bias current, a parameter that is poorly specified: you see typical values of a picoamp or less, but hedged with maximum values a thousand times larger. And, of course, these are leakage currents, which rise exponentially with temperature. But not to worry – the typical room-temperature input currents are sufficiently low that the corresponding R_n values are generally a gigaohm or greater, as seen clearly in Figure 8.58.

I. I_B and i_n for JFET and CMOS op-amps

The input bias-current, I_B, has typical and maximum values. Usually the i_n shot noise of I_B is calculated in the most optimistic fashion, from the *typical* value of I_B. For BJT op-amps the maximum I_B is generally no more than 2× to 3× its typical value. But the situation is quite different for JFET and CMOS op-amps, where the maximum-to-typical I_B ratios are much greater, e.g., 60× for the TLC4501A or 800× for the LMC6442A.[78] This is especially true for low-cost parts with impressively low I_B specs. *A caution*: beware, the input current may be strongly affected by common-mode voltage, and the listed spec is usually with V_{in} at mid-supply.

Another issue: Table 8.3's listed I_B values are for an optimistic 25°C. See §5.10.7 and Figure 5.38 for the dramatic effect of elevated temperatures on JFET and CMOS op-amps (caused both by ambient and in-box heating, as well as by the op-amp's self-heating, see below). Because of these uncertainties in estimating bias current, you can't say with any precision what the input noise current i_n will be for JFET or CMOS op-amps. You can say, though, that (except at high temperatures) it will be quite a bit less than that of BJT-input op-amps, as revealed in the lower plots of Figures 8.60 and 8.61.

J. I_B and junction temperature

As we point out in §5.7.2 (and in further detail in §*4x.10*), BJT op-amps with bias-current cancellation are very effective at high temperatures (see Figure 5.6) and under some conditions they may offer the lowest i_n performance. This is especially true for low-supply-current op-amps, whose input-stage collector currents are a fraction of their already low I_Q. For example, the LT6010A still enjoys under 50 pA typical bias current at 100°C (see the datasheet's plot), much lower than the 1200 pA suffered by the OPA134 CMOS op-amp, which looked superior at 25°C. However, even in this case the *noise* current i_n of the hot JFET op-amp still wins over the hot BJT op-amp, by about a factor of 5. That is because the BJT's low bias current is the result of excellent bias cancellation, which is fine when you care about dc input current, but which does not reduce the *noise* current of the much larger uncancelled bias current.

When thinking about elevated temperatures, keep in mind that in-box and on-board temperatures are often significantly higher than the ambient exterior, especially if power circuitry is involved. Some op-amps also experience significant additional self-heating. For example, our favorite OPA627 JFET op-amp dissipates 210 mW when powered with ±15 V supplies; in the SOIC-8 package this causes the junction temperature to rise 34°C above ambient. Unlike some other parts, the OPA627 and '637 are tested realistically, after a warm-up delay. This is partly responsible for their higher cost.

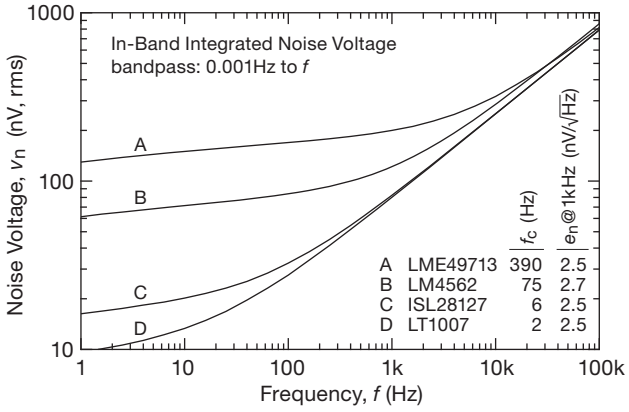
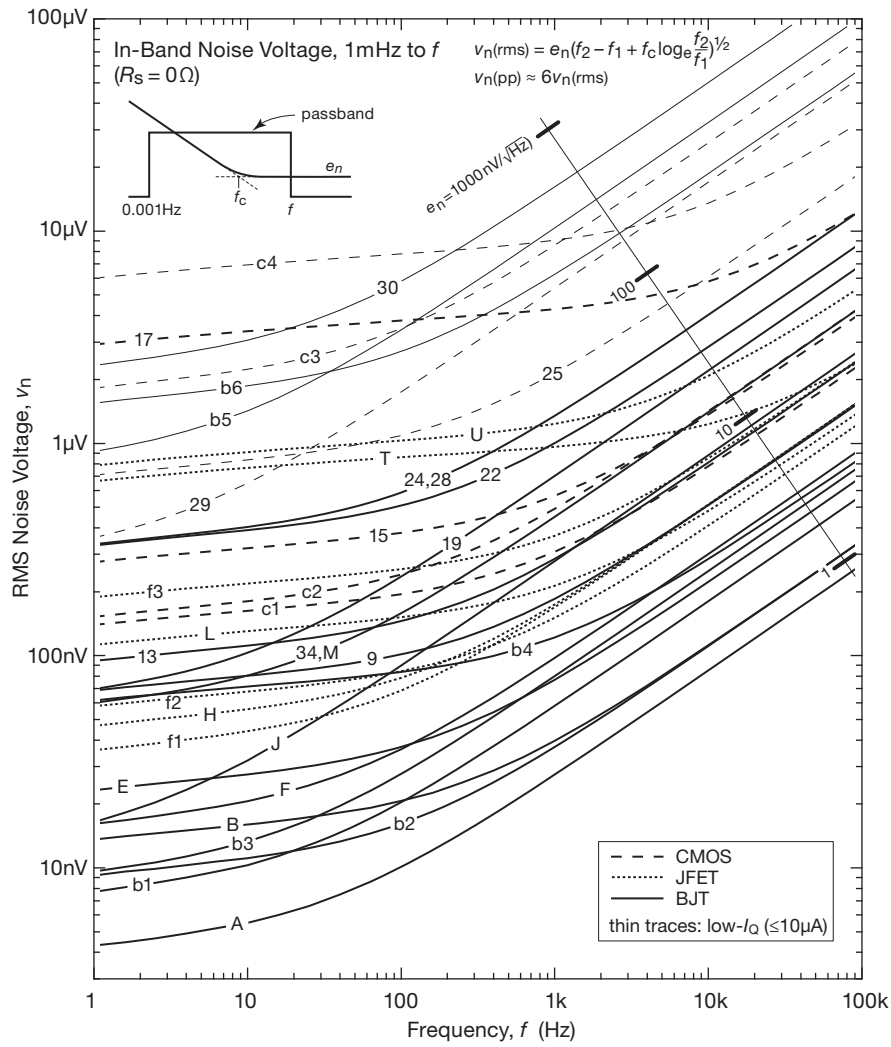


Figure 8.62. A desirably low value of e_n at the usual 1 kHz frequency listed in datasheets doesn't tell the whole story. An op-amp with a high $1/f$ corner frequency f_c will have greater integrated in-band noise voltage, as seen in these four op-amps of comparable 1 kHz e_n values. Note that the effect extends far above the $1/f$ corner frequency.

K. 1/*f* noise and the 1/*f* corner frequency f_c

Figures 8.60 and 8.61 show increasing noise at low frequencies. This is called $1/f$ noise, discussed in various circuit contexts, e.g., in §§5.10.6, 8.1.3, 8.3, and 8.13. Table 8.3 includes $1/f$ voltage-noise corner frequencies f_c, calculated from datasheet noise parameters via eq'n 8.60. Watch out for op-amps that may have attractively low e_n (as specified at the usual 1 kHz), but that suffer from high $1/f$ corner frequencies. If your circuit's bandwidth is less than $\sim 10 f_c$ you should consider its effects.

[78] For very low input-current op-amps you'll often see an overly conservative (high) maximum I_B spec, dictated by the manufacturer's desire for fast and inexpensive automatic testing. Only high-priced parts get treated to slow and expensive low-current tests.



Bipolar (BJT)		e_n (typ, 1kHz) (nV/√Hz)	f_c (Hz)
high voltage			
J	OPA188,2188	8.8	0.4
E	OPA209,2209	2.2	16
b2	OPA211,2211	1.1	10
F	OPA227,2227	3	4
13	OPA277,2277	18	20
24	LM358	40	10
22	LM741	28	20
D	LT1007	2.5	2
M	LT1012	14	2.5
19	LT1013	22	1.3
A	LT1028,LT1115	0.85	3.5
9	LT1468	5	27
b6	LT1495	185	10
b1	AD4004	1.8	2.5
b4	LM4562	2.7	75
34	LT6010,11	14	3.6
B	ADA8597	1.1	22
medium voltage			
b5	LT6003	325	1
b3	LMP7731	2.9	1.4
low voltage			
28	LMV321,358	39	8
30	TLV2401,02	500	3
JFET			
high voltage			
U	OPA129B	17	310
f3	OPA134,2134	8	80
H	OPA140,2140	5.1	12
L	OPA627,637	4.5	90
f1	OPA1641	5.1	7
f2	LT1792	4	30
medium voltage			
T	OPA656	7	1300
CMOS			
medium voltage			
29	LMC6442	170	0.5
17	LMC6482	37	900
c4	ICL7612	100	520
low voltage			
c1	OPA376	7.5	50
c3	LPV521	260	7
25	LMP2231,32	60	20
c2	MAX4236A	14	17
15	LTC6081	13	65

Figure 8.63. Integrated in-band noise voltage for a selection of low-noise and popular op-amps, based on datasheet plots and tabulated values of noise-voltage density. The op-amp labels correspond to those in Figures 8.60 and 8.61 and the listings in Table 8.3. See also Figure 5.54, which includes auto-zero op-amps and extends down to 0.001 Hz.

L. Noise, "integrated noise"

Figure 8.62 shows the *integrated noise* for four low-noise op-amps, the result of adding up (i.e., integrating) the noise density (sometimes called "spot noise") over the operating bandwidth. It has units of voltage (e.g., μVrms), in contrast to noise density, which has units of voltage divided by square root of bandwidth (e.g., nV/$\sqrt{\text{Hz}}$). Sometimes integrated noise is loosely called the "noise voltage," or just the "noise." Integrated noise may be written as v_n, or V_n, but never e_n. In Figure 8.62 the four op-amps have comparable specified e_n *at 1 kHz*, but widely varying $1/f$ corner frequencies f_c, ranging from 2–400 Hz, producing dramatically different integrated noise when used entirely below 1 kHz. In fact, the ghost of the low-frequency noise is seen lurking up to ∼10 kHz.

Figure 8.63 plots integrated noise for three dozen op-amps. In general JFET op-amps are worse than BJT op-amps at low frequencies, and CMOS op-amps are worse still. The plots were calculated from the table's e_n and f_c values, according to eq'n 8.59. At high frequencies, where white noise predominates, the plots simplify to $v_n = e_n\sqrt{f_2}$.

Chopper and auto-zero op-amps (not shown in the figure) do not suffer from $1/f$ noise, but they suffer from higher voltage noise, and generally excessive current noise.

See Table 5.6 and Figure 5.54; the latter illustrates how chopper op-amps are generally noisier than low-noise BJT op-amps above 1–10 Hz.

M. 0.1–10 Hz peak–peak voltage noise

The 0.1–10 Hz peak–peak voltage-noise specification $V_{n(pp)}$ in Table 8.3 is of importance when you are choosing op-amps for low-frequency applications that are sensitive to $1/f$ noise. The manufacturer usually determines this spec from 10-second oscilloscope measurements of a filtered output (see §8.13 and Figure 8.64), but it can also be estimated from eq'n 8.59 (which idealizes to a brick-wall bandpass characteristic). We have seen manufacturers using single-pole, two-pole *RC* pairs, and two- or three-pole Butterworth filters. Comparing datasheet values for a few hundred op-amps shows that multiplying by a factor of six matches the brick-wall equation's $V_n(\text{rms})$ value to the datasheet's filtered peak–peak value. We used this approach to check the datasheet values, or to obtain calculated values where the manufacturer had none.

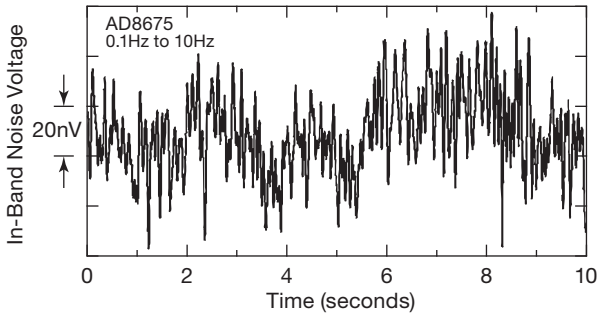


Figure 8.64. Datasheets sometimes provide low-frequency noise voltage snapshots, like this one for the AD8675. The corresponding noise spectrum is similar to curve 8 in Figure 8.61. (Adapted with permission of Analog Devices, Inc.)

N. Input capacitance C_{in}

Op-amp input capacitance is a serious issue when you are considering $e_n \cdot C_{in}$ noise in transimpedance amplifiers (TIAs); see §8.11. For some high-impedance sensors, input capacitance acts both as an additional high-frequency load, and it provides a way for high-frequency noise on the power-supply rails to be coupled to the input as noise current. Some op-amps provide both common-mode and differential capacitance values; for these we have taken the larger value for the table (as always, we suggest careful study of the datasheet before embarking on a design).

Quite often, op-amps with low e_n use large area input transistors, with correspondingly greater capacitance, as can be seen in the e_n–vs.–C_{in} scatterplot of Figure 8.65. Ignoring BJT op-amps and noisy op-amps, the trend shows an increasing noise penalty for low input capacitance. But some parts are exceptions, and we consider them to be quite valuable. An excellent JFET op-amp is the LT1793 with a specified C_{in} of 1.5 pF; even with its relatively high e_n of 5.8 nV/$\sqrt{\text{Hz}}$ it still enjoys the lowest $e_n \cdot C_{in}$ figure of merit, FOM=9 nV-pF.[79]

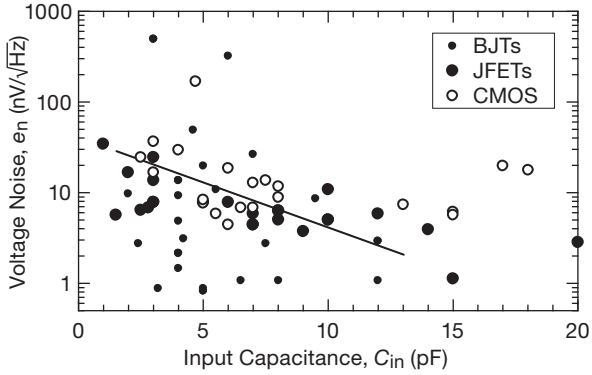


Figure 8.65. Op-amps with low voltage noise tend to have higher capacitance; this is especially true of low-noise JFET-input op-amps, with their large-area input transistors.

Often we have a high-capacitance sensor, or some input coax cable, etc., so in addition to an $e_n \cdot C_{in}$ op-amp FOM (with $C_{ext}=0$), we create additional effective $e_n \cdot C_{in}$ current noise from the external shunt capacitance of, say, 25 pF. Now a low value of e_n becomes more important, and the OPA827, OPA627, and LT1792 are the new winners, with the LT1793 and OPA365 remaining as contenders. The AD743 with its e_n of 2.9 nV/$\sqrt{\text{Hz}}$ *could be* the overall winner for $C_{ext}>25$ pF, but sadly it's entered the land of NRND ("Not Recommended for New Designs"). We recommend that you grab 'em if you can find 'em! Finally, for high-C_{ext} applications, consider discrete designs (see §8.3 and Figure 8.66) that can outperform the best IC op-amps.

Op-amps with small-geometry low-capacitance input transistors nonetheless suffer from the necessary added capacitance of their input protection devices. Truly low-capacitance op-amps may be more susceptible to static damage during handling. Bipolar transistors are less susceptible to static damage and tend to have much lower capacitance than low-noise JFET and CMOS op-amps, but their high I_B and i_n values will often preclude their use in

[79] Other parts to consider are the OPA124, OPA121, AD8067, and OPA656, the latter with an f_T of 230 MHz. For CMOS op-amps consider the OPA365, with a 0.2 pA I_B (typ) spec.

high-impedance circuits. An exception would be for transimpedance amplifiers above, say, 1–10 MHz, but for these applications you'll need to look at the table of high-speed op-amps, Table 8.3c; those parts aren't evaluated here.[80] If a bipolar op-amp's C_{in} is unstated, you can often assume that it's in the range of 2–5 pF.

8.9.2 Power-supply rejection ratio

Quite apart from noise sources *within* the op-amp, any noise (or interfering signals) on the power-supply rails will couple to the output, attenuated by the power-supply rejection ratio (PSRR). Typical op-amps have pretty good PSRR at low frequencies (roughly $1/G_{OL}$, so 80–140 dB), but the PSRR drops as $1/f$ at higher frequencies, allowing substantial supply-noise coupling. Often the high-frequency PSRR is particularly poor with respect to one or the other supply rail because of coupling by the internal compensation capacitor; see §*4x.7*. For example, at frequencies above 10 Hz, the LT1012 positive-rail PSRR is 25 dB worse than the negative-rail PSRR; and the micropower LT6003's negative-rail PSRR plummets to less than 10 dB at 1 kHz! Simple *RC* filtering for the sensitive stages (or a capacitance multiplier, see §8.15.1) can largely solve this problem. Be sure to read the datasheet, though, or you may not even know you *have* a problem.

8.9.3 Wrapup: choosing a low-noise op-amp

In summary, when choosing an op-amp for a low-noise application, begin by restricting your attention to op-amps that meet your other needs, such as accuracy, speed, power dissipation, supply voltage, input and output swing, and the like. Then choose among this subset, based upon their noise parameters. Generally speaking, *you want op-amps with low i_n for high signal impedances, and op-amps with low e_n for low signal impedances.* As we've seen, the total input-referred squared noise-voltage density is just

$$v_n^2 = 4kTR_{sig} + e_n^2 + \left(i_n R_{sig}\right)^2 \quad \mathrm{V^2/Hz}, \tag{8.42}$$

where the first term is due to Johnson noise and the last two terms are due to op-amp noise voltage and current.[81] Clearly the Johnson noise sets a lower bound to the input-referred noise. Refer back to Figure 8.58 for an easy graphical look at v_n (at 1 kHz) as a function of R_{sig} for a selection of low-noise op-amps from Table 8.3;[82] these span a representative range of e_n and i_n, from which you can interpolate for op-amps not shown explicitly.

This plot illustrates graphically the tradeoff between low noise *voltage* (where the BJT parts – solid lines – are king), and low noise *current* (where the CMOS parts – dashed lines – are the winners). A good low-noise JFET op-amp like the OPA140 combines the best (nearly) of both worlds. But note that the voltage noise of even a simple (and noisy) CMOS part like the LMC6482 is irrelevant when driven by a source resistance anywhere between 1 MΩ and 10 GΩ, where the op-amp's contribution to the total noise density is negligible.

See Figures 8.60 and 8.61 for op-amp spectral noise plots. For low-frequency amplifier applications see Figure 8.63, with its plots of total (integrated) rms noise v_n versus bandwidth.

Two cautions: (a) The total noise density plots (v_n versus R_s) of Figure 8.58 characterize the performance only at 1 kHz. So an op-amp with a high $1/f$ noise corner will look considerably worse at, say, 10 Hz. Taking the example of the LMC6482 (curve #17), its e_n at 10 Hz is $\sim 170\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$, aligning it with curve #29. (b) Likewise, at *high* frequencies you care about input capacitance (especially in combination with GΩ-scale input impedances), and you care about the noise current generated by the op-amp's noise voltage in combination with the input capacitance ($i_{nC}=e_n\omega C_{in}$). A quiet JFET op-amp like the OPA627 has nearly triple the input capacitance (because of its large-area input JFET) of its somewhat noisier (and less expensive) OPA656 cousin.

8.9.4 Low-noise instrumentation amplifiers and video amplifiers

In addition to the low-noise op-amps, there are some nice low-noise IC *instrumentation amplifiers* and low-noise *preamplifiers*. Unlike general-purpose op-amps, these generally have fixed voltage gain, or provision for attaching an external gain-setting resistor. Instrumentation amplifiers, intended for precision differential applications, are discussed extensively in Chapter 5 (§5.13). Those categorized as"video amplifiers" often have bandwidths into the tens of megahertz, though they can be used for low-frequency applications as well. Examples are the TI/Burr–Brown INA103 instrumentation amplifier, and the TI/National

[80] Table 8.3c features a new e_nC_{in} figure-of-merit, useful for TIA designs.

[81] In addition, as described in §8.11.3, in the inverting configuration an amplifier's input noise *voltage* creates a noise *current* through the input capacitance, to the tune of $i_n=e_n\omega C_{in}$.

[82] For comparison we included our jellybean JFET LF411, the legacy LM741 and LM358, and a few other popular op-amps that do not qualify as "low noise"; those have numeric labels.

LMH6517 or Linear Technology LTC6400-series video amplifiers. These amplifiers typically have e_n down in the $1\,nV/\sqrt{Hz}$ territory, achieved (at the expense of high input noise *current*, i_n) by running the input transistor at relatively high collector current.

8.9.5 Low-noise hybrid op-amps

Back in §8.9.1E we hinted at the possibility of a hybrid low-noise design, in which a low-noise BJT or JFET differential front-end could combine with a conventional op-amp to provide superior noise performance. The simplified circuits in Figures 8.66 and 8.67 show how this can be done, with BJTs and JFETs, respectively. These amplifiers are also a form of composite amplifier, a subject discussed in §*4x.5*.

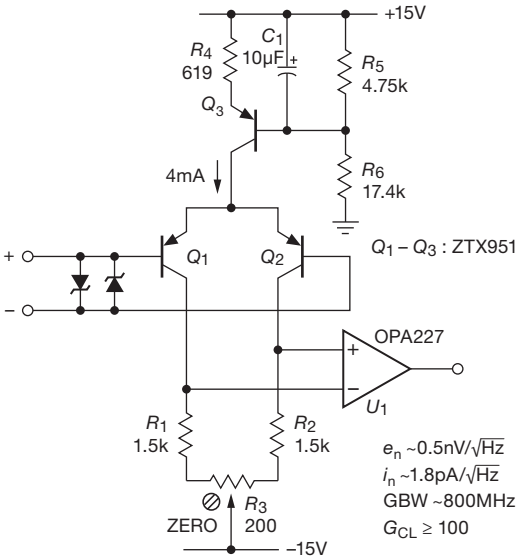


Figure 8.66. Combining the best of both worlds: a BJT hybrid wideband low-noise op-amp.

A. General design issues

Common-mode range In both cases we designed the circuit to provide a substantial common-mode input signal range (at least ± 10 V), intending this to serve as a general-purpose low-noise op-amp substitute. For that reason the current sources in the emitter (or source terminal) are biased for compliance to within 2.5 V of their respective rails, and the collector load resistors (R_1 and R_2) in the BJT design are sized to drop only 3 V (thus accommodating negative input swing to at least -12 V). The JFET design is similarly biased; however, a volt or two is lost in the cascode transistors (Q_3 and Q_4) and in the negative V_{GS} of the input stage, reducing the positive common-mode input range to approximately +10 V.[83]

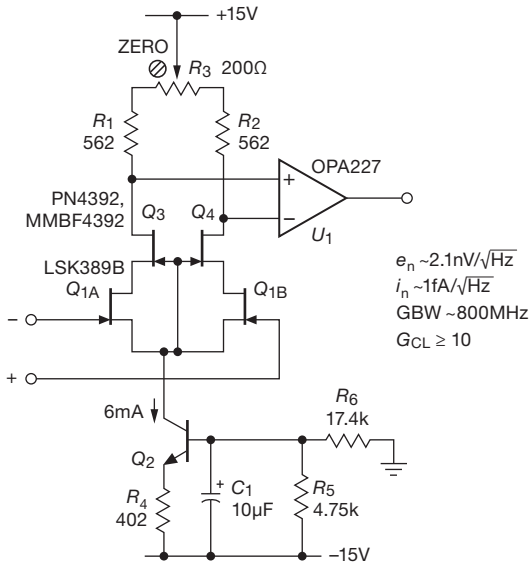


Figure 8.67. Another hybrid op-amp, this time with differential JFET cascode front-end.

Offset trim In the BJT circuit we used one of the quietest available transistors we found in our tests (the Zetex ZTX951); it is not available as a matched pair, so we provided an offset trim R_3, with a relatively large adjustment range ($\pm 6\%$). For the JFET circuit we used the LSK389 "matched" dual, but its worst-case offset of 20 mV mandates an even larger trim range, here $\pm 17\%$. Ironically, the discrete BJT pair's V_{BE} mismatch is likely to be less than that of the matched JFET dual; at least that's been our experience; see, for example, the histogram of measured V_{BE}'s in Figure 8.44.

It's important to realize that a simple offset-trim circuit that unbalances the load resistances, as shown here, greatly compromises the common-mode rejection ratio of the differential input. Additionally, the unbalanced load greatly compromises attenuation of noise in the current source (which would be at least $\times 50$ with 1% load resistors). A better method involves balancing the currents seen at the collector (or drain) load resistors; see for example Figure 8.80 later in the chapter.

Minimum closed-loop gain Because there's voltage

[83] If the cascode runs out of compliance it simply disappears, becoming a low-R_{on} switch.

gain in the transistor front-ends of these hybrid amplifiers (about $\times 120$ for the BJT, $\times 12$ for the JFET), the overall loop gain is substantially greater than that of the second-stage op-amp alone (which by itself is quite high, roughly 160 dB at low frequencies). To ensure stability (given that the OPA277 is unity-gain stable), these hybrid amplifiers should be configured with a closed-loop gain of at least $\times 100$ and $\times 10$, respectively.[84]

B. Some detailed design issues

Choice of transistors Although its noise voltage is unspecified, we found excellent noise performance (and good consistency) with some Zetex bipolar transistors, notably their ZTX851 (*npn*) and ZTX951 (*pnp*). The latter were slightly better, with a measured input noise voltage corresponding to an admirably low base resistance of $r_{bb'} \sim 1.2\,\Omega$ (thus, e.g., an e_n of $0.17\,nV/\sqrt{Hz}$ at 10 mA collector current), compared with $r_{bb'} \sim 1.4\,\Omega$ for the *npn* ZTX851.[85] For the JFET circuit we chose the low-noise LSK389B from Linear Integrated Systems; it specifies $e_n = 0.9\,nV/\sqrt{Hz}$ (typ) and $1.9\,nV/\sqrt{Hz}$ (max) at a drain current of 2 mA.[86] It's a large-geometry JFET (for low noise), with a correspondingly large capacitance of $C_{iss} = 25$ pF. It has pleasantly high transconductance, specified as 20 mS (min) at 3 mA drain current.

Input current The BJT has the expected high input current, of course, given the relatively high collector current needed to keep e_n low; with the specified $\beta = 200$ (typ) the input current is $10\,\mu A$. That's the price you pay for the lowest e_n. By comparison, the JFET circuit's input current is down in the 1 pA territory, judging from the graph of "excess gate current" shown for its predecessor 2SK389 (the LSK389 provides no analogous graphs or specification), and given the very low V_{DS} enforced by the cascode.

Overall noise voltage These are differential input stages, so you need to multiply the single-transistor e_n's by $\sqrt{2}$, which results in the values shown in the figures. For the JFET version, with its lower front-end gain, there is a small contribution also from the op-amp's input noise ($3\,nV/\sqrt{Hz}$), but when referred to the input that amounts to just $0.3\,nV/\sqrt{Hz}$, negligible in the total when combined with the larger front-end squared noise-voltage density.

Note that these are quiet amplifiers – so you need to present their inputs with low signal-source impedances to retain their low e_n. For the BJT version, for example, that implies a resistor value down around $10\,\Omega$ in the lower leg of the feedback divider.

Differential input voltage If your application can expose the differential BJT hybrid op-amp to more than ± 5 V, add a pair of protective back-to-back diodes to prevent base-emitter breakdown and consequent transistor degradation. The differential JFET hybrid op-amp can be safely exposed to full-rail differential-input voltage. These op-amp input stages are running at rather high currents, so a substantial input voltage difference, maintained for more than a few milliseconds, will cause unbalanced heating and consequent input offset voltage. It's a good idea to thermally couple the input pair, and perhaps thermally isolate them from air currents as well.

8.10 Signal transformers

As mentioned earlier in §8.5.6, you're up against the wall when trying to minimize amplifier noise with a signal of very low source impedance Z_s, say less than $100\,\Omega$. The noise voltage of a $50\,\Omega$ resistor, for example, is just $0.9\,nV/\sqrt{Hz}$, which puts you down at the limit of the quietest op-amps. And there are signal transducers with considerably lower resistance, for example magnetic coil pickups. If you are interested only in ac signals (as you would be, for a magnetic coil), you can use a transformer to raise the signal level (by the turns ratio, n:1), simultaneously raising the source impedance seen at the amplifier's input by that ratio squared; i.e., the amplifier sees a signal of source impedance $n^2 Z_s$.

High-quality signal transformers are available from companies such as Jensen Transformers and Signal Recovery. For example, a signal in the audio band (say from 100 Hz to 10 kHz) with source impedance of $100\,\Omega$ would be a poor match for an amplifier like the SR560, whose lowest noise figure occurs for signals of source impedance around $500\,k\Omega$ (see Figure 8.27). The problem is that the amplifier's voltage noise is much larger than the signal source's Johnson noise; the resultant noise figure for that signal connected directly to the amplifier would be 12 dB. By using an external step-up transformer like the

[84] The OPA277 has an excellent phase margin, about 60°, to well past its 1 MHz GBW. If you need a lower closed-loop gain, you can add a series R+C across the op-amp's inputs, effectively reducing the load resistance and open-loop gain at high frequencies.

[85] The *pnp* has lower noise than its *npn* sibling, but much poorer Early voltage, as Table 8.1a shows: $V_A = 120$ V versus 410 V. If the (relatively low) Early voltages of the *pnp* pair are not matched, the CMRR will be degraded. If this is an issue, the circuit could be recast with an *npn* pair, or with an added cascode.

[86] Our measurements landed right in the middle, at $1.34\,nV/\sqrt{Hz}$ ($1.9\,nV/\sqrt{Hz}$ total for the differential pair).

Jensen JT-115K-E with its 1:10 turns ratio (150Ω:15kΩ impedances), the signal level is raised (along with its source impedance), thus overriding amplifier noise voltage. At this signal impedance the amplifier's noise figure is about 0.4 dB; however, the resistance of the transformer's windings[87] contributes to produce an overall noise figure of about 1.5 dB.

There's a tendency to associate transformers with mediocre performance; but well-engineered signal transformers like this one are pretty good: response ±0.15 dB over the audio band (20 Hz–20 kHz), down 3 dB at 2.5 Hz and 90 kHz, distortion less than 0.1% even at 20 Hz,[88] and, as a bonus, common-mode rejection of 110 dB at 60 Hz. The Model 1900 signal transformer from Signal Recovery (originally from Princeton Applied Research) offers turns ratios of 1:100 and 1:1000. It has less bandwidth than the JT-115K-E, but it achieves a minimum noise figure of ∼0.5 dB for source resistances in the 0.8–10 Ω range owing to its low winding resistance of 0.04 Ω. This is somewhat better noise performance than the paralleled-BJT amplifier of §8.5.9; because of transformer limitations, however, it does not work over the full audio band.

At *radio* frequencies (e.g., beginning around 100 kHz) it is extremely easy to make good transformers, both for tuned (narrowband) and broadband signals. At these frequencies it is possible to make broadband "transmission-line transformers" of very good performance. It is at the very low frequencies (audio and below) that transformers become problematic.

Three comments:

(a) The voltage rises proportional to the turns ratio of the transformer, whereas the impedance rises proportional to the square of the ratio. Thus a 2:1 voltage step-up transformer has an output impedance four times the input impedance (this is mandated by conservation of energy).

(b) Transformers aren't perfect. They are susceptible to magnetic pickup, and they have trouble at low frequencies (magnetic saturation) and at high frequencies (winding inductance and capacitance), as well as losses from the magnetic properties of the core and from winding resistance. The latter is a source of Johnson noise, as well. Nevertheless, when dealing with a signal of very low source impedance, you may have no choice, and transformer coupling can be very beneficial, as the preceding example demonstrates. Exotic techniques such as cooled transformers, superconducting transformers, and SQUIDs (superconducting quantum interference devices) can provide good noise performance at low impedance and voltage levels. With SQUIDs you can measure voltages of 10^{-15} V!

(c) Again, a warning: don't attempt to improve performance by adding a resistor in series with a low source impedance. If you do that, you're just another victim of the noise-figure fallacy.

8.10.1 A low-noise wideband amplifier with transformer feedback

Lest we leave the reader discouraged as to the merits of transformers in signal applications, we hasten to point out that some very impressive performance can be gotten with careful engineering. Figure 8.68 shows a nice example: a wideband low-noise amplifier with high input impedance,[89] with a transformer providing lossless (therefore noiseless) feedback to set the voltage gain via the turns ratio (here ×10). The transformer nicely solves also the problem of distributing identical feedback to multiple paralleled input JFETs, each of which is biased to the same 10 mA drain current.

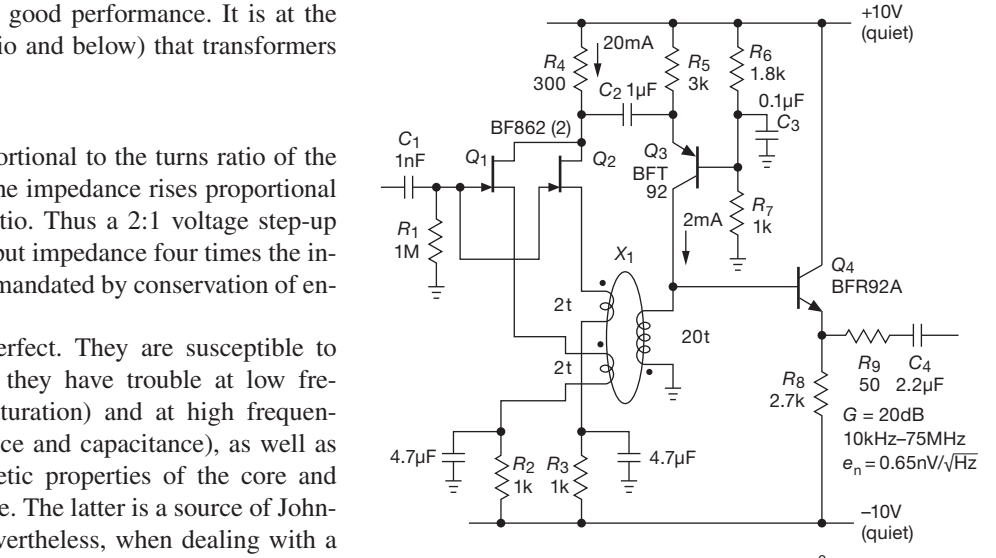


Figure 8.68. Transformer feedback to paralleled JFETs achieves 650 pV/√Hz input noise voltage in this wideband amplifier.

[87] At low frequencies this is simply the dc resistance of the copper wire; but at higher frequencies the effective noise-generating resistance rises owing to the skin effect; see Appendix H (§H.1.4) and §*1x.1*.

[88] Low frequencies are a transformer's Achilles' heel; at 1 kHz this particular transformer's distortion is just 0.001%.

[89] From J. Belleman at CERN; see http://jeroen.home.cern.ch/jeroen/tfpu.

The transformer shown is a small (6.5 mm O.D.) high-mu tape-wound toroid, with just a few turns in the windings to set the high end of the operating band to 75 MHz. That choice limits the low frequency end to frequencies above ∼10 kHz, but the nearly 10,000:1 frequency range is mighty impressive. The source pull-down resistors R_2 and R_3 set the drain current of the BF862 input JFETs to 10 mA (their specified minimum I_{DSS}), where their voltage noise is $0.9\,\text{nV}/\sqrt{\text{Hz}}$; the parallel pair improves that by $\sqrt{2}$, thus $0.65\,\text{nV}/\sqrt{\text{Hz}}$. Bipolar *pnp* Q_3 forms an inverted cascode, with its r_e of 12 Ω diverting essentially all of the input transistors' drain signal. Both Q_3 and Q_4 are wideband transistors (5 GHz); Q_3's collector current is chosen to cancel the dc magnetization induced in the transformer by the input transistors to prevent core saturation. The supply rails need to be quiet, best achieved with a capacitance multiplier (§8.15.1). A caution: this feedback circuit has two low-frequency breakpoints (from C_2 and from the transformer's rather low magnetizing inductance), leading to potential instability and a low-frequency *motorboating*-type oscillation. This is prevented here by the large capacitance of C_2, which puts its breakpoint far below that of the transformer.

8.11 Noise in transimpedance amplifiers

Transresistance amplifiers (or *transimpedance* amplifiers, "TIA," or sometimes just "current amplifiers") produce a voltage output in response to a *current* input. Their gain is therefore V_{out}/I_{in}, with units of ohms, hence their name.[90] We introduced transresistance amplifiers in Chapter 4 (§4.3.1), we evolved a TIA design for an inexpensive lightning detector (§8.5.7, Figure 8.36); and we look at issues of their feedback *stability* in §*4x.3*. As a reminder, the basic circuit, which we've seen before, is shown in Figure 8.69. Assuming ideal components, a current I_{sig} produces an output $V_{out} = -I_{sig}R_f$, so the gain is simply $-R_f$. They are used extensively in circuits where the input is a current, for example from a photodiode or photomultiplier, charged particle detector, tunneling microscope, or patch-clamp amplifier.

The emphasis in this chapter is *noise*, which can arise in several ways in the TIA. The amplifier itself will have its input e_n and i_n; and the feedback resistor creates a Johnson noise voltage that equates to an input

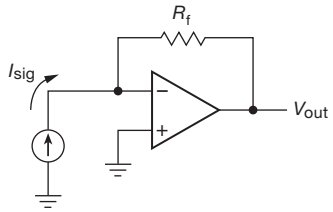


Figure 8.69. Transresistance amplifier: current in, voltage out.

noise current $i_{nR} = \sqrt{4kT/R_f}$ (thus favoring high values of R_f). Capacitance also plays an important role, not only in stability and bandwidth, but also in converting the amplifier's noise *voltage* into a noise current. There's also likely to be what we might call *signal noise*: shot noise in the signal current, Johnson noise in the signal source's resistance, and other forms of signal fluctuation.[91] Let's see how this all plays out.

8.11.1 Summary of the stability problem

As we saw back in §4.3.1C (and which we discuss in greater detail in §*4x.3* in Chapter *4x*), capacitance to ground at the input (e.g., from a current-output sensor and its connecting cable), in combination with the (usually large) feedback resistance, produces a lagging phase shift in the feedback path. This is unstable when combined with the op-amp's 90° (or more) lagging phase shift. You cure this by putting a small capacitor C_f across the feedback

[90] This is the inverse of a device like a JFET, where an input *voltage* produces an output *current*. There the gain is I_{out}/V_{in}, with units of Ω^{-1} (siemens, previously called mhos), hence the name trans*conductance* amplifiers.

[91] Noise in the signal source itself may, in some situations, dominate over the noise contribution of a well-designed amplifier. Optical detectors, for example, are characterized by a "noise-equivalent power" (NEP), which is their electronic noise output stated in terms of optical input power (expressed as optical noise power density, generally in the range of $\text{fW}/\sqrt{\text{Hz}}$). The NEP arises from the detector's dark current (a perfect detector would have zero NEP), a "drift current" that can be thought of as a form of leakage. You get such current when photodiodes are operated with a deliberate reverse-bias voltage (the so-called *photoconductive mode*), in which the capacitance is reduced, and the speed, linearity, and long-wavelength conversion efficiency are improved. If you want to detect very low light levels, and are willing to sacrifice speed, you should operate the photodiode with zero bias (the so-called *photovoltaic mode*), where the greatly reduced NEP arises mostly from a "diffusion current" related dark resistance, powered by the op-amp's offset voltage. For this purpose the photodiode's datasheet usually assumes a 10 mV offset (i.e., detector bias), but you can almost always do better than that. We've measured dark-adapted NEP levels four orders of magnitude better than specified, when driving an op-amp with 100 μV offset in a cool laboratory setting. Taking account of detector noise is an essential part of any system design whose ultimate job is to "convert photons into volts."

resistor (see §*4x.3* for details), which, however, seriously reduces the usable bandwidth. As we saw, you therefore need to use an op-amp of surprisingly high bandwidth in order to get even a modest circuit bandwidth. To put numbers on it, the circuit's usable bandwidth f_c is roughly the geometric mean of the op-amp's GBW (or f_T) and the rolloff frequency, $f_{RC} = 1/2\pi R_f C_{in}$, of the input capacitance; that is

$$\mathrm{GBW} = f_c^2/f_{RCin}\,. \tag{8.43}$$

8.11.2 Amplifier input noise

A transimpedance amplifier input stage, whether discrete or op-amp, will have some noise voltage and current, characterized by densities e_n and i_n; these will exhibit the, by-now, usual $1/f$ rise at low frequencies. And, in addition to the low-frequency noise tail, the current noise will depend on the dc input current (bias current, for BJTs; leakage current for JFETs or MOSFETs), because at mid-frequencies the input noise current is simply shot noise of the dc current.

How do e_n and i_n affect the TIA's overall input noise? We'll see presently (§8.11.3) that e_n generates a noise current through the capacitance seen at the input, and in fact this can easily become the dominant noise term, particularly at higher frequencies. Before worrying about that effect, note simply that the input e_n flows through the feedback resistor, generating a noise-current density $i_n{=}e_n/R_f$ (and if the signal source has a finite source resistance, replace R_f with $R_f\|R_s$). These terms are usually small compared with the corresponding Johnson-noise currents, but they can grow to significant levels at low frequencies where the $1/f$ behavior may raise e_n by factors of $10\times$ to $50\times$ over its mid-frequency ("white") value.[92]

The current noise i_n of the input stage needs no conversion – it contributes directly to the TIA's equivalent input noise current (as square root of sum of squares). Low-bias op-amps with FET inputs (as well as discrete JFETs) will generally have quite low input-noise currents, down in the fA/$\sqrt{\mathrm{Hz}}$ territory. But watch out for rising $1/f$ noise: the datasheet for the low-e_n JFET AD743, for example, shows i_n rising from its mid-frequency 7 fA$\sqrt{\mathrm{Hz}}$ value to 100 fA$\sqrt{\mathrm{Hz}}$ at 1 Hz; that would correspond to the shot noise from a dc bias of 30 nA! (Its typical room temperature input current is specified as 0.15 nA.) Watch out, also, for rising i_n (from rising leakage current in FETs) at elevated temperatures: the AD743 specifies i_n only at 25°C (where it's 7 fA$\sqrt{\mathrm{Hz}}$ value is consistent with shot noise in 0.15 nA dc input current); but you can use their plot of dc input current versus temperature to find that $i_n{=}40$ fA$\sqrt{\mathrm{Hz}}$ at 80°C (where the plot shows a dc input current of 5 nA) and rising further to 400 fA$\sqrt{\mathrm{Hz}}$ at 125°C.

Temperature isn't the only contributor to JFET input current; there's the impact-ionization effect (§3.2.8), which can cause a devastating rise in input current (and noise) when a JFET is operated with drain-to-source voltages greater than a few volts. With a discrete JFET design you can prevent this by operating at low drain voltage (e.g., with a cascode); low-noise op-amps usually are designed with this effect in mind, but you will see increased input current (and current noise) as the input voltage approaches one of the rails (e.g., the positive rail for the LT1792 or ADA4627; the negative rail for the AD8610).

8.11.3 The e_nC noise problem

In addition to the input stage's contributions of i_n (directly) and e_n (flowing through the feedback and input resistances), input *capacitance* (already an annoyance in terms of stability and bandwidth) creates interesting noise problems as well. For example, you might at first think that amplifier *voltage* noise is of little concern in an amplifier whose input is a current; because its feedback looks like a voltage follower, it would seem to produce at most an additive output voltage-noise contribution just equal to the input-voltage noise e_n (equivalent to an input-current-noise contribution of $i_n = e_n/R_f$). But you would be wrong! To see what happens, look at Figure 8.70, where the

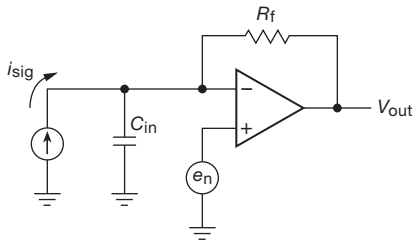


Figure 8.70. The capacitance at the input causes the amplifier's voltage noise e_n to create an input *current* noise $i_n{=}e_n\omega C_{in}$. Neither this "e_nC" current noise nor the amplifier's own input current noise is shown.

[92] For example, the voltage noise of the AD743 (quietest JFET op-amp, with $e_n{=}2.9$ nV/$\sqrt{\mathrm{Hz}}$ at 10 kHz) grows to 23 nV/$\sqrt{\mathrm{Hz}}$ at 1 Hz. The LT1792 grows to 30 nV/$\sqrt{\mathrm{Hz}}$, the OPA627 to 33 nV/$\sqrt{\mathrm{Hz}}$, and the ADA4627 to 42 nV/$\sqrt{\mathrm{Hz}}$. And the speedy OPA656 grows to 75 nV/$\sqrt{\mathrm{Hz}}$ at 10 Hz! Not to be outdone, the otherwise-admirable AD8610 and 8620 JFET op-amps grow from 6 nV/$\sqrt{\mathrm{Hz}}$ to about 200 nV/$\sqrt{\mathrm{Hz}}$. And when it comes to CMOS op-amps, whoa!

op-amp's internal differential-noise voltage e_n is modeled as a voltage in series with the noninverting terminal. Feedback forces the inverting terminal (with its capacitance C_{in} to ground) to follow, creating an actual input current $i_n(t)=C_{in}\,dv_n(t)/dt$ (where $v_n(t)$ is the op-amp's input noise voltage); expressed as a noise *density*, we get

$$i_n=e_n\omega C_{in}=2\pi e_n C_{in} f. \tag{8.44}$$

That is, the amplifier's voltage noise creates a noise current proportional to the capacitance at the input, and rising proportional to frequency. We refer to this input-noise current, produced by the amplifier's internal noise voltage, as "e_nC noise."[93]

8.11.4 Noise in the transresistance amplifier

Let's apply this kind of thinking to figure out the noise performance of the transresistance amplifier we looked at in §*4x.3*. We've redrawn it in Figure 8.71, with the op-amp's input-noise voltage e_n, noise current i_n, and shunt feedback capacitor C_f shown explicitly. In a typical high-speed (small-area) photodiode application you might have $C_{in}\sim$10–20 pF (but more if connected through a shielded cable), $R_f\sim$1–10 MΩ, and, for a FET-input op-amp, $e_n\sim$3–10 nV/$\sqrt{\text{Hz}}$ and $i_n\sim$1–10 fA/$\sqrt{\text{Hz}}$.

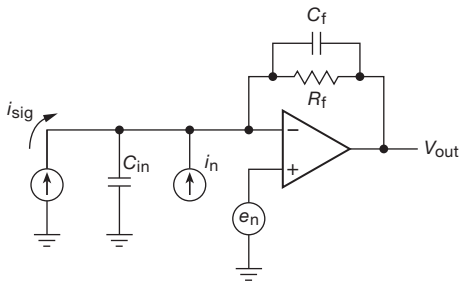


Figure 8.71. Noise in a photodiode amplifier. For the calculations we've used R_f=1 MΩ. Here C_{in} is the total capacitance seen at the input (amplifier, wiring, and input device capacitances).

We'll calculate the noise contributions (from e_n, i_n, and from Johnson noise in the feedback resistor) as effective current noise (versus frequency) *at the input*; after all, that's where the input current signal that we care about enters. For the time being we ignore the capacitor C_f, and use typical circuit values of R_f=1 M and C_{in}(circuit)=10 pF.

As we saw earlier, Johnson noise is flat with frequency, with a noise voltage $e_n=\sqrt{4kTR}$ volts per square root hertz; this translates to a short-circuit noise current of e_n/R; that is, $i_n=\sqrt{4kT/R}$ amps per square root hertz. So for a 1 MΩ feedback resistor at 25°C,

$$i_n=\left(\frac{4kT}{R_f}\right)^{\frac{1}{2}}=1.28\times10^{-10}R_f^{-\frac{1}{2}}=0.128\,\text{pA}/\sqrt{\text{Hz}}.$$

This is one to two orders of magnitude larger than the amplifier's input current noise, which we can therefore ignore. The last contribution is from the amplifier's e_n, which, as we remarked earlier, looks like an input current noise $i_n=2\pi e_n C_{in} f$. It rises proportional to frequency, becoming dominant over resistor Johnson noise at some crossover frequency that we'll call f_X. By equating the Johnson-noise current with the e_nC current, you can find[94]

$$f_X=\frac{\sqrt{4kT/R_f}}{2\pi e_n C_{in}}. \tag{8.45}$$

It would continue to rise forever, except for the effect of the parallel capacitance C_f, which causes the noise current to flatten off at a frequency $f_c=1/2\pi R_f C_f$ (caused by the R_fC_f pole cancelling the +6 dB/octave rising e_nC noise). If we further choose C_f so that f_c equals the geometric-mean frequency[95]

$$f_{GM}=\sqrt{f_{RCin}f_T}, \tag{8.46}$$

we produce a slight peaking at f_c, with the pair of poles causing the e_nC noise to fall at -6 dB/octave (i.e., $\propto 1/f$) at higher frequencies, as seen in Figure 8.72.

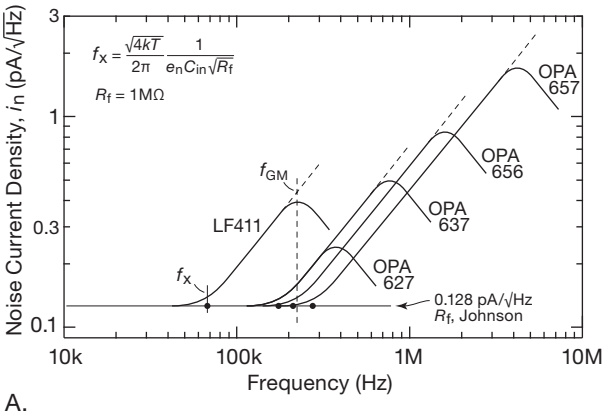
In Figure 8.72 we've plotted i_n(input) for the transresistance amplifier of Figure 8.71, with R_f=1M, for the op-amp choices and corresponding data shown in the figure's table. The plot shows clearly the reduction in input-referred total noise current that you get by choosing an amplifier with low input capacitance and low noise voltage, assuming of course that you are planning to limit the output bandwidth with a later stage of lowpass filtering.

It's interesting to note that, if all other things were equal, an op-amp with greater amplifier bandwidth would not reduce the input-referred noise current; it would simply extend the bandwidth of the transresistance amplifier. However, because faster op-amps tend to have less input capacitance, there is some noise advantage, particularly when the external input capacitance is as low as we've assumed here.

[93] In the literature you'll see descriptions like "noise gain peaking at higher frequency" and "a complex response to the op-amp's input noise voltage." But we like "e-sub-n-C": it's easy to say, and easy to remember.

[94] For R_f=1M this becomes f_X(Hz)=$2\times10^7/e_nC_{in}$, where e_n and C_{in} are in units of nV/$\sqrt{\text{Hz}}$ and pF respectively.

[95] See the stability criterion in §*4x.3*.



A.

Calculated f_C and f_X, with $C_{external}$ = 10pF and R_f=1M

Op-amp	e_n (nV/√Hz)	C_{amp} (pF)	$C_{in(total)}$ (pF)	f_T (MHz)	f_C (kHz)	f_X (kHz)
LF411	25	2	12	4	230	67
OPA627	4.5	15	25	16	320	178
OPA637	4.5	15	25	80	715	178
OPA656	7	3.5	13.5	230	1650	212
OPA657	4.8	5.2	15.2	1600	4100	274

B.

Figure 8.72. A. Input-referred noise-current spectrum for the photodiode amplifier in Figure 8.71. The crossover frequency f_X at which "e_nC" noise becomes dominant is marked with a dot for each configuration. For the LF411 example both f_X and the turnover frequency f_c are marked. In each case it is assumed that the compensation capacitor C_f is chosen so that f_c is equal to $f_{GM}=\sqrt{f_{RCin}f_T}$, thereby obtaining maximum amplifier bandwidth consistent with only modest peaking (damping ratio $\zeta=0.7$); otherwise there will be a region of flat noise density, as in Figure 8.73. B. Input-referred current-noise parameters for the photodiode amplifiers of Figure 8.71, assuming an external input capacitance of 10 pF.

8.11.5 An example: wideband JFET photodiode amplifier

Continuing along this path, in Figure 8.73 we've plotted the transimpedance gain, noise gain, and effective input-noise current for a transimpedance amplifier made from an OPA656 (f_T=230 MHz, e_n=7 nV/√Hz), with a 1M feedback resistor and a conservative value of shunt capacitor (2 pF: f_c=76 kHz) to ensure stability with input capacitances as high as 1000 pF.

Note that there are two "gains" here: the *transimpedance gain* (top plot) is the ratio of output signal voltage to input signal current, a plot that is nominally flat up to the rolloff at f_c, but with the additional constraint imposed by the finite (and falling) op-amp open-loop gain G_{OL}. The *noise gain* (middle plot) is the ratio of output signal voltage to input noise voltage, with its characteristic rising "e_nC" slope proportional to frequency. It flattens at f_c, but (for op-amps of modest open-loop gain[96]) is further limited by G_{OL}.

Finally, the effective input-referred noise current seen at the output (bottom plot) is the sum of the e_nC noise and the feedback resistor's Johnson noise, as shaped by the amplifier's rolloff. Those noise-current terms (as seen at the input) are $e_n\omega C$ and $\sqrt{4kT/R_f}$, respectively. Here the Johnson noise dominates at low frequencies: a 1 MΩ resistor generates a (short-circuit) white-noise current of 0.13 pA/√Hz.

We measured this amplifier's output noise spectrum with the four values of input capacitance; we measured also the op-amp's input voltage noise (e_n). Figure 8.74 is a screen capture (from an SRS785 spectrum analyzer), labeled with corresponding input-referred i_n and e_n scales. The measured data are in reasonable agreement with the predictions of Figure 8.73, with the exception of some excess noise at the low-frequency end of the 10 nF plot. But Figure 8.73 assumed a constant (white-noise) value of 6 nV/√Hz for e_n, whereas the measured voltage noise exhibits the usual $1/f$ rise at low frequencies, roughly triple its high-frequency value at 100 Hz.

8.11.6 Noise versus gain in the transimpedance amplifier

In the preceding discussion we casually took a round-number value of 1 MΩ for the feedback resistor R_f, without much consideration of the consequences for noise and bandwidth. From Figures 8.72, 8.73, and 8.74 it is easy to see that the low-frequency current noise floor is set by R_f's Johnson noise; so larger values of R_f would appear to be "better."

Not so fast! For a given input capacitance, larger values of R_f correspond to a lower input rolloff frequency ($\omega = 1/R_fC_{in}$), requiring more aggressive compensation

[96] Open-loop gain limitations may be encountered with very wideband op-amps (e.g., parts like the OPA655/6/7, with GBW values in the GHz region), but such limitations are rare for lower-frequency op-amps such as the OPA637.

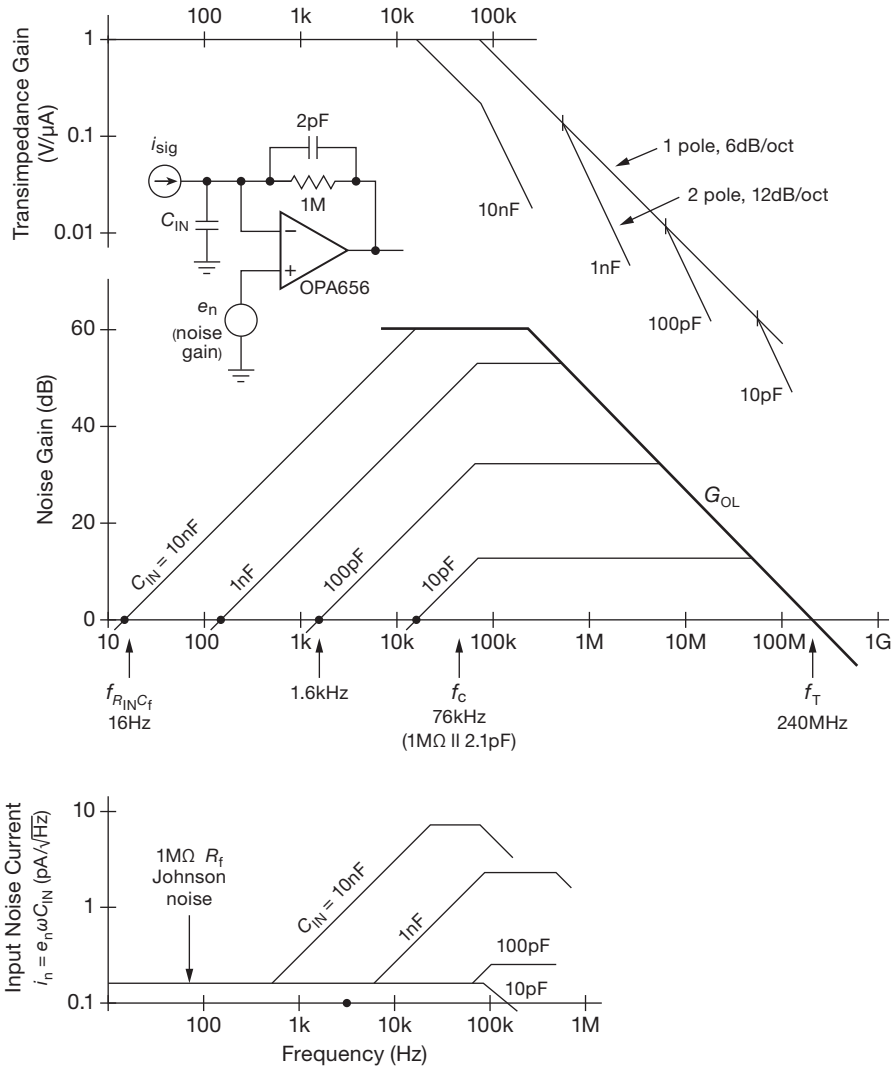


Figure 8.73. Charting the noise in a transimpedance amplifier. A 230 MHz op-amp is used to obtain ∼75 kHz bandwidth for input capacitances from 10 pF to 1 nF. The transimpedance bandwidth is not much affected by input capacitance up to 1 nF, but a 10 nF capacitance reduces the bandwidth substantially. It does this without creating instability in this case, because of the OPA656's limited open-loop gain (65 dB).

(i.e., lower bandwidth f_c). If your noise is dominated by Johnson noise in R_f, and you don't care about bandwidth but want to minimize the low-frequency noise, then a larger R_f is good. But in a wideband photodiode amplifier in which the e_nC noise is dominant and in which you want to improve the bandwidth, it's better to reduce R_f.

But, *not too much!* Here's the reason: a goal in any transimpedance amplifier design should be to ensure that the amplifier adds insignificant noise to the inherent shot noise of the input signal. As we reduce R_f, however, its Johnson-noise current $\sqrt{4kT/R_f}$ increases, ultimately dominating over the input signal's irreducible shot-noise current $\sqrt{2qI_{in}}$. By equating these noise currents we get the condition $I_{in}R_f = 2kT/q = 50\,\text{mV}$. That is, to avoid adding amplifier noise, the feedback resistor should not be chosen so small that the voltage drop produced by the input (which could be a dc component of the input signal) is less than, say, 100 mV.

This noise–bandwidth tradeoff in the choice of feedback resistance is illustrated in Figure 8.75, where reducing

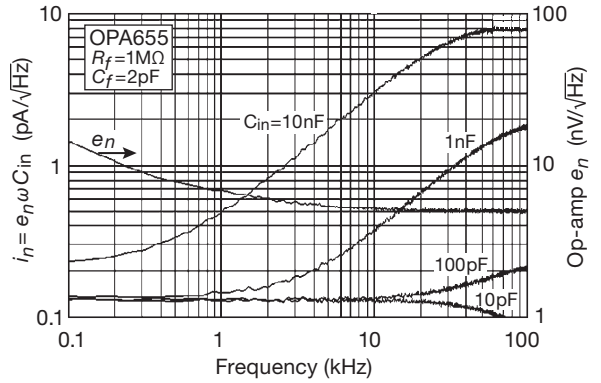


Figure 8.74. Measured input-referred current noise in the amplifier of Figure 8.73. The separate spectra have been here overlaid, with some helpful decorative legends.

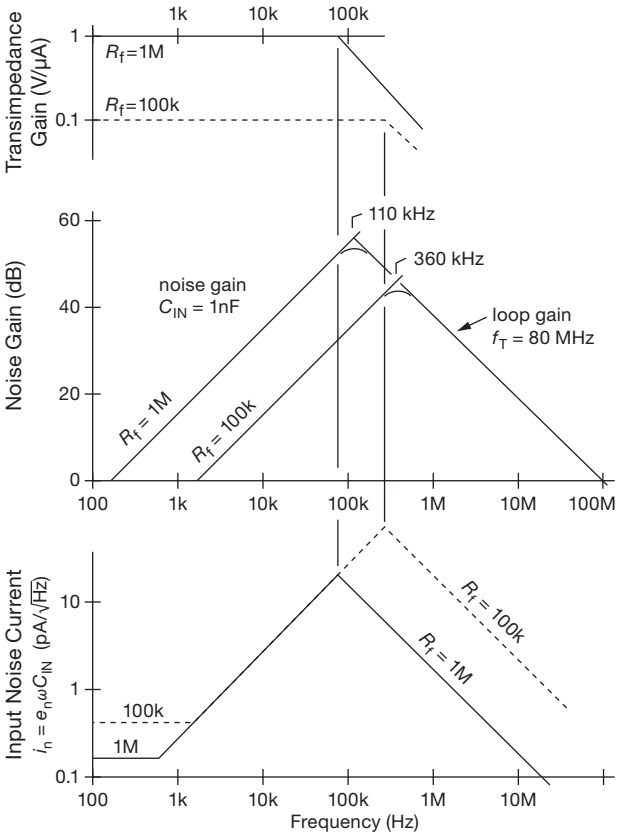


Figure 8.75. In a fast transimpedance amplifier, where e_nC noise dominates over R_f Johnson noise, you can increase the usable bandwidth by reducing R_f; recover the lost gain in a second-stage voltage amplifier.

R_f from 1M to 100k moves the input rolloff upward by a factor of 10, allowing an increase in amplifier bandwidth by a factor of roughly 3 (more precisely, $\sqrt{10}$). The price you pay is $\times 10$ reduced gain (easily recovered with a downstream voltage amplifier stage), and increased low-frequency noise floor (still negligible compared with the e_nC high-frequency noise contribution). Of course, the increased bandwidth comes with a continuing increase in e_nC noise, as seen above 100 kHz in the bottom plot of Figure 8.75.

Happily, there some nice tricks you can use to mitigate these high-frequency e_nC noise effects. One method greatly reduces the effective capacitance at the input by "bootstrapping" the return side of the signal source, e.g., photodiode (and cable shield); another uses a common-base transistor stage (a cascode) to isolate the photodiode capacitance from the transimpedance stage. We'll describe these shortly, after a brief discussion of an important (and too often overlooked) issue: the need for a bandwidth-limiting lowpass filter at the output of the transimpedance stage.

8.11.7 Output bandwidth limiting in the transimpedance amplifier

If you're designing a real-world transimpedance amplifier with significant e_nC noise from capacitance at the input, it's important to add a lowpass filter section at the output. Look again at the noise gain plots in Figure 8.73. Because C_f was chosen for stability with the largest anticipated input capacitance, there is a broad band of noise at high frequencies, particularly for lesser values of input capacitance. This noise extends beyond the amplifier's bandwidth, and can contribute the lion's share of total output noise (remember that logarithmic frequency plots tend to conceal the fact that most of the bandwidth is at the high end).

To see this more clearly, look at Figure 8.76, an analogous set of plots for the current-input amplifier of Figure 8.77.[97] Here we've used a decompensated low-noise JFET op-amp, with C_f chosen such that the amplifier is stable with input capacitances up to 1000 pF. Noise gain and effective input noise current are plotted, for both 1000 pF and 100 pF at the input. The solid lines show U_1's output, and the dashed lines show the signal after a simple *RC* lowpass. Because the amplifier is overcompensated when C_{in}=100 pF, there's a substantial region (the dotted area) of

[97] A simplified version of the exemplary photodiode amplifier discussed in §*4x.3*.

out-of-band noise that is eliminated by the R_1C_1 filter. (The effect is minimal when the amplifier is critically compensated, with f_c nearly equal to f_{GM}, as seen in the 1000 pF plots, because the op-amp's open-loop gain rolloff accomplishes the same thing.)

In this circuit we added an output $\times 10$ gain stage, so that the overall gain is 1 V/μA. That provides another opportunity for an additional section of lowpass filtering, via C_2. This might seem extreme, but remember that a single-section *RC* has a gentle rolloff, such that equivalent white-noise bandwidth extends well beyond its characteristic frequency ($1.57 f_{3dB}$, to be exact; see §8.13).

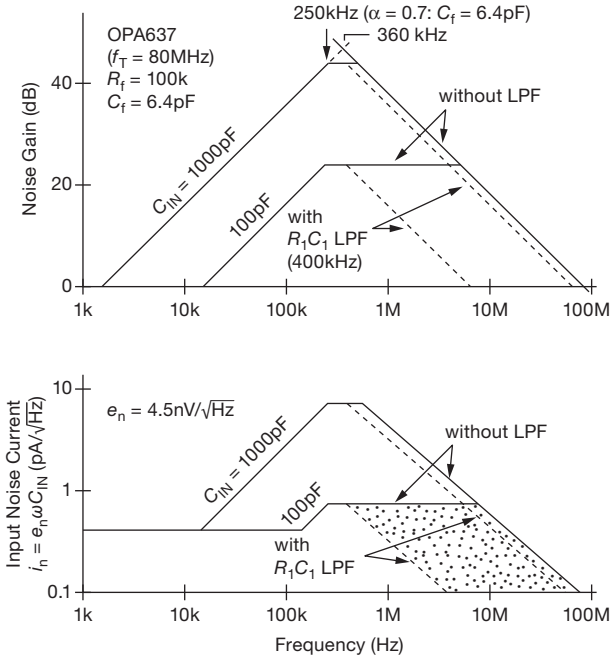


Figure 8.76. Noise plots for the current amplifier of Figure 8.77. Adding an output filter at or slightly above f_c greatly reduces the output noise.

8.11.8 Composite transimpedance amplifiers

When choosing an op-amp for use in a sensitive transimpedance amplifier, you want very low input noise current, thus a JFET or CMOS type. And if you want plenty of speed it's important to select an op-amp with low input noise voltage (to minimize the high-frequency e_nC_{in} noise current it produces), and especially so if the input capacitance is substantial. Finally, because the input noise current produced by the gain-setting feedback resistor R_f decreases

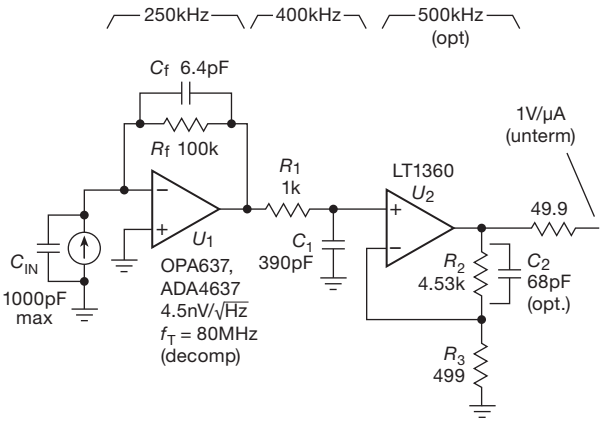


Figure 8.77. Current-input amplifier, BW=250 kHz, compensated for stability with input capacitances up to 1000 pF. The lowpass filters (R_1C_1 and R_2C_2) minimize out-of-band noise, particularly when C_{in} is less than the maximum.

as $1/\sqrt{R_f}$, a low-noise TIA requires a large-value feedback resistor.

But a large R_f produces a low input rolloff frequency ($f_{RCin}=1/2\pi R_fC_{in}$), just what you *don't* need when you're aiming for lots of bandwidth. We'll see presently (in §8.11.9) techniques, such as bootstrapping and cascoding, that can be used in some situations for greatly reducing the effective input capacitance in a transimpedance amplifier. But another approach is simply to select an op-amp with high-enough bandwidth f_T to produce the required TIA bandwidth ($\sim \sqrt{f_{RCin}f_T}$), as we illustrated in §*4x.3.3*.

This is a reasonable approach – but the performance you get with available op-amps often falls short. Table 8.3 shows that most fast op-amps (say $f_T \geq 350$ MHz) have relatively high noise voltage (e_n of $6\,nV/\sqrt{Hz}$ or greater), and quiet op-amps tend to be slow, like our low-noise winner, the AD743 ($e_n=2.9\,nV/\sqrt{Hz}$ and $f_T=4.5$ MHz). From the table you can see also that some of the fastest op-amps are low-voltage parts, for example the OPA657 with $f_T=1500$ MHz, whose total supply voltage range is restricted to 9–13 V. That matters, because the larger values of feedback resistor needed for reduced noise produce more gain, therefore larger output swings (and static dc levels, when there's a nonzero dc component to the input current), favoring op-amps that can run on ± 15 V.

So, what can be done? A nice approach is to separate the input-stage performance from that of the output stage, to optimize overall noise and speed capability. This can be done with a "composite amplifier," a powerful technique seen, for example, in Figures 5.47 and 13.48, and treated in some detail in §*4x.3*.

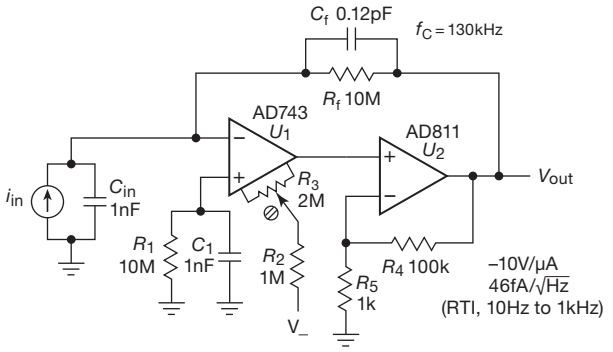


Figure 8.78. Low-noise wideband transimpedance amplifier, exploiting the combination of desirable properties of a pair of op-amps configured as a composite amplifier.

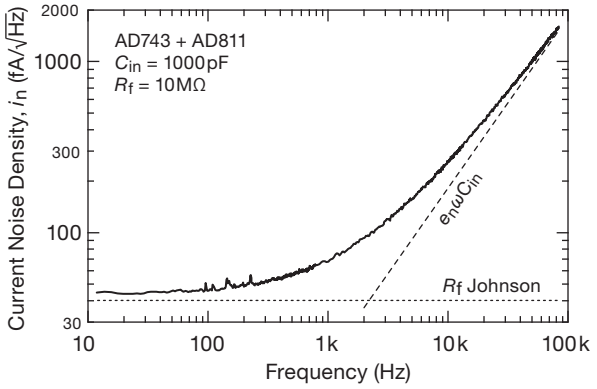


Figure 8.79. Measured current noise versus frequency for the amplifier of Figure 8.78, with a 1000 pF capacitor across the input. The dashed line shows the value of $e_n\omega C_{in}$ for $e_n = 2.9\,nV/\sqrt{Hz}$ (the AD743 datasheet's specified typical e_n), and the dotted line is $\sqrt{4kT/R_f}$, the Johnson-noise current of the 10M feedback resistor (after Kretinin & Chung, used with permission).

Figure 8.78 shows an example,[98] a TIA intended for low-level current signals burdened with a relatively high capacitance of 1000 pF. Here an AD743 low-noise input stage has been mated with a ×100 wideband AD811 output stage, thus maintaining the low e_n while boosting the composite amplifier's f_T by a factor of 100, to 450 MHz. Although the output stage has considerable bandwidth (it's a current-feedback amplifier intended for video applications), the AD743's meager 4.5 MHz bandwidth (with additional poles around that frequency and above, as evidenced by the datasheet's open-loop gain and phase plots) causes the composite amplifier to exhibit a −12 dB/octave rolloff above 5 MHz, reaching a 180° phase shift around 20 MHz. This sounds dangerous, but it's OK because that's well above our f_c frequency (see §*4x.3.2*), and it's above the intercept of the flattened noise gain with the open-loop gain (see Figure 8.73). So the configuration is stable, and in fact the second-stage bandwidth is more than needed.

The performance is quite good, with $i_n \approx 50\,fA/\sqrt{Hz}$ from 10 Hz to 2 kHz when driven with an input capacitance of 1000 pF, and with bandwidth out to 100 kHz. But the input-noise voltage ($2.9\,nV/\sqrt{Hz}$) in combination with the large input capacitance causes a rapidly rising e_nC_{in} noise current, reaching nearly $2000\,fA/\sqrt{Hz}$ at 100 kHz (Figure 8.79). And the low-frequency noise floor is dominated by the Johnson-noise current of the feedback resistor R_f. Given that the AD743 is the quietest available JFET op-amp, we seem to have hit the limit of noise performance in a transimpedance amplifier driven with a highly capacitive source.

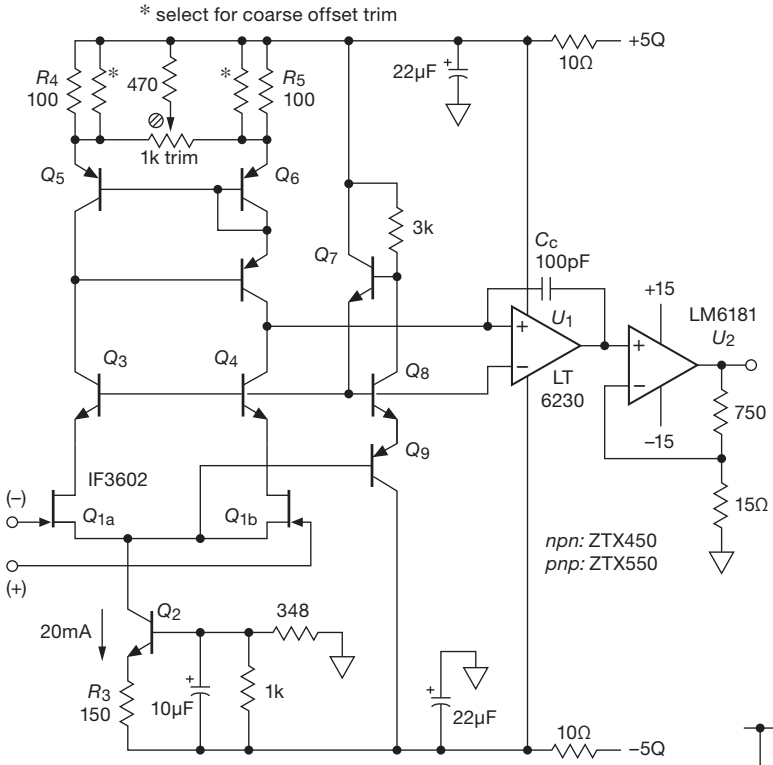
A. Hybrid composite transimpedance amplifier

We *could* do better, though, if we could find a way to reduce the already low e_n (which contributes the rising e_nC_{in} above around 10 kHz) while maintaining a large gain–bandwidth product. The technique that does the trick here is the use of a hybrid input stage to exploit the very low e_n of a large-area JFET pair like InterFET's IF3602 ($e_n = 0.3\,nV/\sqrt{Hz}$, typ, at 100 Hz). Of course, large-geometry JFETs have lots of capacitance (this JFET's C_{iss} is a whopping 300 pF), but that is not terribly serious when you've got a signal source that's burdened already with 1000 pF of capacitance. And we'll stick with the composite amplifier topology, to get the high open-loop amplifier bandwidth that we need in order to achieve reasonable TIA bandwidth with a highly capacitive input.

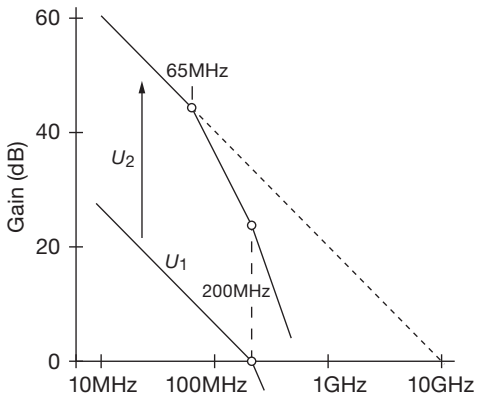
Figure 8.80 shows such a design. The "op-amp" is a 3-stage composite amplifier whose input stage Q_{1ab} is a common-source differential JFET with current mirror drain load Q_5Q_6, isolated with a cascode Q_3Q_4. The overall GBW is 10 GHz, as seen in the Bode plot of Figure 8.80B; this composite op-amp is not stable (!) when used as a voltage amplifier (at least for closed-loop gains <45 dB), but when configured as a transimpedance amplifier (Figure 8.80C) with an effective compensation capacitance of 0.032 pF (set by R_2 in the circuit shown[99]) across the 20M

[98] A. Kretinin and Y. Chung, "Wide-band current preamplifier for conductance measurements with large input capacitance," arXiv:1204.2239v1 (2012).

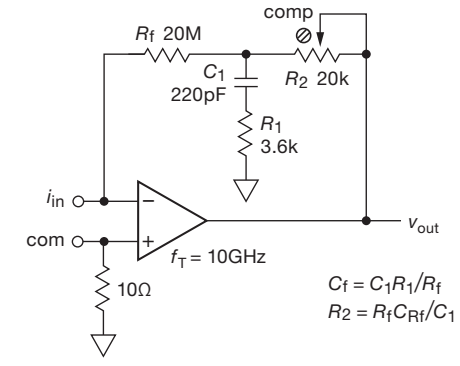
[99] Alternatively you can make R_f from a series string of four or five resistors (spaced out and away from ground) to reduce the total capacitance of their individual ∼0.15 pF parasitic shunt capacitances.



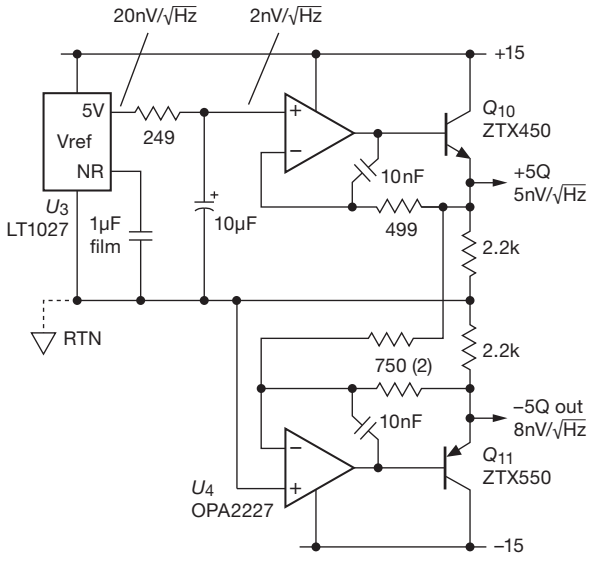
A. "Op-amp"



B. Bode plot



C. TIA block diagram



D. Quiet ±5V supply

Figure 8.80. Hybrid transimpedance amplifier, optimized for lowest noise with capacitive inputs of order ~ 1000 pF by the use of large-area input-stage JFETs with extraordinarily low noise voltage ($e_n = 0.35\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$). The composite amplifier (A) boosts the gain–bandwidth product f_T to 10 GHz, (B), to maintain an overall TIA bandwidth of ~ 250 kHz in spite of the large R_f and C_{in} (20 MΩ and 1000 pF). The transimpedance configuration (C) sets the gain at 20V/µA, with an effective C_f of 32 fF (created by the pole-zero network $C_1R_1R_2$, needed to cancel R_f's excessive self-capacitance (which we'll call C_{Rf}); see §*4x.3*, setting a dominant pole at 200 kHz to ensure stability. A low-noise voltage reference (D) is used to create quiet ± 5 V supply rails.

gain-setting feedback resistor, it is stable.[100] The input stage has a noise voltage density of $e_n \approx 0.6\,\text{nV}/\sqrt{\text{Hz}}$. The second op-amp (LT6230) is a wideband (200 MHz) low-noise ($1.1\,\text{nV}/\sqrt{\text{Hz}}$) op-amp, powered by a low-noise ±5 V power supply[101] (Figure 8.80D); with the JFET input stage, the combined GBW of these two stages is $f_T = g_m/2\pi C_c$, or about 200 MHz with a C_c of 100 pF. The last stage has a gain of 50 and an f_T of 65 MHz, boosting the composite amplifier's GBW to 10 GHz.

When configured as a transimpedance amplifier (i.e., Figures 8.80A and C), this circuit has 5× less noise voltage (thus 5× less $e_n C_{in}$ noise) than the composite TIA of Figure 8.79. It also has greater bandwidth, thanks to its 20× higher f_T, even with the larger value of feedback resistor (20M, chosen to reduce its Johnson-noise current, given the quieter front-end). Look ahead to Figure 8.81 to see this comparison graphically.

Some details We measured $e_n = 0.35\,\text{nV}/\sqrt{\text{Hz}}$ for an IF3602 JFET operating at 10 mA, and $0.3\,\text{nV}/\sqrt{\text{Hz}}$ at 25 mA; here we chose the lower drain current of 10 mA for lower dissipation and reduced $1/f$ noise. Because of the IF3602's very large capacitances (300 pF for C_{iss}, 200 pF for C_{rss}) we clamped the drains with a cascode, setting $V_{DS} = V_{BE}$ independently of V_{GS} and input common-mode level. This further reduces JFET power dissipation (to about 6 mW each), and prevents excessive "impact-ionization" gate current (§3.2.8).

The IF3602 pair is not well matched (specified $V_{os} = 100$ mV max), which is especially troublesome in view of its very high transconductance (we measured $g_m = 130$ mS at 10 mA, see the plot of measured JFET transconductances in §*3x.2*). If we try to trim the offset by unbalancing the drain resistors (with a circuit like that of Figure 8.67), we quickly find from $\Delta I_D/I_D = 0.5\,g_m V_{os}/I_D$ that we need a 39% drain-current unbalance to null a worst-case input offset. Ouch! So we went with the scheme shown, in which a degenerated low-noise Wilson current mirror with selected fixed trim resistors sets the required unbalanced drain currents. The current mirror's noise-voltage contribution with 100 Ω emitter resistors R_4 and R_5, when referred to the JFET input terminals, is $e_n = \sqrt{4kT/R_4} \cdot 1/g_m$, or $0.1\,\text{nV}/\sqrt{\text{Hz}}$, an insignificant 4% contribution when the squared amplitudes are combined. Finally, a comment on the differential pair current sink Q_2: normally a differential pair's current sink noise is not terribly critical, because a reasonably well-balanced differential stage cancels the noise (e.g., 30× for a 3% balance). But here we face a possibly severe current imbalance to trim the offset voltage, so we've used a low-noise voltage regulator ($<10\,\text{nV}/\sqrt{\text{Hz}}$), with further *RC* quieting of the current sink.

Exercise 8.3. Calculate the noise densities of the ±5Q supplies in Figure 8.80D. What is their rms noise?

B. Composite versus single-stage transimpedance amplifiers

The plots of transimpedance gain and input noise current in Figure 8.81 are a good way to see the performance enhancement you get with a composite TIA in the difficult situation of high input capacitance. The improved performance comes from the greater amplifier bandwidth f_T, and from the ability to create a quiet input stage.

Comparing first the single-stage (i.e., not composite) configurations A and B, you see the improved bandwidth gained from the faster OPA637, with the tradeoff of greater noise. Adding a (composite) second stage to the quieter configuration A gets you a lot of extra bandwidth with no increase of noise (configuration C, the circuit of Figure 8.78). The composite amplifier with an even lower-noise discrete input stage (configuration D, the circuit of

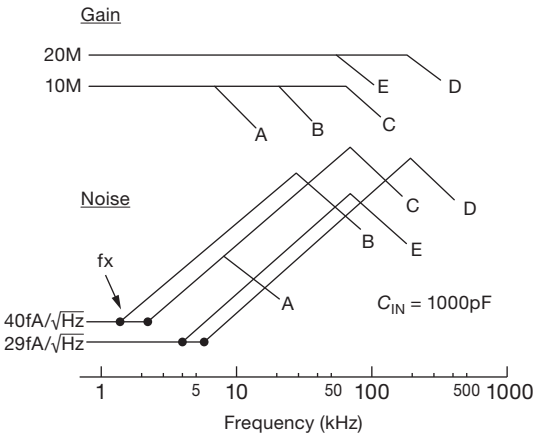


Figure 8.81. Comparing the performance of five transimpedance amplifiers when driven with a current signal of 1000 pF capacitance. A, AD743; B, OPA637; C, AD743 composite (Figure 8.78); D, IF3602 composite (Figure 8.80); E, OPA637 with BF862 bootstrap (similar to Figure 8.82); The feedback resistor R_f is 10 MΩ for circuits A–C, and 20 MΩ for D and E.

[100] With $C_{in} = 1000$ pF, $f_{RCin} \approx 8$ Hz and $f_T = 10$ GHz, so $f_{GM} \approx 280$ kHz; a 0.032 pF effective capacitance across R_f makes $f_c = 200$ kHz.

[101] Where we've started with an LT1027 reference, which uses a buried zener for low $1/f$ noise, quieted further with a bypass capacitor at the noise-reduction pin. Adding a resistor to ground from the upper op-amp's inverting input lets you change the quiet split-supply voltages, e.g., to ±6 V, or whatever.

Figure 8.80) has yet more bandwidth, and can tolerate the larger R_f needed to fully exploit the amplifier's lower e_n. Finally, if you have the luxury of bootstrapping the input capacitance (see for example §8.11.9, Figure 8.82), the greatly lowered C_{in} lets you revert to the simple single-stage OPA637 configuration without a great sacrifice in performance.

This example illustrates once again a lesson we've learned over many decades of designing low-noise amplifiers, namely that circuit complexity increases rapidly as improvements are added to reach the limits of low-noise circuit performance.

8.11.9 Reducing input capacitance: bootstrapping the transimpedance amplifier

Input capacitance has not been our friend. As Figures 8.73, 8.74, and 8.76 make abundantly clear, input capacitance is the root of all evil, in terms of both noise and bandwidth. Large-area photodiodes are plagued with lots of capacitance (as much as 1000 pF or more); and, if the detector is at the end of a length of shielded cable, you can figure on an additional ∼30 pF/ft of cable (this value is not arbitrary; see Appendix H).

As we hinted earlier, there are tricks by which you can greatly reduce the effective capacitance. Figure 8.82 shows an elegant solution,[102] namely an ac-coupled bootstrap of the return side of the capacitive input device (plus shielded cable, if any). In this circuit the JFET follower Q_1 drives the low side of the photodiode with a replica of any signal at the summing junction; Q_1's high transconductance (∼25 mS) ensures a gain close to unity (output impedance ∼40 Ω), thus a reduction of the photodiode's effective input capacitance (as seen at the summing junction) by at least a factor of 10.

But now we have to worry about noise introduced by Q_1 and associated circuitry. You might think first about Q_1's gate current, and also the capacitance C_{iss} that it adds to the input. For this particular JFET, things look pretty good: it has low gate input current (∼1 pA, as long as you keep $V_{DS} < 5$ V), and low feedback capacitance (∼2 pF). But the JFET's noise is another matter, because its e_n voltage noise generates an e_nC noise current in combination with the *unbootstrapped* photodiode (and cable) capacitance. This is in contrast to the op-amp's e_n, which now sees the much lower bootstrapped capacitance; both of these noise currents must come through R_f, generating output noise.

[102] See Linear Technology's LTC6244 datasheet and their DN399 Design Note by Glen Brisebois.

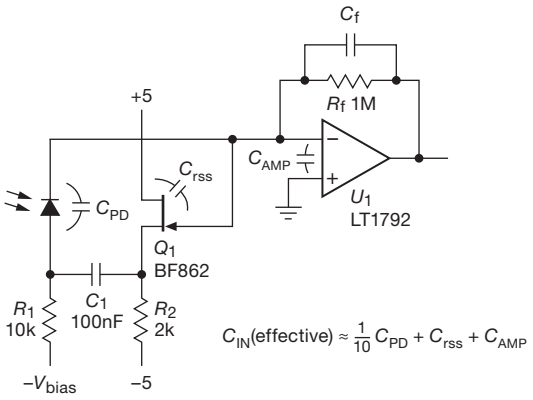


Figure 8.82. Follower Q_1 bootstraps the photodiode at signal frequencies, reducing its effective capacitance by a factor of 10 or more. The BF862 is especially well suited to this task, with its low capacitance and sub-nV/$\sqrt{\text{Hz}}$ noise. This lets us use a less-expensive op-amp (with less bandwidth and somewhat higher e_n) for U_1. We often add an emitter follower (not shown for simplicity). To find the break frequency of the e_nC_{in} noise, use eq'n 8.45 on page 539, with the "reduced" value of C_{in}; see text.

For this reason it's essential to choose a JFET with very low noise voltage, ideally much lower than the e_n of the op-amp. The BF862 is an excellent choice, with its impressively low[103] e_n∼0.9 nV/$\sqrt{\text{Hz}}$. Even so, its noise contribution is larger than that of the op-amp (the latter benefitting, of course, from the JFET's successful reduction of effective photodiode capacitance), because the LTC1792 is a quiet (4.2 nV/$\sqrt{\text{Hz}}$) op-amp to begin with. For example, with a 1000 pF photodiode the op-amp's 4.2 nV/$\sqrt{\text{Hz}}$ acts on the (reduced) ∼100 pF, while the JFET's 0.9 nV/$\sqrt{\text{Hz}}$ acts on the full 1000 pF of the photodiode. Making that quantitative, at 100 kHz the noise-current contributions are 0.26 pA/$\sqrt{\text{Hz}}$ and 0.57 pA/$\sqrt{\text{Hz}}$, respectively, for a combined noise current of 0.63 pA/$\sqrt{\text{Hz}}$. Although the JFET's noise dominates, we're still doing quite a bit better than the unbootstrapped configuration, with its noise current of 2.6 pA/$\sqrt{\text{Hz}}$.

We discuss transimpedance amplifiers in further detail (particularly issues of stability) in §*4x.3*. There we include a design for an input-capacitance-tolerant photodiode amplifier that exploits bootstrapping; the circuit is reproduced here in Figure 8.83. In this design we used the excellent (and not inexpensive) OPA637 (GBW=80 MHz,

[103] Inexpensive, too, about $0.50 in qty 25. And, according to Phil Hobbs, they can be paralleled without source ballasting (especially when taken from the same reel), because "they go from cutoff to I_{DSS} in about 400 mV. JFETs running near I_{DSS} have a low tempco as well. I use boatloads of them – they're Good Medicine."

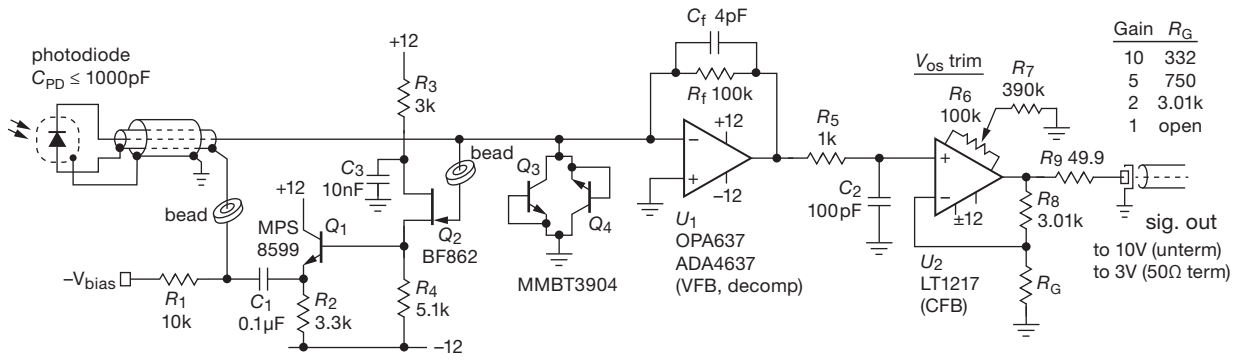


Figure 8.83. A complete photodiode amplifier, suitable for input capacitances up to 1000 pF. Input bootstrapping greatly reduces the effective photodiode and cable capacitance, for enhanced speed and reduced noise.

e_n=4.5 nV/$\sqrt{\text{Hz}}$). In Figure 8.84 we plot the noise and bandwidth, under the conservative estimate of a tenfold reduction of input capacitance. Bootstrapping improves the bandwidth, reduces the noise, and leaves the transimpedance gain unchanged.[104] Not bad!

Exercise 8.4. Design a TIA with an OPA637 op-amp and a BF862 JFET bootstrap follower, for input signals with C_{in}=1 nF. Use R_f=20 MΩ, and assume the BF862 has a noise voltage of 0.85 nV/$\sqrt{\text{Hz}}$ and a voltage gain (when driving the bootstrap terminal) of G_V=0.95. Evaluate your circuit's noise and gain performance, which should be not unlike that of curve E in Figure 8.81.

8.11.10 Isolating input capacitance: cascoding the transimpedance amplifier

Bootstrapping reduces the effective input capacitance (typically by an order of magnitude), allowing reduced noise and greater bandwidth from a capacitive current-output sensor such as a photodiode. But we can do even better: it's possible to isolate the input capacitance entirely, by interposing a common-base (cascode) stage between the input signal and the transimpedance amplifier.

There are some bumps along the way, so we'll take it in stages. We learned about this from Philip Hobbs, whose articles[105] make good reading for those interested in pursuing these techniques.

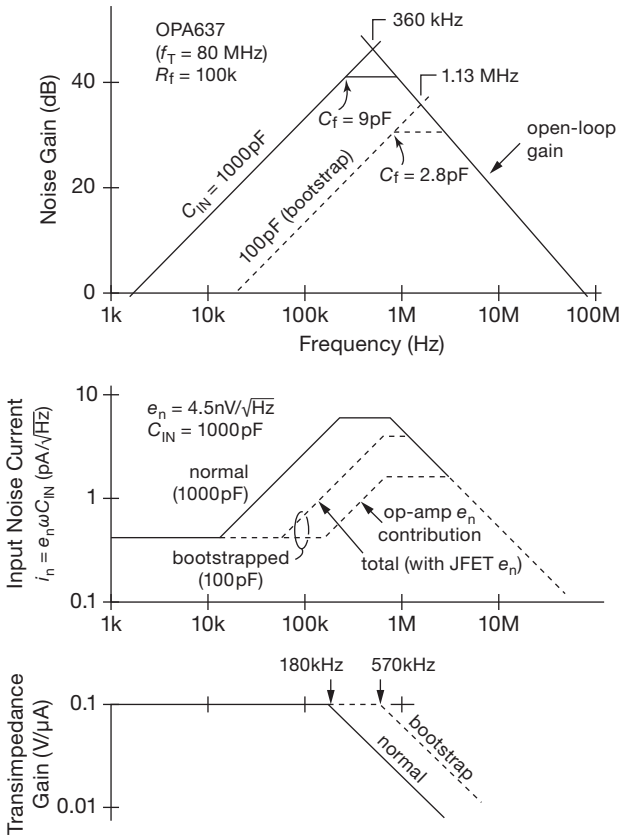


Figure 8.84. Noise, bandwidth, and gain for the bootstrapped photodiode amplifier. The reduced effective capacitance improves both the noise (via reduced e_nC) and bandwidth (via smaller stabilizing feedback capacitance C_f).

A. Creating a capacitance-isolating cascode

Step 1: unbiased cascode

Figure 8.86A shows the core idea, in which the current-sinking signal source drives the emitter of grounded-base

[104] Here the LT1792 costs $4.85 versus $18 for the OPA637 or $12 for the similar ADA4637, and the BF862 costs just $0.67. You often can do even better, with bootstrap-assisted op-amps of adequate performance down in the $2 territory.

[105] For a good first reading assignment, try his "Photodiode front ends – the REAL story," *Opt. Photon. News*, **12**, 42–45 (April 2001).

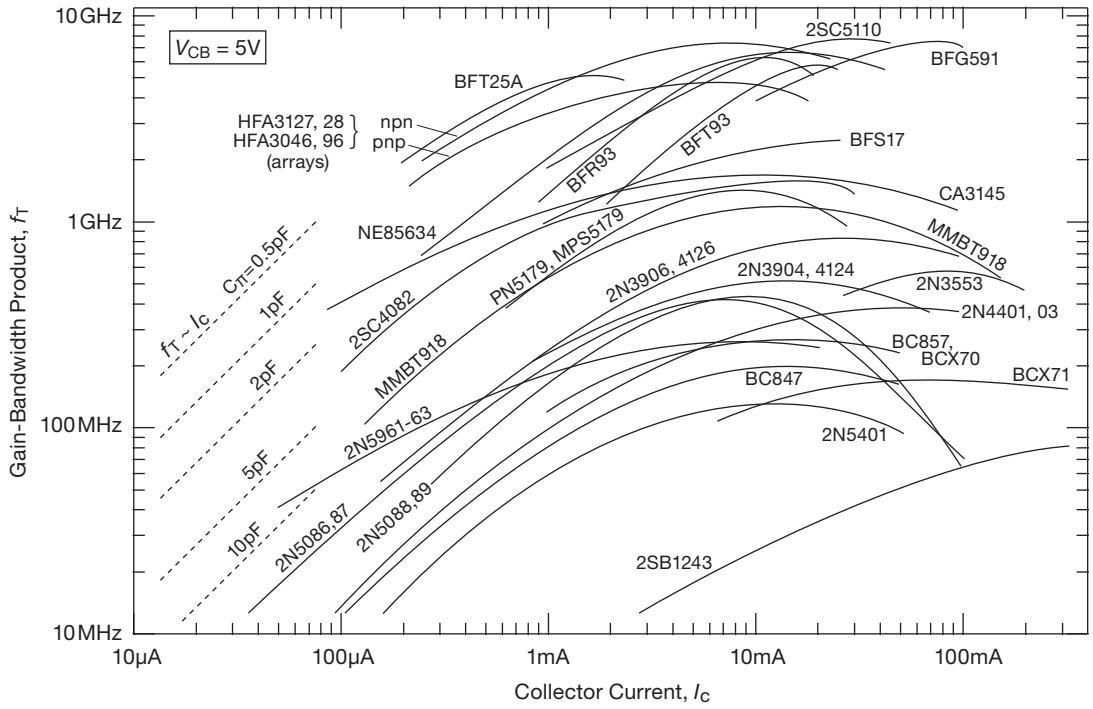


Figure 8.85. Gain-bandwidth product, f_T, versus collector current for selected bipolar transistors, as shown in the manufacturers' datasheets. Some wideband transistors are clear off the top.

npn transistor Q_1 (this is often called a "common-base amplifier"). Assuming reasonable transistor beta, most of that current appears at the collector – but with only the small collector output capacitance of Q_1, typically just a few picofarads.

A minor flaw here is that the input sits a diode drop below ground. But there are more serious snakes lurking in the Garden of Eden. We want good performance at very low signal currents, say fractions of a microamp. Down there we have to worry about falling transistor beta. Worse still, the input impedance r_e seen at the emitter is rising inversely with input current; at 1 μA it's 25 kΩ (recall r_e=25 Ω/I_C[mA]). So the signal current is shunted to ground by the input capacitance at a frequency at which the reactance of C_{in}=25k or less; that is, the input rolls off at $f_{3dB} = 1/2\pi r_e C_{in}$, thus for example around 6.4 kHz for a 1μA signal with C_{in}=1000 pF.[106]

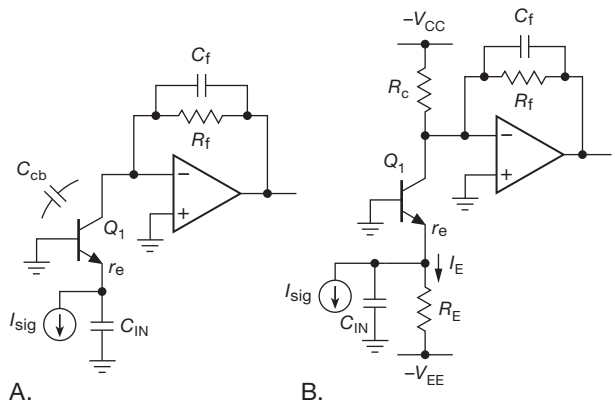


Figure 8.86. Isolating input capacitance with a common-base (cascode) input stage. A. Transistor Q_1 passes the signal current to the transimpedance amplifier, which sees only the small collector-to-base capacitance C_{cb}. B. Adding dc bias reduces r_e, greatly increasing the bandwidth.

[106] You also have to worry about the transistor's falling f_T at low currents; the reduced beta diverts emitter current into the base at high frequencies. For discussion and measurements see the section "BJT Bandwidth and f_T" in Chapter 2*x*, and see Figure 8.85. In the latter you can see that the 2N5089 has an (extrapolated) f_T down around 2 MHz at 10μA, from which you can estimate that about 10% of the emitter current is diverted at 200 kHz.

Step 2: biased cascode

We can drive r_e down (and also preserve reasonable beta) by biasing Q_1, as shown in Figure 8.86B. With ± 15 V supplies, we might choose R_C=R_E=60 kΩ to set I_C at 250 μA quiescent current; thus r_e=100 Ω. This raises the r_eC_{in} rolloff to a quite respectable 1.6 MHz.

But now we've created three new problems. First, this substantial quiescent collector current means that R_C has to be critically chosen so that Q_1's collector operating point (when disconnected from the virtual ground) would be at ground; otherwise the transimpedance stage (if configured for high gain, say 1 V/μA; i.e., R_f=1 MΩ) will have a large output dc offset. Second, in this circuit the noise currents of R_C, R_E, and R_f are all combined at the summing junction; and the required lower R_C and R_E resistor values generate correspondingly larger Johnson-noise currents at the transimpedance input (recall that a resistor of value R generates a short-circuit Johnson-noise current of $i_n = \sqrt{4kT/R}$). Third, Q_1's higher operating current means higher base current and thus higher shot noise ($i_{n,\text{base}} = \sqrt{2qI_B}$), which is another noise-current contribution at the transimpedance input.[107]

So here's the situation so far: with the circuit of Figure 8.86B we have achieved better bandwidth with a given input capacitance, compared with the previous transimpedance designs, but at the cost of (a) dc offset, (b) an input node that is a diode drop below ground, and (c) a noise–speed tradeoff in the choice of Q_1's quiescent current.

Can anything further be done? Read on…

Step 3: bootstrapped biased cascode

Yes! Combine the capacitance-reducing bootstrap trick with the biased cascode. Figure 8.87 shows how, this time as a fleshed-out design showing the specifics of component values and types.

To bring this into the real world, we chose a gallium phosphide UV photodiode that we use in the lab; its measured capacitance at 5 V of reverse bias is 460 pF (it is not specified on the datasheet). The photodiode's low side is bootstrapped by JFET follower Q_2 (as in the earlier Figure 8.82), reducing its effective capacitance tenfold, to ∼50 pF.

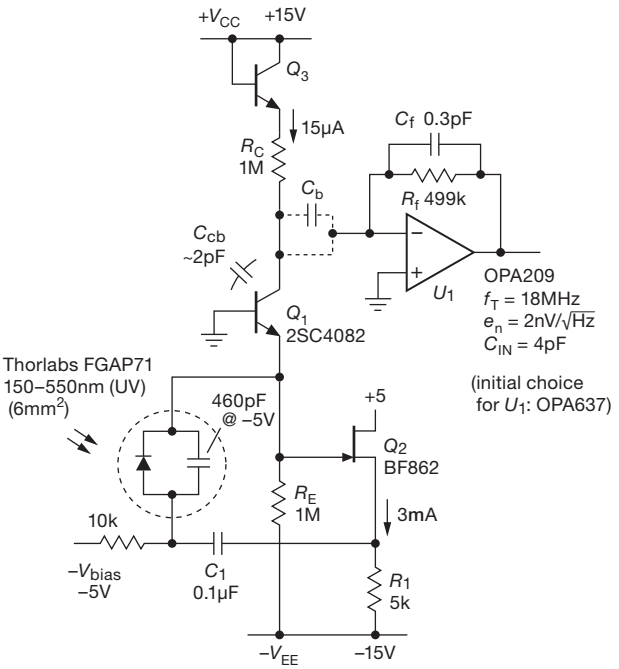


Figure 8.87. Bootstrapping the cascoded transimpedance amplifier to achieve wideband (1 MHz) low-noise performance with a photodiode of rather high capacitance. Optional blocking capacitor C_b can be used to eliminate dc offset in ac-coupled applications.

The next step is to choose the operating current for cascode transistor Q_1, such that its r_e is small enough so that the input r_eC_{in} rolloff[108] does not compromise the bandwidth f_c set by the transimpedance stage U_1. We figure f_{GM} (eq'n 8.46) as usual (see §*4x.3*): the capacitance at U_1's input is the sum of Q_1's collector capacitance (2 pF) and U_1's own input capacitance (15 pF for our initial choice of an OPA637, whose low noise and wide bandwidth make it a favorite choice as an input stage for photodiode amplifiers); the 17 pF total puts U_1's input rolloff f_{RfCin} of 18.7 kHz, given our choice[109] of transimpedance gain (0.5 V/μA). The OPA637's decompensated f_T of 80 MHz gives us an f_{GM}=$\sqrt{f_{RfCin}f_T}$=1.22 MHz; we choose a value of feedback capacitor C_f to set the critical bandwidth f_c to be 0.7 MHz (a conservative ζ=1.2, to ensure stability), hoping for a transimpedance bandwidth f_b of 1 MHz. That's 0.46 pF, of which the resistor itself provides ∼0.1 pF. You can use a pair of PCB traces, an adjustable pot with C_f=0.5pF (see Figure *4x.18*), or a "gimmick" (a few twists of a pair of insulated wires) to add the remaining 0.36 pF.

[107] There's a subtle point here: a current such as I_E that is generated by a voltage drop across a metallic conductor does not exhibit shot-noise current (the charge carriers do not act independently; see §8.1.2). But the base current *does* have the "full" shot noise given by the formula. That noise current manifests itself at the collector, because I_C=I_E(noiseless)−I_B(noisy).

[108] At high enough currents, transistor f_T is given by f_T=$1/2\pi r_eC_{in}$.

[109] Arrived at through an analogous process of speed–noise tradeoff, the details of which we will spare the reader.

B. Iteration: better op-amp choice

Something's amiss here: the OPA637 is a fine op-amp (with a price to match!), but it uses large-area JFETs to achieve its admirably low e_n, resulting in a high input capacitance of 15 pF. That's not a worry when you have, say, an input signal accompanied by lots of capacitance (e.g., a photodiode and some shielded cable). But here it's at a low capacitance node (Q_1's collector) where it dominates the rolloff, forcing the high f_T requirement.

Let's redo the calculation, assuming we use an op-amp with lower capacitance, say C_{in}=4 pF. Now f_{RfCin}=53 kHz, and for an f_b of 1 MHz with ζ=1 (so f_c=0.7 MHz) we need an op-amp with $f_T \geq 9$ MHz. Our effort is rewarded, for now there are many possibilities. We can choose the OPA209 from Table 8.3, with its f_T of 18 MHz and low cost of $2.27.[110] That gives us f_{GM}=0.98 MHz, so we can set $f_b = f_{GM}$=1 MHz, and with ζ=1 (critical damping) we have f_c=0.7 MHz and C_f=3 pF.

Knowing that the transimpedance stage has a usable bandwidth of 1 MHz, we next set the quiescent current of cascode transistor Q_1 high enough so that the rolloff of r_e, combined with the 50 pF (bootstrapped) effective photodiode capacitance, is at a somewhat higher frequency. Here an I_C of 15μA gives us an r_e (the impedance the signal sees at the emitter) of 1.7 kΩ and thus an input rolloff frequency of 1.9 MHz. The reduced C_{in} produced by the bootstrap allows us to run Q_1 at this low current, with the benefit that the Johnson noise current contributed by Q_1's 1 MΩ collector and emitter resistors is no larger than that of R_f. The base current shot noise is similarly reduced; here it contributes an input-referred noise current of 0.1 pA/$\sqrt{\text{Hz}}$ (to be compared with the 0.26 pA/$\sqrt{\text{Hz}}$ combined contribution of R_C, R_E, and R_f).

Some details:[111]

(a) Bootstrap follower Q_2 must drive the low side of the photodiode with an output impedance that is much lower than Q_1's input impedance (i.e., r_e, here 1.7 kΩ); that's satisfied here, where Q_2's high transconductance ensures a low output impedance of $Z_o = 1/g_m \sim 40\,\Omega$.[112]

(b) We chose first a 2N5089 for the cascode transistor Q_1, because of its high beta at low currents (β=400 min at I_C=100 μA) and low output capacitance (2 pF), both of which promote low noise. But then we consulted Figure 8.85 and realized that it had too little GBW: f_T is just 2 MHz at 15μA. So we looked next at an MMBT918 (2N918), with a healthy f_T=13 MHz at 15μA. Its base would rob about f/f_T, about 8% at f_c and thus an insignificant contributor to the circuit's 3 dB rolloff at 1 MHz. But its beta is anemic, less than 40 at 15μA. So we settled finally on the 2SC4082 shown in the figure: it has f_T=20 MHz at 10μA, and a pleasantly flat beta curve with β=90 at 100μA (and likely not much lower at 15μA).

(c) There's an additional noise voltage contribution, in the form of the transistor's input noise voltage ($e_n = \sqrt{4kT[r_{bb'} + 0.5r_e]}$) generating an "$e_n C$" noise current through the photodiode's effective input capacitance. These parameters don't look unfavorable for the 2SC4082.[113]

C. A final trick: "Regulated Cascode" (RGC)

Figure 8.88 shows a circuit configuration that's achieved considerable popularity in the world of photonics, where you're dealing with data rates of *giga*bits per second. Transistors Q_1 and Q_2 form a tight local feedback loop, with cascode transistor Q_1's base "regulated" by Q_2. (Q_1 and Q_2 can be replaced by MOSFETs, as indicated.) The regulated cascode improves on the simple cascode of Figure 8.86 in two important ways. (a) The circuit's input impedance (at Q_1's emitter) is reduced by a factor of Q_2's voltage gain, greatly increasing the $R_{in}C_{in}$–limited bandwidth; and (b) the input noise voltage is set by Q_2, rather than by Q_1, so the latter can be run at desirable low currents without the usual noise penalty.

To expand on this a bit, remember that input capacitance in the simplest transimpedance amplifier (Figure 8.70) is both a bandwidth killer (via R_fC_{in}) and a noise enhancer

[110] The OPA209 may even be *too good* for the job, with its e_n of 2.2 nV/$\sqrt{\text{Hz}}$. We require only that the op-amp's e_n be significantly less than that of Q_1 multiplied by the ratio of the reduced (bootstrapped) input capacitance to the capacitance at the op-amp's input node (here that works out to 7.5 nV/$\sqrt{\text{Hz}}$), and also that it be significantly less than the e_n of Q_2 multiplied by the ratio of the full input capacitance to the capacitance at the op-amp's input node (which usually works out to an even larger value of maximum e_n). In other words, we could have chosen an op-amp not intended for low-noise applications from the tables in Chapter *4x*.

[111] For a great read, see Phil Hobbs' book *Building Electro-Optical Systems, Making It All Work*, 2nd ed., Wiley (2009); a fine collection of tricks for designing cascode photodiode amplifiers, including series-peaking inductors and *T*-coils to extend the bandwidth, noise cancellers, and more.

[112] We usually add an emitter follower running at a few milliamps to further stiffen the guard signal, especially if it may be driving the external shield of a coax line where it is susceptible to pickup of AM radio stations, etc.; see Figure *4x.19*.

[113] The 2SC4082 has $r_{bb'}C_c$=6 ps, which (with C_{ob}=0.9 pF) implies an $r_{bb'}$ less than 10 Ω.

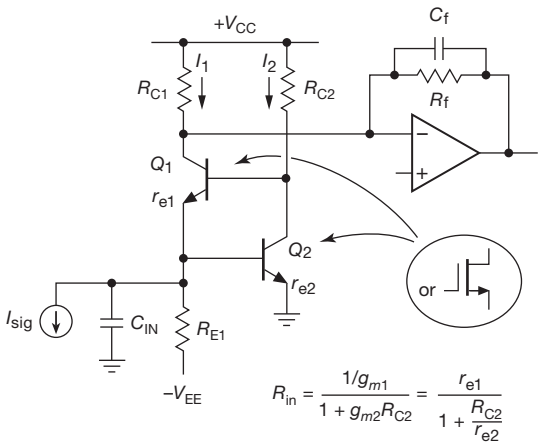


Figure 8.88. A "regulated cascode" input stage to transimpedance amplifier U_1 allows capacitance-isolating cascode Q_1 to operate at low current without a corresponding e_n penalty. It also reduces the impedance seen at the capacitive input (by a factor of G_{V2}), for enhanced bandwidth.

(via e_nC). We struggled with this, and the stability issues it presents, first by adding enough feedback compensation to ensure stability, then piling on op-amp speed to recover some bandwidth. Then we fiddled with circuit configurations to address directly the problem of input capacitance: first we bootstrapped the low side of the photodetector to *reduce* the effective input capacitance; then we added a cascode input stage to *isolate* the input capacitance, but that raised the circuit's input impedance (sacrificing bandwidth); so we biased the cascode to reduce R_{in}; and finally we bootstrapped the biased cascode.

All well enough. But the result was an input stage that had to run at significant collector current, generating a difficult-to-tame dc offset that seriously limits the amount of gain in the transimpedance stage. This is not a good situation, particularly if you want to sense small input currents.

The regulated cascode elegantly addresses these problems, by allowing the cascode stage (Q_1) to run at low current (so the circuit works well with small-signal currents), while circumventing the bandwidth and noise penalties (by driving down R_{in}, and by allowing Q_2 to replace Q_1's voltage noise). To minimize input *current* noise you should operate Q_1 at low collector current, but make sure you don't reduce the bandwidth excessively;[114] see the discussion in the "BJT bandwidth and f_T" section of Chapter *2x*. A variation of this configuration, in integrated form, is used in most contemporary fiber-optic receivers.[115]

8.11.11 Transimpedance amplifiers with capacitive feedback

There is a way to eliminate completely the Johnson-noise contribution of the gain-setting feedback resistor R_f, namely to eliminate the resistor itself. Feedback is then provided by the capacitor C_f alone, forming an integrator. The output signal must be differentiated to recover an output proportional to the input signal current; and of course both the integrator and differentiator must be reset (interrupting its operation) often enough to prevent saturation. To preserve the low noise of the input stage, the voltage noise of the differentiator op-amp must be significantly less than that of the input (integrating) op-amp.

Although capacitance feedback may seem an unusual approach, in fact, for ordinary TIAs operating at modestly high frequencies with high-value feedback resistors (say 100 MΩ and above), the self-capacitance of the resistor conceptually turns the amplifier into an integrator, with the resistor playing something akin to a reset role. Once one starts thinking along these lines, the idea of deliberately raising the feedback capacitance to ∼1 pF is no longer so scary. See §*4x.8* for an example circuit and discussion.

This technique is often used in "patch-clamp" amplifiers[116] and other low-level current detectors, for example, chilled germanium or silicon x-ray detectors (called IGX and Si(Li) respectively), in which the integrator is reset by an optical pulse from an LED (thus avoiding switch leakage effects).

A more common application of such an integrating transimpedance amplifier is for the readout of an imaging detector, where the quantity of interest is the total charge delivered during the brief readout, rather than its current-versus-time waveform. For this application you need to know only the *change* in integrator output voltage caused

[114] The worry, of course, is that Q_1 will have poor f_T at low collector current. But these designers know how to make impressively good microwave transistors in their ICs.

[115] See, for example, E. Säckinger and W. Guggenbühl, "A high-swing, high-impedance MOS cascode circuit," *IEEE J. Solid-State Circuits* **25**, 1 (1990); S. M. Park, "1.25-Gb/s regulated cascode CMOS transimpedance amplifier for gigabit ethernet applications," *IEEE J. Solid-State Circuits* **39**, 1 (2004); or Z. Lu et al., "Broad-band design techniques for transimpedance amplifiers," *IEEE Trans. Circuits Sys.* **54**, 3 (2007).

[116] For example, the Axon 200B patch-clamp amplifier is claimed to achieve an open-input noise current of just $0.2\,\text{fA}/\sqrt{\text{Hz}}$ at 150 Hz when operated in capacitance-feedback mode and thermoelectrically cooled; that's equivalent to the shot noise of 0.1 pA of leakage current.

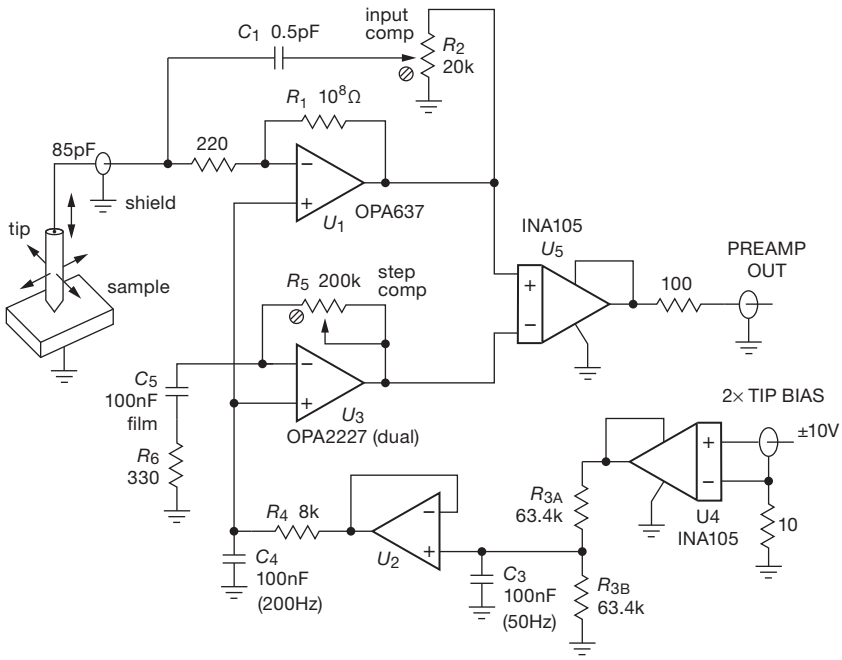


Figure 8.89. A low-noise transimpedance preamp for a scanning tunneling microscope (STM), where tip current signals are in the nanoamp range. The tip-to-sample voltage is set by the "tip-bias" input; the frequency response goes to ∼10 kHz, and the gain is 0.1 V/nA. Trimmer R_2 adjusts the effective feedback capacitance, to compensate the 100 pF (or more) of capacitance seen at the shielded input line emerging from the cryogenic vacuum chamber.

by the charge delivery. This technique is known as "correlated double-sampling," and dates back to the 1950s; see the footnote on page 571.

8.11.12 Scanning tunneling microscope preamplifier

Scanning tunneling microscopy (STM), dating to Binnig and Rohrer's work in the early 1980s,[117] allows one to image the surface topography (and other properties) of a sample at the atomic level. Figure 8.89 shows a sharp metallic tip hovering above a sample in a vacuum chamber, connected to a current-measuring preamp. When the tip is close to the surface (∼1 nm, about ten atom diameters) and biased to ∼1 V, a current of order 1 nA flows via quantum-mechanical "tunneling" through the potential barrier. The current is very sensitive to tip-to-sample spacing, varying exponentially such that the current typically increases by an order of magnitude when the spacing is reduced by 0.1 nm.[118]

[117] See for example their article "Scanning tunneling microscopy," *Helvetica Physica Acta* **55**, 726–735 (1982).

[118] See for example J. A. Golovchenko, "The tunneling microscope: a new look at the atomic world," *Science* **232**, 48–53 (1986).

To form a topographic image, the tip is held at a fixed voltage and scanned across the sample, with its spacing controlled by a piezo actuator such that the measured tip current is held constant. The piezo drive voltage is then a measure of variations in the surface height, with a vertical resolution better than an atomic diameter. If the tip's point is made sufficiently small (no more than a few atoms wide), the horizontal resolution is likewise of atomic scale. Figure 8.90 shows an example, an STM image of a silicon surface.

The STM preamp is a transimpedance amplifier, usually mounted at a flange on the vacuum chamber, where it receives the tip's tunnel current signal through a length of shielded line. That adds input capacitance, typically in the range of 50–200 pF. Because the preamp is within the tip spacing servo loop, it needs plenty of bandwidth, say 20 kHz, to permit rapid scanning, thus requiring careful loop compensation (see §*4x.3*). The preamp also needs to apply the tip-bias voltage.

In Figure 8.89 the transimpedance stage U_1 is an OPA637, an 80 MHz (decompensated) JFET op-amp with 16 pF of input capacitance and 4.5 $\text{nV}/\sqrt{\text{Hz}}$ input voltage noise (see Table 8.2 on page 516). That's rather a lot of

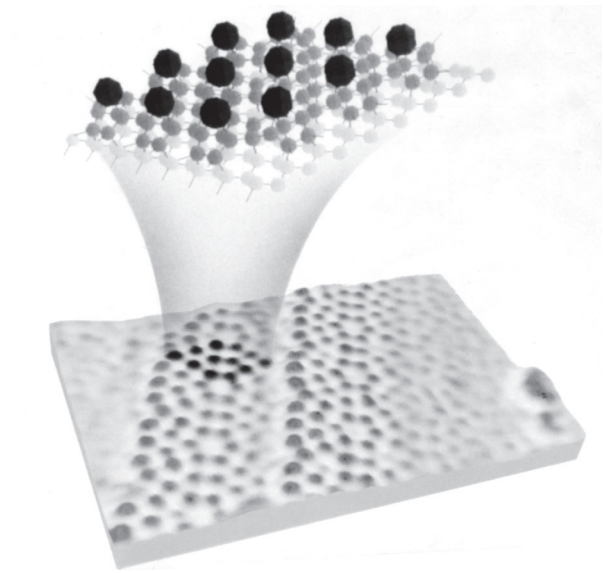


Figure 8.90. Structure of a 10×10 nm square region of a silicon crystal surface, imaged at the atomic level by an STM. The graphic model (expanded above) elucidates the observed surface structure (known as a *reconstruction*, because the surface atoms adopt a different arrangement compared with the bulk; here several atomic planes are seen, in a characteristic "7×7" reconstruction). This first image of an atomic step is adapted from the cover illustration accompanying the article referenced in Footnote 118 on page 553. (Courtesy J. Golovchenko)

input capacitance, but here it's insignificant compared with the cable capacitance, and in exchange we get a quiet wideband amplifier with low input current. Tip bias is set by the dc voltage at U_1's noninverting input; the op-amp's output is offset by the bias voltage, which we remove with difference amplifier U_5 (see §5.14 and Table 5.7).[119]

Exercise 8.5. Calculate the preamp's bandwidth for C_1=0.1 pF with C_{in}=100 pF, and also calculate the maximum allowed input capacitance. Show your stability criteria. Make a graph of the total effective input-noise-current spectral density, with curves for several values of C_{in}, including the maximum value (don't forget R_1's Johnson noise). Calculate the corresponding f_x breakpoint frequencies.

This exercise demonstrates that e_nC_{in} noise is a serious limitation in overall speed and performance. Note that it's not just the op-amp's e_n that matters here – it's combined with noise in the bias voltage applied to U_1's noninverting input. To keep the bias voltage quiet, we use a difference amplifier U_4 to isolate ground path noise in its computer-controlled DAC, and we add a pair of *RC* lowpass filters in the bias path through U_2; the filter at U_2's output is needed because otherwise the voltage noise from U_2 would make a significant contribution to that of U_1, as the next exercise demonstrates. Op-amp U_3 buffers the bias voltage into the 25k input impedance of U_5.

Exercise 8.6. Calculate and draw a graph of the allowed spectral noise for U_2 and for the computer's bias-voltage DAC signal (input to U_4), assuming that the noise voltage of the final bias applied to U_1's noninverting input is no more than 30% of that contributed by U_1 itself.

If the STM tip scanning is momentarily stopped, the bias voltage can be changed in a ramp or a series of steps, creating an *I–V* scan that can be used to determine additional information about the atom immediately under the tip. One approach is to repeatedly stop and make *I–V* scans. In this way the STM can provide not only an elevation map of the surface, but also a map of elemental composition. But changing the tip-bias voltage causes a pulse of current $i=C_{in}\,dV/dt$ that is seen and amplified by the TIA op-amp U_1. A nice way to deal with this effect is to add R_5 and C_5 such that $R_5C_5=R_1C_{in}$. This creates a cancelling pulse at the inverting input of U_5, allowing an accurate tunneling current to be measured before the voltage has completely settled. This greatly speeds up the full *I–V* scan.

This kind of circuitry is useful in other current-input applications, such as patch-clamp amplifiers in neurophysiology. It's basically a form of source-measure unit (SMU), a handy device we used to make the JFET transconductance measurements in §*3x.2* and the BJT beta plots in Figure 8.39.

8.11.13 Test fixture for compensation and calibration

To adjust the input compensation (R_2 in Figures 8.80, 8.89 and *4x.17*) you'd like a clean source of nanoamp-scale square waves supplying a calibrated current at the input connector; with the input cable in place you'd then adjust R_2 for best step response. You might imagine rigging up a 1 V square wave in series with a 1 GΩ resistor. The problem is that the resistor's parasitic shunt capacitance of ∼0.1 pF causes input current spikes at each transition of

[119] A higher-gain difference-amplifier could be used for U_5 (e.g., an INA106 with G=10), as long as the tunneling currents are no more than a few nanoamps. STM preamps are often operated at very low current levels, with $R_1=10^9\,\Omega$. The summing-junction node connections (both sides of the 220 Ω resistor, along with the op-amp's inverting input) are often placed on Teflon standoffs to prevent PCB leakage currents.

the square wave; it takes only 0.01 V/μs of input slewing to produce 1 nA current pulses.

Figure 8.91 shows two solutions to this problem. In circuit A a tunable input series *RC* (lowpass, or "pole") compensates the current-scaling resistor R_2's parallel parasitic capacitance C_p (highpass, or "zero"). For this simple model of parasitic capacitance, cancellation requires $R_1C_1=R_2C_p$. The optional isolated input connector suppresses ground path current spikes between the signal generator and the preamp and oscilloscope. Circuit B takes a different approach, using a small series capacitor (into the summing junction at the output) as a differentiator; $i=C_2dV_{in}/dt$ predicts a $\pm$1 nA output square wave for a 0.5 Vpp input triangle wave at 1 kHz. This circuit is the simpler, but its performance depends sensitively on the quality of the triangle wave at its turning points; we've gotten good results from Agilent 33120A (and later models) synthesized function generators. Your mileage may vary.

8.11.14 A final remark

We remind the reader that we've loosely referred to transimpedance amplifiers as "photodiode amplifiers"; that is, of course, just one application (though an important one) among many.

Our treatment in this chapter is concerned primarily with noise in transimpedance amplifiers, assuming familiarity with the basics. Be sure to visit §*4x.3* for an introduction to transimpedance amplifiers, a treatment of stability and bandwidth, and a healthy collection of important and useful tricks.

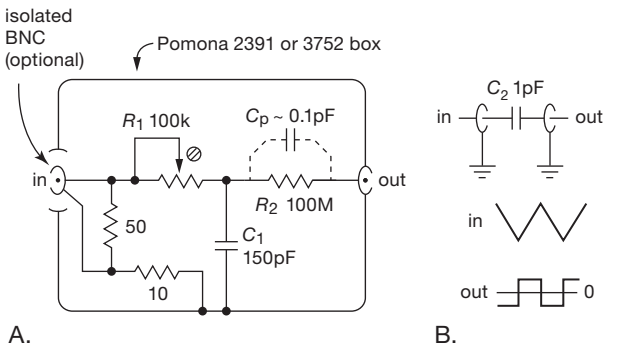


Figure 8.91. Test fixture for producing a nanoamp square-wave input for compensating and calibrating a transimpedance amplifier. A. Adjustable R_1C_1 (a "pole") cancels the peaking caused by R_2's shunt capacitance (a "zero"), so that a 0.1 V square wave creates a clean 1 nA output square wave of current into a summing junction. B. Capacitive differentiator converts a 500 mV (peak-to-peak) 1 kHz triangle voltage wave into a $\pm$1 nA square current wave at the output summing junction.

8.12 Noise measurements and noise sources

It is a relatively straightforward process to determine the equivalent noise voltage and current of an amplifier, and from these the noise figure and signal-to-noise ratio for any given signal source. That's all you ever need to know about the noise performance of an amplifier. Basically the process consists of putting known noise signals across the input, then measuring the output-noise signal amplitudes within a certain bandwidth. In some cases (e.g., a matched input-impedance device such as a radiofrequency amplifier) an oscillator of accurately known and controllable amplitude is substituted as the input signal source.

Later we discuss the techniques you need to do the output voltage measurement and bandwidth limiting. For now, let's assume you can make rms measurements of the output signal with a measurement bandwidth of your choice.

8.12.1 Measurement without a noise source

For an amplifier stage made from an FET or transistor and intended for use at low to moderate frequencies, the input impedance is likely to be very high. You want to know e_n and i_n so that you can predict the SNR with a signal source of arbitrary source impedance and signal level, as we discussed earlier. The procedure is simple.

First, determine the amplifier's voltage gain G_V by actual measurement with a signal in the frequency range of interest. The amplitude should be large enough to override amplifier noise, but not so large as to cause amplifier saturation.

Second, short the input and measure the rms noise output voltage, e_s. From this you get the input noise voltage per root hertz from

$$e_n = \frac{e_s}{G_V B^{\frac{1}{2}}} \quad \mathrm{V/Hz^{\frac{1}{2}}} \tag{8.47}$$

where B is the bandwidth of the measurement (see §8.13).

Third, put a resistor R across the input and measure the new rms noise output voltage, e_r. The resistor value should be large enough to add significant amounts of current noise, but not so large that the input impedance of the amplifier begins to dominate. (If this is impractical, you can leave the input open and use the amplifier's input impedance as R.) The output you measure is just

$$e_r^2 = [e_n^2 + 4kTR + (i_nR)^2]BG_V^2, \tag{8.48}$$

from which you can determine i_n to be

$$i_n = \frac{1}{R}\left[\frac{e_r^2}{BG_V^2} - (e_n^2 + 4kTR)\right]^{\frac{1}{2}}. \quad (8.49)$$

With some luck, only the first term in the square root will matter (i.e., if current noise dominates both amplifier voltage noise and source resistor Johnson noise).

Now you can determine the SNR for a signal V_s of source impedance R_s, namely,

$$\begin{aligned}\mathrm{SNR(dB)} &= 10\log_{10}\left(\frac{V_s^2}{v_n^2}\right)\\ &= 10\log_{10}\left[\frac{V_s^2}{[e_n^2 + (i_nR_s)^2 + 4kTR_s]B}\right], \quad (8.50)\end{aligned}$$

where the numerator is the signal voltage (presumed to lie within the bandwidth B) and the terms in the denominator are the amplifier noise voltage, amplifier noise current applied to R_s, and Johnson noise in R_s. Note that increasing the amplifier bandwidth beyond what is necessary to pass the signal V_s only decreases the final SNR. However, if V_s is broadband (e.g., a noise signal itself), the final SNR is independent of amplifier bandwidth. In many cases the noise will be dominated by one of the terms in the preceding equation.

8.12.2 An example: transistor-noise test circuit

For the measured noise data in Table 8.1a we used the circuit in Figure 8.92. It's basically an elaboration of the simple single-ended grounded-emitter amplifier of Figure 8.42, with provision for gain calibration, and sockets and test points for component substitution. The single-ended configuration requires a large input blocking capacitor. Follower Q_2 biases the device under test (DUT) to the nominal quiescent voltage point by way of R_8, with choice of R_C setting the quiescent current. Optional resistor R_B permits measurement of both β and i_n. The unusually long RC time constants throughout (capacitance multiplier, input and output blocking networks) are needed to suppress excess low frequency noise; the circuit takes many seconds to settle.

This circuit could be improved in several ways (but it was not, owing to operator exhaustion).[120]

8.12.3 Measurement with a noise source

The preceding technique of measuring the noise performance of an amplifier has the advantage that you don't need an accurate and adjustable noise source, but it requires an accurate voltmeter and filter, and it assumes that you know the gain versus frequency of the amplifier, with the actual source resistance applied. An alternative method of noise measurement involves applying broadband noise signals of known amplitude to the amplifier's input and observing the relative increase of output noise voltage. Although this technique requires an accurately calibrated noise source, it makes no assumptions about the properties of the amplifier, because it measures the noise properties right at the point of interest, at the input.

Again, it is relatively straightforward to make the requisite measurements. You connect the noise generator to the amplifier's input, making sure that its source impedance R_g equals the source impedance of the signal you ultimately plan to use with the amplifier. You first note the amplifier's output rms noise voltage, with the noise-source attenuated to zero output signal. Then you increase the noise source rms amplitude V_g until the amplifier's output rises 3 dB (a factor of 1.414 in rms output voltage). The amplifier's input noise voltage in the measurement bandwidth, for this source impedance, equals this value of added signal. The amplifier therefore has a noise figure

$$\mathrm{NF(dB)} = 10\log_{10}\left(\frac{V_g^2}{4kTR_g}\right). \quad (8.51)$$

From this you can figure out the SNR for a signal of any amplitude with this same source impedance, using eq'n 8.14:

$$\mathrm{SNR\,(dB)} = 10\log_{10}\left(\frac{V_s^2}{4kTR_s}\right) - \mathrm{NF}(R_s). \quad (8.52)$$

There are nice calibrated noise sources available, most

[120] Some details, for the curious: the circuit sets both I_C and V_{CE} for the DUT by means of a collector-to-base feedback loop. Lowpass filter R_5C_3 attenuates negative feedback that would unduly lower the transistor's common-emitter gain. The job of the large-value capacitor C_2 is to ground the base at all signal frequencies (as low as 4 Hz for our measurements) so we can measure $r_{bb'}$ (from its Johnson noise), and also the noise voltage created across $r_{bb'}$ by the low-frequency $1/f$ tail of base-current noise. But capacitor C_2, in combination with bias-setting pot R_8, adds a second lagging phase shift (a second pole) within the feedback loop, creating a low-frequency bias instability; confusingly, this mimics $1/f$ noise in the DUT at the low end of the spectrum. We could address this by introducing a compensating zero (e.g., a small resistor in series with C_3), but we were reluctant to tamper further with the circuit. The dominant pole is R_8C_2, and for low-frequency noise measurements down in the sub-nV/$\sqrt{\mathrm{Hz}}$ territory we found ourselves increasing C_2, to 0.15 F and even 0.35 F (!), and waiting for the modest instabilities to settle. This defeated our original goal, which was to avoid the large emitter bypass capacitors normally used with other non-feedback bias configurations – but, whatever, we were able to get reliable noise data for the plots and tables. The reader is welcome to experiment further.

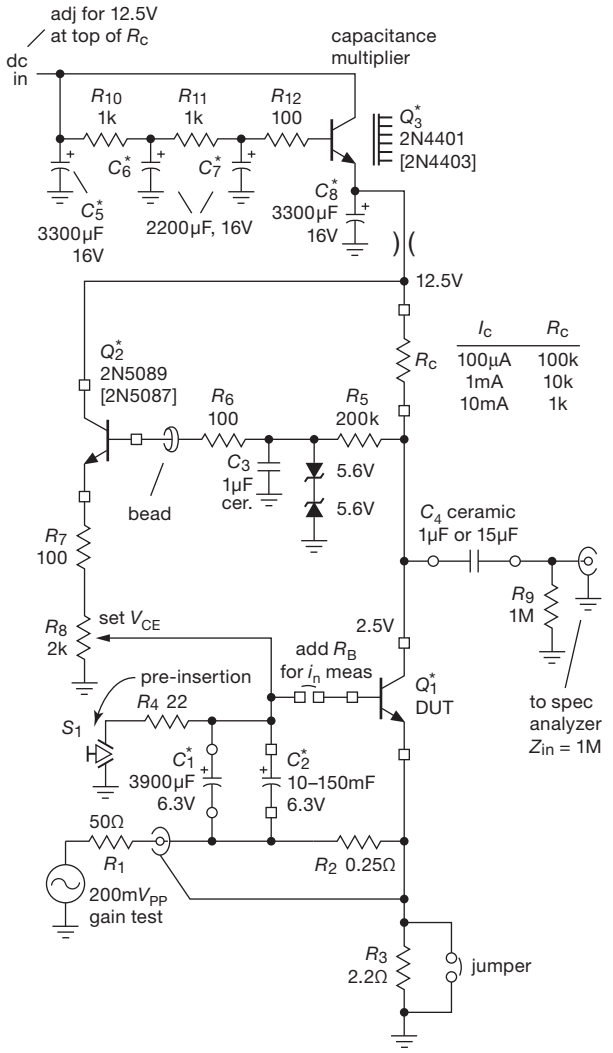


Figure 8.92. Transistor noise test circuit. Q_3 and associated components form a noise-eliminating capacitance multiplier. Polarities shown are for an *npn* device under test (DUT); for *pnp* reverse the polarity of components marked with asterisks. The gain-calibrating input signal is replaced by a 0 Ω shorting plug during noise measurement.

of which provide means for attenuation to precise levels in the microvolt range. Note: once again, the preceding formulas assume $R_{in} \gg R_s$. If, on the other hand, the noise-figure measurement is made with a *matched* signal source, i.e., if $R_s = Z_{in}$, then omit the factors "4" in the preceding expressions.

Note that this technique does not tell you e_n and i_n directly, just the appropriate combination for a source of impedance equal to the driving impedance you used in the measurement. Of course, by making several such measurements with different noise source impedances, you could infer the values of e_n and i_n.

A nice variation on this technique is to use resistor Johnson noise as the "noise source." This is a favorite technique used by designers of very low noise radiofrequency amplifiers (in which, incidentally, the signal source impedance is usually 50 Ω and matches the amplifier's input impedance). It is usually done the following way: a dewar of liquid nitrogen holds a 50 Ω "termination" (a fancy name for a well-designed resistor that has negligible inductance or capacitance) at the temperature of boiling nitrogen, 77°K; a second 50 Ω termination is kept at room temperature. The amplifier's input is connected alternately to the two resistors (usually with a high-quality coax relay), and the output noise power (at some center frequency, with some measurement bandwidth) is measured with an RF power meter. Call the results of the two measurements P_C and P_H, the output noise power corresponding to cold and hot source resistors, respectively. It is then easy to show that the amplifier's noise temperature, at the frequency of the measurement, is just

$$T_n = \frac{T_H - YT_C}{Y - 1} \quad \text{Kelvin}, \tag{8.53}$$

where $Y = P_H/P_C$, the ratio of noise powers. The noise figure is then given by eq'n 8.16, namely,

$$\text{NF(dB)} = 10\log_{10}\left(\frac{T_n}{290} + 1\right). \tag{8.54}$$

Exercise 8.7. Derive the foregoing expression for noise temperature. *Hint*: begin by noting that $P_H = \alpha(T_n + T_H)$ and $P_C = \alpha(T_n + T_C)$, where α is a constant that will shortly disappear. Then note that the noise contribution of the amplifier, stated as a noise temperature, *adds* to the noise temperature of the source resistor. Take it from there.

Exercise 8.8. Amplifier noise temperature (or noise figure) depends on the value of signal source impedance, R_s. Show that an amplifier characterized by e_n and i_n (as in Figure 8.28) has minimum noise temperature for a source impedance $R_s = e_n/i_n$. Then show that the noise temperature, for that value of R_s, is given by $T_n = e_n i_n/2k$.

If you don't want to bother with liquid nitrogen and are interested only in relatively low-frequency amplifiers, you can exploit the curious fact that a BJT's input noise voltage (at low currents, where $r_{bb'}$ effects are insignificant) is equal to the Johnson noise of a real resistor of value $r_e/2$. For example, if you ground the base, tie the collector to +5 V, and pull down the emitter through 10k to −5 V, you'll see at the emitter a noise signal with 50 Ω source impedance and 150K noise temperature. Add a blocking

capacitor, and alternate this noise source with a real 50 Ω resistor (use a coax relay, not a CMOS switch!), and you've got a simple (and seriously inexpensive!) two-temperature noise calibrator.[121]

A. Amplifiers with matched input impedance

This last technique is ideal for noise measurements of amplifiers designed for a matched signal-source impedance. The most common examples are in radiofrequency amplifiers or receivers, usually meant to be driven with a signal source impedance of 50 Ω, and that themselves have an input impedance of 50 Ω. See Appendix H for an explanation of this departure from our usual criterion that a signal source should have a small source impedance compared with the load it drives. In this situation, e_n and i_n are irrelevant as separate quantities; what matters is the overall noise figure (with matched source) or some specification of SNR with a matched signal source of specified amplitude.

Sometimes the noise performance is explicitly stated in terms of the *narrowband* input signal amplitude required for obtaining a certain output SNR. A typical radiofrequency receiver might specify a 10 dB SNR with a 0.25 μV rms input signal and 2 kHz receiver bandwidth. In this case the procedure consists of measuring the rms receiver output with the input driven by a matched sinewave source initially attenuated to zero, then increasing the (sinewave) input signal until the rms output rises 10 dB, in both cases with the receiver bandwidth set to 2 kHz. It is important to use a meter that reads true rms voltages for a measurement where noise and signal are combined (more about this later). Note that radiofrequency noise measurements often involve output signals that are in the audiofrequency range.

8.12.4 Noise and signal sources

Broadband noise can be generated from the effects we discussed earlier, namely Johnson noise and shot noise. The shot noise in a vacuum diode is a classic source of broadband noise that is especially useful because the noise voltage can be predicted exactly; zener diode noise is also widely used in noise sources, as are gas discharge tubes. These extend from dc to very high frequencies, making them useful in audiofrequency and radiofrequency measurements. Look ahead to Figure 13.121 for an example of a "true random" noise generator.

Versatile signal sources are available with precisely controlled output amplitude (down to the microvolt range and below) over frequencies from a fraction of a hertz to many gigahertz, programmable via GPIB, USB, or LAN. An example is the Agilent model E8257D synthesized signal generator, with output frequencies from 0.25 MHz to 20 GHz, calibrated amplitudes from 40 nV to 1 V rms, modulation and frequency sweeping modes, handsome digital display and bus interface, and nifty accessories that extend the frequency range to 500 GHz. This is a bit more than you usually need to do the job. And at +45 dB$ it likely costs a bit more than you're interested in paying. For a tenth the price (and a hundredth the bandwidth) you can get one of Tektronix's "Arbitrary Function Generators," for example the 2-channel AFG3102C (about $6k), which provide a Gaussian noise output in addition to standard waveforms (sine, square, pulse, triangle, etc.) and arbitrary (programmed) waveforms. And if it's only a *noise* waveform you want, you can get broadband noise sources like the Noisecom NC346 (10 MHz–18 GHz).

Some versatile noise sources can generate pink noise as well as white noise. Pink noise has equal noise power per *octave*, rather than equal power per hertz. Its power density (power per hertz) drops off at 3 dB/octave (though to the ear it sounds more like spectrally flat random noise than does white noise). It is widely used as a source signal for audio systems; in a typical application you would equalize a speaker system, driven with a pink noise source, by taking measurements with a handheld real-time spectrum analyzer (e.g., the Smaart® acoustic software tools). Because an *RC* filter's attenuation falls at 6 dB/octave, a more complicated filter is necessary to generate a pink spectrum from a white-noise input. Read the next subsection to see how this is done.

A. Pseudorandom noise source

We can make an interesting noise source by using digital techniques, in particular by connecting long shift registers with their input derived from a modulo-2 addition of several of the later bits (see §§11.3.1 and 13.14). The resultant output is a pseudorandom sequence of 1s and 0s that, after lowpass filtering, generates an analog signal of white spectrum up to the lowpass filter's breakpoint, which must be well below the frequency at which the register is shifted. These things can be run at very high frequencies, generating noise up to many megahertz or more. The "noise" has the interesting property that it repeats itself exactly after a time interval that depends on the register length (an n-bit maximal-length register goes through $2^n - 1$ states before repeating). Without much difficulty, that time can be made to be very long (years, or millennia), although most often a period of a second is long enough. For example, a 50-bit

[121] We learned this trick from Phil Hobbs, who suggests the use of a diode like an MBD301 (which doesn't suffer from high-level injection at 500 μA) instead of a BJT.

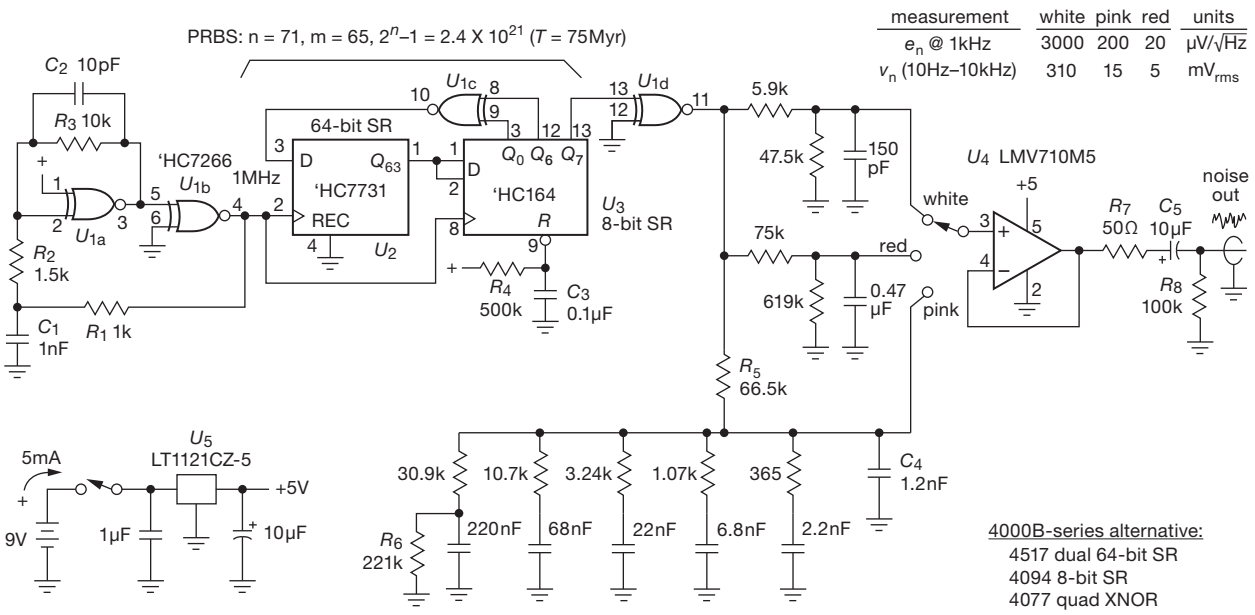


Figure 8.93. Pseudorandom noise source providing three colors of analog noise from 10 Hz to 100 kHz, with measured values of output noise density (at 1 kHz) and band-limited noise voltage (4-pole Butterworth, 10 Hz–10 kHz).

register shifted at 10 MHz will generate white noise up to 1 MHz or so, with a repeat time of 3.6 years.

The circuit shown in Figure 8.93 uses a 71-bit maximal-length shift register, clocked at 1 MHz, to generate a digital pseudorandom waveform that is spectrally flat (±0.07 dB) out to 100 kHz. The pseudorandom bit sequence is ridiculously long – with a 1 MHz clock it repeats in about 75 million years. Creating either white or red noise is easy: the "white" selection simply filters the raw 2-level waveform through an *RC* lowpass at 200 kHz, above the band of interest, to suppress the (wideband) clocking edges. The "red" selection inserts instead a lowpass filter at 5 Hz, below the band of interest, so the output rolls off at the usual 6 dB/octave.

Pink noise is trickier; you need a filter that reduces the white-noise amplitude by a factor of $1/\sqrt{2}$ (rather than 1/2) for every doubling of frequency. The usual analog method is to use a parallel set of series *RC* sections (as in Figure 8.93), with the characteristic frequency of each successive section increasing by a fixed ratio (here ×10, i.e., one decade) with an impedance that decreases by the square root of that same ratio (here $\sqrt{10}$). Even with such generous decade spacing you do remarkably well, as seen in the SPICE results of Figure 8.94, where the deviation from ideal −3 dB/octave behavior is just ±0.25 dB over the 5-decade frequency range of 10 Hz to 1 MHz.[122]

Simulations are fun, but the "rubber hits the road" when you actually build something and measure its performance. We did that, and Figure 8.95 shows the measured spectra of the circuit of Figure 8.93, which, to within the thickness of the (wiggly) lines, performs as advertised.

A few notes on the circuit

We cannot resist a few comments on the circuit design. As always, there are several choices at each step: a good design balances issues of performance, cost, complexity, availability of components, power, reliability, and (dare we say it?) *elegance*. For example, the ∼1 MHz clocking signal could be supplied from a $2 crystal-oscillator module; but for this application it needn't be accurate or stable, and so we chose to use two unused sections of the quad XNOR to make a simple *RC* relaxation oscillator. (We haven't seen this particular implementation used elsewhere, but it is completely straightforward: it is the topology of Figure 7.5, with U_{1a} configured as a noninverting Schmitt buffer, followed by the U_{1b} inverter. The hysteresis is set by R_2R_3, to 0.65 V, with small speed-up capacitor C_2. The frequency is set by R_1C_1.)

[122] A bit of candor here: to achieve this performance we did a bit of fine tuning to arrive at the circuit's choice of 1% resistor values.

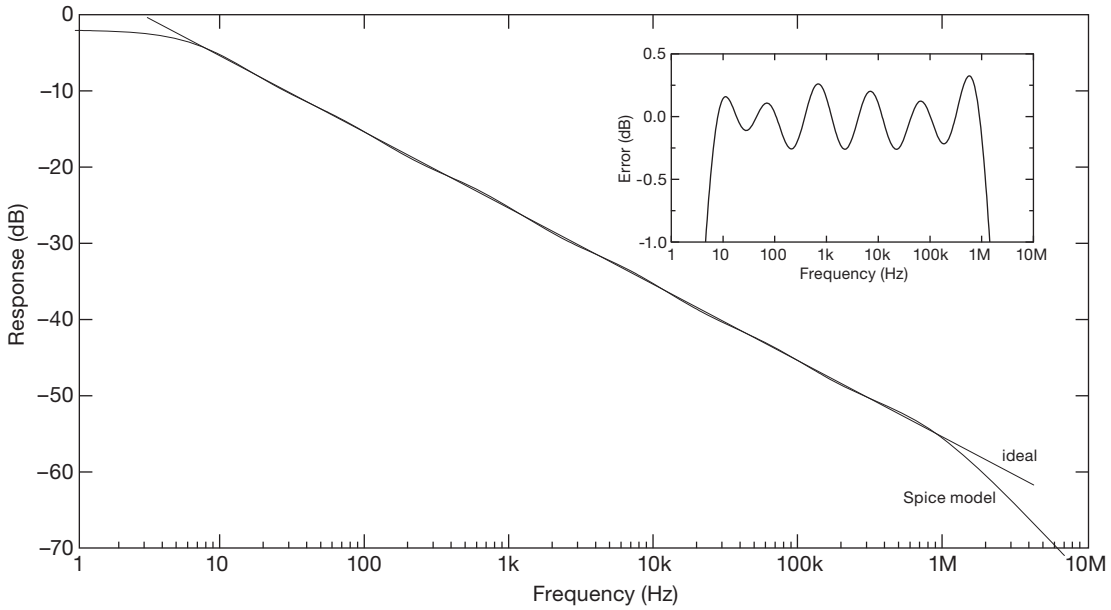


Figure 8.94. Simulation in SPICE of the pink-noise filter (R_5 and components below) in Figure 8.93.

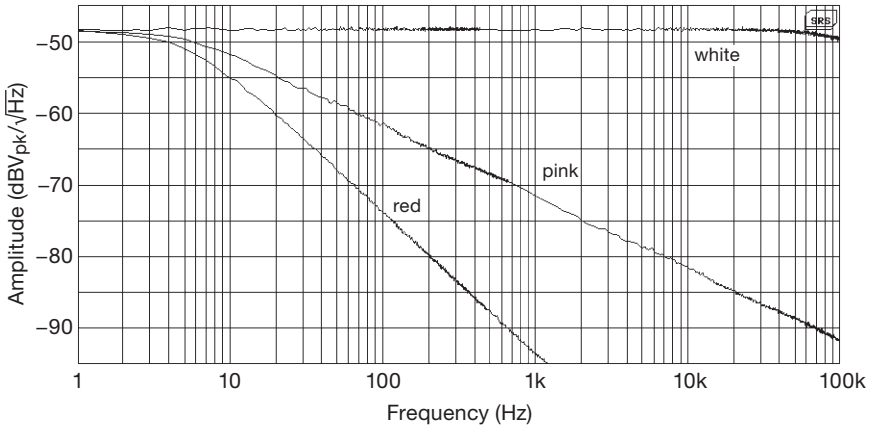


Figure 8.95. Measured spectra of the circuit of Figure 8.93. Each plotted spectrum is made from two 800-point FFT spectra to span the full 100,000:1 frequency range. The measured total noise voltage v_n and noise densities e_n are listed in Figure 8.93.

We chose the 74HC logic family because parts are readily available, easy to prototype (through-hole DIP), plenty fast, and of low power.[123] There are standard logic functions for what we needed, so we chose that path (rather than the alternatives: a cPLD, FPGA, or microcontroller). Available shift registers were either too small (8 stages), or outrageously large (the 'HC7731 we used is 256 stages, as four separate 64-stage shift-register banks). We went with the latter, using only one bank, and appending a 'HC164 8-stage parallel-out shift register so we could get at the needed taps (here m=65 and n=71; see Table 13.14). Note the use of exclusive-NOR (rather than exclusive-OR), to prevent the stuck state of all zeros and thus guarantee startup.

The output buffer is a nice BiCMOS rail-to-rail input/

[123] We could have used instead the "high-voltage" 4000B CMOS parts listed on the diagram, running directly from the 9 V battery. But there would be tradeoffs. (a) We would want to clamp the output to a stable amplitude, requiring some extra components; and (b) we would be clocking at a frequency uncomfortably close to the specified upper limit at low battery, foreclosing on any flexibility for going to a faster clock, say 10 MHz.

output op-amp, with 5 MHz GBW, low input current (4 pA, typ), plenty of output drive capability (±20 mA), and modest power consumption (1.2 mA); it's noisy ($20\,\text{nV}/\sqrt{\text{Hz}}$, with a high $1/f$ corner frequency), but that hardly matters when you're amplifying noise anyway. The series resistor R_7 ensures stability into capacitive loads and (if you care) provides source termination into 50 Ω cable.

For the 5 V regulator (U_5) we initially reached for something from the legacy LP2950 or LM2931 series of low-dropout linear regulators. But the LP2950 is intolerant of reversed input polarity (a feat easy enough to accomplish when replacing a 9 V battery); and both regulators are somewhat fussy when it comes to the output capacitor. In particular, they require some minimum amount of equivalent series resistance (ESR),[124] which is worrisome – how is one expected to deal with the low-ESR ceramic bypass capacitors that litter the rest of the circuit? Happily, there are nice LDOs that do not burden the designer with such worries: the LT1121-5 we chose is stable with output capacitors of zero ESR, and it's tolerant of reverse input (to −30 V); it also has the usual overcurrent and overtemperature protection.

There's more extensive discussion of pseudorandom noise, along with another pseudorandom noise source design (Figure 13.119), in §13.14.

8.13 Bandwidth limiting and rms voltage measurement

8.13.1 Limiting the bandwidth

All the measurements we have been talking about assume that you are looking at the noise output only in a limited frequency band. In a few cases the amplifier may have provision for this, making your job easier. If not, you have to hang some sort of filter on the amplifier output before measuring the output noise voltage.

***RC* filter** The easiest thing to use is a simple *RC* lowpass (or bandpass) filter, with 3 dB point(s) set at roughly

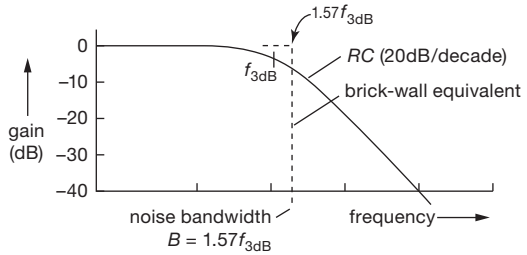


Figure 8.96. Equivalent brick-wall noise bandwidth for *RC* lowpass filter.

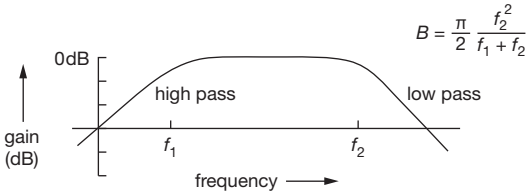


Figure 8.97. Equivalent brick-wall noise bandwidth for *RC* bandpass filter. For the case $f_1 = f_2$ the midband gain is −6 dB.

the bandwidth you want. For accurate noise measurements, you need to know the "equivalent noise bandwidth" (ENBW), i.e., the width of a perfect "brick-wall" filter that lets through the same noise voltage (Figure 8.96). This noise bandwidth is what should be used for B in all the preceding formulas. It is not terribly difficult to do the mathematics, and you find, for an *RC* lowpass filter,

$$B = \frac{\pi}{2} f_{3\text{dB}} = 1.57 f_{3\text{dB}}. \tag{8.55}$$

For a pair of cascaded lowpass *RC* sections (buffered so they don't load each other), the magic formula becomes $B = 1.22 f_{3\text{dB}}$. For the Butterworth lowpass filters discussed in §§6.2.6 and 6.3.2, the noise bandwidth is

$$\begin{aligned} B &= 1.57 f_{3\text{dB}} = \tfrac{1}{4RC} \quad (1\text{ pole}),\\ B &= 1.11 f_{3\text{dB}} \approx \tfrac{1}{5.6RC} \quad (2\text{ poles}),\\ B &= 1.05 f_{3\text{dB}} \approx \tfrac{1}{6RC} \quad (3\text{ poles}),\\ B &= 1.025 f_{3\text{dB}} \approx \tfrac{1}{6.1RC} \quad (4\text{ poles}). \end{aligned}$$

If you want to make band-limited measurements up at some center frequency, you can use a pair of *RC* filters (Figure 8.97), in which case the noise bandwidth is as indicated. You may wish to use higher-order Butterworth filters for more precise bandpass characteristics. In that case you'll need to know the corresponding equivalent noise bandwidths; don't panic – they're listed in Table 8.4 on page 564.

***RLC* filter** Another way to make a bandpass filter for noise measurements is to use an *RLC* circuit. This is

[124] From the LP2950 datasheet: "Ceramic capacitors whose value is greater than 1000 pF should not be connected directly from the LP2951 output to ground. Ceramic capacitors typically have ESR values in the range of 5 to 10 mΩ, a value below the lower limit for stable operation (see curve Output Capacitor ESR Range). The reason for the lower ESR limit is that the loop compensation of the part relies on the ESR of the output capacitor to provide the zero that gives added phase lead. The ESR of ceramic capacitors is so low that this phase lead does not occur, significantly reducing phase margin. A ceramic output capacitor can be used if a series resistance is added (recommended value of resistance about 0.1 Ω to 2 Ω)."

better than a pair of cascaded highpass and lowpass *RC* filters if you want your measurement over a bandpass that is narrow compared with the center frequency (i.e., high Q). Figure 8.98 shows both parallel and series *RLC* circuits and their exact noise bandwidths. In both cases the resonant frequency is given by $f_0 = 1/2\pi\sqrt{LC}$. You might arrange the bandpass filter circuit as a parallel *RLC* collector (or drain) load, in which case you use the expression as given. Alternatively (recall Figure 1.107), you might interpose the filter as shown in Figure 8.99; for noise-bandwidth purposes the circuit is exactly equivalent to the parallel *RLC*, with $R = R_1 \| R_2$.

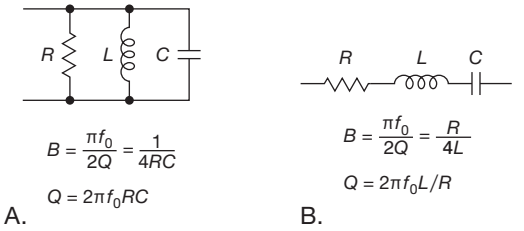


Figure 8.98. Equivalent *brick-wall* noise bandwidth for *RLC* bandpass filter. For the parallel circuit (A) the source signal is a current, and the output is the voltage across the terminals; for the series configuration (B) the input is a voltage applied across the circuit, and the output is the resulting current.

Averaging As we mentioned in §8.1.5, another way to accomplish lowpass filtering of a slow signal (a dc voltage, say, in the presence of additive white noise) is simply to average it over some time interval T; this is the same averaging operation as is done by an integrating ADC (e.g., in a digital voltmeter, see §13.8.3). In that case the equivalent noise bandwidth[125] is $B{=}1/2T$. Thus, for example, an averaging duration of 1 second admits a noise bandwidth of 0.5 Hz. This is a simple lowpass "filter" – but one that is less sharp even than a single-section *RC* (the latter falls off at 6 dB/octave, compared with 3 dB/octave for the time average).

[125] It is important to distinguish the operation of an averaging time T when measuring a static signal, as just described, from the use of a time "window" (again of duration T) to bound the measurement interval of an ac signal. In the latter case the use of such a *rectangular window* imposes a resolution bandwidth $B{=}1/T$ (i.e., twice that of the dc measurement). In digital signal processing the business of windowing plays an important role; see for example Harris, F. J., "On the use of windows for harmonic analysis with the discrete Fourier transform," *Proc. IEEE* **66** 51–83 (1978). We'll see this same theme, in connection with *synchronous signal averaging*, in §8.14, where synchronous (or "lock-in") detection of signals over some time duration T produces the same bandwidth reduction: $B{=}1/T$.

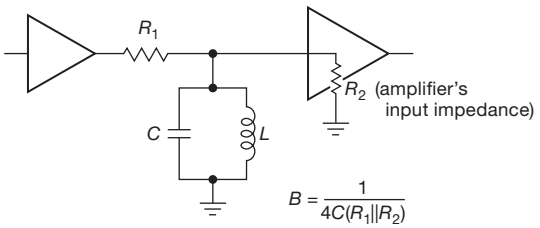


Figure 8.99. Interstage *RLC* bandpass filter.

Digital filter Digital signal processing (DSP) is an effective way to implement extremely well-defined filter functions, with their characteristics easily modified by changing the stored numerical coefficients. See §6.3.7 for more detail.

Frequency translation Imagine you want to make measurements of the "spot noise voltage" in some narrow bandwidth (say 10 Hz) centered at some relatively high frequency f_{in}, where the latter may be up at tens to hundreds of kilohertz, or perhaps even a megahertz or more. That is, the ratio $Q{=}f_{\text{in}}/\Delta f$ is very large, say greater than a thousand. Such a high-Q bandpass filter is pretty difficult to implement! But there's a good way to do the measurement without filter heroics.

The trick is to shift (translate) the frequency band of interest down to a much lower frequency, where it's easy to make a narrowband filter. The technique is called *heterodyning*, and it's a basic technique in most radiofrequency communications systems. It's easiest to understand if you think initially of a single input frequency. Figure 8.100 shows the basic scheme, in which an input signal voltage (at frequency f_{in}) is multiplied by a sinusoidal voltage (at frequency f_{LO}) from a *local oscillator* ("LO"), creating a pair of sinusoidal signals at frequencies $f_{\text{in}} \pm f_{\text{LO}}$.[126] The multiplier is called a *mixer*, and its output is filtered to eliminate one of the mixing products. Mixers can be in the form of an active circuit (a "4-quadrant multiplier"), or, for use at higher frequencies, a passive transformer-coupled diode arrangement called a *balanced mixer*.

In a communications application there may be several stages of frequency translation, passing through several intermediate-frequency ("IF") stages where amplification and filtering takes place.[127] For a simple noise-measuring

[126] From the identity $\cos x \cos y = \frac{1}{2}\cos(x+y) + \frac{1}{2}\cos(x-y)$, with $x{=}2\pi f_{\text{in}}$ and $y{=}2\pi f_{\text{LO}}$.

[127] For example, in an FM radio the LO is set 10.7 MHz below the desired station, with the 10.7 MHz IF signal amplified and demodulated. Radiofrequency and communications techniques are discussed in greater detail in Chapter 13 of this book's second edition.

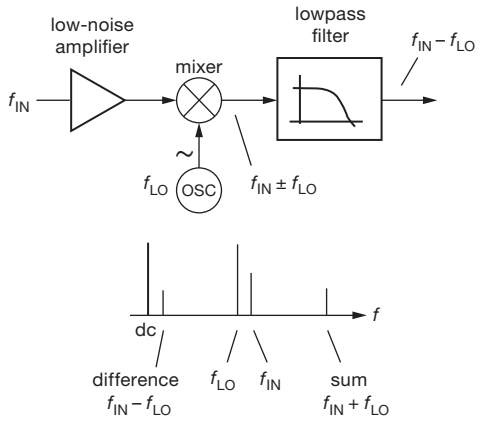


Figure 8.100. Heterodyne frequency shifting: the mixer produces sum and difference frequencies; only the latter survives the lowpass filter.

application it may be adequate to mix directly to "baseband," by tuning the LO to the frequency at which the spot noise voltage is to be measured. This is the same process used in *synchronous detection*, described later in §§8.14 and 13.13.6D. Commercial synchronous (lock-in) amplifiers let you make narrowband spot noise voltage measurements at any frequency in their range, extending to 100 kHz (for typical lock-in amplifiers) to as high as 200 MHz (e.g., for the SR844).

8.13.2 Calculating the integrated noise

Let's begin with the simple single-section *RC* lowpass band-limiting filter. From its equivalent brick-wall bandwidth (eq'n 8.55), the integrated noise-*voltage* output from filtering a white-noise input of noise *density* e_n is found by taking the square root of

$$v_n^2 = e_n^2 B = e_n^2 \frac{\pi}{2} f_{3dB} \quad \mathrm{V^2(rms)} \tag{8.56}$$

and analogously for noise *current*.

Things get more complicated if the noise density e_n depends on frequency, as it does, for example, with $1/f$ ("pink") noise. In such a case you must integrate, over frequency, the squared noise density $e_n^2(f)$ times the filter's spectral power bandpass characteristic. And it's common to use a *bandpass* filter, with both lower and upper frequency limits (call them f_1 and f_2). Then an ideal "brick-wall" (or should it be "squash court"?!) bandpass filter has unit response between f_1 and f_2, and a noise bandwidth of $B = f_2 - f_1$.

Brick walls are hard to implement in analog technology, and a simple expedient, as suggested earlier, is simply to use a cascaded pair of *RC*'s, with 3 dB frequencies of f_1 and f_2. With a bit of mathematical finesse you could demonstrate proficiency by performing the appropriate integrals, thus figuring the output noise voltage for the "standard noises" (white, or $1/f$, or even $1/f^2$). A shortcut is to exploit the brilliance of Wolfram's impressive *Mathematica* program. A colleague[128] did this for us, and Table 8.4 on the next page summarizes the results, for those three colors of noise, and for four styles of bandpass filters (brick-wall, simple *RC* (single-pole; that is, a pair of cascaded *RC* sections), 2-pole Butterworth, and *m*-pole Butterworth.

The expressions listed give you the integrated squared noise voltage v_n^2, for given lower and upper -3 dB frequency limits f_1 and f_2, and noise voltage density e_n (Figure 8.101); the rms noise voltage is then gotten by taking the square root: $V_{n(rms)} = \sqrt{v_n^2}$. For pink noise ($1/f$ in power) and red noise ($1/f^2$ in power) the noise density e_n is a function of frequency; for these expressions we have used for the overall multiplying factor the noise density at the high band end; that is, $e_n^2(f_2)$ (abbreviated as e_{n2}^2 in the table), which is the squared noise density at $f = f_2$, in units of V^2/Hz.

Note that you can set $f_1 = 0$, in the expressions for white noise, to get the dc-to-f_2 (lowpass-limited) noise voltage. If you do so, you'll get the expressions shown on page 561. This does not work for pink or red noise, though, because the integral diverges at zero frequency, conveyed in these expressions by the embarrassment of a zero denominator in the log's argument (pink) or, worse, in the result itself (red). This is why $1/f$ noise is typically measured with a 0.1 Hz to 10 Hz limited bandwidth, etc.; see Figures 8.102 and 8.103.

Noise *current* is treated just the same: substitute everywhere the noise current density i_n in place of e_n, to find the integrated noise current $I_{n(rms)}$.

A. The high end matters most

To calculate the integrated noise voltage over some bandpass, you need to integrate the noise power density (e_n^2) over frequency, taking account of the filter's passband response (call that $H(f) \equiv V_{out}/V_{in}$, and don't worry about phase); that is,

$$v_n^2 = \int_{f_1}^{f_2} e_n^2 H^2(f)\, df \quad \mathrm{V^2(rms)}; \tag{8.57}$$

and then you take the square root: $V_{n(rms)} = \sqrt{v_n^2}$.

So, looking at log–log plots of filter passbands like

[128] Jason Gallicchio, again!

Table 8.4 Noise Integrals[a]

	Noise Spectral Shape		
	white	pink	red
Filter Type	$(e_n = \text{const})$	$(e_n \sim 1/\sqrt{f})$	$(e_n \sim 1/f)$
brickwall	$e_n^2\ (f_2 - f_1)$	$e_{n2}^2\ f_2 \log_e \frac{f_2}{f_1}$	$e_{n2}^2\ \frac{f_2}{f_1}(f_2 - f_1)$
1-pole (*RC*)	$e_n^2\ \frac{\pi}{2}\ \frac{f_2^2}{f_1 + f_2}$	$e_{n2}^2\ \frac{f_2^3}{f_2^2 - f_1^2}\ \log_e \frac{f_2}{f_1}$	$e_{n2}^2\ \frac{\pi}{2}\ \frac{f_2^3}{f_1(f_1 + f_2)}$
2-pole Butterworth	$e_n^2\ \frac{\pi}{2\sqrt{2}}\ \frac{f_2^4}{(f_1 + f_2)(f_1^2 + f_2^2)}$	$e_{n2}^2\ \frac{f_2^5}{f_2^4 - f_1^4}\ \log_e \frac{f_2}{f_1}$	$e_{n2}^2\ \frac{\pi}{2\sqrt{2}}\ \frac{f_2^5}{f_1(f_1 + f_2)(f_1^2 + f_2^2)}$
m-pole Butterworth	$e_n^2\ \frac{\pi/2m}{\sin(\pi/2m)}\ \frac{f_2 - f_1}{1 - (f_1/f_2)^{2m}}$	$e_{n2}^2\ \frac{f_2}{1 - (f_1/f_2)^{2m}}\ \log_e \frac{f_2}{f_1}$	$e_{n2}^2\ \frac{\pi/2m}{\sin(\pi/2m)}\ \frac{f_2 - f_1}{1 - (f_1/f_2)^{2m}}$

Notes: (a) Values of the mean-squared noise voltage, v_n^2, in a frequency band from f_1 to f_2, as bandpass limited by the indicated filter type. The noise voltage density e_{n2} is the value of e_n at $f = f_2$.

those in Figure 8.101, you might at first think that a bandpass filter with symmetrical and sharp rolloffs at both low and high ends is needed. But – surprise – the v_n^2 integral weights the high end disproportionately, as you can see from Figure 8.102, where we've plotted the value $e_n^2 H^2(f)$ for white noise (e_n=1) over a 100:1 bandpass, on *linear* frequency and amplitude scales (because that's what integrals do for a living). The integral is the area under the curve, lazily accumulating lots of unwanted spectrum for a single-section *RC* at the high end, but acquiring some discipline with higher orders; by contrast, the filter order at the low end matters hardly at all. And this general behavior persists, even with a narrower 10:1 passband (Figure 8.103).

We've taken the simple case of white noise, with its uniform spectral density. But the situation is not much changed even when the noise density is rising at low frequencies (e.g., pink noise, with $e_n^2 \propto 1/f$): the combination of a response $H^2 \propto f^2$, integrated over a small frequency span at low frequency, kills off the modestly increasing noise density.

8.13.3 Op-amp "low-frequency noise" with asymmetric filter

Because "the high end matters most," the *low-frequency noise-voltage* specifications (0.1–10 Hz) reported on many op-amp datasheets are measured with an asymmetrical filter, often a single-section highpass at 0.1 Hz cascaded with a 2-section (or sometimes higher-order) lowpass at 10 Hz. Here are the equivalent noise bandwidths for bandpass filters that are first-order (*RC*) highpass at f_1 and second-order Butterworth lowpass at f_2; as before, for pink or red noise the multiplying factor is the noise density at the upper frequency limit f_2, i.e., "e_{n2}"$\equiv e_n(f_2)$:

$$
\begin{aligned}
v_n^2 &= e_n^2 \frac{\pi}{4} \frac{\sqrt{2} f_1^2 f_2^3 - 2 f_1 f_2^4 + \sqrt{2} f_2^5}{f_1^4 + f_2^4} \quad \text{(white)}, \\
v_n^2 &= e_{n2}^2 \frac{\pi f_1^2 f_2^3 + 4 f_2^5 \log_e(f_2/f_1)}{4(f_1^4 + f_2^4)} \quad \text{(pink)}, \\
v_n^2 &= e_{n2}^2 \frac{\pi}{4} \frac{\sqrt{2} f_1^3 f_2^3 - \sqrt{2} f_1 f_2^5 + 2 f_2^6}{f_1^5 + f_1 f_2^4} \quad \text{(red)}.
\end{aligned}
\tag{8.58}
$$

Op-amp datasheets usually specify a 0.1 Hz to 10 Hz bandwidth for their listed low-frequency noise voltage; as often as not, it is defined with an asymmetric filter. But, curiously, they tend to list a peak-to-peak value (rather than rms), taken from a 10-second 'scope capture (similar to the bottom trace in Figure 8.4). It's common to estimate the rms noise voltage with this rule of thumb: $v_{n(\text{rms})} \approx v_{n(\text{pp})}/6$.

A. Op-amp low-frequency noise voltage

Operational amplifiers (with the exception of auto-zeroing op-amps) exhibit the, by now, familiar noise-density characteristic: flat at higher frequencies (call it e_{nH}), but rising approximately as $e_n \propto 1/\sqrt{f}$ (pink noise) for frequencies below the $1/f$ noise corner (call that frequency f_c). If you know f_c and e_{nH}, you can use the expressions in Table 8.4 to estimate the integrated noise voltage over any span of bandpass f_1 to f_2.

There are three possibilities. (a) The bandpass is entirely

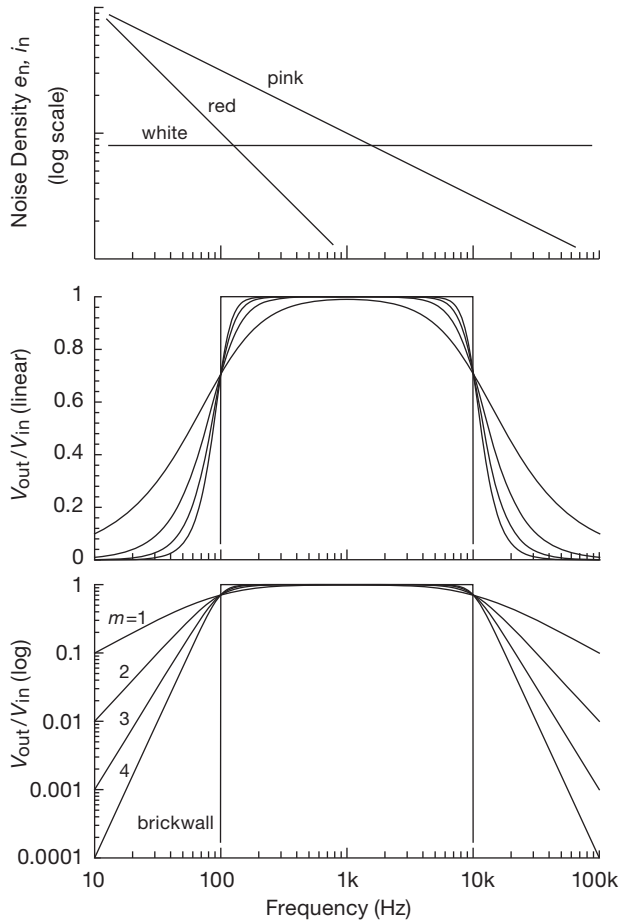


Figure 8.101. Spectral shapes of the three classic noises, and the bandpass filters used to evaluate their integrated noise voltage or current.

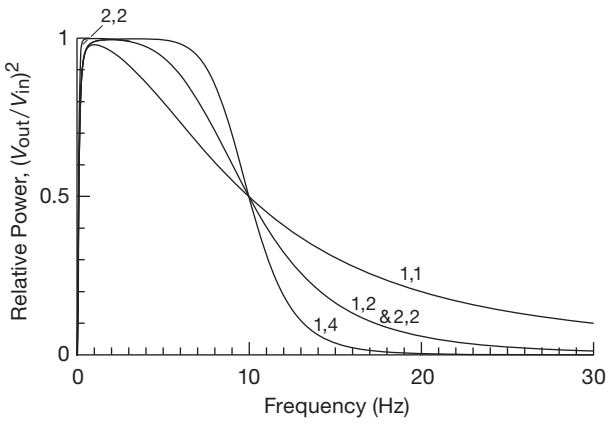


Figure 8.102. Power spectrum of filtered white noise, for 0.1–10 Hz Butterworth bandpass filters of the indicated orders (at their "low,high" ends; thus a "1,2" bandpass filter consists of a first-order highpass RC at f_1 cascaded with a second-order Butterworth lowpass at f_2).

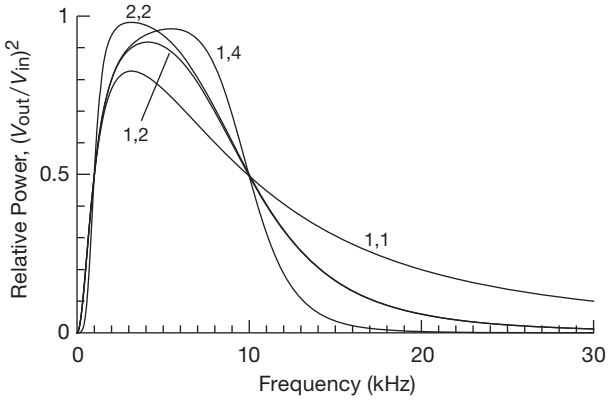


Figure 8.103. Power spectrum of filtered white noise, for 1 kHz–10 kHz Butterworth bandpass filters of the indicated orders ("low,high" ends).

in the white noise region, i.e., $f_1 > f_c$; (b) the bandpass is entirely in the pink noise region, i.e., $f_2 < f_c$; or (c) the bandpass straddles the $1/f$ corner frequency. For the first two cases, use the corresponding expression from Table 8.4 corresponding to the filter characteristic in use. For case (c), just calculate the separate white and pink v_n^2 contributions over the full passband (f_1 to f_2), and take their sum. In the idealized case of a brick-wall bandpass filter, this process gives an integrated noise voltage of

$$v_n^2 = e_{nH}^2 \left(f_2 - f_1 + f_c \log_e \frac{f_2}{f_1} \right) \quad \mathrm{V}^2(\mathrm{rms}). \tag{8.59}$$

This is what we've done to create the integrated noise curves in Figure 5.54 (in §5.11.1), based on the datasheet noise densities plotted in Figure 5.37 (in §5.10.6) and Figure 8.63 (from the data plotted in Figures 8.60 and 8.61).[129] Figure 8.104 shows an example, using the AD8671's datasheet e_n curve ($1/f$ corner at 5 Hz) to find the integrated-noise voltage v_n as a function of upper cutoff frequency (it's necessary to choose a nonzero low-frequency limit f_1 to prevent divergence).

[129] A spreadsheet program provides a handy way to make such calculations, and to plot the results.

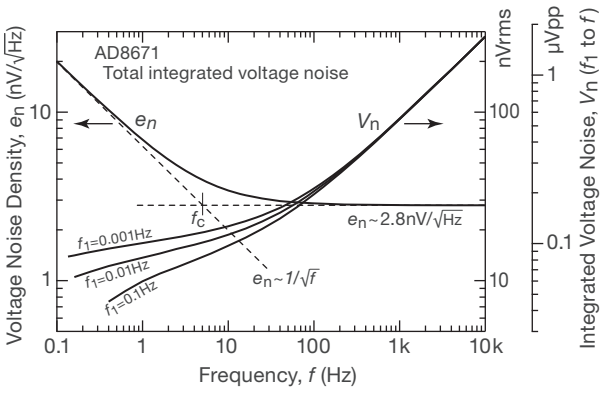


Figure 8.104. This op-amp's voltage noise density e_n has its $1/f$ corner frequency f_c at 5 Hz. Integrating the noise power ($\sim e_n^2$) from a low frequency f_1 up to a cutoff frequency $f_2{=}f$ gives the squared integrated noise voltage v_n^2, from which these curves of v_n result. If f_1 were set to zero, the v_n integral would diverge.

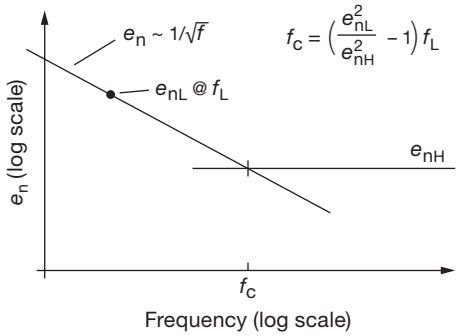


Figure 8.105. For the textbook noise density ($1/f$ at low frequencies, flat at high frequencies) you can get the corner frequency from the e_n values at two points.

8.13.4 Finding the 1/*f* corner frequency

If you're looking at a plot of noise density versus frequency, you can try to "eyeball" the $1/f$ corner frequency f_c. But it's nice to be able to find f_c from a pair of tabulated values when there's no graph available (often e_n is specified at 10 Hz and at 1 kHz in a datasheet's tabulated data); and anyway some folks are impressed by an equation or two in a not-inexpensive book like this one. You can figure it out, guided by Figure 8.105; the result is

$$f_c = \frac{e_{nL}^2 - e_{nH}^2}{e_{nH}^2} f_L = \left(\frac{e_{nL}^2}{e_{nH}^2} - 1\right) f_L \quad \text{Hz}, \tag{8.60}$$

where e_{nL} is the noise density at some frequency f_L that is below the corner frequency, and e_{nH} is the noise density well above f_c.

A. Noise voltage over the decades

A white-noise source (i.e., e_n constant), filtered to a bandwidth B (for example with a brick-wall bandpass $f_2{-}f_1{=}B$), has integrated noise voltage $v_n{=}e_n\sqrt{B}$. So, looking at successive decades of bandpass (0.1–1 Hz, 1–10 Hz, 10–100 Hz, etc.), we expect to see the noise voltage increasing by factors of $\sqrt{10}$; and that's why, of course, "the high end matters most." The situation for pink noise is different: the $1/f$ dependence of e_n^2 compensates for the increasing bandwidths, yielding a constant noise voltage, as you can verify by looking at the expression for brick-wall-filtered pink noise on page 564. These dependencies are illustrated nicely in the integrated noise plots of Figure 5.54 and Figures 8.62 and 8.63, where the low-frequency $1/f$ characteristic of conventional op-amps causes their integrated noise voltage to flatten at the low-frequency end; by contrast, auto-zero op-amps exhibit constant e_n at low frequencies, so their integrated noise voltage continues to decrease by 10 dB/decade as the bandwidth is reduced.

Do bench measurements confirm this behavior? Look at the 'scope traces in Figure 8.106, which are single-shot waveforms of band-limited input noise current in an LT1012 BJT op-amp. The noise-current amplitude decreases about 10 dB from the top decade to the next decade down, but it appears to level off and then *increase* again at the lowest decade. That suggests a current noise density i_n that rises faster than the $1/\sqrt{f}$ of pink noise; and indeed the measured current noise spectrum of the LT1012, shown in Figure 8.111, is "steeper than pink." In contrast, an op-amp whose low-frequency noise conforms to an ideal pink-noise spectrum would exhibit approximately constant noise amplitude per decade of bandwidth, once well below the $1/f$ corner frequency.

B. Forever 1/*f* ?

You often hear talk about low-frequency noise power conforming to a "$1/f$ law," as if there's some statutory requirement involved. You might at first think that this cannot possibly be true, because (you say to yourself) a $1/f$ power spectrum cannot continue forever, since it would imply unlimited noise amplitude. If you waited long enough, the input offset voltage (or input current, in this case) would become unbounded. In fact, the popular mythology of a low-frequency noise catastrophe (to which your thinking would have fallen victim) is quite without merit: even if the noise power *density* continues as $1/f$ all the way down to zero frequency, its total noise power (i.e., the *integral* of noise power density) diverges only logarithmically, given that $\int f^{-1} df = \log_e f$. To put some numbers to it, the total noise power in a pure $1/f$ spectrum between 1 *micro*hertz and

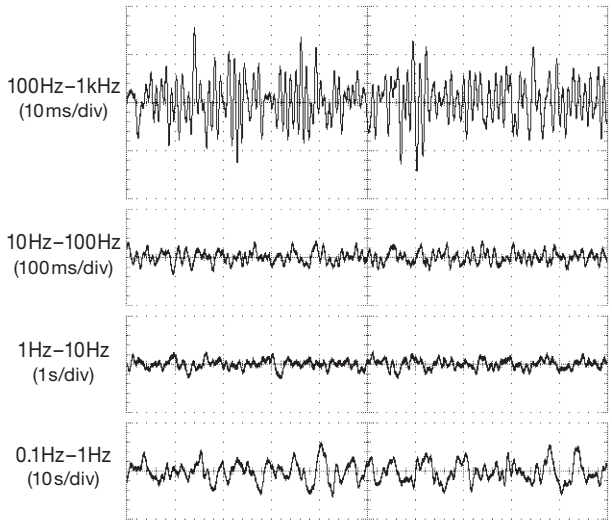


Figure 8.106. LT1012 noise current versus time, for successive decade bandpasses. Vertical: 5 pA/div. Horizontal: scaled to bandpass, as indicated.

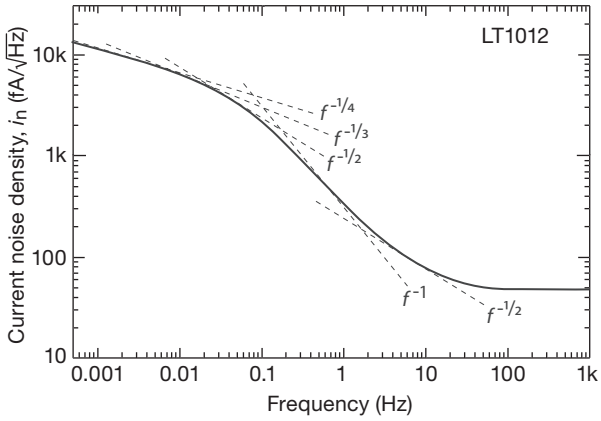


Figure 8.107. Measured LT1012 noise current spectrum, extending down to 500 microhertz. Several slopes are shown, for your amusement.

10 Hz is only 3.5 times greater than that between 0.1 Hz and 10 Hz; going down another six decades (to 10^{-12} Hz), the corresponding ratio grows only to 6.5. Put another way, the $1/f$ total noise power, going all the way down to a frequency that is the reciprocal of 32,000 years (when Neanderthals still roamed the planet, and there were no op-amps), is just six times greater than that of the usual datasheet 0.1–10 Hz "low-frequency noise." So much for catastrophes.

To find out whether the low-frequency noise of real op-amps continues to conform to a $1/f$ spectrum, we measured the current noise spectrum of an LT1012 op-amp all the way down to 0.5 *milli*hertz,[130] with the result of Figure 8.107. As we remarked above, this op-amp is unusual in that its current noise density i_n rises faster than the usual $1/\sqrt{f}$ (pink noise) for a decade around 1 Hz; but even so it settles back to the canonical pink noise, and ultimately becomes something closer to "pale white" ($f^{-1/4}$ or slower).

You could conclude that this demonstrates the unphysical nature of $1/f$ behavior all the way down to zero. But there's another possible explanation, namely that this op-amp is afflicted with some mild burst noise. That would be consistent with the "faster-than-pink" slope around 1 Hz (recall the burst-noise spectrum in Figure 8.6), and it would also lead you to incorrectly attribute a "slower-than-pink" slope at the low frequency end of the spectrum in Figure 8.107.

If this latter explanation is correct, measurements at even lower frequencies (to 0.00001 Hz, say) would confirm a continuing $1/f$ (pink) slope. But it takes a full day to reach 10 μHz, which is why reliable long timescale data are hard to find. An interesting datapoint is provided by Daire's measurement of the "Spectral Noise Distribution" of a Keithley 6430 high sensitivity (0.05 fA resolution) source-measure instrument,[131] which displays a $1/f$ character all the way down to a microhertz (corresponding to a time scale of several weeks). With this in mind, it is quite possible that the low-frequency flattening seen in Figure 8.107 is indeed an artifact of a burst-noise plateau. Or maybe not – there's no statutory requirement that extremely-low-frequency noise must obey a $1/f$ (pink-noise) spectrum.

8.13.5 Measuring the noise voltage

There is a class of test instruments, called *spectrum analyzers* or *dynamic signal analyzers*, that measure and display the frequency spectrum of an input signal. One style is optimized for low-frequency and audio use, generally up to ∼100 kHz or so, with spectral calculations done with a discrete Fourier transform; examples are the Stanford Research Systems SR780/5 and the Agilent U8903A. At the other end you find RF and microwave spectrum analyzers whose upper frequency limits range from ∼3 GHz

[130] We averaged 100 power spectra, accumulating a 2000-second time series for each one: not a quick experiment!

[131] Adam Daire, "Counting electrons: how to measure currents in the attoampere range," Keithley Instruments, Inc., September 2005. Available as a pdf from www.keithley.com.

to upward of 50 GHz; these use an internal swept oscillator and mixer scheme (often augmented with a digital Fourier transform backend) to map out the spectrum sequentially. A popular configuration accommodates the frequency range from 9 kHz to 3 GHz, often with an internal "tracking generator" that lets you sweep the responses of filters or amplifiers; examples are the Agilent E4403 and the Rohde & Schwarz FSL3.

These instruments are quite flexible, with a wide range of input gain settings, frequency spans, display scaling, and so on. The low-frequency instruments have 1 MΩ input impedance, convenient for circuit measurements (e.g., op-amp or voltage-reference noise spectra), whereas the RF analyzers present the standard 50 Ω (or 75 Ω, for video applications) input impedance. To measure an op-amp noise spectrum, for example, just use the circuit of Figure 8.108, with R_s=0; you'll get a spectrum like the lower curve in Figure 8.109. Voltage-noise spectra for some op-amps thus measured are shown in Figure 8.110.

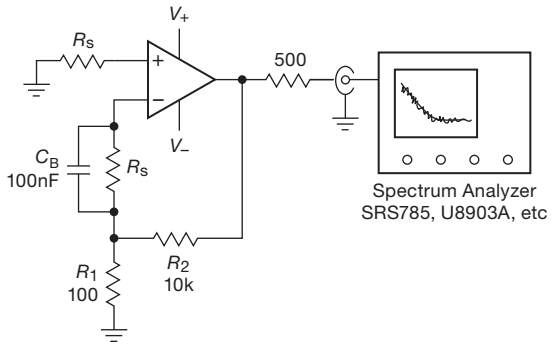


Figure 8.108. Measuring op-amp noise voltage and current spectra. For e_n, set R_s=0; for i_n, choose R_s substantially larger than the amplifier's noise resistance ($R_n = e_n/i_n$). Use a shielded box with filtered dc feedthroughs.

The most accurate way to make integrated output noise voltage measurements is to use a true rms voltmeter. These operate either by measuring the heating produced by the signal waveform (suitably amplified) or by using an analog squaring circuit followed by averaging. If you use a true rms meter, make sure it has response at the frequencies you are measuring; some of them go up to only a few kilohertz. True rms meters also specify a "crest factor," the ratio of peak voltage to rms that they can handle without great loss of accuracy. For Gaussian noise, a crest factor of 3 to 5 is adequate.

You can use a simple averaging-type ac voltmeter instead, if a true rms meter is unavailable. In that case, the values read off the scale must be corrected. As it turns out,

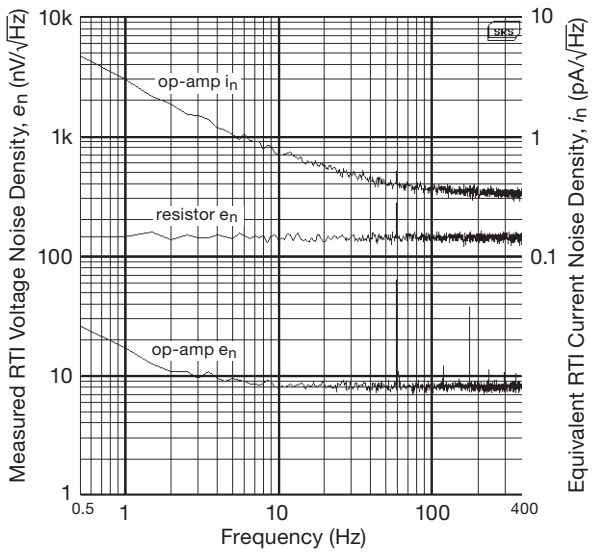


Figure 8.109. Voltage and current noise spectra of an OPA277 op-amp, measured with the circuit of Figure 8.108. The Johnson noise of the 1 MΩ resistor R_s used for the i_n measurements sets a "noise floor," here seen to be well below the op-amp's measured i_n.

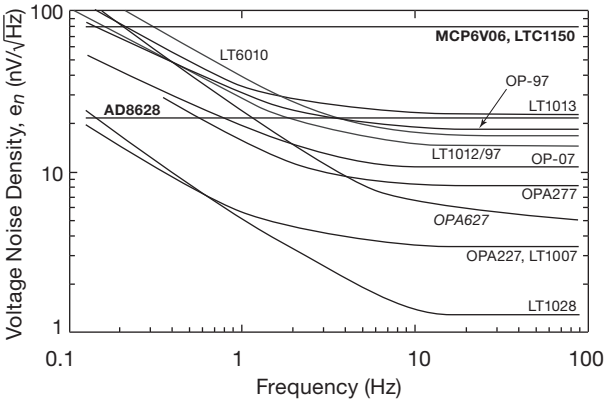


Figure 8.110. Measured voltage-noise-density spectra of a selection of op-amps. The **boldface** parts are auto-zero amplifiers, which circumvent the $1/f$ "flicker-noise" demon by repetitive offset correction; the *italic* OPA627 is a JFET-input op-amp. See also Figures 8.60 and 8.61.

all averaging meters (VOMs, DMMs, etc.) already have their scales adjusted, so what you read isn't actually the *average*, but rather the rms voltage *assuming a sine-wave signal*. For example, if you measure the powerline voltage in the United States, your meter will read something close to 117 V. That's fine, but if the signal you're reading is Gaussian noise, you have to apply an additional correction. The rule is as follows: to get the rms voltage of Gaussian noise,

multiply the "rms" value you read on an averaging ac voltmeter by 1.13 (or add 1 dB). Warning: this works fine if the signal you are measuring is pure noise (e.g., the output of an amplifier with a resistor or noise source as input), but it won't give accurate results if the signal consists of a sinewave added to noise.

A third method, not exactly world famous for its accuracy, consists of looking at the noise waveform on an oscilloscope: the rms voltage is 1/6 to 1/8 of the peak-to-peak value (depending on your subjective reading of the pp amplitude). It isn't very accurate, but at least there's no problem getting enough measurement bandwidth.

8.13.6 Measuring the noise current

An easy way to measure input noise current in an op-amp is to use the circuit of Figure 8.108, with a large input resistor R_s. Its value has to be large enough so that the noise voltage generated across it by the op-amp's input noise current is at least comparable to (and preferably much larger than) the op-amp's noise voltage: $i_n R_s \gtrsim e_n$. Another way to put it is to say that $R_s \gtrsim R_n$, the op-amp's noise resistance.

We're not done yet. It's also necessary that $i_n R_s$ dominates over the resistor's Johnson-noise voltage density: $i_n R_s \gtrsim \sqrt{4kTR_s}$. That is, R_s looks like a current noise source of $i_n = \sqrt{4kT/R_s}$, so you have to choose a large-enough value so that the op-amp's input current noise dominates. We find it easiest to remember Johnson noise voltage and current values for a round-number resistance, and then scale it according to the square root of R. So, take note: the Johnson noise of a 1 MΩ resistor is e_n=127 nV/$\sqrt{\text{Hz}}$ (open-circuit), scaling as $\sqrt{R}$; and i_n=127 fA/$\sqrt{\text{Hz}}$ (short-circuit), scaling as $1/\sqrt{R}$.

Figure 8.111 shows input current noise spectra for a selection of op-amps (most of which have BJT inputs with bias-current cancellation), measured with the circuit of Figure 8.108 with R_s=100 MΩ (for which the resistor's Johnson-noise voltage is equivalent to a current-noise density of 12.7 fA/$\sqrt{\text{Hz}}$), and was so measured with an OPA627 JFET-input op-amp (whose voltage- and current-noise contributions are negligible by comparison). The two auto-zero (chopper-stabilized) parts exhibit a flat low-frequency noise spectrum, in contrast to the usual "pink" $1/f$ rising noise power density of conventional op-amps. But auto-zeros usually have nasty spectral peaks at higher frequencies, caused by the clocked switching at the input (those for the AD8628A are around 15 kHz, off the right-hand end of the graph; but Figure 5.52 shows them handsomely, for the MCP6V06).

Table 8.5 Auto-zero Noise Measurements

	Noise Voltage			Spectral pk	Noise Current		
	e_n		meas peak ampl (μV)	(kHz)	i_n		meas peak ampl (pA)
	spec[t] ($\frac{\text{nV}}{\sqrt{\text{Hz}}}$)	meas ($\frac{\text{nV}}{\sqrt{\text{Hz}}}$)			spec[t] ($\frac{\text{fA}}{\sqrt{\text{Hz}}}$)	meas ($\frac{\text{fA}}{\sqrt{\text{Hz}}}$)	
AD8551	42	46	5	5.2	2	21	20
AD8572	51	55	a	a	2	16	a
AD8628	22	22	8	15	5	53	100
LMP2021	11	18	8	25	350	120	400
LTC1049	100	90	150	2.0	2	100	200
LTC1050	90	70	80	3.8	1.8	130	8000
LTC1150	90	92	2	0.6	1.8	70	15000[s]
MAX4239	30	28	20	18	-	24	b
MAX9617	42	42	20	60	100	74	50
MCP6V06	82	80	40	8.9	0.6	38	800
OPA335	55	52	5	11	20	12	50
OPA734	135	120	5	18	40	27	8000[s]
OPA2188	8.8	8.5	-	-	7	750	-
TLC4501A[c]	*70/12*[e]	*60/11*[e]	*1/2/5*[f]	-	*0.6*	*0.6*[g]	-
MAX4236A[d]	*23/14*[e]	*24/16*[e]	*1/3/8*[f]	-	-	*0.45*[g]	-

Notes: (a) broad spectrum, slight rise at 2.2kHz and harmonics. (b) output noise dominated by V_n, unable to measure I_n separately. (c) auto-zero at power-up, no corrections thereafter. (d) precision conventional CMOS op-amp, for comparison. (e) at 10Hz/1kHz. (f) no spectral features; peak amplitudes for bandwidths to 1kHz/10kHz/100kHz. (g) at 1Hz. (s) spikey waveform, value listed is the peak amplitude. (t) typical.

A caution: datasheet values for input current noise are sometimes seriously in error, evidently because the manufacturer did not measure it, believing that it was accurately predicted by a shot noise calculation based on dc input current. We were curious, and measured the input noise of a dozen auto-zero op-amps, with the results in Table 8.5 on the current page. And how interesting! The datasheet values for *voltage* noise were correct; but for some of the parts, the specified *current* noise values were far too optimistic, sometimes by as much as a factor of fifty. Interestingly, some datasheets even admit what they've done; for example, the i_n entry for each of the three LTC parts is accompanied by a footnote saying "Current noise is calculated from the formula $i_n=\sqrt{2qI_B}$, where q=1.6·10^{-19} coulombs." This same error afflicts some BJT-input op-amps, in particular those with input bias cancellation, where the (incorrect) current noise in the datasheet evidently was calculated from the shot noise corresponding to the *net* (i.e., cancelled) input current, rather than from the much larger uncancelled input current. See §§*4x.10* and 5.10.8 for details.

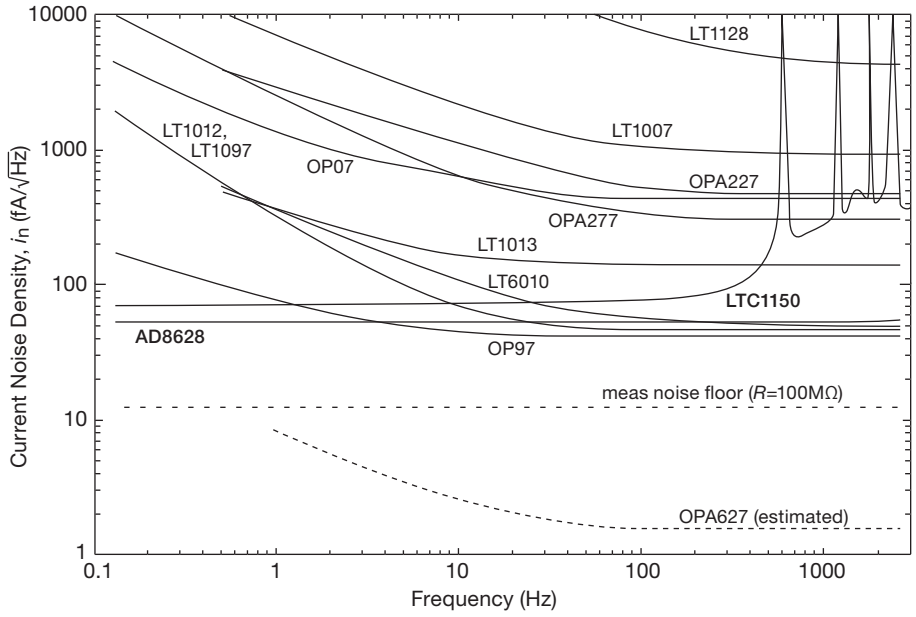


Figure 8.111. Measured current noise density spectra for most of the op-amps of Figure 8.110. The **boldface** parts are auto-zero amplifiers, which circumvent the $1/f$ "flicker-noise" demon by repetitive offset correction, but exhibit switching-induced clock noise at higher frequencies (see the expanded plot in Figure 5.52). See also Figures 8.60 and 8.61.

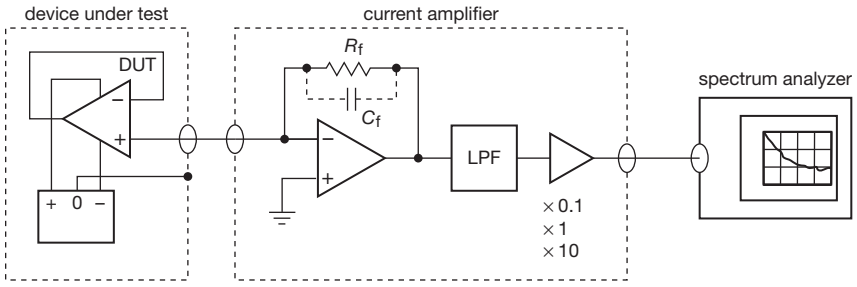
A. Some limitations: bandwidth, stability, dc offset

This simple i_n measurement scheme – letting the device under test amplify its own input current noise, as a voltage developed across a large input resistor – has some drawbacks, which seriously limit the ability to measure noise currents down in the low fA/$\sqrt{\text{Hz}}$ range. As just discussed, you've got to use sufficiently large values of R_s to overcome both the amplifier's e_n and the resistor's Johnson noise. For a noise current of 1 fA/$\sqrt{\text{Hz}}$, for example, that requires R_s at least 10 GΩ (equivalent i_n contribution is 1.3 fA/$\sqrt{\text{Hz}}$). But now you've got to worry about the dc voltage produced by the input bias current: a current of 10 pA, for example, causes 100 mV of dc input, thus saturation at the output after G=100. You've also got to worry about instability, because it doesn't take much feedback capacitance from the output pin to the noninverting input to make the amplifier into an oscillator. That can be tamed by a small shunt capacitor to ground, but the added capacitance reduces the (already small) bandwidth: with our R_s=10 GΩ, for example, a mere 1 pF of input capacitance limits the measurement bandwidth to 16 Hz! And in our measurements we needed additional shunt capacitance to prevent low-frequency oscillation in our (socketed) test jig.

B. Improved bandwidth with a current amplifier

The lesson here is that it's not easy to make measurements of low-level input current noise in good amplifiers. You can do better, though, by using a carefully designed *external* current amplifier (which presents a low impedance input, i.e., a virtual ground) connected directly to the noninverting input of the device under test, as in Figure 8.112. For example, the DL Instruments[132] Model 1211 "current preamplifier" has a specified input current noise of 1.5 fA/$\sqrt{\text{Hz}}$ (corresponding to a 10 GΩ feedback resistor to its input summing junction) and a bandwidth of 400 Hz; it is an "electrometer" configuration (dc negative feedback to an input summing junction), so it holds its input within 0.2 mV of ground at full-scale input current. Some other suppliers of low-noise-current amplifiers and current-measuring instruments are Laser Components (for example their Model DDPCA-300, with switchable gains from 10^4 V/A up to 10^{13} V/A, and with noise as low as 0.2 fA/$\sqrt{\text{Hz}}$ in the most sensitive ranges), and Keithley (for example their Model 428, with switchable gains from 10^3 V/A up to 10^{11} V/A).

[132] Formerly the instrumentation division of Ithaco, Inc.



Range (A/V)	Ithaco 1211 i_n ($f_A/\sqrt{Hz}$)	Ithaco 1211 BW (Hz)	DDPCA-300 i_n ($f_A/\sqrt{Hz}$)	DDPCA-300 BW (Hz)
10^{-8}	15	13k	15	150
10^{-9}	5	4k	15	150
10^{-10}	2.5	800	1.3	20
10^{-11}	1.5	400	1.3	20
10^{-12}	-	-	0.2	1
10^{-13}	-	-	0.2	1

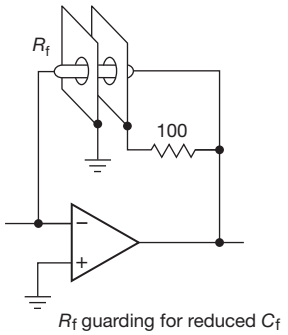


Figure 8.112. Measuring op-amp current noise with a sensitive external current amplifier.

8.13.7 Another way: roll-your-own fA/√Hz instrument

Commercial high-performance current amplifiers can cost quite a bundle – you're talking numbers like 30–40 dB$. It's nice to have such a general-purpose bench instrument on hand, of course. But if all you want is to measure low-level op-amp input-current noise, and you're willing to make a special-purpose test jig, you can wire up something like the circuit in Figure 8.113. With the values shown you can measure noise currents down to 0.1 fA/$\sqrt{\text{Hz}}$ (or less, with a larger R_s), with a component cost of just a few dollars.

This circuit takes an unusual approach, and is worth a bit of discussion. Our first idea was to eliminate entirely the noise-producing feedback resistor: recall that a resistor of value R has a Johnson noise-voltage density $e_n=\sqrt{4kTR}$, thus an equivalent noise-current density $i_n=\sqrt{4kT/R}$; so, for example, to keep that i_n contribution less than 1 fA/$\sqrt{\text{Hz}}$ requires an R_f greater than 16,000 MΩ (with consequent problems of bandwidth, offset, and stability). So here we use instead a feedback *capacitor!*[133] This makes an integrator, here implemented as a composite amplifier in which the device-under-test (DUT) (configured as a follower) drives an inverting op-amp stage whose unity-gain bandwidth is limited (by C_{comp}, in combination with R_1) to ∼16 kHz. We don't need any more bandwidth, and this overcompensation ensures stability.

Ignoring for the moment the DUT's dc input leakage current, this integrator converts input current noise to output voltage noise according to $v_n=i_n/\omega C_f$. So, for example, a flat (white) i_n spectrum produces a v_n whose spectral amplitude falls as $1/f$, or, equivalently, its noise power density falls as $1/f^2$ (red noise). Of course, there's also voltage noise e_n in both op-amps, which gets added to v_n (in the usual square root of sum of squares manner of uncorrelated noise); by choosing a smaller value for C_f we can reduce the effect of the e_n's, because smaller values of C_f produce larger current-to-voltage "gain." (Another way to put it is that the effective current noise produced by the input noise voltage e_n is $i_n=e_n\omega C_f$. Equivalently, the current-noise measurements extend only up to a frequency $\omega=i_n/e_nC_f$,

[133] This isn't as nutty as it sounds. In fact, it is the method of choice ("correlated double-sampling") used in certain low-noise applications, for example photodiode or CCD imager readout amplifiers, where the thermal noise produced by a conventional feedback resistor is unacceptable. (An engineer would say that a capacitor has no Johnson noise; a physicist would say that a capacitor may have kT worth of thermal energy on average, but, being uncoupled from the thermal bath, that energy does not fluctuate.) This technique may have been originated by Garwin, who used it in the 1950s for high-energy physics instrumentation, and then in 1969 for image detector readout (where Garwin instructed "read twice to cancel kT").

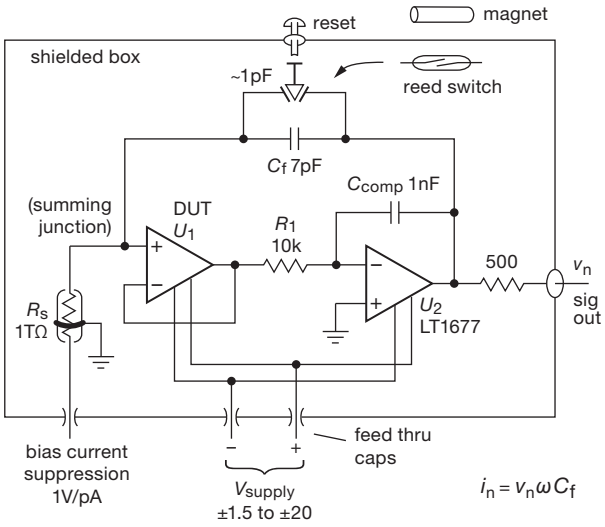


Figure 8.113. Measuring i_n down to 0.1 fA/$\sqrt{\text{Hz}}$, with bandwidths out to ~1 kHz, using a composite integrator. The suppression resistor R_s (made by Welwyn) has an integral metallic guard band, used to intercept leakage currents on the outside of its glass envelope (hey, this is a *really* high-value resistor: 1 TΩ is a million megohms!); the 7 pF value for C_f includes wiring and switch capacitances. The reset switch presented educational challenges (see text).

above which the amplifier's voltage noise dominates.) For this circuit we chose the LT1677 for U_2, because it combines pretty low voltage noise (3.2 nV/$\sqrt{\text{Hz}}$) with low bias current, wide operating voltage range (±1.5 V to ±20 V), and rail-to-rail output.

Figure 8.114 shows a screen capture of the voltage-noise spectrum of the circuit of Figure 8.113, when an LTC1049 auto-zero op-amp is plugged into the DUT socket. It conforms accurately to $1/f$, evaluating to i_n=100 fA/$\sqrt{\text{Hz}}$ over this frequency range, agreeing with the i_n measurement we made using the circuit of Figure 8.108 (and reported in Table 8.5 on page 569). There's no hint of a leveling off at the high frequency end, consistent with a predicted measurement bandwidth of ~10 kHz for the i_n and e_n of this op-amp.

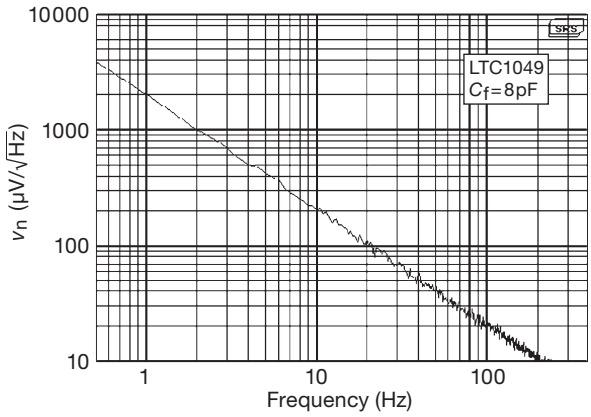


Figure 8.114. Measured voltage-noise density v_n from the circuit of Figure 8.113, for an LTC1049 auto-zero op-amp. The $1/f$ amplitude spectrum of the integrator's output (v_n) corresponds to a flat input current noise spectrum over this range of frequencies, with a value i_n=100 fA/$\sqrt{\text{Hz}}$.

Having calibrated the technique (and having confirmed the calibration by measuring also the AD8628 auto-zero op-amp), we used it to measure the tough cases – the conventional op-amps listed in Table 8.5, with current-noise values down in the ~1 fA/$\sqrt{\text{Hz}}$ territory – with the results listed. For those measurements the bandwidth extends only to ~1 kHz, above which the op-amp's voltage noise e_n, in combination with the input capacitances to ground (both within the op-amp, and externally as wiring, etc.), creates an "e_nC" equivalent input current noise (more precisely, $e_n\omega C_{in}$) that dominates the small current noise of the amplifier itself.

We also measured an LMC6081, which should have even lower current noise, given its specified typical input current I_b of 10 fA; for this op-amp the measured v_n corresponds to an i_n of 0.15 fA/$\sqrt{\text{Hz}}$. That sounds pretty good – but the expected current noise, calculated as I_B shot noise, should be quite a bit lower, about 0.06 fA/$\sqrt{\text{Hz}}$. So now we have to confess that the resistorless purity of this measurement technique has been compromised by the

"current-suppression" resistor R_s, needed to cancel the dc input current. We used a very large value (1 TΩ, that's 10^6 megohms!) to reduce the added noise current, which here amounts to $i_n=\sqrt{4kT/R_s}=0.13\,\text{fA}/\sqrt{\text{Hz}}$. So for this op-amp what we've actually measured is the noise introduced by the bias-suppression resistor! We could do better, say with a 100 TΩ resistor (you can get these things, for example the 3810-series from Welwyn), which would push the added noise down by a factor of ten. We could do better, that is, if you really believe that you can keep all other insulation resistances (moisture, fingerprints, etc.) that high; a tall order.[134]

A. An amusing complication: the reset switch

Here's a little story about reality, and some considerable confusion on the way to ultimate enlightenment. Right "out of the box" this circuit worked nearly perfectly. But there was one curious complication: when the integrator RESET switch was pressed, the output went obediently to zero; but when the switch was released, the output took a large jump – typically to a value in the range of +1.5 V to +2 V. What could cause *that?* Most likely, some sort of electrostatic charging caused by the movement of the plastic parts, in a pushbutton switch series of which we've grown quite fond (Panasonic EVQ2130x–EVQ2150x series, which are compact, inexpensive, pleasantly tactile, and nearly free of annoying contact "bounce").

We pondered this, and decided to substitute a magnetically operated *reed switch* – a small sealed tubular glass envelope containing a pair of contacts, with a wire sticking out each end, and which is actuated by bringing a small permanent magnet in proximity (these things are widely used, most familiarly for door and window switches in home alarm systems). The reed switch has good properties: when OFF its capacitance is ∼0.3 pF and its resistance is greater than $10^{12}\,\Omega$. And, with the switch inside the shielded aluminum box and the magnet outside, there would be no opportunity for accumulation of static charge. Better still, with no dc-actuated coil (as in a conventional relay, or a reed relay) there would be no magnetic or electric coupling to the sensitive circuit.

This worked considerably better. But still not as expected: imagine our surprise when the output took a −50 mV step when the switch was opened. What could be causing *that?* We guessed at first that it might be due to a flux rearrangement when the (magnetic) contacts opened, injecting a pulse of charge equal to the change in flux that was linked by the circuit wiring. Easy to check: just move the switch in close proximity to the op-amp, reducing greatly the circuit's enclosed area. We did that, and, encouragingly, the output jump changed sign (now a positive jump, of about the same magnitude). OK, we thought, move the switch halfway back to where it was and the effect should go to zero. It didn't: still +50 mV.

This was really weird! On an inspiration we guessed that we might have reversed the ends when we made the first change, so we flipped the switch over and – aha! – the jump reversed sign (back to −50 mV).

So the reed switch seemed to have an asymmetry, and a memory of its own, such that it injected some charge (not much: $Q=C_f\Delta V=0.35$ pC) when opened. What could be causing that? We spoke with some wise colleagues, who suggested we think about the difference of "work function" between the two contacts in the switch (which might plausibly be chosen of different metals, to prevent cold-weld sticking), and who mumbled terms like "Fermi levels," "contact potential difference," and the "Kelvin probe" technique for measuring the latter.

OK, so we tried a reed switch of different design, and also one from a different manufacturer (for the original Hamlin MDRR-4 we substituted first a Hamlin MDSR-10, then a Coto RI-01BAA), both of which exhibited the same curious effect, though with different amplitudes (15 mV and 100 mV, respectively).

Time to call for help! The Hamlin datasheet proclaims hospitably "For details on electrical specifications, contact Hamlin." So we did. We learned that the contact pairs are identical, in fact they just make a zillion of them, all the same, and take any two for each switch. There goes all that Fermi-level theory!

Finally we figured it out.[135] Glass is a darned good insulator, and it's easy to deposit charge onto the reed switch's glass envelope just by handling it (removing it from its plastic bag, etc.). There's a fancy name for this: it's called the *triboelectric effect*. A simpler name is static electricity. Glass is one of the classic materials, and in courses on

[134] Indeed! We tested this proposition by breathing heavily into the box, then remeasuring the bias suppression through R_s: it increased 20-fold, owing to the external leakage path evidently created by moisture condensation, returning to its dry value a minute later. That is why these "High-value Glass-sealed" resistors from Welwyn are outfitted with a conductive guard ring, which can be connected to ground (as shown) to shunt off such external leakage currents: the portion of the case between the guard ring and the summing junction then has no voltage drop across it (more precisely, a voltage equal to the op-amp's offset voltage, a millivolt or less); thus no leakage current flows. We confirmed this with an encore of the breathing test.

[135] Isn't it curious that it's always the *last* conjecture that proves to be correct?

electricity you see demonstrations of glass rods being rubbed with cat's fur.[136] With some charge stuck on the glass, the electric field that it produced caused the electrodes of the reed switch to acquire a small amount of opposing charge when separated.

Here's what we did to confirm this conjecture: we took an inch of woven metal cloth,[137] grounded one end with a clip lead, wrapped it around the reed switch, and slid it along the length several times, to provide an opportunity for any static charge on the glass surface to take a hike. And it did: after one treatment the output-voltage step was reduced to a negligible $\sim$2 mV.

The moral: sensitive measurements (here we're talking femtoamps and picofarads) can reveal effects that are so small that you've never thought about them. They can seriously disrupt your work... but there's a redeeming delight in discovering them for yourself. And then eliminating them.

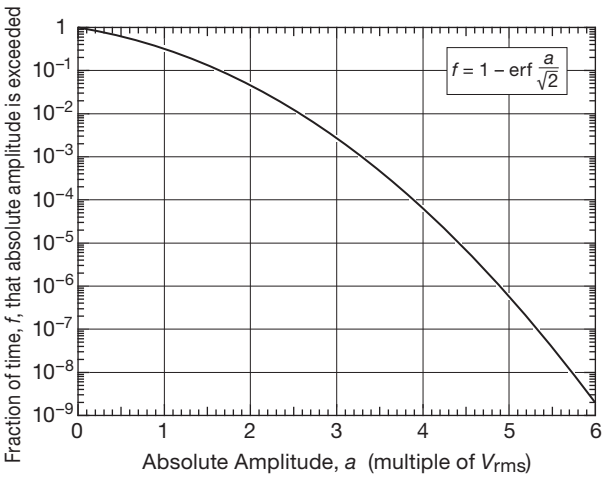


Figure 8.115. Relative occurrence of amplitudes in Gaussian noise. Potentially useful for estimating false trigger rates, required "crest factor" in rms measurements, and the like.

8.13.8 Noise potpourri

Herewith a collection of interesting, and possibly useful, facts.

1. The averaging time required in an indicating device to reduce the fluctuations of a rectified noise signal to a desired level for a given noise bandwidth is

$$\tau \approx \frac{1600}{B\sigma^2} \text{ seconds}, \tag{8.61}$$

where τ is the required time constant of the indicating device to produce fluctuations of standard deviation σ percent at the output of a linear detector whose input is noise of bandwidth B.

2. For band-limited white noise, the expected number of maxima per second is

$$N = \sqrt{\frac{3(f_2^5 - f_1^5)}{5(f_2^3 - f_1^3)}}, \tag{8.62}$$

where f_1 and f_2 are the lower and upper band limits. For $f_1 = 0$, $N = 0.77 f_2$; for narrowband noise ($f_1 \approx f_2$), $N \approx (f_1 + f_2)/2$.

3. rms-to-average (i.e., average magnitude) ratios:

Gaussian noise: $\text{rms/avg} = \sqrt{\pi/2} = 1.25 = 1.96\,\text{dB}$,
sinewave: $\text{rms/avg} = \pi/2^{\frac{3}{2}} = 1.11 = 0.91\,\text{dB}$,
square wave: $\text{rms/avg} = 1 = 0\,\text{dB}$.

4. Relative occurrence of amplitudes in Gaussian noise. Figure 8.115 plots the fractional time that a given amplitude level is exceeded by a Gaussian noise waveform of unit rms amplitude.
5. Positive-threshold crossing rate of lowpass-filtered Gaussian white noise of unit rms amplitude is

$$\text{TCR} = \frac{\text{BW}}{\sqrt{3}} \exp\left(-V_{\text{th}}^2/2\right) \text{ crossings/second}, \tag{8.63}$$

where V_{th} is the positive threshold voltage and BW is the brick-wall lowpass bandwidth.[138]

6. Standard deviation of noise resulting from quantization error is

$$\sigma_n = \frac{\text{LSB}}{\sqrt{12}} \approx 0.3\,\text{LSB}. \tag{8.64}$$

8.14 Signal-to-noise improvement by bandwidth narrowing

As luck would have it, the signals you often want to measure are buried in noise (where the "noise" may include other signals nearby in frequency, i.e., interference), frequently to the extent that you can't even see them on an

[136] Or is it a cat being rubbed with glass wool?

[137] Wrapped around some foam, this handy stuff is used to make flexible conductive shielding gaskets. Check out the self-stick "fabric-over-foam" gasket materials from Laird Technologies, e.g., their rectangular 4046 or D-shaped 4283.

[138] The crossing rate through a pair of symmetrical thresholds (i.e., the rate of crossings of *magnitude* greater than V_{th}) is double that obtained from the formula. We thank Phil Hobbs for this fact, found along with many others in his fine book referenced on page 551.

oscilloscope. Even when external noise isn't a problem, the statistics of the signal itself may make detection difficult, as, for example, when counting nuclear disintegrations from a weak source, with only a few counts detected per minute. Finally, even when the signal is detectable, you may wish to improve the detected signal strength in order to make a more accurate measurement. In all these cases some tricks are needed to improve the signal-to-noise ratio. They all amount to a narrowing of the detection bandwidth in order to preserve the desired signal while reducing the total amount of (broadband) noise accepted.

The first thing you might be tempted to try when thinking of reducing the bandwidth of a measurement is to hang a simple lowpass filter on the output, in order to average out the noise. There are cases where that therapy will work, but most of the time it will do very little good, for a couple of reasons. First, the signal itself may have some high frequencies in it, or it may be centered at some high frequency. Second, even if the signal is in fact slowly varying or static, you invariably have to contend with the reality that the density of noise power usually has a $1/f$ character, so as you squeeze the bandwidth down toward dc you gain very little. Electronic and physical systems are twitchy, so to speak.

In practice, there are a few basic techniques of bandwidth narrowing that are in widespread use. They go under names like signal averaging, transient averaging, boxcar integration, multichannel scaling, pulse-height analysis, lock-in detection, and phase-sensitive detection. All of these methods assume that you have a repetitive[139] signal; that's no real problem, since there is almost always a way to force the signal to be periodic, assuming it isn't already. Here we discuss an important one of these techniques, known as "lock-in" or "synchronous" detection.

8.14.1 Lock-in detection

This is a method of considerable subtlety. It consists of two steps. (1) Some parameter of the source signal is *modulated*; for example, an LED might be driven with a square wave at a fixed frequency. (2) The detected (and noisy) signal is *demodulated*, for example by multiplying it by a fixed-amplitude reference signal at the same modulating frequency. The modulation moves the source signal spectrum up to the modulating frequency, above noisy $1/f$ low-frequency backgrounds, and away from other noise sources (such as ambient-light fluctuations in the case of the LED example). The demodulation step creates a dc output proportional to the signal, which can be lowpass filtered (a simple *RC* filter may be adequate) to narrow the detected bandwidth.

To understand the method, it is necessary to take a short detour into the phase detector, a subject we first take up in §13.13.2.

A. Phase detectors

In §13.13.2 we describe phase detectors that produce an output voltage proportional to the phase difference between two digital (logic-level) signals. For purposes of lock-in detection, you need to know about *linear* phase detectors, because you are nearly always dealing with analog voltage levels.

The simplest circuit[140] is shown in Figure 8.116. An analog signal passes through a linear amplifier whose gain is reversed by a square-wave "reference" signal controlling a FET switch (see Table 3.3 or 13.7 for candidates). The output signal passes through a lowpass filter, *RC*. That's all there is to it. Let's see what you can do with it.

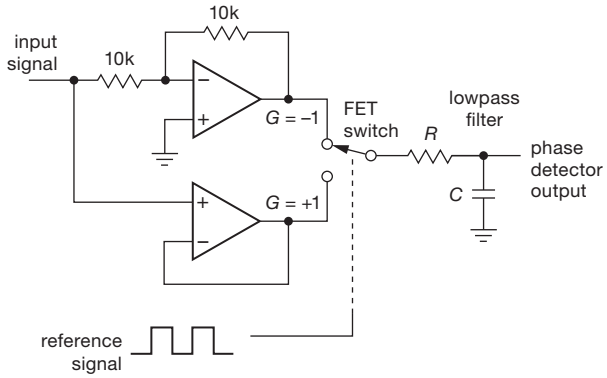


Figure 8.116. Phase detector for linear input signals. Most simply, you can implement this with a dual op-amp and a CMOS switch IC. This scheme is used in the monolithic AD630.

Phase-detector output

To analyze the phase-detector operation, let's assume we apply a signal

$$E_s \cos(\omega t + \phi)$$

to such a phase detector, whose reference signal is a square wave with transitions at the zeros of $\sin \omega t$, i.e., at

[139] Or, more generally, a *known* signal variation to which the measured signal can be correlated.

[140] But less than ideal: the square-wave modulation causes response at odd harmonics. The use of an analog multiplier like the AD633 or AD734, driven by a sinewave reference, eliminates this deficiency.

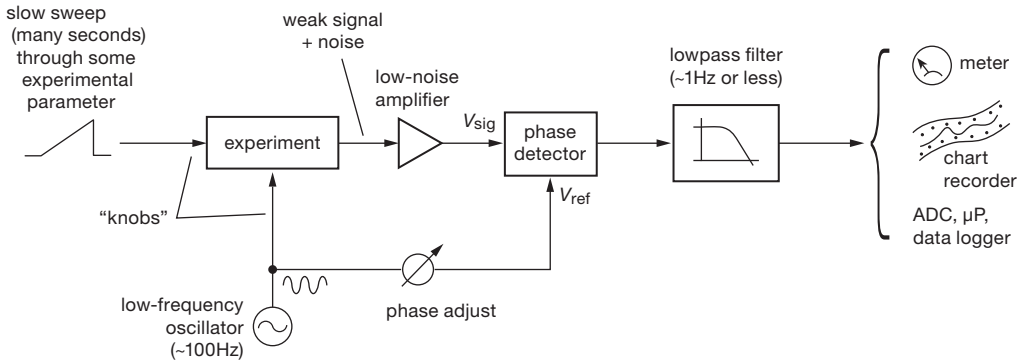


Figure 8.117. Lock-in amplifier detection.

$t = 0, \pi/\omega, 2\pi/\omega$, etc. Let us further assume that we average the output, V_{out}, by passing it through a lowpass filter whose time constant is much longer than one period:

$$\tau = RC \gg T = 2\pi/\omega.$$

Then the lowpass filter output is

$$\langle E_s \cos(\omega t + \phi)\rangle|_0^{\pi/\omega} - \langle E_s \cos(\omega t + \phi)\rangle|_{\pi/\omega}^{2\pi/\omega},$$

where the angle brackets represent averages and the minus sign comes from the gain reversal over alternate half-cycles of V_{ref}. As an exercise, you can show that

$$\langle V_{out}\rangle = -(2E_s/\pi)\sin\phi.$$

Exercise 8.9. Perform the indicated averages by explicit integration to obtain the preceding result for unity gain.

Our result shows that the averaged output, *for an input signal of the same frequency as the reference signal*, is proportional to the amplitude of E_s and sinusoidal in the relative phase.

We need one more result before going on: what is the output voltage for an input signal whose frequency is close to (but not equal to) the reference signal? This is easy, because in the preceding equations the quantity ϕ now varies slowly, at the difference frequency:

$$\cos(\omega + \Delta\omega)t = \cos(\omega t + \phi) \quad \text{with } \phi = t\Delta\omega,$$

giving an output signal that is a slow sinusoid:

$$V_{out} = (2E_s/\pi)\sin(t\Delta\omega),$$

which will pass through the lowpass filter relatively unscathed if $\Delta\omega < 1/\tau = 1/RC$ and will be heavily attenuated if $\Delta\omega > 1/\tau$.

B. The lock-in method

Now the so-called lock-in (or phase-sensitive) amplifier should make sense. First you make a weak signal periodic, as we've discussed, say at a frequency in the neighborhood of 100 Hz. The weak signal, contaminated by noise, is amplified and phase detected relative to the modulating signal. Look at Figure 8.117. In many cases you'll want to measure the weak signal as some experimental condition is varied – you'll have an experiment with two "knobs" on it, one for fast modulation to do phase detection, and one for a slow sweep through the interesting features of the signal (in NMR, for example, the fast modulation might be a small 100 Hz modulation of the magnetic field, and the slow modulation might be a frequency sweep 10 minutes in duration through the resonance). The phase shifter is adjusted to give maximum output signal, and the lowpass filter is set for a time constant long enough to give a good signal-to-noise ratio. The lowpass-filter rolloff sets the bandwidth, so a 1 Hz rolloff, for example, gives you sensitivity to spurious signals and noise only within 1 Hz of the desired signal. The bandwidth also determines how fast you can adjust the "slow modulation," because now you must not sweep through any features of the signal faster than the filter can respond; people use time constants of fractions of a second up to tens of seconds or more.[141]

Note that lock-in detection amounts to *bandwidth narrowing*, with the bandwidth set by the post-detection lowpass filter. Another way to reduce the detection bandwidth is with the technique of *signal averaging*, in which the results of repetitive measurements (e.g., frequency sweeps) are accumulated; this is a common option on instruments such as spectrum analyzers. In either case the effect of the modulation is to center the signal at the fast modulation

[141] In the old days the slow modulation was done with a geared-down clock motor turning an actual knob on something.

frequency, rather than at dc, in order to get away from $1/f$ noise (flicker noise, drifts, and the like).

C. Two methods of "fast modulation"

There are several ways to do the fast modulation: the modulation waveform can be either a very small sinewave or a very large square wave compared with the features of the sought-after signal (line shape versus magnetic field, for example, in NMR), as sketched in Figure 8.118. In the first case the output signal from the phase-sensitive detector is proportional to the *slope* of the line shape (i.e., its derivative), whereas in the second case it is proportional to the line shape itself (providing there aren't any other lines out at the other endpoint of the modulation waveform). This is the reason all those simple NMR resonance lines come out looking like dispersion curves (Figure 8.119).

For large-shift square-wave modulation there's a clever method for suppressing modulation feedthrough, in cases where that is a problem. Figure 8.120 shows the modulation waveform. The offsets above and below the central value kill the signal, causing an on–off modulation of the signal at *twice* the fundamental of the modulating waveform. This is a method for use in special cases only; don't get carried away by the beauty of it all!

Large-amplitude square-wave modulation is a favorite with those dealing in infrared astronomy, where the telescope's secondary mirrors are rocked to switch the image back and forth on an infrared source. It is also popular in radio astronomy, where it's called a Dicke switch.

Commercial lock-in amplifiers have a variable-frequency modulating source and tracking filter, a switchable-time-constant post-detection filter, a good low-noise wide-dynamic-range amplifier (you wouldn't be using lock-in detection if you weren't having noise problems), and a nice linear phase detector. They also let you use an external source of modulation. The phase shift is adjustable, so you can maximize the detected signal. The whole item comes packed in a handsome cabinet, with a meter or digital display showing the output signal. Typically these things cost a few thousand dollars. Stanford Research Systems has a nice selection of lock-in amplifiers, including several that use digital signal-processing methods for enhanced linearity and dynamic range. In these the amplified input signal is accurately digitized (to 20 bits), the "oscillator" is a computed lookup table of (quadrature) sines and cosines, and the "mixer" is a numeric multiplier. Ordinarily lock-in amplifiers are of rather limited signal bandwidth, typically 100 kHz. But by using the radiofrequency "heterodyne" technique (input frequency band translation via linear mixing with a "local oscillator"), the lock-in method can be extended into high radio-frequencies. For example, the SR844 goes to 200 MHz; it uses a hybrid of analog techniques (input filtering and downconversion) and digital signal processing (baseband digitizing and synchronous detection).

To illustrate the power of lock-in detection, we set up a small demonstration for our students. We use a lock-in to modulate a small LED of the kind used for panel

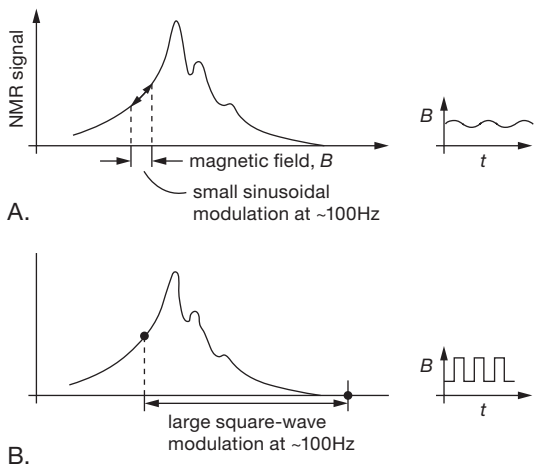


Figure 8.118. Lock-in modulation methods. A. Small sinusoid. B. Large square wave.

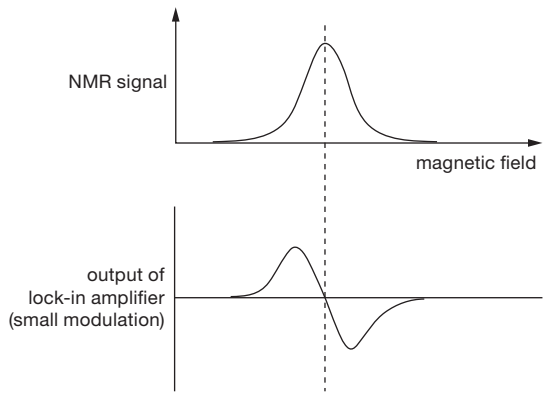


Figure 8.119. Line-shape differentiation resulting from lock-in detection.

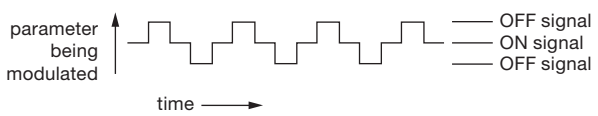


Figure 8.120. Modulation scheme for suppressing modulation feedthrough.

indicators, with a modulation rate of a kilohertz or so. The current is very low, and you can hardly see the LED glowing in normal room light. Six feet away a phototransistor looks in the general direction of the LED, with its output fed to the lock-in. With the room lights out, there's a tiny signal from the phototransistor at the modulating frequency (mixed with plenty of noise), and the lock-in easily detects it, using a time constant of a few seconds. Then we turn the room lights on, at which point the signal from the phototransistor becomes just a huge messy waveform, jumping in amplitude by 50 dB or more. The situation looks hopeless on the oscilloscope, but the lock-in just sits there, unperturbed, calmly detecting the same LED signal at the same level. You can check that it's really working by sticking your hand in between the LED and the detector. It's darned impressive.

At the other end of the cost spectrum, synchronous detection is used to accomplish the same rejection of ambient light in some inexpensive light-beam-detection components, for example the S6809/46 and S6986 from Hamamatsu. These ICs come in a clear plastic case (several available package styles) containing an integrated photodiode, preamp, and synchronous detector with logic-level output; included also is the internal oscillator and output driver for the external LED light source. They cost about $6 in small quantities.

8.15 Power-supply noise

Amplifier circuits that do not have a high degree of power-supply rejection are susceptible to noise (and signals) on the dc power supplies. If the dc rails are noisy, the output will be too, so you've got to keep them quiet. The problem is not as bad as it could be, though, because supply noise appears unamplified at the output, with the amplifier's signal gain of, say, $\times 100$ signal gain. Still, dc power supplies are rarely quiet even to the level of $100\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$ (100 times a reasonable input noise target of, say, $1\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$). That is why the dc rails in many of the circuits in this chapter are marked "quiet."

How noisy are typical dc bench power supplies? You'll see specs like "0.2 mVrms, 2 mVpp," which sounds respectable enough until you realize that signal levels in a sensitive circuit may be far smaller. For example, the output noise level over the 20 kHz audio band of the preamp of Figure 8.42 is just $1\,\mu$Vrms ($v_{\mathrm{n(out)}} = G_{\mathrm{V}} e_{\mathrm{n(in)}} \sqrt{\Delta \mathrm{f}}$). So such a supply's specified noise is 46 dB above the output noise floor.

Specs are one thing, actual performance is another. To get a measure of the power supply noise scene we measured two dozen dc supplies from our lab's collection, with the resulting spectra of Figure 8.123.[142]

The spread in noise performance is stunning – the outstanding performer (#4) turned out to be a "precision power source" half a century old (we bought it in 1967), with an ovenized compartment for the zener reference and (discrete) error amplifier. By comparison, a contemporary bench supply with elegant digital readout like #12 is nearly 100 times (40 dB) as noisy. Not to be outdone, the real screamers turn out to be a simple switchmode cellphone charger (#22), and an unregulated wall-wart (#23) whose powerline-tracking output wanderings dominate the low-frequency spectrum. At the other end, nothing beats a lead-acid battery (#3, right down at our measurement noise floor) for the utmost in inherently quiet dc sources.[143]

What accounts for these large differences? Switching supplies are inherently noisy, of course. But even among the linear supplies there is a $100\times$ spread (40 dB) in noise voltage. A quiet dc regulated supply must have plenty of loop gain, implemented with low-noise amplifiers. And it's critically important to select a low-noise voltage reference (particularly down at low frequencies, where it can't be quieted with filters); see the discussion and plots in §9.10.

8.15.1 Capacitance multiplier

A nice trick for cleaning up a noisy supply is a "capacitance multiplier" circuit (Figure 8.121). We introduced this back in §8.5.9, where we explored the properties of a BJT input stage with $e_{\mathrm{n}} \sim 0.07\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$. Here we choose R small enough so that there is at most a volt or so drop at maximum load current, then choose C for a long-enough time constant RC to attenuate adequately the portion of the noise spectrum you care about. For our ribbon-mic amplifier we elaborated on this a bit, with a 2-stage RC filter of $\sim$2 s time constant; it's shown in the top portion of the BJT noise-measuring circuit of Figure 8.92. There we increased RC until the output noise spectrum was down to the analyzer's noise floor.

Note that the capacitance multiplier compromises the dc output regulation – there's no feedback from its output pin. However, this technique is especially effective when

[142] These supplies were tested "as is," with no effort to confirm their operation within original specifications. The reader is cautioned not to rely on these data when making purchasing decisions.

[143] The battery was being trickle charged at the same current as its load for curve #3. If the battery is being discharged (without replenishment) by the load, the slight downward voltage "tilt" appears as a low-frequency excess, seen here in curve #2.

added upstream of the regulating pass element (i.e., after the rectifier and storage capacitor). We have applied this simple modification to commercial power supplies and scientific instruments with great success. It's a far easier way to achieve 100× lower output ripple than the alternative of raising the control-loop gain and bandwidth. If you do this, be sure there's enough drop across Q_1 to handle the full ripple amplitude – in this location the 120 Hz (or 100 Hz) ac ripple at full load current may be as much as a few volts. You can add a resistor from the base to ground to increase the dc drop across Q_1; or you can use a MOSFET, whose larger operating V_{GS} may provide adequate headroom.

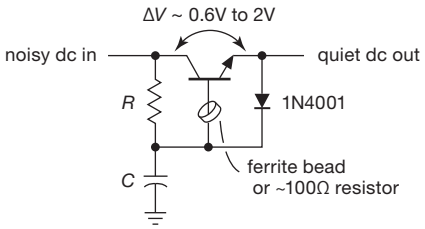


Figure 8.121. A "capacitance multiplier" for filtering incoming dc consists of an emitter follower biased with a smoothed (lowpass-filtered) replica of the noisy input. A bead or small resistor prevents oscillation.

Figure 8.122 shows the effect of Figure 8.92's capacitance multiplier, measured with two of the power supplies that appear in Figure 8.123's crowd. For each supply we measured (a) the spectrum directly from the output terminals (solid curves), (b) the spectrum downstream of the capacitance multiplier (dashed curves), and, for comparison, (c) the spectrum with 10,000 μF across the power supply's terminals (dotted curve). In all cases the measured output was loaded with 100 Ω. The capacitance multiplier is stunningly effective in eliminating power supply noise,[144] and it's a far better use of 10,000 μF than the brute-force approach of putting the same capacitance directly across the dc terminals.

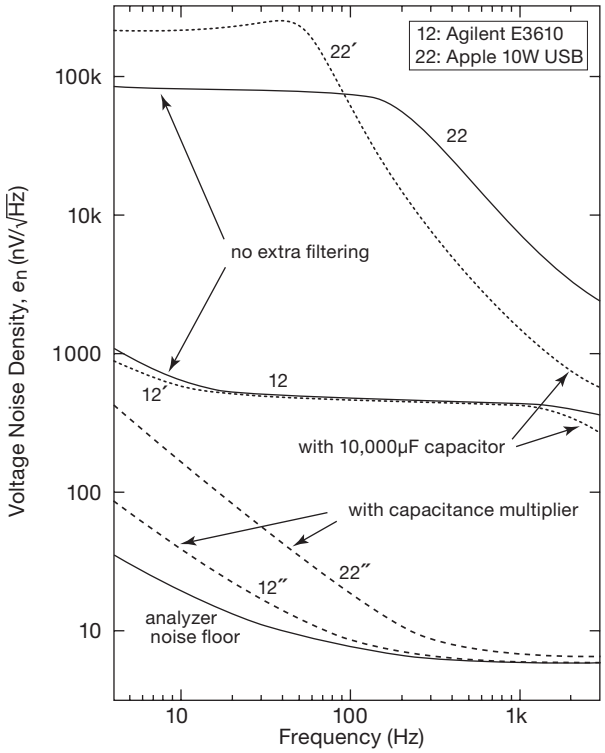


Figure 8.122. A capacitance multiplier is highly effective in eliminating power-supply noise, as shown in these measured noise spectra of two of the supplies whose spectra are included in Figure 8.123 (with same label numbers). Solid curves, power supply alone; dotted curves, (single-prime label), with added 10,000 μF capacitor across the dc output terminals; dashed curves (double-prime label), with capacitance multiplier shown in Figure 8.92.

8.16 Interference, shielding, and grounding

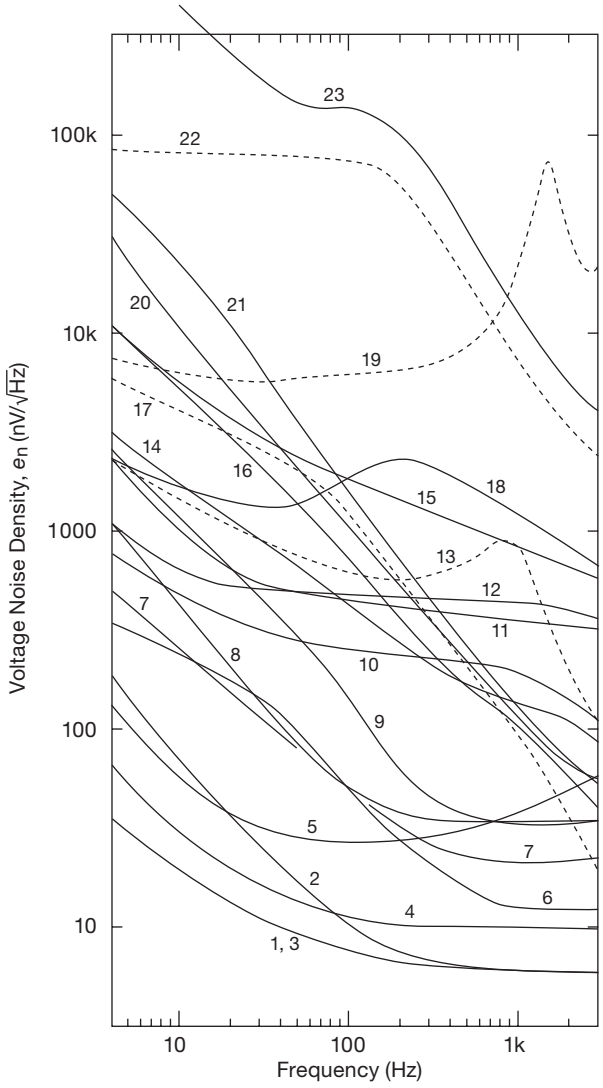
"Noise" in the form of interfering signals, 60 Hz pickup, and signal coupling via power supplies and ground paths can turn out to be of far greater practical importance than the intrinsic noise sources we've just discussed. These interfering signals can all be reduced to an insignificant level (unlike thermal noise) with proper circuit design, layout, and construction. In stubborn cases the cure may involve a combination of filtration of input and output lines, careful layout and grounding, and extensive electrostatic and magnetic shielding. In these subsections we offer some suggestions that may help illuminate this dark area of the electronic art.[145]

8.16.1 Interfering signals

Interfering signals can enter an electronic instrument through the powerline inputs or through signal input and output lines. In addition, signals can be capacitively

[144] Except at very low frequencies, where there's no substitute for a power supply with a stable voltage reference.

[145] For more extensive advice see the popular classics: R. Morrison, *Grounding and Shielding: Circuits and Interference*, Wiley–IEEE Press (2007), and H. Ott, *Noise Reduction Techniques in Electronic Systems*, Wiley–Interscience (1988).



Trace	Description	Peak[a] (@ *n*·60Hz) dB[b]	µVrms[c]
1	SRS785 spectrum analyzer noise floor	-	-
2	12V 7Ah sealed lead-acid battery	-	-
3	same, with matching 120mA trickle charge	-	-
4	Power Designs 4010 40W precision bench linear	54	1.2
5	HP6114A 40W precision bench linear	39	2
6	Fluke 343 DC voltage calibrator	42	2
7	Analogic AN3100 DC voltage calibrator	51	14
8	Agilent E3630 35W triple bench linear	23	0.6
9	HP6002A 200W bench linear	50	18
10	Keithley 230 programmable voltage source	33	11
11	HP6612C 40W bench linear	28	10
12	Agilent E3610 30W bench linear	12	1.8
13	Cui 9V 20W desk-wart switcher	40	56
14	Tektronix PS283 75W triple bench linear	46	70
15	HP66312 40W bench linear	27	10
16	same, with 33,000uF capacitor across output	26	2.2
17	HP6024A 200W bench switcher	40	100
18	Jerome RYD313F-001 15W triple desk-wart linear	53	640
19	Corsair VX550W 550W computer "ATX" switcher[d]	44	400
20	Leader PS152 70W triple bench linear	24	15
21	HP6216B 8W bench linear	47	280
22	Apple 10W "USB Power Adapter" switcher	12	400
23	Cui DV9500 unregulated 9V 500mA wall-wart	6	-

a. Highest amplitude powerline-related tonal (60, 120, or 180Hz).
b. In a 1Hz bandwidth, relative to smoothed baseline at 120Hz.
c. Absolute, filtered with 1Hz bandwidth.
d. While operating in a desktop PC.

Figure 8.123. All dc power supplies are not created equal! Measured noise voltage spectra of many varieties of supplies found in our lab. Dashed lines are switching supplies; the rest are linear. A 100 Ω load was used in most cases. Spectral peaks ("tonals") at powerline frequency and harmonics have been edited out for clarity; they range from 5 dB to more than 50 dB above the smoothed baseline, as listed. A drifting output voltage produces a rising low-frequency tail.

coupled (electrostatic coupling) onto wires in the circuit (the effect is more serious for high-impedance points within the circuit), magnetically coupled to closed loops in the circuit (independent of impedance level), or electromagnetically coupled to wires acting as small antennas for electromagnetic radiation. Any of these can become a mechanism for coupling of signals from one part of a circuit to another. Finally, signal currents from one part of the circuit can couple to other parts through voltage drops on ground lines or power-supply lines.

A. Eliminating interference

Numerous effective tricks have been evolved to handle most of these commonly occurring interference problems. Keep in mind the fact that these techniques are all aimed at reducing the interfering signal or signals to an acceptable level; they rarely eliminate them altogether. Consequently, it often pays to raise signal levels, just to improve the signal-to-interference ratio. Also, it is important to realize that some environments are much worse than others; an instrument that works just dandy on the bench may perform

miserably on location. Some environments worth avoiding are those (a) near a radio or television station (RF interference), (b) near a subway (impulsive interference and powerline garbage), (c) near high-voltage lines (radio interference, frying sounds), (d) near motors and elevators (powerline spikes), (e) in a building with triac lamp dimmers and heater controllers (powerline spikes), (f) near equipment with large transformers (magnetic pickup), and (g) near arc welders (unbelievable pickup of all sorts). Herewith a gathering of advice, techniques, and black magic.

B. Signals coupled through inputs, outputs, and powerline

The best bet for powerline noise is to use a combination of RF line filters and transient suppressors on the ac powerline. You can achieve 60 dB or better attenuation of interference above a few hundred kilohertz this way, as well as effective elimination of damaging spikes.

Inputs and outputs are more difficult, because of impedance levels and the need to couple desired signals that may lie in the frequency range of interference. In devices like audio amplifiers you can use lowpass filters on inputs *and outputs* (much interference from nearby radio stations enters via the speaker wires, acting as antennas). In other situations shielded lines are often necessary. Low-level signals, particularly at high impedance levels, should always be shielded. So should the instrument cabinet.

C. Capacitive coupling

Signals within an instrument can get around handsomely via electrostatic coupling: some point within the instrument has a 10 volt signal jumping around; a high-Z input nearby does some sympathetic jumping, too. The best things to do are to reduce the capacitance between the offending points (move them apart), add shielding (a complete metal enclosure, or even close-knit metal screening, eliminates this form of coupling altogether), move the wires close to a ground plane (which "swallows" the electrostatic fringing fields, reducing coupling enormously), and lower the impedance levels at susceptible points, if possible. Op-amp outputs don't pick up interference easily, whereas inputs do. More on this later.

D. Magnetic coupling

Unfortunately, low-frequency magnetic fields are not significantly reduced by metal enclosures. A dynamic microphone, audio recorder, low-level amplifier, or other sensitive circuit placed in close proximity to an instrument with a large power transformer will display astounding amounts of 60 Hz pickup. The best therapy here is to avoid large enclosed areas within circuit paths and try to keep the circuit from closing around in a loop. Twisted pairs of wires are quite effective in reducing magnetic pickup, because the enclosed area is small, and the signals induced in successive twists cancel.

When dealing with very low level signals, or devices particularly susceptible to magnetic pickup (tape heads, inductors, wire-wound resistors), it may be desirable to use magnetic shielding. "Mu-metal shielding" is available in preformed pieces and flexible sheets. If the ambient magnetic field is large, it is best to use shielding of high permeability (high mu) on the inside, surrounded by an outer shield of lower permeability (which can be ordinary iron, or low-mu shielding material), to prevent magnetic saturation in the inner shield. Of course, moving the offending source of the magnetic field is often a simpler solution. It may be necessary to exile large power transformers to the hinterlands, so to speak. Shielded inductors (e.g., pot cores) are configured so that the magnetic material (usually ferrite) provides a closed magnetic path. Toroidal transformers have smaller fringing fields than the standard frame types, and a single reverse turn (or arranging the winding pattern to return back to where it begins) cancels the one-turn effective area of the toroid's winding.

E. Circuit boards and cables

Both capacitive and magnetic coupling occurs, also, between traces on circuit boards or between line pairs in cables and ribbons. Magnetic coupling is sometimes called *inductive* coupling, to distinguish it from capacitive coupling. This business of "crosstalk" is a rich subject, dealt with in good detail in references such as Johnson and Graham's *Handbook of Black Magic*.[146] Perhaps unintiutively, it turns out that the magnitudes of inductive and capacitive crosstalk are generally comparable, but they behave differently as seen at the two ends of a pair of closely spaced lines: forward-going crosstalk is proportional to the signal's slew rate, but the capacitive and inductive components are of opposite sign and tend to cancel. For reverse-going crosstalk the coupled signal looks like a flat-topped pulse (of width equal to the round-trip travel time), with capacitive and inductive components reinforcing. If the driving impedance does not match the line's characteristic impedance (see Appendix H), that "near-end crosstalk" (aka *NEXT*) signal will reflect and become forward going, contributing to (and often dominating) the "far-end crosstalk" (or *FEXT*).

[146] H. W. Johnson and M. Graham, *High-Speed Digital Design – A Handbook of Black Magic*, Prentice-Hall (1993). See also their follow-on, *High-Speed Signal Propagation – Advanced Black Magic*, Prentice-Hall (2003).

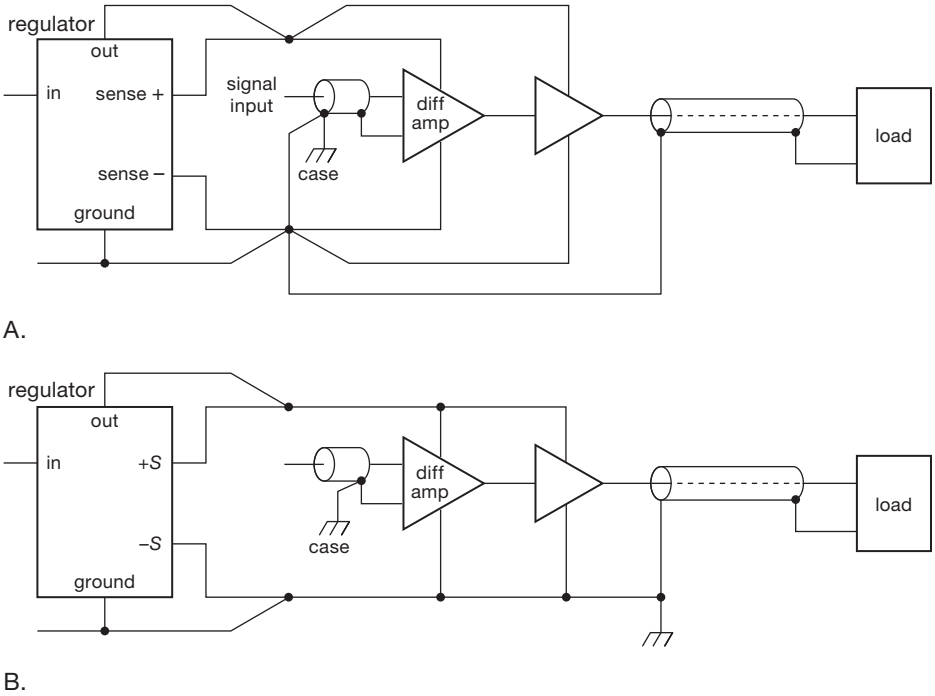


Figure 8.124. Ground paths for low-level signals. A. Right. B. Wrong.

See also additional discussion in Chapter 12 (§12.9, "Digital signals and long wires").

F. Radiofrequency coupling

RF pickup can be particularly insidious, because innocent-looking parts of the circuit can act as resonant circuits, displaying enormous effective cross-section for pickup. Aside from overall shielding, it is best to keep leads short and avoid loops that can resonate. Ferrite beads may help, if the problem involves very high frequencies. Doing good deeds can sometimes create havoc. For example, you may want to use several ceramic capacitors to improve power-supply bypassing; but in combination with the inductance of the connecting supply rails they can form a lovely parasitic tuned circuit somewhere in the HF to VHF region (tens to hundreds of megahertz), nudged by active circuits into ringing and even oscillation.[147]

8.16.2 Signal grounds

Ground leads and shields can cause plenty of trouble, and there is a lot of misunderstanding on this subject. The problem, in a nutshell, is that currents you forgot about that are flowing through a ground line can generate a signal seen by another part of the circuit sharing the same ground. The technique of a ground "mecca" (a common point in the circuit to which all ground connections are tied) is often seen, but it's a crutch; with a little understanding of the problem you can handle most situations intelligently.

A. Common grounding blunders

Figure 8.124 shows a common situation. Here a low-level amplifier and a high-current driver are in the same instrument. The first circuit is done correctly: both amplifiers tie to the supply voltages at the regulator (right at the sensing leads), so *IR* drops along the leads to the power stage don't appear on the low-level amplifier's supply voltages. In addition, the load current returning to ground does not appear at the low-level input; no current flows from the ground side of the low-level amplifier's input to the circuit mecca (which might be the connection to the case near the BNC input connector).

In the second circuit there are two blunders. Supply-voltage fluctuations caused by load currents at the high-level stage are impressed on the low-level supply voltages. Unless the input stage has very good supply rejection, this can lead to oscillations. Even worse, the load current

[147] One cure is to include some aluminum electrolytic capacitors – their lossy series resistance damps the resonance.

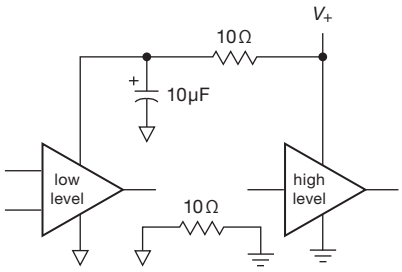


Figure 8.125. Decoupling the dc rail supplying low-level stages.

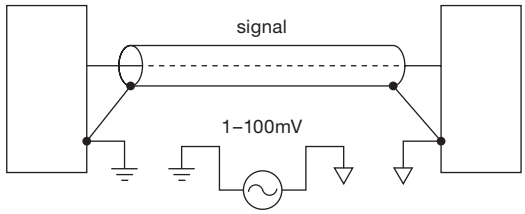


Figure 8.126. When connecting signals between instruments, you may see 100 mV (or more) differences (at powerline frequency) between their local grounds, even when they are bridged as shown.

returning to the supply makes the case "ground" fluctuate with respect to power-supply ground. The input stage ties to this fluctuating ground, a very bad idea. The general idea is to look at where the large signal currents are flowing and make sure their *IR* drops don't wind up at the input. In some cases it may be a good idea to decouple the supply voltages to the low-level stages with a small *RC* network (Figure 8.125). In stubborn cases of supply coupling it may pay to put a zener or 3-terminal regulator on the low-level-stage supply for additional decoupling.

8.16.3 Grounding between instruments

The idea of a controlled ground point within one instrument is fine, but what do you do when a signal has to go from one instrument to another, each with its own idea of "ground"? Some suggestions follow (and see also the extensive treatment of the transmission of *digital* signals in §12.9).

A. High-level signals

If the signals are several volts, or large logic swings, just tie things together and forget about it (Figure 8.126). The voltage source shown between the two grounds represents the variations in local grounds you'll find on different powerline outlets in the same room or (worse) in different rooms or buildings. It consists of some 60 Hz voltage, harmonics of the line frequency, some radiofrequency signals (the powerline makes a good antenna), and assorted spikes and other garbage. If your signals are large enough, you can live with this.

B. Small signals and long wires

For small signals this situation is intolerable, and you will have to go to some effort to remedy the situation. Figure 8.127 shows some ideas. In the first circuit, a coaxial shielded cable is tied to the case and circuit ground at the driving end, but it is kept isolated from the case at the receiving end (use a Bendix 4890-1 or Amphenol 31-010 insulated BNC connector). A differential amplifier is used to buffer the input signal, thus ignoring the small amount of "ground signal" appearing on the shield. A small resistor and bypass capacitor to ground is a good idea to limit ground swing and prevent damage to the input stage. The alternative receiver circuit in Figure 8.127A shows the use of a "pseudodifferential" input connection for a single-ended amplifier stage (which might, for example, be a standard noninverting op-amp connection, as indicated). The 10 Ω resistor between amplifier common and circuit ground is large enough to let the signal source's reference ground set the potential at that point, because it is much larger than the impedance of the source's ground. Any noise present at that node, of course, appears also at the output. However, this becomes unimportant if the stage has sufficiently high voltage gain, G_V, because the ratio of desired signal-to-ground noise is reduced by G_V. Thus, although this circuit isn't truly differential (with infinite CMRR), it works well enough (with effective CMRR $= G_V$). This pseudo-differential ground-sensing trick can be used also for low-level signals *within* an instrument, when ground noise is a problem.

In the second circuit (Figure 8.127B), a shielded twisted pair is used, with the shield connected to the case at both ends. Because no signal travels on the shield, this is harmless. A differential amplifier is used as before on the receiving end. If logic signals are being transmitted, it is a good idea to send a differential signal (the signal and its inverted form), as indicated. Ordinary differential amplifiers can be used as input stages, or, if the ground interference is severe, special "isolated amplifiers" are available from manufacturers like Analog Devices, Inc. and TI/Burr–Brown. The latter permit kilovolts of common-mode signals. So do optoisolator modules (§12.7), a handy solution for digital signals in some situations.

At radio frequencies, transformer coupling offers a convenient way of removing common-mode signal at the

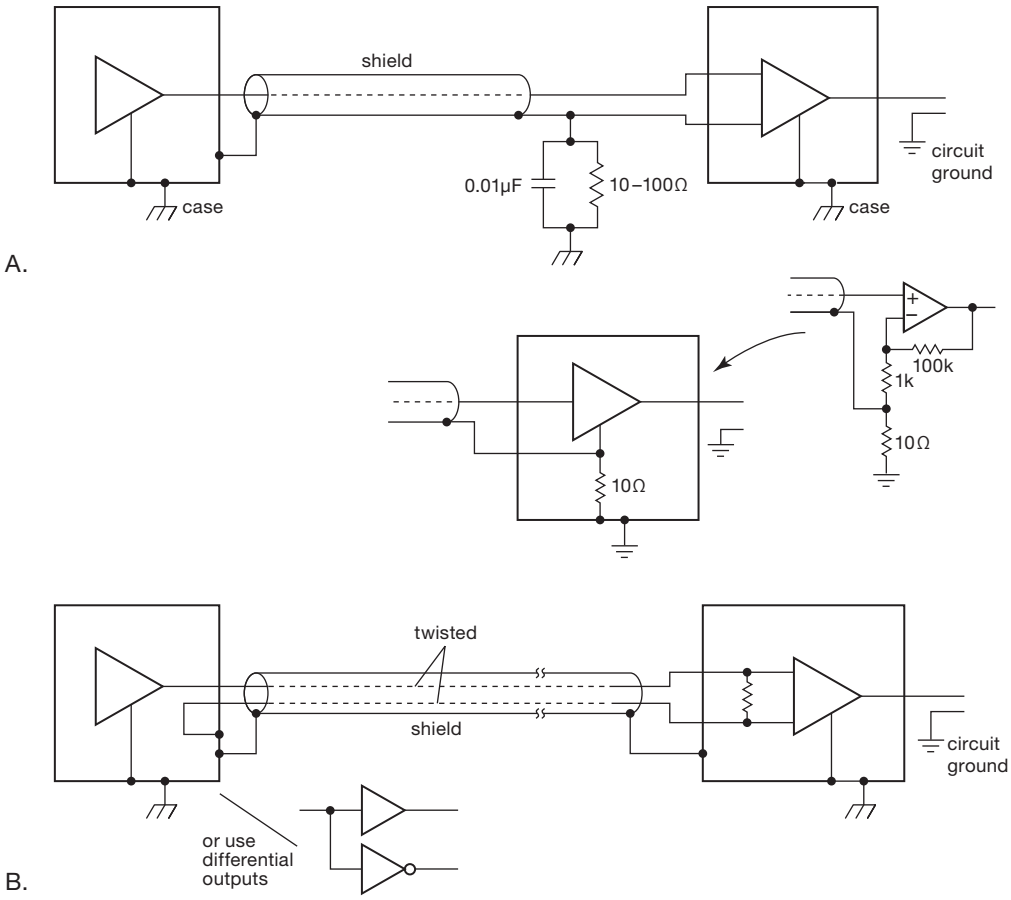


Figure 8.127. Ground connections for low-level signals through shielded cables.

receiving end; this also makes it easy to generate a differential bipolarity signal at the driving end. Transformers are popular in audio applications as well, although they tend to be bulky and lead to some signal degradation, as described in §8.10.

For very long cable runs (measured in miles) it is useful to prevent large ground currents flowing in the shield at radio frequencies. Figure 8.128 suggests a method. As before, a differential amplifier looks at the twisted pair, ignoring the voltage on the shield. By tying the shield to the case through a small inductor, the dc voltage is kept small while preventing large radiofrequency currents. This circuit also shows protection circuitry to prevent common-mode excursions beyond ± 10 V.

Figure 8.129 shows a nice scheme to save wires in a multiwire cable in which the common-mode pickup has to be eliminated. Because the signals all suffer the same common-mode pickup, a single wire tied to ground at the sending end serves to cancel the common-mode signals on each of the n signal lines. Just buffer its signal (with respect to ground at the receiving end) and use it as the comparison input for each of n differential amplifiers looking at the other signal lines.

The preceding schemes work well to eliminate common-mode interference at low to moderate frequencies, but they can be ineffective against radiofrequency interference, owing to poor common-mode rejection in the receiving differential amplifier.

One possibility here is to wrap the whole cable around a ferrite toroid (Figure 8.130). That increases the series inductance of the whole cable, raising the impedance to common-mode signals of high frequency and making it easy to bypass them at the far end with a pair of small bypass capacitors to ground. The equivalent circuit shows why this works without attenuating the differential signal: you have a series inductance inserted into both signal lines

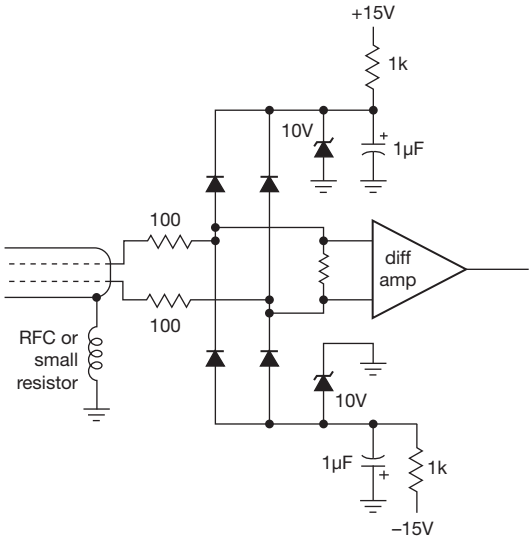


Figure 8.128. Input-protection circuits for use with very long lines.

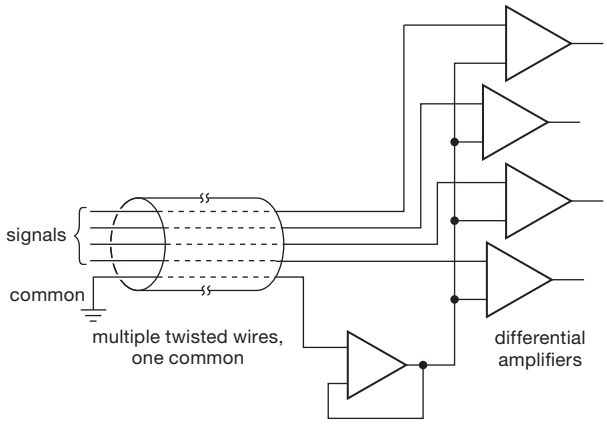


Figure 8.129. Common-mode interference rejection with long multiwire cables.

and the shield, but they form a tightly coupled transformer of unit turns ratio, so the differential signal is unaffected. This can be thought of as a "1:1 transmission-line transformer" (see §13.10 in the second edition of this book).

C. Floating signal sources

The same sort of disagreement about the voltage of "ground" at separated locations enters in an even more serious way at low-level inputs, just because the signals are so small. An example is a magnetic tape head or other signal transducer that requires a shielded signal line. If you ground the shield at both ends, differences in ground potential will appear as signal at the amplifier input. The best

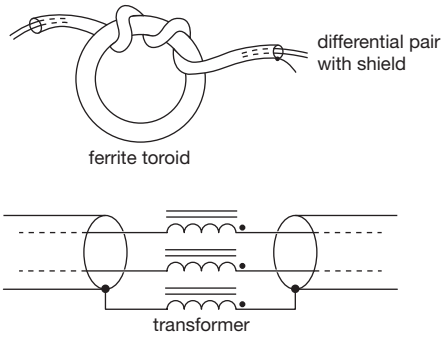


Figure 8.130. Wrapping a coaxial or multi-wire cable around a ferrite core for high-frequency common-mode suppression.

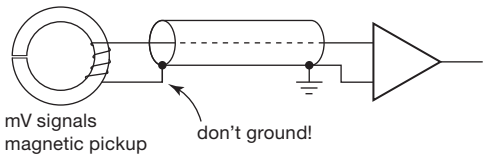


Figure 8.131. Preventing sneak ground currents: tie the shield to ground at the receiving end only.

approach is to lift the shield off ground *at the transducer* (Figure 8.131).

D. Isolation amplifiers

Another solution to serious ground-contention problems is the use of an "isolation amplifier." Isolation amplifiers (isoamps) are commercial devices intended for coupling an analog signal (with bandwidth clear down to dc) from a circuit with one ground reference to another circuit with a completely different ground (Figure 8.132). In fact, in some bizarre situations the "grounds" can differ by many

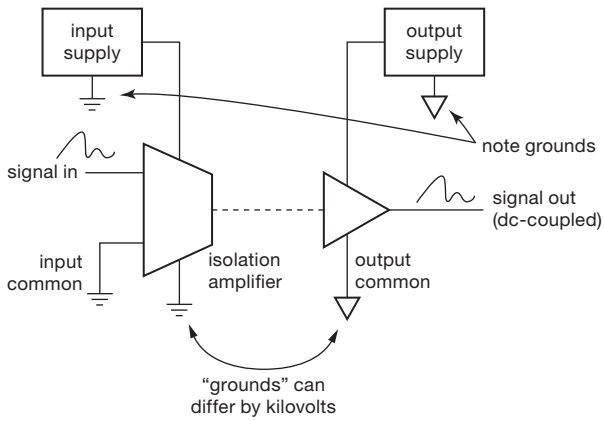


Figure 8.132. Isolation amplifier concept.

kilovolts! Isolation amplifiers are mandatory for medical electronics in which electrodes are applied to human subjects, in order to isolate completely those connections from any instrument circuits powered directly from the ac powerlines. Currently available isolation amplifiers use one of three methods.

1. Transformer isolation of a high-frequency carrier signal, which is either frequency modulated or pulse-width modulated with the relatively low bandwidth signal (dc to 10 kHz or so) to be isolated (Figure 8.133). This method is used in isolation amplifiers from Analog Devices, Inc. Transformer-isolated iso-amps have the convenient feature of requiring dc power only on one side; they all include a transformer-coupled dc-to-dc converter in the package. Transformer-coupled iso-amps can isolate up to 1.5 kV and have typical bandwidths of 5 kHz, though some units (e.g., the AD215) go to 120 kHz. The latter has 0.015% maximum nonlinearity, and costs about $80 in single-piece quantities.
2. Optically coupled signal transmission via an LED at the sending end and photodiode at the receiving end. This technique is typified by the ACPL-C79 series from Avago. These devices use a delta–sigma modulation and demodulation scheme, and achieve a bandwidth from dc to 200 kHz. This series is inexpensive (about $5 in modest quantity), with maximum nonlinearity of 0.06%. An interesting alternative, which involves no clocking signal or carrier frequency, is to couple an LED at the sending end to a photodiode at the receiving end. To achieve good linearity you use feedback from a second matched photodiode at the transmitting side that receives light from the same LED, thus cancelling nonlinearities in both LED and photodiode (Figure 8.134; see also Figure 12.90 in Chapter 12). Some additional analog optoisolation techniques are discussed in §12.7.4, and related digital isolated couplers in §12.7.2.

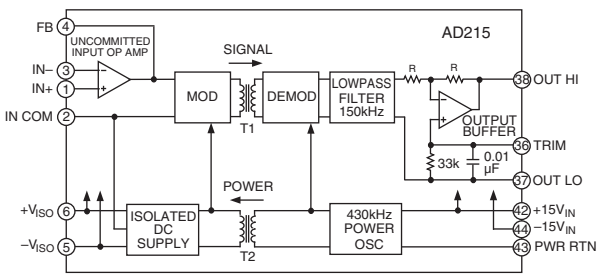


Figure 8.133. AD215 transformer-coupled isolation amplifier. (Adapted from the AD215 datasheet, courtesy Analog Devices, Inc.)

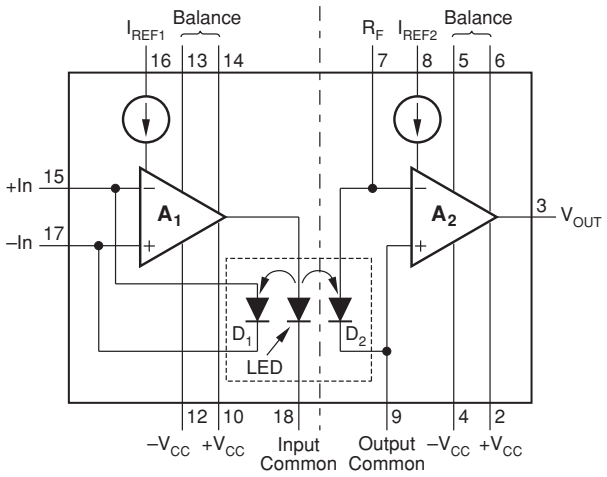


Figure 8.134. Unmodulated optical coupling eliminates clocking noise, with feedback from a matched detector to achieve reasonable linearity. (Adapted from the ISO100 datasheet, courtesy of Texas Instruments.)

3. Capacitively coupled isolation of a high-frequency carrier signal, which is modulated with the signal to be isolated (Figure 8.135). This technique is typified by the ISO122 and ISO124 from TI/Burr–Brown (Figure 8.136). There is no feedback, as with transformer isolation, but for most models you need power supplies at both ends. This usually isn't a problem, as you are likely to have electronics at both ends, generating and using the signal. If not, you can get an isolated dc–dc converter to use with the iso-amp. The ISO124 isolates to 1.5 kV, with 0.01% maximum nonlinearity and

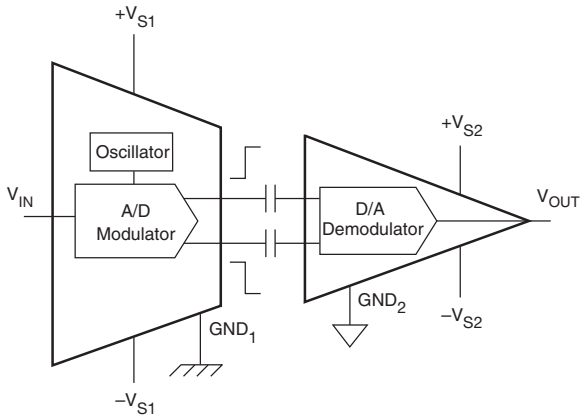


Figure 8.135. Capacitively-coupled isolation amplifier. (Adapted from the Burr–Brown Application Bulletin AB-047, courtesy of Texas Instruments.)

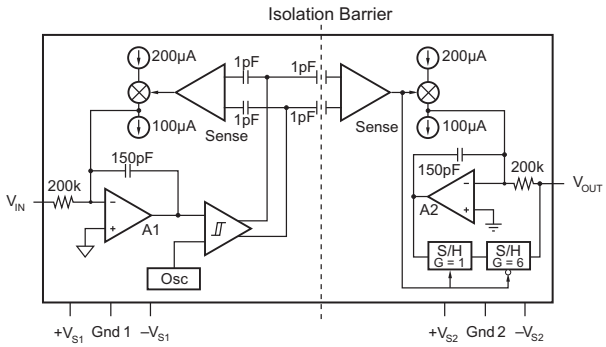


Figure 8.136. ISO124 capacitively-coupled isolation amplifier. (Adapted from the ISO124 datasheet, courtesy of Texas Instruments.)

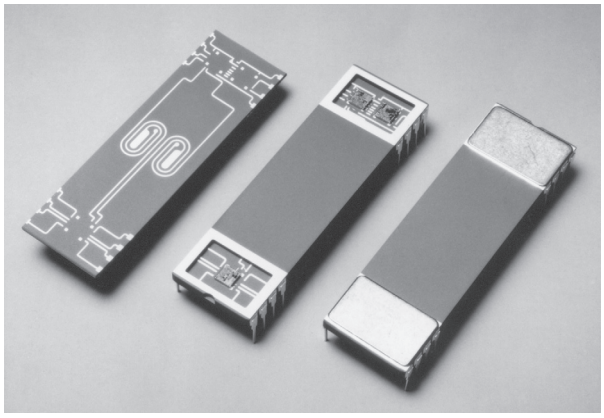


Figure 8.137. Burr–Brown (TI) ISO106 isolation amplifier, courtesy of Texas Instruments.

50 kHz bandwidth; it costs $18 in single-piece quantities. Figure 8.137 shows nicely the innards of one of these capacitively coupled devices.

These isolation amplifiers are all intended for *analog* signals. The same sorts of ground problems can arise in digital electronics, where the solution is simple and effective: optically coupled isolators ("optoisolators") are available, with plenty of bandwidth (10 MHz or more), isolation of several kilovolts, and low cost (a dollar or two). They are discussed extensively in Chapter 12.

A caution: isolation amplifiers can introduce noise of their own, particularly those that use some form of signal modulation (which is most of them!). For the latter you've got a residue of the modulating clock frequency; and all isolation amplifiers (whether clocked or not) introduce broadband noise of the usual sort. One remedy for clock noise is to add an analog lowpass filter at the output of the receiving side. For more detail see the helpful application document "Noise Sources in Applications Using Capacitive Coupled Isolated Amplifier" (Application Bulletin SBOA028, available at the Texas Instruments website, www.ti.com).

E. Signal guarding

A closely related issue is signal *guarding*, an elegant technique to reduce the effects of input capacitance and leakage for small signals at high impedance levels. You may be dealing with signals from a micro-electrode or a capacitive transducer, with source impedances of hundreds of megohms. Just a few picofarads input capacitance can form a lowpass filter, with rolloffs beginning at a few hertz! In addition, the effects of insulation resistance in the connecting cables can easily degrade the performance of an ultra-low input current amplifier (bias currents less than

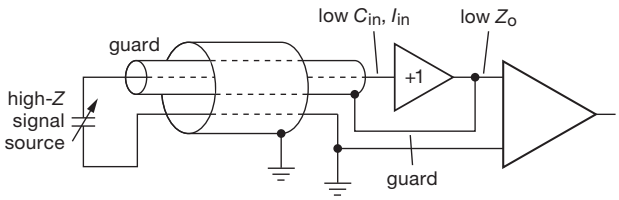


Figure 8.138. Using a guard to raise input impedance.

a picoamp) by orders of magnitude. The solution to both these problems is a *guard electrode* (Figure 8.138).[148]

A follower bootstraps the inner shield, effectively eliminating leakage current and capacitive attenuation by keeping zero voltage difference between the signal and its surroundings. An outer grounded shield is a good idea, to keep interference off the guard electrode; the follower has no trouble driving that capacitance and leakage, of course, given its low output impedance.

You shouldn't use these tricks more than you need to; it would be a good idea to put the follower as close to the signal source as possible, guarding only the short section of cable connecting them. Ordinary shielded cable can then carry the low-impedance output signal out to the remote amplifier.[149]

F. Coupling to outputs

Ordinarily the output impedance of an op-amp is low enough that you don't have to worry about capacitive signal coupling. In the case of high-frequency or fast-switching

[148] We saw this earlier, in §8.11.9.

[149] Discussed in more detail, in connection with high-impedance microelectrodes, in §15.08 of the second edition of this book.

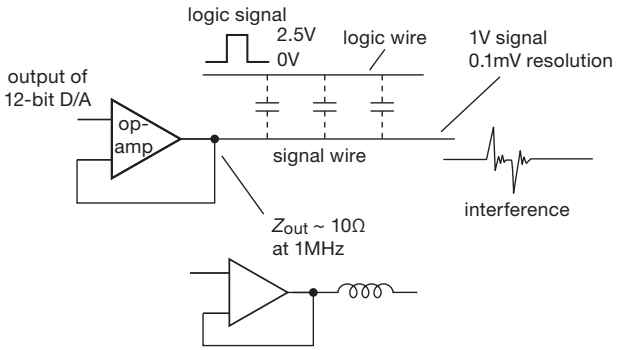


Figure 8.139. Digital cross-coupling interference with linear signals.

interference, however, you have just cause for alarm, particularly if the desired output signal involves some degree of precision. Consider the example in Figure 8.139. A precision signal is buffered by an op-amp and passes through a region containing digital logic signals jumping around with slew rates of 0.5 V/ns. The op-amp's closed-loop output impedance rises with frequency, typically reaching values of 10 to 100 Ω or more at 1 MHz (see §5.8.3). How large a coupling capacitance is permissible, to keep coupled interference less than the analog signal's resolution of 0.1 mV? The surprising answer is a maximum of 0.02 pF.

There are some solutions. The best thing is to keep your small analog waveforms out of the reach of fast-switching signals. A moderate bypass capacitor across the op-amp's output (with perhaps a small series resistor, to maintain op-amp stability) will help, although it degrades the slew rate. You can think of the action of this capacitor as lowering the frequency of the coupled charge bundles to the point where the op-amp's feedback can swallow them. A few hundred picofarads to ground will adequately stiffen the analog signal at high frequencies (think of it as a capacitive voltage divider). Another possibility is to use a low-impedance buffer such as the LT1010, or a power op-amp such as the LM675. Don't neglect the opportunity to use shielding, twisted pairs, and proximity to ground planes to reduce coupling.

Additional Exercises for Chapter 8

Exercise 8.10. Prove that SNR $= 10\log_{10}(v_s^2/4kTR_s) -$ NF(dB) (at R_s).

Exercise 8.11. A 10 μV (rms) sinewave at 100 Hz is in series with a 1M resistor at room temperature. What is the SNR of the resultant signal (a) in a 10 Hz band centered at 100 Hz, and (b) in a 1 MHz band going from dc to 1 MHz?

Exercise 8.12. A transistor amplifier using a 2N5087 is operated at 100μA collector current and is driven by a signal source of impedance 2000 Ω. (a) Find the noise figure at 100 Hz, 1 kHz, and 10 kHz. (b) Find the SNR (at each of listed frequencies) for an input signal of 50 nV (rms) and an amplifier bandwidth of 10 Hz.

Exercise 8.13. Measurements are made on a commercial amplifier (with $Z_{in} = 1$M) in order to determine its equivalent input noise e_n and i_n at 1 kHz. The amplifier's output is passed through a sharp-skirted filter of bandwidth 100 Hz: a 10 μV input signal results in a 0.1 V output. At this level the amplifier's noise contribution is negligible. With the input shorted, the noise output is 0.4 mV rms. With the input open, the noise output rises to 50 mV rms. (a) Find e_n and i_n for this amplifier at 1 kHz. (b) Find the noise figure of this amplifier at 1 kHz for source resistances of 100 Ω, 10k, and 100k.

Exercise 8.14. Noise measurements are made on an amplifier using a calibrated noise source whose output impedance is 50 Ω. The generator output must be raised to $2\,\text{nV}/\sqrt{\text{Hz}}$ in order to double the output noise power of the amplifier. What is the amplifier's noise figure for a source impedance of 50 Ω?

Exercise 8.15. Your boss tells you she's working on a super-sensitive pressure-change instrument that uses a diaphragm-type pressure sensor with a 350 Ω strain-gage bridge. She tells you there's a 10 Hz measurement bandwidth, either from a filter or an integrating ADC, and she asks you to select an op-amp for the input stage. First consider auto-zero amplifiers (see §5.11 Table 5.6, and Figure 5.54). (a) Is a bipolar op-amp better than a JFET op-amp? Hint, don't forget i_n. (b) The bridge has a standard 2 mV/V full-scale output, and your boss is planning a 2.5 V excitation voltage. What will be your system's noise floor as a fraction of full scale? (c) Should you suggest to your boss that she consider running the ADC faster and averaging the results? (d) A fellow engineer (who suffers the indignities of the same boss) suggests that a silicon strain-gauge pressure sensor might be better, because they have higher output, 2.5 mV/V. You find sensors with $R_s = 1.4$ kΩ to 3 kΩ. Choose a good op-amp for this case, and calculate the system performance.

Exercise 8.16. Digital oscilloscope input stage (*this is a difficult problem!*). Your assignment is to design an input stage for a low-cost "oscilloscope app," to be powered by a single +3.3 V supply from the mobile device's lithium-ion battery. The 'scope's architecture calls for attenuating its probe signals to 1 mV per division via a range switch, then amplifying and digitizing this signal, with a goal of peak-to-peak noise no more than 5% of a division. The screen shows ten vertical divisions, and you need to be able to shift it vertically through ±5× over-range, plus there's a 40 mV budget for "software offset." (a) What is the total input range, and what is your rms input noise level goal? (b) If a 14-bit ADC is used, what is the LSB resolution?

An ADS7946 14-bit ADC has been selected; it draws 0.5 mA from +3.3V when converting at 100 kSPS and meets cost goals at

$6 in qty 1k. (c) What is a reasonable ADC input voltage range, and how much gain will you need to provide?

The ADC is capable of 2 Msps, for a 1 MHz bandwidth, but there are plans to use a faster ADC in a later revision; hence your design-spec bandwidth is 10 MHz. (d) What is your spectral noise-density goal?

Standard oscilloscopes are dc coupled, have a 1 MΩ||15 pF input impedance, and do not exhibit visible dc shifts with source impedance. (e) Create a design that meets all your specs. Low power consumption is a big plus.

Review of Chapter 8

An A-to-Q summary of what we have learned in Chapter 8. This summary reviews basic principles and facts in Chapter 8, but it does not cover application circuit diagrams and practical engineering advice presented there.

¶A. Noise Basics.

See §8.1. The random noise you care about is characterized by its *density* (rms noise amplitude in a 1 Hz band of frequency), §8.2.1. Noise *voltage* density is called e_n, and has units like $\text{nV}/\sqrt{\text{Hz}}$. Likewise, the symbol for noise *current* density is i_n, with units like $\text{fA}/\sqrt{\text{Hz}}$. A noise current at an amplifier's input flows through the signal's source resistance, creating its own noise voltage density $e_n = i_n R_s$. Independent noise sources combine as the square root of the sum of their squares: $e_{n(\text{total})} = \sqrt{e_{n1}^2 + e_{n2}^2 + \cdots}$. If a noise source is uniform over frequency, it's called "white noise," and the rms noise voltage (as contrasted with noise voltage *density*) contained within a bandwidth B is just $v_n = e_n\sqrt{B}$. Knowing that, you can go to Table 8.3a–c on page 522, which lists e_n and i_n for a wide selection of op-amps, to figure out how much noise is added in an op-amp amplifier stage. Compare this with the input signal's noise level; or multiply by the amplifier's voltage gain and, voilà, you've got the output noise voltage density.

¶B. Noise Spectra.

Whatever its source, noise density can vary with frequency, see §8.1. White noise (e_n constant over frequency, up to some cutoff) is common, for example a resistor's Johnson noise (¶E, below) or the shot noise fluctuations of a steady current (¶F, below). Also prevalent is "$1/f$ noise," sometimes called flicker noise, or pink noise; it is characterized by a $1/f$ *power* spectrum (equal power per octave, or per decade), thus a voltage noise density $e_n(f)$ proportional to $1/\sqrt{f}$. Most electronic circuits (and many other physical phenomena) exhibit $1/f$ noise, often characterized by the "$1/f$ corner frequency" at which the $1/f$ noise component is equal to the white noise component. Finally, the term *red noise* refers to a noise density e_n proportional to $1/f$ (thus a $1/f^2$ power spectrum); see for example Figure 8.95. The rms noise voltage v_n in some bandwidth B (extending from f_1 to f_2) is gotten by integrating $e_n^2(f)$ over frequency, then taking the square root: $v_n = \sqrt{v_n^2}$, where $v_n^2 = \int_{f_1}^{f_2} e_n(f)^2\, df$. For a white spectrum this reduces to the simple $v_n = e_n\sqrt{B}$.

Actual circuit noise need not conform to these idealized spectra, which, however, are useful in characterizing the noise of real devices over chosen regions of frequency; see for example Figure 8.107. In real life, spectra may show a noise "shelf" (e.g., curve Z in Figure 8.52), or a noise peak (e.g., Figures 5.52 or 8.72).

¶C. Noise Amplitude Distribution.

Quite apart from its spectrum, one can characterize the amplitude distribution of noise; that is, the distribution of instantaneous amplitudes sampled in time. Most noise sources obey a Gaussian distribution (Figure 8.2), a fact that is of lesser concern than the important properties of noise spectrum and amplitude. A notable exception is *burst* noise (also called *popcorn* noise, *bistable* noise, or *telegraph* noise), which jumps randomly between several voltage levels (Figure 8.5). Burst noise was prominent in the early decades of semiconductor technology, but it has been largely banished in contemporary products. One suspects some low-level remnants remain, though, as evidenced for example in a comparison of the measured burst-noise spectrum of Figure 8.6 with the measured JFET noise spectra of Figure 8.51.

¶D. Sources of Noise.

The dominant sources of noise in electronic circuits (detailed in paragraphs ¶E–H below) are:

Johnson noise. Thermal fluctuations generate a noise voltage in a resistor.

Shot Noise. The discrete nature of electric charge creates fluctuations in a "steady" current.

Excess Noise. Various semiconductor phenomena contribute additional noise (often $\sim 1/f$ in power density) at low frequencies.

Amplifier Noise. Transistors (both BJTs and FETs) add noise, traceable to the above sources (e.g., Johnson noise of the base resistance $r_{bb'}$, shot noise of the collector current, and excess noise terms).

¶E. Johnson Noise.

Thermal fluctuations cause a self-generated noise voltage across the terminals of an unloaded resistor; this is *Johnson noise*, of (white-) noise voltage density $e_n = \sqrt{4k_B TR}$, where k_B is Boltzmann's constant. Don't bother remembering the latter; just remember the value of e_n for a round-number resistance (we like $1.28\,\text{nV}/\sqrt{\text{Hz}}$ for $R = 100\,\Omega$, or $4\,\text{nV}/\sqrt{\text{Hz}}$ for $R = 1\text{k}\Omega$), and scale by the square root of R. If short-circuited, a resistor generates a Johnson noise current $i_n = e_n/R$; that is, $i_n = \sqrt{4k_B T/R}$. See the graph in Figure 8.1 and the mini-table in §8.1.1.

Johnson noise is a fundamental physical phenomenon, and does not depend on the particular construction of the

resistor (or resistance). However, when a steady current flows through a resistance, you may get some additional noise current (which you can think of as due to resistance fluctuations), generally with something approximating a $1/f$ power spectrum. This *excess noise* varies with resistor construction, being worst in a granular "carbon composition" type, but insignificant in a wirewound resistor.

¶F. Shot Noise.
The finiteness of the charge quantum (electron charge) causes statistical fluctuations even in a steady current. If the charges act independently, the (white-) noise current density is $i_n=\sqrt{2qI_{dc}}$, where q is the electron charge (1.6×10^{-19} coulombs). As with Johnson noise, it's handy to simply remember the value of i_n for a round-number dc current (e.g., $18\,pA/\sqrt{Hz}$ for $I_{dc}=1\,mA$), and scale by the square root of current.[150] See the mini-table in §8.1.2. *Important*: the shot-noise formula assumes that the charge carriers act independently; the current noise is greatly reduced if there are long-range correlations, as for example in a metallic conductor.

¶G. BJT Noise.
See §8.3. The primary noise terms of a BJT are an input noise voltage e_n in series with the base, combined with an input noise current i_n injected at the base (Figure 8.9). The amplifier's input-referred noise (i.e., ignoring the Johnson noise of the signal source's R_s) is then $e_{a(rms)}=[e_n^2+(R_s i_n)^2]^{\frac{1}{2}}$. As you increase the collector current, e_n decreases and i_n increases, so there's an operating-current tradeoff. The ratio e_n/i_n has units of resistance; it's called the *noise resistance*, and it's a useful quantity in circuit design; see ¶I below. Table 8.1a lists measured values of e_n for many low-noise BJT candidates.

Voltage noise, e_n. In the simplest model, the noise voltage term arises from two sources: collector shot-noise current flowing through the transistor's emitter resistance r_e, and Johnson noise in the transistor's internal base resistance $r_{bb'}$. Combining these independent noise terms (eq'n 8.20), we get the total input-referred squared noise voltage, $e_n^2=2qI_C r_e^2+4kTr_{bb'}=4kT(r_e/2+r_{bb'})$. Put another way, the input noise voltage is equal to the combined Johnson noise of the transistor's base resistance ($r_{bb'}$) and a fictional resistance equal to half its intrinsic emitter resistance ($r_e/2$). The latter is inversely proportional to collector current, so a BJT's noise voltage decreases with increasing I_C, ultimately limited by its internal base resistance; see for example Figure 8.12. Thus to minimize BJT *voltage* noise, choose a part with a low $r_{bb'}$, and operate at relatively high collector current. A more refined model includes the effects of base current shot noise, important at low frequencies and high collector currents; see Figure 8.11 and eq'n 8.24.

Current noise, i_n. The primary noise current term is shot noise in the dc base current, $i_n=\sqrt{2qI_B}$ (eq'n 8.21). Taking this term alone, you minimize BJT *current* noise by operating at low collector current. A more refined model includes a rising $\propto 1/\sqrt{f}$ noise current term at low frequencies, and a rising $\propto f$ noise current term at high frequencies caused by falling beta; see eq'n 8.22 and Figure 8.13.

¶H. JFET Noise.
See §8.6. The low-noise choice for signals of high source impedance are FETs, owing to their very low input current noise; JFETs are quieter than MOSFETs, and, unlike the latter, they are available as discrete small-signal parts (see Table 8.2).

Voltage noise, e_n. The dominant noise voltage term is Johnson noise in the channel resistance (eq'n 8.34), $e_n^2\approx4kT(\frac{2}{3}\frac{1}{g_m})$. Put another way, the noise voltage is equivalent to Johnson noise in a resistor of value $R=\frac{2}{3}\frac{1}{g_m}$; see Figure 8.46. To minimize JFET voltage noise, choose a JFET of high transconductance, and operate it at a relatively high drain current (note, however, that e_n falls slowly, only as the fourth root of I_D). As with BJTs, JFETs exhibit a rising $1/f$-like noise tail at low frequencies (Figure 8.52), with enormous variation among types, and manufacturers.

Current noise, i_n. At low frequencies the current noise is low, just shot noise of the gate (leakage) current: $i_n=\sqrt{2qI_G}$, see Figure 8.48. To frame this in numbers, a typical gate leakage current of 10 pA has a noise current density i_n of just $1.8\,fA/\sqrt{Hz}$, generating only $1.8\,nV/\sqrt{Hz}$ of noise voltage through a 1MΩ source resistance. That's down at the low end of transistor noise voltage, and it's completely dwarfed by the $128\,nV/\sqrt{Hz}$ of Johnson noise produced by the source resistance itself. At rising frequencies there are some additional sources of input noise current. For example, in a transimpedance amplifier the gate input is a summing junction, at which the FET's *voltage* noise generates a noise current through

[150] Because both shot-noise current and Johnson-noise current go as $1/\sqrt{R}$, it's easy to derive this handy factoid: if the dc drop across a resistor is greater than 50 mV, shot noise dominates over the resistor's own Johnson noise.

the input capacitance, of magnitude $i_n = e_n \omega C_{in}$, see ¶N (below) and §8.11.

¶I. Noise Figure, Noise Temperature, and Noise Resistance.

See §8.2. Noise figure (NF) is a popular measure of added amplifier noise. It is the ratio (in dB) of the amplifier's output noise to the output of a noiseless amplifier of the same gain, each driven with a source resistance R_s: $NF = 10\log_{10}(1 + v_n^2/4kTR_s)$, where v_n^2 is the mean squared noise voltage per hertz contributed by the amplifier when a noiseless (cold) resistor of value R_s connected across its input (see eq'n 8.13 and Figure 8.7). Another (and equivalent) way to quantify the excess noise contributed by an amplifier when driven by a signal of source resistance R_s is to state its noise *temperature* (T_n, see Figure 8.8 and eq'n 8.16).

An amplifier's noise figure (and its noise temperature) is minimum when driven by a signal of source resistance equal to its noise resistance, i.e., when $R_s = R_n = e_n/i_n$. Transistor datasheets sometimes provide contours of NF versus operating current and source resistance at a given frequency (e.g., Figure 8.22); for amplifiers where you have no control over operating current you'll find instead contours of NF versus frequency and source resistance (e.g., Figure 8.27). *Caution*: do not make the mistake of adding a series resistor to a signal of low source resistance to improve the noise figure (§8.4.6). Instead choose an amplifier that provides the noise figure you need with that input source resistance; in some situations (e.g., very low source resistance) you can use a signal transformer to make a lossless match to the amplifier's optimum source resistance.

¶J. Noise Sources and Measurements.

See §8.12. At low to moderate frequencies you can determine the noise properties of a transistor amplifier of known voltage gain by making two measurements of its rms output noise voltage in a known bandwidth, first with shorted input (to get e_n) and then with an appropriately chosen resistor across its input (to get i_n); see §8.12.1. Because you're measuring the integrated noise voltage, you need to know the equivalent noise bandwidth; see §8.13. A more general technique, applicable to amplifiers that require a matched source impedance (e.g., 50 Ω RF amplifiers) and insensitive to actual measurement bandwidth, is to drive the input with a calibrated noise source while observing the rms noise output; see §8.12.3. For frequencies up to a few tens of megahertz you can make your own "pseudorandom" noise source with a feedback shift register (§8.12.4A), or you can use an off-the-shelf noise source (a function generator, or noise diode; see §8.12.4) for frequencies well up into the gigahertz range. Resistor Johnson noise itself is the noise source for the "hot-load/cold-load" method, useful for low-noise microwave amplifiers; see eq'n 8.53 in §8.12.3.

¶K. Low-noise Design with Op-amps.

See §8.9. Just as with BJTs and FETs, op-amps exhibit input-referred voltage noise e_n and current noise i_n, whose magnitudes are characteristic of their input-stage transistor type. What's different, from a circuit designer's point of view, is that you don't have control over the stage's operating current. Instead you choose the op-amp type: FET-input for lowest i_n (for signals of high source impedance), BJT-input for lowest e_n (for signals of low source impedance). Among the FET-input types, the CMOS parts exhibit lower i_n but considerably higher e_n, compared with JFET types; the latter thus combine the best (nearly) of both worlds, see §8.9 the extensive Table 8.3 on page 522ff, and the noise plots in Figures 8.60, 8.61, and 8.63. A helpful graphical tool is the plot of *effective noise density* (v_n) versus source resistance, in which the op-amp's noise contributions (e_n, $i_n R_s$) are combined and plotted along with the source's Johnson noise, the latter setting a lower noise bound; i.e., $v_n^2 = 4kTR_s + e_n^2 + (i_n R_s)^2$, see Figure 8.58. Op-amp noise (with the exception of auto-zero op-amps) exhibits a typical rising $1/f$ low-frequency tail, characterized by a *corner frequency*, f_c. Auto-zero op-amps have no $1/f$ tail, but they have considerably higher broadband noise (both e_n and i_n), along with spectral noise peaks and chopping artifacts. Be sure to evaluate all noise sources in an op-amp design: a poor choice of component values may compromise the noise performance (e.g., Johnson noise from high-value resistors).

When choosing an op-amp for a low-noise application, begin by restricting your attention to op-amps that meet your other needs, such as accuracy, speed, power dissipation, supply voltage, input and output swing, and the like. Then choose among this subset, based upon their noise parameters.

¶L. Low-noise Design with BJTs.

See §8.5. Compared with op-amps, circuit design with discrete transistors gives you more control over noise parameters, but the price you pay is the additional labor associated with details of biasing and the like. A nice compromise is a *hybrid* approach, with a discrete front-end prepended to an op-amp, see §8.9.5. As with op-amps, a plot of total input noise versus source resistance is a useful graphical tool, see Figure 8.32 in §8.5.2, and Figure 8.41. The flexibility you

get from control of operating current is illustrated nicely in Figure 8.33, where the curves shift downward with increasing collector current (lower e_n, higher i_n). Table 8.1a on page 501 lists datasheet and measured noise parameters for an extensive selection of low-noise BJTs; see also measured curves of current gain in Figure 8.39.

¶M. Low-noise Design with FETs.
See §8.6. You can't beat FETs for lowest input current noise, and for *discrete* amplifier circuits (or hybrid front-ends) you are limited to JFETs. In terms of device parameters, JFETs with large geometry have lowest e_n and highest g_m, but the large geometry means higher capacitance and leakage current (thus higher i_n), see Table 8.2 on page 516. If the JFET's input is a summing junction, the product of noise voltage and input capacitance creates a rising noise current $i_n = e_n \omega C_{in}$ (¶H, ¶N). As with BJTs, raising the operating current reduces a JFET's noise voltage, though not dramatically ($e_n \propto I_D^{-0.25}$). Watch out for some chronic problems with many JFETs: loose I_{DSS} and $V_{GS(th)}$ specs (a 5:1 range is typical), low transconductance, low output resistance, sharply rising gate current (thus i_n) at high temperatures and at elevated drain voltages (see §3.2.8), and poor low-frequency noise performance (see Figure 8.52).

¶N. Noise in Transimpedance Amplifiers.
See §8.11. Transimpedance amplifiers (TIA) convert an input current to an output voltage, with current feedback through a resistor R_f to the input summing junction, as introduced in §4.3.1, and elaborated in §*4x.3*. The noise sources are the amplifier's input i_n and e_n, and Johnson noise in R_f. Taking these in reverse order, the feedback resistor generates a Johnson noise *current* of $i_n = \sqrt{4kT/R_f}$; the amplifier's voltage noise e_n generates through the capacitance at the input a noise current of magnitude $e_n \omega C_{in}$, and through the feedback resistor a noise current of magnitude e_n/R_f; to these must be added the amplifier's own input noise current i_n. The choice of R_f is a compromise: the noise terms are minimized by choosing a high resistance, but, because of feedback stability requirements, the bandwidth suffers; see §§8.11.1, 8.11.4–8.11.6, and *4x.3*. These competing constraints can be mitigated with more complex configurations, used singly or in combination: a composite (2-stage) TIA (§8.11.8); or the use of a discrete first stage (i.e., a hybrid, §8.11.8A); or reduced capacitance by use of bootstrapping (§8.11.9) or a cascode connection (§8.11.10). The noise contribution of the feedback resistor can be eliminated entirely by replacing it with a *capacitor* (§8.11.11), forming an integrator; differentiating the output then recovers an output proportional to input current. Capacitor feedback is used in low-level detectors, and in the technique known as *correlated double-sampling*.

¶O. Lock-in Amplifiers.
See §8.14. For the detection of slowly-varying signals it is desirable to minimize added noise by narrowing the measurement bandwidth; and because of typical $1/f$ rising noise at low frequencies, amplification of signals should take place at frequencies above the $1/f$ corner frequency. In the elegant technique of *lock-in detection*, the slowly-varying signal is modulated (typically at a few hundred hertz), amplified, then demodulated with a phase detector, and finally lowpass filtered with a bandwidth appropriate to the original signal; see Figure 8.117.

¶P. Power-supply Noise.
See §8.15. All power supplies are not created equal, as stunningly illustrated by the measured spectra of Figure 8.123. When designing a dc supply, be sure to use a quiet voltage reference, and adhere to good low-noise design practice. Switchmode power supplies are inherently noisy, but attention to current paths and the use of filters can reduce conducted and radiated noise levels. With power supplies of any design, the addition of an outboard *capacitance multiplier* can reduce noise by 40 dB or more (Figure 8.122). See also the extensive treatment of dc power regulation in Chapter 9.

¶Q. Shielding and Grounding.
See §8.16. To paraphrase a cliché,[151] "One man's signal is another man's noise." In the real world, unintentionally coupled signals can easily overwhelm the subtle forms of noise that dominate this chapter. Mitigation techniques include wiring and trace layout (to minimize both capacitive and inductive coupling), raised signal levels, careful attention to grounding (both within and between instruments), shielding, filtering, balanced lines, common-mode chokes, guarding, signal transformers, and isolation amplifiers.

[151] "One man's trash is another man's treasure," perhaps deriving from Lucretius' 1st-century BCE epic poem *De Rerum Natura*, where he writes "Ut quod ali cibus est aliis fuat acre venenum" (What is food to one, is to others bitter poison).